Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator

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November 13, 2026

Paper release: r2067 **Released:** November 13, 2026

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Abstract

Observer Patch Holography describes particles through complete preparations, transport and detector records. On a separately declared gauge carrier, a protected flux register has an exact dictionary to Standard Model centre classes; individual species require additional observables. A declared generation analyzer at phase 2/9 gives charged-lepton ratios 206.7703 and 3477.47, realized as free Dirac gaps with one electron calibration. A positive native-current action fits nucleon, charged-pion, neutral-vector and Delta-assigned energies, the axial coefficient and deuteron binding with seven empirical inputs. The weak current is unchanged. A common semileptonic source contains three charged and three neutral Dirac branches, an exact electron threshold and certified nucleon energy windows. A separate complete neutral-current apparatus realizes calibrated oscillation laws with total-variation error below 4.1×10^{-9}. Native outgoing beta records have a certified parity-odd momentum contrast; a meson memory equation gives a controlled positive infinite-line emission probability. Full occupations, failures, quantum references and apparatus work accompany these results. Nuclear, spin and graph controls expose failed neutral-pion and charge-radius comparisons, while the complete nuclear census excludes extra bound channels on the seven-input source. Exact finite identities and disclosed calibrations establish consequences of each source; they do not identify a complete observed particle spectrum or physical decay lifetimes.

1 The particle question

Observer Patch Holography (OPH) proposes a self-referential complete history: observers infer the specification of their world from shared records and realize it through operations within that same history. The executable version specifies a carrier Hilbert space, accessible algebras, repair instruments, a scheduler, memory, feedback and preparation services [4, 8]. These are permitted design choices. Operational agreement concerns accepted shared data; state selection and endogenous implementation of response transformations are additional stated principles, recorded with the exact theorem premises in the common registry [35].

The source specification is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \{\Phi_{e,r}\}, \sigma, \mathcal{M}, \mathcal{F}, \mathcal{P}),$$

where $\Phi_{e,r}$ is the operation on seam e with retained outcome r, σ selects operations, $\mathcal{M}$ is memory, $\mathcal{F}$ is feedback, and $\mathcal{P}$ is the preparation domain. A particle assertion concerns the joint and conditional laws of actual preparations, interventions and detector outputs of $\mathcal{S}$. Two equal spectra do not identify the states, currents, clocks or instruments that produced them, and Section 9 proves this in several exact forms.

Hypothesis 1.1 (Matter as localized defects)**.** *Stable massive matter admits a localized excitation of a continuing source reference, with an invariant preserved by the admitted dynamics. Its creation, transport, interactions and detection are operations of the same source. Its total excess energy and propagation use one energy and clock dictionary. Unstable species may be metastable or internal excitations; massless radiation need not carry a defect invariant.*

The hypothesis does not ask for a distinct conserved topological charge on every mass eigenstate. Flavor mixing and decay can connect internal states of a defect family, and a protected label cannot change at an admitted vertex without another field or record carrying the compensating charge. Section 4.5 gives the exact constraint.

1.1 Main results

Table 1 states what this paper establishes. Three kinds of statement occur and we keep them apart. Some results follow from the unchanged source with every outcome retained. Some are exact on a chosen design, whose inputs are stated. Some compare a specified branch with measurement, and these are conditional postdictions.

The finite code and flux classes have an exact dictionary, which does not decode arbitrary defects into individual physical species. The specified generation register supplies three coefficients; its free-field compiler and electron anchor give conditional muon and tau values. Section 10 states the additional attachments required for a physical spectrum.

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Table 1: Particle results, their source domains and calibration inputs.

Result	**Inputs and boundary**
Flux-class dictionary $6Y$ mod 6, spin lift and conserved reactions (Section 2)	Declared code and lift; centre classes do not identify species
Charged-lepton ratios 206.7703, 3477.47 and exactly three charged-lepton band minima (Section 2.6)	Declared analyzer, phase 2/9 and finite-range regulator; electron anchor for GeV
Proton, neutron, charged-pion, deuteron, neutral-vector and Delta-assigned energies with native $\|g_A\| = 1.2753$	Seven empirical inputs on one positive hyperfine action; complete spectra and failures
Complete weak-dressed lines and three-dimensional beta angular records (Section 6.8)	Actual source transfer; point-contact coefficients and full-cube error bounds, without a physical neutron lifetime
Massive weak survival intervals, $\Gamma_\mu = 3.00876 \times 10^{-19}$ GeV and tau leptonic ratio 0.972563836	Calibrated weak normalization; unused tau-ratio comparison; positive-frequency source
Neutral masses $(0.001, 0.00871206, 0.0501398)$ eV and complete vacuum/matter records (Section 6.9)	Six fitted oscillation inputs, chosen lightest mass; explicit forward-current medium
Common positive matter source, longitudinal Gauss law, electron threshold and moving records (Section 6.9)	Explicit reference and self-energy convention; compact transverse extension has controlled rotor tails, without a common light/matter continuum
Strictly positive infinite-line meson emission probability with all feedback and finite-volume errors	Native current, resonance-selected box and source parameter; no irreversible rate
Complete nuclear spin/parity controls and one shared quadrupole/string length	No additional bound nuclear channel in the seven-input census; calibrated scale, failed neutral-pion and radius comparisons retained
Complete spectral instruments, localized currents and finite detector errors (Sections 5.3 and 7)	Specified interacting actions and preparation services
Exact population and reversible-bath records	Finite class populations and declared bath; no cosmic abundance identification

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2 Defect classes and particle representations

The finite runs of the executable observer carry a protected register on the faces of every carrier. A face holds a flux class $k \in \mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$, the product of a colour-centre charge and a weak-centre parity. Accepted register twists create conjugate pairs of classes, the pairs move, fuse and annihilate, and the classes on one closed carrier always add to zero modulo six. This section compares these classes with Standard Model representation labels, states the statistics under the declared quantum-code premises, and reads the native run census. A face-flux class supplies no particle decoder, generation, chirality or mass.

2.1 The class rule

The internal space of one quantum is the exterior algebra of the five-level code $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$ of Section 4. A pattern has $n_3 \in \{0, \dots, 3\}$ colour levels and $n_2 \in \{0, 1, 2\}$ weak levels occupied, so

there are 32 patterns, and

$$6Y = 3n_2 - 2n_3, \qquad Q_{\mathrm{em}} = T_3 + Y.$$

Theorem 2.1 (Centre classes of the declared code)**.** *On every one of the* 32 *patterns the centre element* $(\omega I_3, -I_2, e^{2\pi i Y})$ *of* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, $\omega = e^{2\pi i/3}$, *acts as the identity, and a flux of class* f *acts by the character* $e^{2\pi i f \cdot 6Y/6}$. *The class seen by a pattern is*

$$k \equiv 6Y \pmod 6, \qquad k \bmod 3 = n_3 \bmod 3 \ (\textit{triality}), \qquad k \bmod 2 = n_2 \bmod 2 \ (\textit{weak centre}).$$

The classes have multiplicities $(4, 6, 6, 4, 6, 6)$ *for* $k = 0, \ldots, 5$ *and carry the centre charges of the Standard Model representations in Table 2.*

Proof. On a pattern with n_3 colour and n_2 weak levels, ωI_3 acts by ω^{n_3}, $-I_2$ by $(-1)^{n_2}$ and $e^{2\pi i Y}$ by $e^{i\pi(3n_2-2n_3)/3} = (-1)^{n_2}\omega^{-n_3}$; the product is one. A flux of class f acts by $e^{2\pi i f \cdot 6Y/6}$, which depends on $6Y$ only modulo six. Reducing $6Y = 3n_2 - 2n_3$ modulo 3 and 2 gives the triality and the weak centre, and counting patterns gives the multiplicities. The executable certificate checks all 32 cases in exact arithmetic [1]. □

Table 2: Defect classes and field representations with those centre charges. The class is $6Y$ mod 6; the table distinguishes the chiral representation labels and includes their conjugates. A run class alone supplies no chirality or species assignment.

k	$\mathbb{Z}_3$	$\mathbb{Z}_2$	Field representations	Native class records
1	ω	-1	left-handed quarks $Q = (u_L, d_L)$	mobile; annihilates with 5
5	ω^2	-1	their antiquarks $\bar{Q}$	as class 1
4	ω	$+1$	right-handed quarks u_R, d_R	frozen after sweep 15
2	ω^2	$+1$	their antiquarks u^c, d^c	as class 4
3	1	-1	lepton doublet L, its conjugate, the Higgs doublet	frozen
0	1	$+1$	e_R, ν_R; photon, gluons, W, Z	no defect

The identification is exact at the level of centre classes. It decides which flux each particle sees: a quantum of class k taken around a flux of class f acquires the Aharonov–Bohm phase $e^{2\pi i f k/6}$. Colour flux ($f = 2, 4$) is seen only by quarks, weak flux ($f = 3$) only by the weak doublets, the left-handed fermions and the Higgs, and the generating fluxes $f = 1, 5$ by every class except zero. The electrically charged e_R sits in class zero and sees none of these fluxes. The class carries neither the generation nor T_3, so u and d, or e and ν, share a class, as do a colour singlet and an octet. These finer labels sit in the quantum's full pattern and in the generation register of Section 2.6.

2.2 Spin and statistics

The binary icosahedral cover of the carrier rotations has one central involution z. On the code it acts as

$$\rho(z) = \mathrm{diag}(1, 1, 1, -1, -1) = e^{i\pi \cdot 6Y},$$

which equals $(-1)^{n_2}$ on every pattern and the holonomy of a class-three flux. On a port space of flux k the same element acts as $(-1)^k$; the weak levels sit at odd flux, so the half-integer behind odd $6Y$ is the charge–monopole spin of the doublet levels. The construction supplies a spin doublet on each quantum, the stated port-flux sectors and antisymmetric occupations. On its declared Gauss-invariant sector the code, register and link-end factors of z multiply to one, and the rotation by 2π equals $(-1)^N$, with N the number of quanta.

Proposition 2.2 (Statistics of the objects)**.** *With the supplied spin-doublet, Gauss-sector and antisymmetric-occupation premises, exactly one grading compatible with the declared vertices agrees with the 2π rotation. Under it every pattern is a fermion and the defects, the gauge links and the Higgs register are bosons. Three quanta on one carrier have odd fermion parity.*

The vertex constraints of the source alone admit sixteen gradings; on the declared Gauss sector four survive, generated by the quantum-number parity and the coloured-quanta parity, and the rotation selects one. The fermions are therefore the quanta whose pattern carries the class, and the defect of that class is the flux the pattern sees. A carrier holding the pure flux content $\{4, 4, 4\}$ has the centre classes of three quarks and the statistics of a boson; a baryon is three fermionic quanta on the carrier. Antisymmetry of the occupations is forced under the filled-reference hypothesis; this package does not construct that reference, and the tick alone admits both occupation lifts.

2.3 Reactions and closed carriers

Every one of the 16 reaction types of the runs conserves the class modulo six, and the frequent ones carry the centre classes of the Standard Model vertices, with one diquark channel (Table 3). Because a carrier is a closed screen, its classes add to zero: a colour class never sits alone on a carrier.

Table 3: Reaction channels of the level-10 run and the Standard Model vertex with the same centre classes. Event counts are whole-run totals; the conjugate channel is in brackets.

Channel		Events		Vertex
$1 + 5 \to 0$		609,494		quark–antiquark annihilation
$1 + 3 \to 4$	$(3 + 5 \to 2)$	238,523	(239,773)	quark mass vertex $\bar{Q} H q_R$
$1 + 2 \to 3$	$(4 + 5 \to 3)$	315,024	(314,514)	quark Yukawa vertex $Q\, d^c H^\dagger$
$0 \to 3 + 3$		89,909		lepton mass vertex $L\, H^\dagger e^c$
$1 + 1 \to 2$	$(5 + 5 \to 4)$	458,209	(456,755)	two quarks into an antitriplet

The carrier contents of the level-10 run are exactly class-neutral: $\{2, 4\}$ on 69.16 percent of occupied carriers, $\{1, 5\}$ on 14.43 percent, $\{3, 3\}$ on 10.88 percent, $\{2, 2, 4, 4\}$ on 1.93 percent, the triples $\{4, 4, 4\}$ and $\{2, 2, 2\}$ on 1.20 percent each, and rarer neutral mixtures such as $\{1, 2, 3\}$, the centre content of the quark mass term. These contents share centre labels with possible meson, baryon and lepton-pair representations. They do not identify those physical objects; no carrier holds a net colour or weak centre charge.

2.4 Population history of the finite runs

The runs start from independent random loads and a blank register, so the preparation is statistically homogeneous. In the level-10 run every class is created in pairs within the first consistent slice: 124,786 pairs $(1, 5)$, 208,860 pairs $(2, 4)$ and 41,328 pairs $(3, 3)$. Classes 1 and 5 then annihilate (609,494 of 757,258 pair births), while classes 2, 4 and 3 freeze once the loads balance. The conjugate-class counts are 3,053,295 and 3,050,633 for classes 1 and 5. The conjugate triples have counts 86,476 and 86,788, whose difference has magnitude 0.75 times the square root of their sum. This normalization is an arithmetic diagnostic; it supplies no independent Poisson sampling law for correlated run histories or physical matter asymmetry.

Of all defect identities of the level-10 run, the classes $1, 5, 4, 2, 3$ take 24.1, 24.1, 19.6, 19.6 and 12.7 percent, and the shares at levels six and seven agree with these within 0.3 percentage points.

A separate counting control assigns equal weights to the fermionic code components and conditions on nonzero class. Its 28 retained components give 6/28, or 21.4 percent, in each of classes 1, 2, 4 and 5, and 4/28, or 14.3 percent, in class three. The control is identical for the full and degree-restricted fermion menus because their difference lies in class zero. It excludes the Higgs and gauge components and supplies no thermal occupation law.

The run's pair-birth fractions are 50.6, 43.4 and 6.0 percent for $(1,5)$, $(2,4)$ and $(3,3)$. Folding the separate counting control into these pair labels gives 42.9, 42.9 and 14.3 percent. Component multiplicities do not derive pair-production probabilities. The simulation measures class populations and persistence on its declared event clock. It supplies no physical temperature, species decoder, baryon number or photon count, so it determines no baryon-to-photon asymmetry or early-Universe composition. Section 2.12 states the exact code-count inverse problem and its separate thermal comparison.

2.5 Gauge group, couplings and lepton number

The record law is covariant under independent phases on the twelve ports. Read on the protected code, twelve relative port phases per seam generate exactly

$$G = \mathrm{S(U(3)\times U(2))} = \frac{\mathrm{SU(3)\times SU(2)\times U(1)}}{\mathbb{Z}_6},$$

where $\mathbb{Z}_6$ is the kernel of the covering product. The centre of G is the circle $\{\mathrm{diag}(e^{2it}I_3, e^{-3it}I_2)\}$, and the defect register carries its order-six subgroup generated by $e^{2\pi iY}$, which acts on a pattern by $e^{2\pi ik/6}$ with $k = 6Y \bmod 6$. The seam cost of a code state carried around a plaquette, averaged over the code, is $1 - \mathrm{Re}\,\mathrm{tr}_5\, U_P/5$; this action gives equal bare couplings at its declared cutoff, with $\sin^2\theta_W = 3/8$ there, and the gauge waves travel on one cone with two polarizations each [7]. The native transports alone form a finite group; the continuous port phases are the extension that makes the links continuous. Equality of bare couplings does not establish equality of renormalized couplings at the scale of the comparison chain.

Every operation in the declared sector conserves

$$B - L = N - \tfrac{2}{3}\,(N_{c1} + N_{c2} + N_{c3}),$$

a combination of the number of quanta and the three colour-level numbers. It is anomaly free on the sixteen states of one family with ν^c, and it forbids number-violating Majorana pairing in that sector. A Dirac neutrino mass additionally requires the corresponding coupling and a physical species identification; conservation of this charge alone supplies neither a neutrino mass nor that identification.

2.6 The charged-lepton register

The proposed charged-lepton reading assigns three characters of a declared generation register to the electron, muon and tau. The register is a construction on the coherent seam law; a class-three defect alone carries no generation label. Its squared-eigenvalue ratios can be compared with lepton masses, conditional on the stated preparation, analyzer, clock and mass interpretation.

The repair event records its source. On the oriented address (a, b) of a seam, with $p = (|a\rangle + |b\rangle)/\sqrt{2}$ and $m = (|a\rangle - |b\rangle)/\sqrt{2}$, the coherent repair event is the swap of the port qubit (p, m) with the record qubit. Read in its conjugate basis, the record holds the source endpoint relative

to the address. The oriented record V changes the sign of record one at the reversed address; VQ equals the law with the screen address's unitary at both addresses, and it keeps the repair branch that the plain record cancels.

The register. Two seam objects share one face f of the carrier: object C on a seam whose orientation is the screen orientation and object E on a second edge of the same hexagon around the f axis. Each has its load, record and address. A neutral register F on the ports of f carries the third-turn $R = \rho_f$ of the face, with $R^3 = I$. One tick is

$$U(\theta) = K_\theta\, T\, V\, (W_C \otimes W_E), \qquad K_\theta = e^{i\theta J}, \qquad J = |1\rangle\langle 1|_{r_C} \otimes \sigma_z(o_C),$$

where T applies $R^{\pm 1}$ when C repairs in orientation $\pm$ and J is the oriented current. Each object is prepared with activity $1/3$ and its load at the screen tail of its seam. The analyzer Φ is the Fourier zero over the three flags E agrees with C idle, C repairs $+$ with E idle, and C repairs $-$ with E idle.

Theorem 2.3 (Exact compression to the balanced circulant)**.** *The full* 432*-state tick is unitary, and its compression is*

$$M(\theta) = \langle \Phi | U(\theta) | p \rangle = \frac{\sqrt{2}}{3}\, C(\theta), \qquad C(\theta) = \tfrac{1}{\sqrt{6}}\, I + \tfrac{1}{\sqrt{12}}(e^{i\theta} R + e^{-i\theta} R^{-1}).$$

M commutes with R, its eigenvalues on the characters $Rv_k = \omega^k v_k$ are $\lambda_k = \frac{\sqrt{2}}{3}\big(\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{3}} \cos(\theta + 2\pi k/3)\big)$, and $|b|/a = 1/\sqrt{2}$. With masses proportional to λ_k^2,

$$Q = \frac{\sum_k m_k}{\big(\sum_k \sqrt{m_k}\big)^2} = \frac{2}{3}$$

for every θ in the positive chamber $|\theta| \le \pi/12$ modulo $2\pi/3$.

The balance is a count of branch weights. The C object repairs in each orientation with probability $1/12$, and the E object holds with probability $1/6$, so the circulant coefficients satisfy $a^2 : b^2 : b^2 = 1/6 : 1/12 : 1/12$. The phase is a count as well: the register's exclusive read $X = [R_+] \oplus [H]$ has

$$P(X) = \tfrac{1}{12} + \tfrac{1}{6} - 2 \cdot \tfrac{1}{72} = \tfrac{2}{9}.$$

The clock. On every blank-record input with uniform address the seam program satisfies $\langle I - \mathrm{Re}\, U \rangle = P(\text{record } 1)$, so the energy clock reads read probabilities. Inside a tick, N weak rounds of the coupling $J \otimes X$ give the zero-return operator $A_N(t)$ with

$$\| A_N(t) - e^{-itP(X)J} \| \le t^2/N.$$

The declared clock fixes the unit of t: since $I - S$ has eigenvalues 0 and 2, the edge term $(\pi/4)(I - S)$ run for clock time τ reproduces the tick's root-swap gate exactly when $\tau \in 1 + 4\mathbb{Z}$, and the principal value is one unit per tick. At that unit the oriented current carries the phase $\theta = P(X) = 2/9$ per tick; at finite N the phase is $2/9 - 35/(2187N^2) + O(N^{-4})$. A dressing that commutes with J leaves the phase unchanged at every order. On this clock the family mode k has the rate λ_k^2, which is the mass reading of Theorem 2.3.

The proposed electron, muon and tau labels are assigned to the characters ω^1, ω^2 and ω^0 of the face third-turn; the half-turns that carry f to its opposite face reverse the phase and swap

Table 4: Squared-eigenvalue ratios of the declared register at $\theta = 2/9$, compared with charged-lepton masses. No measured mass enters the evaluation of the ratios; the architecture and comparison are not blind. Reference values: CODATA for m_μ/m_e, $m_\tau = 1776.93 \pm 0.09$ MeV.

	Register	Measured	Difference
m_μ/m_e	206.770316	206.7682827(46)	+9.83 ppm
m_τ/m_e	3477.4728	3477.37 ± 0.18	$+0.61\,\sigma$
Q	2/3	0.666665	

the electron and muon labels, leaving the spectrum unchanged. The muon residual of +9.83 ppm has a linear phase-displacement diagnostic of -1.75×10^{-7}. That displacement is not applied. A dressing commuting with the oriented current leaves its phase unchanged. A physical correction requires a derived mass-reading map and its coefficient; neither the complete finite-channel error nor a selected radiative coefficient explains the residual. The fixed comparison and its preparation budget are given in Section 2.10.

The register on the carrier. On the sixty-address carrier the count 2/9 is the exclusive read of two cells with independent address registers, read through the law's own records: in the defect's cell the two signs of the native current are written with weight 1/12 each, and one second cell writes its hold with weight 1/6. One register on one carrier gives only multiples of 1/240. A class-three flux keeps one two-load content, the pair on one port, and the kick of the weak round is the commutator of an oriented toggle with the tick of that pair. For two double-loaded cells the analyzer compression is $\sqrt{5/18}\,C(\theta)$, whereas the specified native-pair analyzer gives $(\sqrt{2}/3)C(\theta)$. The two readings have the same eigenvalue ratios and different absolute rates; their traces $\operatorname{tr} M^\dagger M$ are 5/18 and 2/9, respectively.

The weak round over many ticks. One round is the tick of both cells, the kick $e^{-i(1/N)J\otimes X}$, the inverse tick, one return test per cell and a reset after a negative test; N rounds are one unit of clock time. Every round reads the first-tick weights, and the phase per unit clock time is

$$\theta_N = N \arctan \frac{P \sin(1/N)}{1 - P + P\cos(1/N)}, \qquad P = \tfrac{2}{9}, \qquad \theta_N = \tfrac{2}{9} - \tfrac{35}{2187N^2} + O(N^{-4}).$$

The register was executed on the sixty-address carrier for 10^4 to 2×10^6 rounds. The executed phases follow the closed form at every N and at every elapsed number of rounds, and their limit is 2/9 within 10^{-13}. The marked component of the state has squared projections 4/81, 11/324, 125/1296 and 55/1296 onto the four return-test sectors; these sum to P, not to one. For a current eigenvalue j, put $a = e^{-ij/N} - 1$. The actual four outcome probabilities are $|1 + aP|^2$ and $|a|^2(11/324, 125/1296, 55/1296)$, which sum to one. With every outcome kept and the reset applied, the coherence between two values j, j' of the current is multiplied per round by $1 - P + Pe^{-i(j-j')/N}$, so the phase 2/9 holds on the complete record and no branch is selected. The forward tick alone does not keep the count: it reads 2/9, 0.486, 0.536 and 0.467 at ticks one to four.

Rates and masses. The three characters of the face third-turn are the eigenvectors of the declared native-pair reading. Their rates per unit clock time are

$$y_k = \lambda_k^2 = \tfrac{1}{27}\big(1 + \sqrt{2}\cos(\tfrac{2}{9} + \tfrac{2\pi k}{3})\big)^2, \qquad \textstyle\sum_k y_k = \tfrac{2}{9},$$

and the executed rates agree with this form within 10^{-13}. Table 5 lists them. With the measured electron mass as a scale and the squared-rate interpretation, the register gives the displayed conditional muon and tau values; the physical species and mass attachment is an additional requirement.

Table 5: Rest rates of the three characters at $\theta = 2/9$ and the masses they give with the electron mass as the only measured input. Measured values: CODATA and the Particle Data Group.

Character	Lepton	Rate y_k	Mass (MeV)	Measured (MeV)
ω^0	τ	0.2096935560	1776.985	1776.93 ± 0.09
ω^2	μ	0.0124683656	105.659414	105.6583755(23)
ω^1	e	0.0000603006	input	0.51099895069

The register, its preparation and its analyzer are a construction on the coherent law, and the executable certificate checks the compression, the counts, the round, the clock unit and the carrier statements in exact arithmetic [1]. Open inside it are four items: the operation that brings two loads onto one port, the source event that applies the kick, the rule that one partner is read per unit clock time, and the number N_p of phase rounds, which shifts m_μ/m_e by $-824/N_p$ at leading order. The balanced circulant was first written down from the measured masses [9]; here its balance and its phase are the register's own read counts. A mass in MeV is $m_k = F\,E_{\mathrm{cell}}\,y_k$ with one number F; on the electroweak dictionary F is fixed by the lepton Yukawa weight $y_J \approx 0.0108$, which is open.

2.7 The integer lift of the register

Keeping the integer current of every applied twist unreduced lifts the $\mathbb{Z}_6$ register to integers. On the runs at levels four to seven the lifted value reduces modulo six to the stored class on every face of every stored cut, and the twenty face values of every carrier sum to zero in all 68,515,840 carrier states. As a set of values the lift takes $6Y$ of a Standard Model multiplet on 99.7 percent of the defect faces. It is no conserved label of the member inside a class: the law reads the class only, a value of size four or more does not travel with its defect, and one carrier content occurs with several integer words [1].

2.8 The coupling chain and the electroweak scale

The gauge couplings and the electroweak scale follow one chain of five forms: the cell energy $E_{\mathrm{cell}} = E_* P^{-1/2}$, the unification scale $M_U = E_* e^{-2\pi} P^{1/6}$, one-loop running with slopes $(33/5, 1, -3)$, the transmutation law

$$v = E_{\mathrm{cell}} \exp\Big(-\frac{2\pi}{4\alpha_U}\Big),$$

and the pixel closure $\bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} = P/4$ with heat times $t_i = 4\pi^2\alpha_i$ read at the electroweak anchor $\mu = m_Z(\mu)$. At the fixed point $P = 1.6309721$ the chain gives $1/\alpha_U = 24.3166$ at $M_U = 2.4736{\times}10^{16}$ GeV and the values of Table 6. These are tree values of the chain; the multiloop pole comparison is Table 9.

Four parts of the chain have a source reading. The cell energy is one radian of phase per tick on the cubic arrangement of carriers. The 4 in the transmutation law is the inverse plaquette cost of one class-three flux quantum: the central element $\mathrm{diag}(1, 1, 1, -1, -1)$ has shortest generator $2\pi i T_3$ with $\mathrm{tr}_5\, H^\dagger H = 2\pi^2$, and the five nontrivial classes give the counts $(12/5, 3, 4, 3, 12/5)$. The anchor is the first freeze of the weak currents: the class-three register gives the neutral direction, which contains hypercharge, the largest mass, so the collar law $d_R e^{-tC_2}$ with one coefficient for the three

Table 6: Outputs of the coupling chain against measured values. The chain's forms are declared; the comparison is made after the values were known.

Quantity	Chain	Measured
v (GeV)	246.60	246.22
m_Z, m_W tree (GeV)	91.588, 80.330	91.188, 80.369
$1/\alpha_2(m_Z)$	29.605	29.58
$\alpha_s(m_Z)$	0.11834	0.1180(9)
$\sin^2\theta_W(m_Z)$	0.23074	0.23122
$1/\alpha_{\rm em}(m_Z)$	128.308	127.95
$1/\alpha(0)$	136.995	137.036

weak directions holds down to m_Z, under one stated premise on the collar generator below the anchor. Hypercharge carries no representation entropy because every irreducible representation of U(1) has dimension one, and both factors 2π of the chain are the modular period.

The other parts are declared. The source as built has $16 + \overline{16}$ patterns per copy, twelve gauge currents and one complex Higgs doublet, with one-loop slopes $(81/10, 5/6, -3)$; on the same chain it gives $v = 222.9$ GeV, $\alpha_s = 0.1123$ and $\sin^2\theta_W = 0.2056$. The ratio $(1/\alpha_2 - 1/\alpha_3)/(1/\alpha_1 - 1/\alpha_2)$ at m_Z equals $115/218$ for this content and $5/7$ for the declared slopes; the measured couplings give 0.717. The weak and colour slopes of the declared chain equal those of the source's content with two doublets, and the hypercharge slope does not. The stack's own unification scale is $E_{\rm cell}e^{-2\pi} = 1.7853\times10^{16}$ GeV, which gives $v = 264.6$ GeV; the factor $P^{2/3}$ between it and the declared form has no source reading. The displayed chain uses collar weights d_R and unit shape number as declared inputs. Its quoted numerical precision requires control of higher-loop and threshold terms; one-loop evaluation alone does not supply that control.

The transmutation law has the structure $v = E_{\rm cell}\, b\, \langle A\rangle$, where $\langle A\rangle = \sum_j p_j(-1)^{2j}$ is the class balance of the seam link that the mass step flips. In a leading weak-coupling form $\langle A\rangle \sim Me^{-cS}$, with $S = 2\pi/(4\alpha)$, the proposed law requires $c = 1$ and $bM = 1$. One free link in its one-tick thermal law has leading weak-coupling exponent $c = 1$, with prefactor tending to 2 on the Cartan circle. A link inside a lattice shares the half turn with its surroundings: in the infinite-volume Gaussian limit on the Cartan circle, $c = 1 - 2/d$ on a cubic lattice in d dimensions, and Monte Carlo for SU(2) with the heat-kernel weight gives $c = 0.5001(8)$ on 6^4, 0.3347(12) on 8^3 and 0.3988(5) on the icosahedral space. None of the ensembles measured here has coefficient one; the single-link value requires a pinned plaquette. The sector law that pins the seam link is open [1].

2.9 Lattice ensembles and thermal comparisons

The stored runs measure finite repair counts. The thermal calculations below use separate Euclidean lattice ensembles, with temperature $T = 1/(aN_t)$ in units $\hbar = c = k_B = 1$. The lattice spacing a and its physical calibration are inputs; Euclidean time slices are not identified with the repair clock. These ensemble comparisons do not supply a shared cosmological cooling history.

The electroweak crossover. With the couplings of the declared chain as inputs and the published reduction to the three-dimensional SU(2) and Higgs theory [10], the transition has crossover central value

$$T_c = 161.1298 \text{ GeV}.$$

The Standard Model lattice value is 159.5 ± 1.5 GeV [11]. The same pipeline on Standard Model inputs gives 159.76 GeV against the published 159.58(6) GeV. On the chain the first-order line ends at a Higgs mass of 72.5 ± 0.7 GeV. With the assumed Standard Model thermal census $g_* = 106.75$ and a radiation-dominated Friedmann relation, the crossover corresponds to 9.0×10^{-12} s. Converting this to 1.3×10^{32} repair ticks additionally assumes $\hbar/E_{\mathrm{cell}} = 6.885 \times 10^{-44}$ s per tick. With the declared source content and assumed Higgs and top masses the crossover lies between 145 and 149 GeV.

The reduction uses $y_c = y(T_{\mathrm{lat}}) + y_{\mathrm{fit}} - y_{\mathrm{ref}}$; the reference ensemble enters all three terms. Paired peak replicas are absent from the retained data, so the statistical standard deviation is not identified. Linear propagation of the supplied marginal variances, with independence between distinct lattice sets and arbitrary covariance within the reference set, gives the sharp compatible range $[0.11810, 0.13680]$ GeV. Without the cross-set independence premise the range is $[0, 0.27631]$ GeV. These are uncertainty ranges under a fixed linearized fit, not confidence intervals. The declared one-percent reduction allowance is 1.6113 GeV and has no supplied probability law; it is not combined in quadrature with the statistical range or used to quote a standardized agreement. A separate comparison at matched physical size $Lg_3^2 = 8$ gives 160.8546 ± 0.0720 GeV for a linear cutoff fit. Linear, quadratic and omitted-set variants span a maximum displacement of 0.3490 GeV from that value. This finite-design sensitivity is separate from the reduction estimate and does not bound omitted cutoff powers or the infinite-volume limit. Three points fitted to a constant, linear and quadratic term have zero residual degrees of freedom. The measured susceptibility-volume exponent decreases from 0.227(38) to 0.089(89) across the two largest-size comparisons; the first-order control grows from 2.163(32) to 2.881(88), toward the volume exponent three.

Colour deconfines. Restricted to a colour holonomy W, the seam cost $1 - \mathrm{Re}\,\mathrm{tr}_5\, U/5$ is $(3/5)(1 - \mathrm{Re}\,\mathrm{tr}\, W/3)$, the Wilson plaquette action with $\beta = 6/g^2$. The colour centre $\mathbb{Z}_3$ is the flux classes 2 and 4 of the register, and the classes charged under it are the quark classes $1, 2, 4, 5$. A Monte Carlo study of these links at $N_t = 4$ gives the transition coupling $\beta_c = 5.69105(39)$ from three volumes and 5.69248(71) from the two largest, against the published 5.6925(2) [12]. The susceptibility maximum grows as $N_s^{2.555(29)}$, the histogram of the Polyakov loop has two peaks, and the deconfined runs settle in the three $\mathbb{Z}_3$ sectors: the transition is of first order. With the string tension from the same lattice,

$$T_c/\sqrt{\sigma} = 0.614(18),$$

against the published 0.598(3) at this lattice spacing and 0.629(3) in the continuum limit. The study holds the links alone at one lattice spacing and scans the coupling as an ensemble parameter; continuous links on the seams are a declared step.

Quarks bind. Quenched Wilson quarks on the same links at $\beta = 5.7$ on $12^3 \times 24$, with 128 configurations, give a static potential that rises linearly, $a^2\sigma = 0.1667(17)(135)$, and in the chiral limit

$$m_N/m_\rho = 1.45 \pm 0.13, \qquad m_\rho/\sqrt{\sigma} = 1.333(122), \qquad m_N/\sqrt{\sigma} = 1.932(182).$$

The normalized-mass errors combine same-stream jackknife errors with the largest displacement over crossed hadron and potential fit choices. They do not bound quenching, finite-volume or continuum effects. Keeping the paired jackknife replicas also determines the joint mass covariance. The same-stream contrast is $(m_N - m_\rho)/\sqrt{\sigma} = 0.59903 \pm 0.04286$ statistically, whereas discarding the covariance gives 0.04407 for that statistical error. This contrast does not include fit-systematic variation. The common scale cancels from m_N/m_ρ in every paired replica. The retained covariance

therefore belongs in joint comparisons of the normalized masses. The measured ratio m_N/m_ρ is 1.211. Above the transition the binding melts: the slope of the free energy of a static pair falls from 0.9896(51) per lattice unit at $\beta = 5.6$ to 0.00107(1) at $\beta = 5.85$. Masses in GeV need the light-quark masses and the lattice spacing in GeV, which are open. With the chain's $\alpha_s(m_Z)$ and imported lattice ratios the colour scale is $\Lambda^{(3)} = 0.340$ GeV, the nucleon 0.933(47) GeV and the rho 0.773(42) GeV.

Standing of the three comparisons. The colour and hadron numbers are those of pure and quenched SU(3) with the Wilson action. Their agreement with published lattice values confirms that the model's colour links, once continuous, are that theory. The crossover temperature takes the declared chain's couplings as inputs. The class shares of the stored runs lie near equal occupation per state without equalling it, and the baryon excess and the light-element abundances are not computed.

2.10 Complete cycles and controlled lepton rates

The finite lepton instrument can be composed without discarding a return failure. Prepare each cell's sixty-address register in the uniform state u, and let $S = u \otimes u$. A Householder unitary gives an explicit map from a blank address basis vector to u. The blank double loads, independent fresh cells and this preparation service are declared inputs. After the native forward tick, the exclusive-read kick and the inverse tick, the joint state has the form

$$S \otimes \psi + X \otimes (U_h - I)\psi, \qquad U_h = e^{-ihJ}, \quad J = \mathrm{diag}(-1, 0, 1), \quad \langle S, X\rangle = \|X\|^2 = p = \frac{2}{9}.$$

The two return tests have four outcomes. Their complete Kraus operators are

$$A_{11} = (1-p)I + pU_h, \quad A_{01} = \sqrt{\frac{11}{324}}(U_h - I), \quad A_{10} = \sqrt{\frac{125}{1296}}(U_h - I), \quad A_{00} = \sqrt{\frac{55}{1296}}(U_h - I).$$

The first outcome includes interference with the standing source. The sum of the four effects is identity, and the unconditional register channel is

$$\Phi_h(\rho) = (1-p)\rho + pU_h\rho U_h^*.$$

Exchanging the old source pair with an independently prepared fresh pair resets the active apparatus. The old pair, its flag and all its correlations are exported. This unitary exchange supplies the reset while retaining every outcome; it does not erase an unknown quantum state. It requires fresh storage and a preparation independent of the continuing register and reference.

The mass-read analyzer gives a separate complete channel. In the three Fourier characters its compressed effect is $E = \mathrm{diag}(e_k)$, where

$$e_k = \left[\sqrt{\frac{5}{108}} + 2\sqrt{\frac{5}{216}}\cos\left(\theta + \frac{2\pi k}{3}\right)\right]^2, \qquad \mathrm{tr}\, E = \frac{5}{18}.$$

The pulled-back analyzer vectors X_k satisfy $\langle X_l, X_k\rangle = \delta_{kl}e_k$, by the equal three-character weights and $1 + \omega + \omega^2 = 0$. The sum of their nonreturn Gram matrices is $\mathrm{diag}(e) - ee^{\mathsf{T}}$, positive by weighted Cauchy–Schwarz. It has rank three at $\theta = 2/9$, so a single scalar failure operator does not reproduce the source environment. With $z = e^{-ih}$, the complete mass channel fixes populations and, for $k \neq l$, obeys

$$\Psi_h(\rho)_{kl} = \left[1 + (z-1)e_k + (\bar{z}-1)e_l\right]\rho_{kl}.$$

The double-load rates have common factor 5/4 relative to the native-pair rates of total 2/9. Their ratios agree; their clock normalization is explicit.

For N independent fresh-source cycles with $h = t/N$, finite channel estimates give

$$\frac{1}{2}\|\Phi_{t/N}^{N} - \mathrm{Ad}(e^{-itpJ})\|_{\diamond} \le \frac{14t^2}{81N},$$
$$\frac{1}{2}\|\Psi_{t/N}^{N} - \mathrm{Ad}(e^{-itE})\|_{\diamond} \le \frac{349t^2}{1296N}.$$

The first estimate centers the binomial kick count and uses the spectral diameter $D(J) = 2$. The second compares the Taylor remainders of the source projector, of diameter one, and its compression, whose diameter is at most 5/18. Channel contraction and telescoping extend both estimates to arbitrary untouched references. At $t = 1$, the conservative ideal cycle counts for error at most 10^{-6} are 172,840 and 269,291. Per-cycle channel error ϵ and independent preparation error η add at most $N(\epsilon + \eta)$. Marginal preparation fidelity alone does not bound correlations with the register. Diagonal densities are stationary, including a faithful uniform density; this does not select a unique reference. The exported history grows. No physical duration or mass-pole interpretation is provided by these finite estimates.

The spectrum's parameter dependence can be calculated separately. For the shape $1 + c\cos(\theta + 2\pi k/3)$, the positive chamber gives $Q = 1/3 + c^2/6$. At $(\theta, c) = (2/9, \sqrt{2})$, the Jacobian of the logarithmic muon/electron and tau/electron ratios is

$$\begin{pmatrix} 56.14388 & 32.61133 \\ 51.22679 & 34.45440 \end{pmatrix}, \qquad \det = 263.82944.$$

The two shape parameters are locally identifiable from the two ratios; the common normalization cancels. The CODATA 2022 comparison value $206.7682827 \pm 0.0000046$ leaves a muon/electron residual of +9.83358 parts per million, or 442.02 experimental standard uncertainties. No parameter or round count is adjusted to cancel it. Fixed individual flag marginals 1/12 and 1/6 permit an exclusive phase anywhere in $[1/12, 1/4]$; independence fixes their joint probability to 1/72 and the phase to 2/9. Thus the full preparation, rather than the marginals alone, supplies that phase premise. The executable source arrays, complete instruments and finite-history checks are in `model/lepton_cycle` [1].

The same fixed shape gives $m_\tau/m_e = 3477.47283710$. Dividing the Particle Data Group 2024 average $m_\tau c^2 = 1776.93 \pm 0.09$ MeV by the CODATA 2022 electron energy $0.51099895069 \pm 0.00000000016$ MeV gives 3477.36526191 [36]. The electron and direct muon-ratio inputs are listed in the 2022 CODATA table. The selected tables supply no cross-edition covariance. For $r = t/e$, put $a = u_t/e$ and $b = tu_e/e^2$. All positive-semidefinite covariances give $|a - b| \le u_r \le a + b$ under linear propagation. Here this range is $[0.17612452, 0.17612670]$, so the tau residual is 30.93584 parts per million or $[0.610783, 0.610791]$ experimental standard uncertainties. These reference-only discrepancies supply neither a joint significance nor a physical-theory error. The mass-pole interpretation is unproved. The comparison is retrospective; measured masses and Koide relations were known during architecture development. The executable comparison freezes both ratios and selects its finite round counts from a channel-accuracy target independent of the mass residuals.

Preparation uncertainty has a separate, finite statement. For a fixed analyzer projector and independent source states at trace distance at most η, the compressed effects differ in operator norm by at most η. This follows by applying the variational trace-distance bound to the source effect induced by each unit register vector. Weyl's inequality bounds each ordered rate by η. If

$\eta < e_e$ and twice η is below all spectral separations, the mode labels are preserved and

$$\frac{e_j - \eta}{e_e + \eta} \le \frac{e'_j}{e'_e} \le \frac{e_j + \eta}{e_e - \eta}.$$

These are fixed-analyzer effect-rate bounds. Complete-channel distance and source preparation errors are not adjustable physical mass corrections.

2.11 An exact finite population and its symmetry

A closed triangular cell supplies a finite test of persistent populations. It has three loads in $\{0, \dots, 5\}$ and two oppositely oriented face classes $(k, -k)$ in $\mathbb{Z}_6$. Each attempt selects one of the three oriented seams uniformly and a fair balancing coin. It uses the same native load repair and class-acceptance functions as the federation. The state space has 1296 elements and the complete instrument has 7776 input-event branches, including load-only and no-operation events. The geometry is a declared two-face cell; no transfer of its population to the full federation is asserted.

With class cost $c = (0, 5, 8, 10, 8, 5)$ and $\kappa = 5$, the quantity

$$V(x, k) = \kappa \sum_{a=0}^{2} x_a^2 + 2\big[c(k) + c(-k)\big]$$

does not increase on any branch. An accepted current j changes the load square by $-2j(d - j)$ and the register cost by at most $\kappa j(d - j)$. Rejected twists retain only the nonpositive load contribution. The complete kernel conserves total load and total face class. Its finite harmonic absorption equations determine the persistent occupied-face probability from every initial state. For independent uniform initial loads and zero initial flux, their exact average is

$$\Pr(\text{persistent occupied faces}) = \frac{1007}{13824} = 0.0728443287037\dots.$$

The full distribution after 128 attempts bounds this limiting share in an interval of width less than 5.86×10^{-10}. These probabilities describe operational repair events, without a physical temperature, cooling law or species-abundance dictionary. For every initial distribution on all 1296 states, the probability of remaining outside the terminal set after n attempts is at most $25^{-\lfloor n/32 \rfloor}$, from the exact all-state 32-attempt absorption bound.

The native law also has an exact conjugation. Send every load x to $5 - x$, every face class and integer lift to its negative, and every coin to its complement. The repaired current changes from j to $-j$ while the release $j(d-j)$ and the class-cost comparison are unchanged. Exhaustive enumeration checks all 2592 local load/class/coin branches; the same identity verifies all 7776 branches of the population cell. It extends pathwise on the closed federation with an equivariant seam schedule.

For a finite kernel K and conjugation C, the conditions $K(Cx, Cy) = K(x, y)$ and $\mu(Cx) = \mu(x)$ imply $(\mu K^n)(Cx) = (\mu K^n)(x)$ for every finite n. Every odd observable $f(Cx) = -f(x)$ then has zero expectation. These finite-sum identities are formalized in the Lean dynamics library; the native branch checker supplies the source-specific premises. The signed class differences are odd, though single histories can fluctuate. If the initial symmetry defect is a_0 in total variation and the complete kernel at attempt r has equivariance defect at most ε_r, maximal coupling gives

$$|\mathbb{E} f| \le 1 - (1 - a_0) \prod_{r=1}^{n} (1 - \varepsilon_r), \qquad |f| \le 1.$$

A coin of probability p_r contributes at most $|2p_r - 1|$ to that defect. Conjugation-even selection of probability q can amplify the bound by at most $1/q$; asymmetric selection can produce a conditional imbalance from an exactly symmetric law. The complete records therefore distinguish dynamical asymmetry from selection. Class conjugation alone does not identify baryon number or predict the baryon-to-photon ratio. Reproduction is supplied by `model/native_abundance_epochs` and `model/charge_conjugation_dynamics` [1].

2.12 Species information in a centre-class census

For a declared five-bit exterior-code component, let n_3 and n_2 count colour and weak occupations. Its centre class is $k = (3n_2 - 2n_3) \bmod 6$. The matrix $A_{kj} = 1_{k(j)=k}$ pushes component populations $x \geq 0$ to the six-class census $c = Ax$. The full 32-component menu has multiplicities $(4, 6, 6, 4, 6, 6)$, rank six and nullity 26. Removing exterior degrees zero and five, the two right-neutrino labels, gives 30 components and nullity 24. Adding four Higgs and twelve gauge internal components gives nullities 42 and 40, respectively; physical helicity multiplicities are a separate count.

An aggregate $w \cdot x$ is identifiable from every census precisely when w is constant within each class. Necessity follows by exchanging two unit populations in that class; sufficiency follows by summing the shared coefficient against c_k. More generally the sharp nonnegative bounds are

$$\sum_k c_k \min_{j:k(j)=k} w_j \ \leq w \cdot x \leq \sum_k c_k \max_{j:k(j)=k} w_j.$$

Assigning all class population to a minimizing or maximizing component attains either endpoint. Thus the quark-labelled total $q = c_1 + c_2 + c_4 + c_5$ is identified on this declared component dictionary. With bosons admitted, the lepton total spans $[0, c_0 + c_3]$, the gauge total $[0, c_0]$ and the Higgs total $[0, c_3]$. These are separate marginal bounds; their endpoints need not occur together.

One additional fermion/boson bit within classes zero and three suffices for all four totals. To distinguish also up/down and charged/neutral lepton aggregates requires three label values in the boson extension: the electron, neutrino and Higgs signatures in class three prove the lower bound. Class-dependent signature labels attain it. This is a minimal information theorem for the declared dictionary, not a physical decoder. Native face-flux events, including zero flux, do not by themselves count code quanta or thermal particles.

Uniform component weights give quark/lepton counts $24 : 8$ in the full fermion menu and $24 : 6$ in the restricted one. They supply no occupation probability law. For an explicit comparison, a massless ideal Standard Model gas at one common temperature of 1 TeV, zero chemical potentials, three active-neutrino families and no populated right-handed neutrinos has 90 fermionic and 28 bosonic helicity states. Number densities weight fermions by 3/4 relative to bosons; energy densities use 7/8 [36]. This ideal-gas benchmark uses the thermodynamic conventions of Saikawa and Shirai (2018); it omits their interacting equation-of-state corrections. The effective totals are therefore 191/2 and 427/4. Quark and lepton fractions among all particles are 108/191 and 27/191 by number, versus 252/427 and 63/427 by energy. Their fermion-only split is 80% : 20%. These are ideal-gas reference values with interactions, masses and thermal-mass corrections omitted. No run temperature, equilibration law or early-Universe abundance match is inferred. The rational maps, extremizers, minimum labels and thermal counting receipt are in `model/native_abundance_epochs` [1].

2.13 Scalar response of localized defect modes

A Yukawa coefficient can be defined operationally through the response of a fermionic creation–propagation–detection amplitude to a scalar intervention. In canonical field coordinates, the con-

vention

$$M(h) = M(0) + \frac{h}{\sqrt{2}} Y_{\text{eff}} + O(h^2)$$

defines the zero-scalar-momentum mass-kernel vertex. Its derivation requires the source intervention and field normalizations. Charge and topology constrain allowed tensor channels, whereas their amplitudes depend on the mode profiles and dynamics. This definition supplies no independent interaction coefficient when the complete source specifies its response. An on-shell particle interpretation requires the associated propagation and state.

There is a finite overlap calculation on the specified compact texture with cores at $(0, 0, \pm 1)$, texture amplitude $M = 64$, and radius-three exterior box. Its complete 686-row Hamiltonian has four isolated modes. Let W be an orthonormal frame for their entire spectral subspace and $C = D_{\text{stagger}} i\sigma_2$ its particle–hole pairing map. The source identities give

$$C\overline{W} = WS, \qquad S = W^\dagger C\overline{W}, \qquad S^T = -S, \qquad S^\dagger S = I_4.$$

The unit local scalar insertion at site x has pairing kernel $P_x C$. Its complete band response is

$$B_x = W^\dagger P_x W, \qquad Y_x = W^\dagger P_x C\overline{W} = B_x S.$$

The same spatial factor can multiply a unit symmetric internal Higgs tensor. No family coefficient is inserted in this spatial calculation. All 343 site kernels are retained. A change $W \mapsto WU$ gives $Y_x \mapsto U^\dagger Y_x \overline{U}$, so the singular spectra are independent of a basis choice inside the band.

For a complete orthonormal complement Q, define $L_x = Q^\dagger P_x C\overline{W}$. Every cross-site leakage obeys

$$L_x^\dagger L_y = S^\dagger(\delta_{xy} B_x - B_x B_y) S.$$

Thus a real scalar profile f gives $Y[f] = \sum_x f_x Y_x$ with

$$L[f]^\dagger L[f] = S^\dagger \left(\sum_x f_x^2 B_x - B[f]^2 \right) S, \qquad B[f] = \sum_x f_x B_x.$$

The uniform profile has $Y[1] = S$, four unit singular values, and zero leakage. A uniform interaction on this carrier supplies no spatial hierarchy.

For the specified 27-site intervention $f_x = 1_{\|x-(0,0,1)\|_\infty \le 1}$, the numerical singular values are

$$(0.999999981366,\ 0.999999981366,\ 0.000061031430055,\ 0.000061031430055).$$

The largest-to-smallest ratio is approximately 16385. The weaker channel reaches the patch through the opposite core's spatial tail. The first-hop transfer Gram is $I_2/4$, yielding leading strength $1/(4M^2)$. Complete controls at $M = 32, 64, 128$ give $M^2 y_{\text{weak}} = 0.2499389, 0.2499847, 0.2499962$. These values are numerical diagnostics, with no outward error interval asserted for the ratio. No measured mass selects the profile or source parameters.

The exterior resolvent supplies the normalization explicitly. In core and exterior coordinates, $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$ and an actual band eigenvector is

$$\psi = (u, -(D-E)^{-1} B u), \qquad \|\psi\|^2 = u^\dagger \big[I + B^\dagger (D-E)^{-2} B \big] u.$$

Every band mode is reconstructed with this norm. Complete 1372-row Nambu interventions also retain their complementary Schur response; the nonzero spatial band Hamiltonian is kept. Dividing f by its Euclidean norm rescales every strength together. This coordinate normalization does not determine a physical Higgs kinetic residue.

The result exhibits two different scalar-response strengths, each repeated twice, for modes carrying the same internal current representation. A dynamical scalar Hamiltonian must determine which contraction $Y[f]$ describes its collective mode. The specified profile is an intervention, and the four spatial modes have no three-generation assignment. Physical Yukawa values and relativistic masses are not inferred from these finite overlaps. The complete site kernels, complement identities and spectral controls are reproduced by `model/defect_mode_overlap` [1].

2.14 Dynamical scalar selection and native contact response

On the same compact texture, declare the scalar stiffness and its positive frequency operator

$$K = I + L_{\mathrm{graph}} + M^2 \,\mathrm{diag}(|n_x|^2), \qquad \Omega = \sqrt{K}.$$

The scalar carrier consists of its blank and all site-labelled particle and antiparticle doublets, with one global excitation cap. Its dimension is $1 + 4(343) = 1373$. Each of the 343 spatial scalar modes is retained. For the scalar blank, the literal finite matrix units give

$$D_\phi(z) = (z - \Omega)^{-1} - (z + \Omega)^{-1} = 2\Omega(z^2 - K)^{-1}.$$

Consequently $q = (2\Omega)^{-1/2}\phi$ has response $(z^2 - K)^{-1}$. A unit coupling to a ladder coordinate ϕ_j becomes $\sqrt{2\omega_j}$ times that coupling in the coordinate q_j. This normalization belongs to the blank reference; an interacting ground requires its own response.

The lowest scalar mode is unique, positive and even under exchange of the two cores. Its stiffness eigenvalue is approximately 6.99829101533. Its two doubled overlap strengths are 0.706890884 and 0.706804630, with ratio 1.000122033. Thus the large hierarchy of the specified local probe is absent for this selected scalar mode.

The interaction can also select a scalar response without imposing a classical profile. Fill every negative spatial fermion mode with all fifteen native labels and take the scalar blank. The resulting Slater sea is a global gauge singlet because all twelve native generators are traceless. Retain its conjugate sea as an idle reference. Let P be the complete negative spectral projector and define the spatial pair Gram

$$G_{xy} = D_x D_y \det(P_{xy}).$$

The full unit up, down and lepton tensors have exact internal Gram $\mathrm{diag}(6, 6, 2)$. Wick contraction gives $\|H_Y|s\rangle\|^2 = 56 \,\mathrm{Tr}\, G \simeq 29.01797916$. The complete scalar mode probabilities in this first interaction image put approximately 0.964506587 in the odd bound mode and 0.000411883 in the even lowest mode. The dominant reduced response profile has four equal spatial strengths near 0.706847830. These are first-image records, not a dressed Higgs pole or a selected vacuum expectation.

The orthonormal pair $|s\rangle$, $H_Y|s\rangle/\|H_Y|s\rangle\|$ gives a two-dimensional Rayleigh–Ritz matrix for the full law, retaining all native channels and scalar states. The numerical upper-bound estimate is $E_{\mathrm{ground}} - E_{\mathrm{sea}} \le -2.46785318$. Independently, the complete certified spatial gap and operator bounds prove the strict rational inequality

$$E_{\mathrm{ground}} \le E_{\mathrm{sea}} - \frac{1456224000}{169057134218809}.$$

The ground is taken in the same global singlet domain. The full sea is therefore not its ground. A complete spectral instrument has success operator $H_Y/(28\cdot343)$ and failure effect $I - H_Y^2/(28\cdot343)^2$; every scalar mode and the original failure probability are retained. These statements are reproduced by `model/defect_scalar_modes` [1].

For the one-site texture-free native control, all thirty fermion modes and the five-state scalar are retained. The equation-of-motion insertion

$$\mathcal{K}_{ab} = \{[c_a, H], c_b\}$$

is the exact contact pairing derivative. Contraction with the actual scalar Gram reconstructs all its components from scalar–fermion three-point records. The lepton-only 320-state control has scalar first moment $Z_H = 14/5$. With all quark vertices active, the local native-current singlet ground is different. Writing $\Delta = \sqrt{1+4\lambda}$, $e = (1-\Delta)/2$ and $n = (1-1/\Delta)/2$, it has

$$Z_H = 2 - \frac{7}{4}n - \frac{3}{4}e \simeq 9.58523463744.$$

Here λ is the selected source root, approximately 137.35663135. All 900 fermion pairs give 56 nonzero oriented contact components. The three unit native channels have canonical contact modulus $\sqrt{Z_H} \simeq 3.09600301$. Their complete scalar-pulse responses differ: the normalized pair-pointer slopes are approximately -0.2108914572 for either quark channel and -0.1983104425 for the charged lepton. This is a state-specific contact and intervention result, without an on-shell Yukawa or observed family assignment. The corresponding exact spectral records and finite pulse controls are given by `model/defect_family_vertices` [1].

The same local source also has a complete Higgs-dressed neutral-composite spectrum, despite its elementary neutral contact nullspace. All nine target roots are retained on the 26-dimensional target in each of four spin/addition/removal settings. The resulting 44 original records comprise 36 success energies, four failures and four zero-weight outside outcomes. The minimum success energy lies in the exact interval

$$\left[\frac{6301896140017}{70368744177664}, \frac{25207584560069}{281474976710656}\right],$$

approximately 0.089555330476 source units. This gapped local composite has no physical neutrino species or dispersion assignment.

The common finite construction has full CAR action on thirty native fermion modes per site and retains the global scalar cap. Its texture-free three-site member executes a scalar preparation and pair detection on all ninety fermion modes. The literal supports of $H^n\psi$, for $n = 0, 1, 2, 3$, have $1, 11, 664, 7799$ coordinates, with coincident amplitudes combined in exact arithmetic. If P_R detects a complete right-site pair, then

$$P_R\psi = P_R H\psi = 0, \qquad \|P_R H^2\psi\|^2 = 8(2/3 - \sqrt{2}/2)^2 > 0.$$

The spectral Taylor remainder is bounded by $t^3\|H^3\psi\|/6$. At $t = 10^{-4}$ this gives a rigorous receiver probability between approximately 2.70×10^{-19} and 3.90×10^{-19}, with exact radical endpoints and the complete complementary outcome retained. The dual charged preparation reference is retained throughout. This is a finite interaction event, without a particle mass or finite-speed interpretation; $\sqrt{K}$ directly couples the outer scalar sites. The full source columns and error bounds are reproduced by `model/defect_mass_join` [1].

2.15 Propagation tests and a common energy unit

A periodic extension of the compact texture gives exact supercell momentum labels. For even cell side $L \geq 6$, the hopping norm is at most three and the on-site spectrum has four zero entries and all other entries $\pm M$. Ordered-eigenvalue bounds therefore give exactly four bands in $[-3, 3]$ and a complement with $|E| \geq M - 3$, at every Bloch phase when $M \geq 8$. The complete Bloch

spectra for $L = 6, 8$ and $M = 8, 64$ are retained. Direct construction of a two-cell ring verifies the Fourier intertwiner on all rows. Filled-sea additions and removals, each scaled by the inverse square root of the one-body dimension, give a complete energy and crystal-momentum instrument by the canonical anticommutation identity.

The low bands are nearly flat and depend strongly on direction. In the $L = 6$, $M = 8$ control, positive-band secants of E^2 against p_z^2 are negative between Bloch phases zero and $\pi/2$ along the third axis. These numerical controls do not establish relativistic rest masses from the zero-momentum gaps. The periodic geometry also differs from an isolated moving defect and from the interacting source. The construction and complete complementary spectra are reproduced by `model/defect_dispersion` [1].

For a fixed source clock, a GeV conversion multiplies every energy by one positive constant a. A positive source interval $[x_-, x_+]$ and positive target interval $[y_-, y_+]$ permit exactly

$$a \in [y_-/x_+,\ y_+/x_-].$$

A common scale exists for independent interval anchors precisely when these intervals intersect. This exact feasibility calculation is distinct from a likelihood. The equal native up/down record-zero bins cannot both match the unequal fixed running quark parameters at a common scale through one such conversion. A physical quark comparison also needs its renormalization scheme and scale map.

For the separate fixed lepton effects, the sole electron anchor $0.00051099895069\,\mathrm{GeV}$ gives

$$m_\mu^{\mathrm{conditional}} = 0.10565941450\,\mathrm{GeV}, \qquad m_\tau^{\mathrm{conditional}} = 1.77698497081\,\mathrm{GeV}.$$

As a coefficient assignment alone this supplies no propagation proof. The explicit compiler below supplies a free-field construction on its declared clock. Anchor-only uncertainty is not a theory error. The complete target catalogue retains all nine charged fermions, three neutrino mass eigenstates and five boson types, together with the extra source modes. Empirical running-action parameters, optional neutrino coefficients and tree-level bosonic coefficients are distinguished from source-derived observer rest masses. Global current labels alone supply no propagating gauge bosons. The exact common-scale test and conditional GeV table are reproduced by `model/defect_mass_catalogue` [1].

2.16 Charge-sensitive pole splitting with equal primitive vertices

Equal primitive quark coefficients need not give equal dressed response poles. On the unit local source, the total hypercharge Q_Y commutes with H. Its removal charges are $2/3$ for the up-conjugate channel and $-1/3$ for the down-conjugate channel. A symmetry commuting with Q_Y cannot exchange those two charged operators. The complete spatial equal-spectrum symmetry instead transforms Y to $-Y + B$, preserving the baryon-zero singlet domain and reflecting the probe position. Equality therefore applies to invariant stationary densities and matched reflected probes; it is not an assertion about an undetermined spatial ground sector.

A neutral two-state polarization memory with unit occupation cost gives the declared extension

$$H' = H + N_A + Q_Y X_A, \qquad M_q = \begin{pmatrix} 0 & q \\ q & 1 \end{pmatrix}.$$

On a target of charge q, every original spectral line splits by $\lambda_\pm = (1 \pm \sqrt{1 + 4q^2})/2$, with blank-memory weights $(1 \mp 1/\sqrt{1 + 4q^2})/2$. The exact blank resolvent has selfenergy $\Sigma_q(w) = q^2/(w - 1)$.

Both branches, every original energy residue, and preparation failures are retained. The lower dominant source gaps are approximately 0.4482403502 and 0.6806484710 for the up and down channels. Their ratio 0.6585489711 fails the fixed common-scale target comparison. No parameter was chosen using that target ratio.

The unchanged neutral ground has $Q_Y = 0$. All nine hundred anomalous pairing contacts are unchanged, since $\{[c_i, Q_Y X_A], c_j\} = Y_i\{c_i, c_j\}X_A = 0$. Thus unequal propagator poles can arise with equal primitive pairing vertices. This finite mechanism supplies neither three observed quark families nor renormalized physical quark masses. Exact controls and complete records are reproduced by `model/quark_flavour_resolution` [1].

A separate electrostatic extension embeds the full native source as one charge island on a thirteen-vertex cone, with a compensating reference and a finite flux register. Solving its Gauss constraints in the positive electric energy $\sum_e E_e^2/2$ gives CQ_Y^2, where $C = 26/217$. Every target energy is translated by Cq^2 with its original residue. These are finite source-reference electrostatic energies. They supply no transverse photon dynamics, locally gauged 343-site matter source, or intrinsic rest-mass assignment. The exact Gauss isometry and complete shifted measures are reproduced by `model/quark_gauge_dressing` [1].

2.17 Neutral composite response and a coupled sterile control

The native neutral composite $J_s = \phi_0 c_{s,11} - \phi_1 c_{s,10}$ commutes with all twelve gauge currents and carries lepton number -1. The original unit source conserves lepton number. Consequently its anomalous expectation $\langle J_s(t) J_t(0)\rangle$ vanishes on every invariant reference; a symmetry-preserving elimination cannot generate a lepton-number-two term. This does not imply a gapless normal response. The actual local ground gives nine positive spectral lines. Its normalized cyclic resolvent has an exact eight-mode complementary selfenergy, with positive spectral variance approximately 36.4254564105 and all original success and failure weights retained.

On the full three-site ninety-fermion-mode source, neutral addition and removal preparations give an independently executed transport event. For the matching opposite-spin middle-site receiver,

$$\langle R|\psi\rangle = \langle R|H|\psi\rangle = 0, \qquad \langle R|H^2|\psi\rangle = i/3.$$

The complete third-order support has 15739 coordinates. At $t = 10^{-4}$ the spectral remainder bound encloses receiver probability between approximately 2.5920×10^{-18} and 2.9700×10^{-18}, retaining its full complement and quantum reference. These preparations are distinct from the local ground.

A declared sterile extension adds the left-handed conjugate n^c of a right-handed gauge singlet and the unit interaction

$$V_\nu = n^c_\uparrow J_\downarrow - n^c_\downarrow J_\uparrow + \text{h.c.}$$

The literal thirty-two-mode algebra verifies all gauge and spin Ward identities. With quark vertices disabled, the complete eight-fermion-mode and five-state-scalar control has dimension 1280, with 80 physical Gauss states. Its unique ground energy is -4. The active and sterile neutral probes have positive gaps $1, 5$. Sterile removal residues are $35/72, 1/72$ and active-composite residues are $35/72, 25/72$; addition gives the same respective weights. Every other physical energy, failure and outside-support outcome is retained. The normalized one-branch sterile propagator is

$$G(z) = \frac{z - 44/9}{(z-1)(z-5)}, \qquad \Sigma(z) = \frac{35/81}{z - 44/9}.$$

These two response poles are not counted as two observed neutrino species. A separate unit sterile Majorana interaction has its own recomputed ground and complete spectrum. Activating the quark

vertices leaks out of the lepton control, so neither control ground is transferred to the all-unit full source.

One declared positive energy unit s gives neutral gaps $s, 5s$. For illustration, an empirical normal-ordering scale with zero lightest mass gives $s = 5.01098792655 \times 10^{-11}$ GeV. This unit choice neither assigns an active or sterile species to either pole nor establishes a relativistic rest mass. The active tau antineutrino is a weak-flavour readout, distinct from the sterile coordinate. Three-family mixing and oscillation observables require their own construction. The exact source operators, complete spectra and transport bounds are reproduced by `model/neutrino_mass_response` [1].

2.18 Compiling the lepton effect into a free mass operator

The literal lepton analyzer acts on 3600 addresses and three character coordinates. Executing all 10800 columns gives a unitary U whose selected block K_{00} has squared singular values e_k and $\mathrm{Tr}(K_{00}^\dagger K_{00}) = 5/18$. Independent extraction of the configured charged-family polynomial verifies

$$FE_{\mathrm{native}}F^\dagger = C_{\mathrm{ex}}^\dagger C_{\mathrm{ex}}/3.$$

The exact prepared/native-pair factor is $5/4$. Controls reject an extra square, substitution of the native pair, active-only normalization and the conjugate Fourier orientation. This identity is a coefficient and frame theorem, independent of a physical mass assignment.

Lift U to the entire lepton doublet and its conjugate to the singlet on the full canonical anticommutation algebra. A rank-three selected projector Q gives $W = U^\dagger Q U$. With neutral background norm one, vacuum compression of the literal pairing kick gives the coefficient $Y = K_{00}^\dagger K_{00}$. This is an operator identity on all physical occupations, without replacing the source Hamiltonian by the desired effect. There are 90 physical fermion modes and 324000 enlarged modes per site. The mean interaction norm is $5/9$ per site and the uncompressed kick norm is six per site.

Apply N rounds of duration t/N, measure vacuum versus nonvacuum ancillary occupation, retain the committed classical history and its quantum outputs, and supply fresh ancillary vacuum. A common free Weyl kinetic operator commutes with the address analyzer. Interaction-picture expansion and telescoping give complete unhalved trace-norm error

$$\epsilon \le \frac{108 V^2 t^2}{N}$$

for V sites, before separately adding source-dilation errors. Quantum coherence within each classical history sector is retained. An unmeasured coherent purification has a different, order $N^{-1/2}$ comparison; the recorded bound is not applied to it.

The same native Nambu symbol has three charged free Dirac fields per valley with rest coefficients e_k, together with all eight sine valleys and 39 zero internal coefficients. A full sixteen-state canonical transformation identifies the charged pairing with the Dirac mass Hamiltonian and separately verifies its electric charge. Its complete addition/removal instrument retains 36 records and reads the one-particle energy e_k, without an extra square. A compensated four-effect Ramsey instrument reads probabilities $(1 \pm \cos(e_k t))/4$ and $(1 \pm \sin(e_k t))/4$. Complete-output error ϵ bounds the recovered energy error by $4\epsilon/t$ on the specified unaliased phase window. The positive electron denominator is certified before forming ratios.

One empirical electron anchor gives the three free coefficients

$$(0.00051099895069,\ 0.1056594144959,\ 1.7769849708134)\,\mathrm{GeV}.$$

The fixed phase $2/9$, scalar background and ancillary preparation are construction inputs. The uncompressed norm estimate certifies one-site error $\epsilon = 10^{-12}$ at $t = 1$ with 108×10^{12} rounds; this

round count is not executed. Section 2.19 gives a sharper bound and an exact reduction of the same compiler. Physical clock matching, a selected interacting Higgs vacuum, mirror removal and observed Standard Model identification are not inferred. The construction, complete finite controls and error budgets are reproduced by `model/lepton_mass_compiler` [1].

2.19 Exact active-mode compression and a leakage bound

The same neutral compiler admits an exact reduction of its ancillary coordinates. For character k, let p be the blank address, $v_k = U_k^\dagger p$, $\lambda_k = \langle p, v_k\rangle > 0$ and $e_k = \lambda_k^2 \in (0,1)$. Then

$$a_k = \frac{v_k - \lambda_k p}{\sqrt{1-e_k}}, \qquad B_k = (p, a_k), \qquad v_k v_k^\dagger = B_k \begin{pmatrix} e_k & \sqrt{e_k(1-e_k)} \\ \sqrt{e_k(1-e_k)} & 1-e_k \end{pmatrix} B_k^\dagger.$$

The two spin-oriented pairing blocks per character need twelve ancillary fermion modes in total. Retaining all ninety physical modes gives 102 modes per site. The exterior lift $\Gamma(T)$ uses B_k on a left field and $\overline{B}_k$ on its conjugate right field, fixes the physical modes, and satisfies $H_{\text{original}}\Gamma(T) = \Gamma(T)H_{\text{compressed}}$. The common Weyl hopping preserves these position-independent spans. Every omitted address mode starts and stays in vacuum. This identity holds on the complete physical occupation space and with arbitrary untouched references. Twelve ancillary modes are minimal among number-preserving mode embeddings that fix the physical modes and reproduce this first-order coupling.

The ancillary isometry also intertwines vacuum/nonvacuum flags and reconstructs the full quantum archive after every round. It preserves coherence within each committed history. For one oriented pair with kick $\mathcal{V}$, let P project onto fresh ancillary vacuum. On its physical range the leakage form is

$$P\mathcal{V}^2P - (P\mathcal{V}P)^2 = (1-e^2)I - e(1-e)(n_L + n_R).$$

Its maximum is $1-e^2$. Distinct pairs and sites emit into orthogonal ancillary sectors, so for V sites the full off-diagonal norm obeys

$$B^2 := \|(I-P)HP\|^2 = V\Big(6 - 2\sum_k e_k^2\Big) < 6V.$$

The kinetic term preserves P. Block Duhamel integration therefore gives vacuum-branch operator error at most $B^2h^2/2$ in one round of duration h, independently of the kinetic norm. Telescoping N rounds and retaining every nonvacuum output gives the complete unhalved bound

$$\|\rho_{\text{actual}} - \rho_{\text{ideal}}\|_1 \le \min\{2, 2B^2t^2/N\} \le \min\{2, 12Vt^2/N\}.$$

Per-round half-diamond implementation error η adds $2N\eta$. The sharper one-site bound at $t = 1$ certifies error 10^{-12} with 11724207179903 rounds; this resource count is not an executed campaign. The free masses, Ramsey inversion and positive ratio-denominator guard are unchanged. Complete finite occupation and quantum-archive controls are reproduced by `model/lepton_compiler_compression` [1].

2.20 Complete calibrated quadratic spectrum

A separate reference action uses the explicitly calibrated common-scale Yukawa, gauge, scalar and optional Weinberg inputs. Its literal native Higgs tensors give a 45×45 fermion mass kernel. Keeping both momenta $\pm(0,0,1\,\text{GeV})$ retains 180 physical fermion modes and the full 360-dimensional

Nambu symbol. An analytic mass frame preserves every charge, helicity, colour, momentum and conjugate label, including the small neutral amplitudes. The quadratic dispersion obeys $E^2 - |p|^2 = m^2$ in this declared action.

Differentiating the actual gauge-Higgs action gives tree coefficients approximately

$$(80.8480001663,\ 92.2264476377,\ 131.279873449)\,\mathrm{GeV}$$

for W, Z and the scalar, with zero photon and gluon rest coefficients. A global total-occupation cap two on the 56 boson modes has dimension 1653 and preserves colour rotations. The complete boson instrument retains 375 pointer/energy records, including 362 zero-weight records. Both fermion and boson instruments retain their complementary outcomes and compensating references for the symmetries preserved by the chosen Higgs background.

The catalogue covers seventeen elementary target types, complete antiparticle and mode multiplicities, and separately calibrated composite controls. The quark coefficients are supplied running parameters, the boson coefficients are tree-action quantities, and the optional neutrino input has its own scale-matching obligations. These conventions are not merged with the lepton compiler's free mass coefficients or the quenched hadron fits. Extra sterile, valley, complementary and response modes are classified as source modes, without declaring undiscovered particles. The complete finite construction and input ancestry are reproduced by `model/conditional_particle_spectrum` [1].

2.21 Joint quantum-number and mass records

On the same calibrated two-momentum source, electric charge, colour Casimir, spin about z and momentum have commuting Nambu lifts

$$Q_N = \mathrm{diag}(Q, -Q^T), \quad C_N = \mathrm{diag}(C, C^T), \quad S_N = \mathrm{diag}(S, -S^T), \quad P_N = \mathrm{diag}(P, -P^T).$$

Their commutators with the actual 360-dimensional Hamiltonian vanish. At the declared momenta $\pm(0, 0, 1\,\mathrm{GeV})$, write helicity as $h = S_N P_N / |P_N|$. Grouping by the measured tuple $(Q, C, 2h, P_z, E)$ gives 84 fermion projectors of total rank 180. Colour-triplet components occupy rank-three projectors. Circular polarization similarly gives 28 boson projectors of total rank 56, with rank-eight adjoint gluon groups and no longitudinal massless oscillator. The adjoint Casimir is exactly $3I_8$. Species names enter only as a comparison dictionary after this grouping.

These 112 one-excitation projectors, the vacuum and the entire many-excitation complement sum to identity on the full $2^{180} \times 1653$-dimensional source. A fair mixture of the complete fermion and boson probes, composed with these 114 projectors, retains 570 pointer/projector records. On the declared vacuum, 113 are positive and 457 are zero. Fermion and boson group probabilities are respectively rank/360 and rank/116, and the boson no-operation vacuum has probability 1/58. Charge and colour references are retained. Within a rank-r colour group their conditional output is coherently entangled; dephasing its colour coordinates changes that density by unhalved trace norm $2(1 - 1/r)$.

Exact mass frames attach $E^2 = p^2 + m^2$ to these projectors. Rational square-root enclosures distinguish the optional neutrino energies even when binary64 subtraction would lose their mass differences. At momentum one GeV, energy bins of half-width $10^{-6}\,\mathrm{GeV}$ merge these neutral energies, whereas half-width $10^{-26}\,\mathrm{GeV}$ resolves them in the fixed input packet. This is an arithmetic and instrument definition, without a physical detector-precision claim. Opposite-helicity Majorana records do not add sterile species.

Separately, the compiled lepton's literal sixteen-state Dirac block has fifteen nonzero joint charge/spin/energy projectors. Each family has sixty complete reservoir and detector records;

forgetting spin recovers the original thirty-six exactly. Momentum is zero and no rest-frame helicity is assigned. Both constructions are reproduced by `model/particle_joint_mass_identity` [1]. The calibrated Yukawa packet and electron-anchored compiler retain their separate input ancestry and clocks.

2.22 Family invariants and an interference measurement

The native three-family source has forty-five internal coordinates and ninety spin-resolved fermion modes. Its twelve gauge currents commute with all forty-five generators of family-coordinate $U(3)^5$. With canonical kinetic terms, the charged kernels transform as

$$Y_u \mapsto U_Q^\dagger Y_u U_u, \qquad Y_d \mapsto U_Q^\dagger Y_d U_d, \qquad Y_e \mapsto U_L^\dagger Y_e U_e.$$

This is equivalence under a change of coordinates when preparations, detectors and references transform together. The exact generic tangent rank is 41 on the 54 real charged coefficients, leaving thirteen invariants: nine singular values, three quark mixing angles and one charge-conjugation/parity (CP) phase. A complex symmetric neutral kernel transforms as $M_\nu \mapsto U_L^T M_\nu U_L$. Its generic extension has 66 real coefficients and orbit rank 44, leaving twenty-two invariants. On the rank-two neutral stratum the corresponding count is $64 - 44 = 20$: a zero mass also removes one independent Majorana phase. The supplied normal-ordering, zero-lightest-mass packet consequently has nineteen empirical flavour coordinates and one independent unmeasured Majorana-phase choice.

The native current and Higgs identities admit all twenty-seven complex charged family coefficients: their embedding has positive diagonal Gram matrix, and the full ninety-mode pairing is antisymmetric. They impose no additional linear family restriction. The specified common-feature trial has $[Y_u Y_u^\dagger, Y_d Y_d^\dagger] = 0$. This alignment is not forced by those identities. An exact admitted counterexample uses

$$Y_u = \mathrm{diag}(1,2,4), \quad Y_d = V\,\mathrm{diag}(2,3,5), \quad Y_e = \mathrm{diag}(1,3,7),$$

where V has mixing sines $3/5, 5/13, 8/17$ and phase $\pi/2$. It gives

$$J = \mathrm{Im}(V_{00} V_{11} \overline{V_{01} V_{10}}) = \frac{51840}{830297}.$$

The actual native weak-raising current has this V block in all three colour copies of the mass frame. No measured flavour target selects the counterexample.

A declared two-path current instrument prepares family pairs $(0,1)$ and $(1,0)$ in distinct colour registers and applies $V \otimes V$. For every output pair (r,s), put $a_{rs} = V_{r0} V_{s1}$ and $b_{rs} = V_{r1} V_{s0}$. Its eighteen Kraus rows are

$$K_{rs,+} = \frac{(a_{rs}, -i b_{rs})}{\sqrt{2}}, \qquad K_{rs,-} = \frac{(a_{rs}, +i b_{rs})}{\sqrt{2}}, \qquad \sum_{r,s,\pm} K_{rs,\pm}^\dagger K_{rs,\pm} = I_2.$$

On the equal path superposition, $p_{rs,\pm} = |a_{rs} \mp i b_{rs}|^2/4$ and $J = p_{01,-} - p_{01,+}$. Every other outcome and the path reference are retained. Conjugating V preserves its singular-value and incoherent probability data but reverses this contrast. Thus one real signed contrast supplies information absent from masses and $|V_{ij}|^2$. The instrument is a coherent current service with a common charge reference, without a native pulse compilation or laboratory flavour assignment. Exact orbit calculations, full-source identities and complete records are reproduced by `model/source_flavour_identifiability` [1].

2.23 A finite-frequency amputated scalar–fermion response

The literal 320-state lepton/scalar control permits a full three-point calculation with six fermion fields and two scalar fields. Its ground projector P_0 has rank two and energy -2. Preparing the marginal $I_{320}/320$ of a source/dual-reference entangled state and measuring the complete energy gives $\rho_0 = P_0/2$ with probability $1/160$. All other energy outcomes and the quantum reference are retained. Define $R(z) = (z - H - 2)^{-1}$ and $O = (c_i, c_j, \phi_k^\dagger)$. For $\omega_1 + \omega_2 + \omega_3 = 0$, the complete ordered transform is

$$T_{ij,k} = \sum_{(a,b,c)\in S_3} \mathrm{sgn}_F(a,b,c)\,\mathrm{Tr}[\rho_0 O_a R(i\omega_a) O_b R(-i\omega_c) O_c]\,,$$

where the sign changes exactly when the fermions exchange order. This follows by integrating the two positive Euclidean time intervals. All one-point functions vanish, so the transform is connected. The full resolvent retains every intermediate state, including 198 scalar-occupied dark states.

The complete two-point functions used for amputation are

$$G_{ij}(z) = \mathrm{Tr}\,\rho_0[c_i R(z) c_j^\dagger - c_j^\dagger R(-z) c_i],$$
$$D_{ab}(z) = \mathrm{Tr}\,\rho_0[\phi_a R(z) \phi_b^\dagger + \phi_b^\dagger R(-z) \phi_a].$$

In the native frame, G is diagonal and D is scalar on the doublet. Equal addition/removal matrices give $D(z) = D(-z)$, fixing the conjugate scalar-leg convention. At each specified nonzero imaginary frequency,

$$\Gamma_{ij,k}(\omega_1,\omega_2) = \frac{T_{ij,k}}{G_{ii}(i\omega_1) G_{jj}(i\omega_2) D_{kk}(-i(\omega_1+\omega_2))}.$$

The exact denominators are nonzero. All seventy-two components at $(1,2)$ are retained, of which eight are nonzero. In particular,

$$\Gamma_{04,0}(1,2) = \frac{474925}{2394532}, \qquad \Gamma_{04,0}(2,3) = \frac{11640331925}{26140075569}.$$

These differ from the primitive unit contact. Canonical scalar normalization $\phi/\sqrt{14/5}$ multiplies Γ by $\sqrt{14/5}$.

Summing the four native oriented scalar/pair components and all six orders at $(1,2)$ gives an operator of norm at most eight. Its quotient B by eight commutes with all twelve native currents. The four effects $(I \pm \mathrm{Re}\,B)/4$ and $(I \pm \mathrm{Im}\,B)/4$ are positive and sum to identity. Their declared square-root instrument gives four successful pointer records alongside thirteen failed preparation energies, hence seventeen original records of total weight one. This finite response has no selected physical mass shell, spacetime momentum or renormalized Standard Model coupling. Exact full-source resolvents, signs, propagators and pointer effects are reproduced by `model/frequency_yukawa_vertex` [1].

2.24 A neutral scalar occupation pole

On the full thirty-mode local source, let N be the capped scalar occupation and write the unit-coupling Hamiltonian as $H = V + N$. The literal operators obey

$$N^2 = N, \qquad \{2N - I, V\} = 0, \qquad H^2 - H = V^2, \qquad [N, V^2] = 0.$$

Let λ be the V^2 eigenvalue selected by the exact ground projector and put $\Delta = \sqrt{1+4\lambda}$. The two energies in that invariant pair are $(1 \pm \Delta)/2$. The scalar occupation has ground mean $n = (1-\Delta^{-1})/2$ and connected variance $w = n(1-n) = \lambda/(1+4\lambda)$. Its retarded response is exactly

$$D_N(z) = \frac{2\Delta w}{z^2 - \Delta^2}, \qquad D_q(z) = \frac{1}{z^2 - \Delta^2}, \quad q = \frac{N-n}{\sqrt{2\Delta w}}.$$

The positive pole is approximately 23.4611705885 source units. The elementary charged scalar field has a different response. The neutral operator here is constructed from the capped fields themselves: $\sum_k \{\phi_k^\dagger, \phi_k\} = 4I - 3N$.

An occupation measurement followed by complete energy measurement has probabilities n^2, w for occupied records at energy transfers $0, \Delta$ relative to the initial ground, and $(1-n)^2, w$ for blank records. Each occupation also retains its exact zero-weight outside-pair outcome. These six records sum to one; the measurement supplies mean work $2\Delta w$. This is a local scalar response pole without a spatial propagation or physical Higgs identification. Exact algebraic intervals and the complete instrument are reproduced by `model/native_scalar_density_pole` [1].

2.25 Pion operators and calibrated composite energies

The retained quenched Wilson calculation has degenerate light quarks and no electromagnetic field. Normalized pseudoscalar flavour matrices $M_+ = |u\rangle\langle d|$, $M_- = |d\rangle\langle u|$ and $M_0 = (|u\rangle\langle u| - |d\rangle\langle d|)/\sqrt{2}$ have charges $+1, -1, 0$ under $[Q, M]$, with $Q = \mathrm{diag}(2/3, -1/3)$. Identical light-quark propagators give connected contraction $\mathrm{Tr}(M_a^\dagger M_b)C(t) = \delta_{ab}C(t)$. The neutral disconnected contraction cancels because $\mathrm{Tr}\, M_0 = 0$. Thus this source supplies one isospin-degenerate pseudoscalar decay energy for the three pion charge labels.

The archived fit tunes the light-quark point using the supplied ratio $c = m_\pi/m_\rho = 0.1780$. Choosing the empirical rho anchor $R = 0.77526\,\mathrm{GeV}$ then gives

$$a^{-1} = \frac{R}{am_\rho}, \qquad m_\pi = cR = 0.13799628\,\mathrm{GeV}, \qquad m_N = R\frac{am_N}{am_\rho}.$$

The pion value is fixed by these two inputs. Its conditional fit variance vanishes by shared-scale cancellation, without implying zero physical uncertainty. The independently computed nucleon ratio gives approximately $1.1275\,\mathrm{GeV}$, with paired-stream statistical and fit-sensitivity errors retained. An alternative conventional string scale $\sqrt{\sigma} = 0.44\,\mathrm{GeV}$ gives approximately $0.1054020\,\mathrm{GeV}$ for the degenerate pion. Both conversions are reported, exposing their scale disagreement.

The complete four-kappa table includes the unresolved lightest nucleon estimate. This calculation replays reduced correlator and Wilson-loop tables on $12^3 \times 24$ at $\beta = 5.7$; the raw gauge configurations are absent. Quark loops, electromagnetic and light-quark splitting, finite-volume and continuum control, and transfer to the native observer source and clock are independent physical obligations. The flavour identities, full fit replay and conditional conversions are reproduced by `model/pion_mass_readout` [1].

2.26 Correlated hadron fits and paired GeV uncertainties

The retained thirty-two Wilson streams also support a fit with the full time covariance in each declared window. If j_i is its leave-one-stream correlator mean, the compatible stream mean is $x_i = 32\bar{j} - 31j_i$. For m included streams, the covariance of their mean is

$$C = \frac{1}{m(m-1)} \sum_i (x_i - \overline{x})(x_i - \overline{x})^T.$$

Every outer jackknife replica refits with its own thirty-one-stream covariance. The prescribed rules require enough streams, positive diagonal entries, full correlation rank and condition number at most 10^8. All forty-eight windows and all their replicas pass. No pseudoinverse, shrinkage or discarded replica supplies an accepted fit. The maximum replica condition number is approximately 4.50×10^7. At fixed decay coefficient m, the amplitude is profiled exactly as $A = f^T C^{-1} y/(f^T C^{-1} f)$; a bounded numerical search over $0.02 \le m \le 3$ supplies the fitted coefficient. The fit window and search rule do not use empirical hadron masses.

Paired stream labels propagate through the chiral extrapolation and potential-scale conversion. The table gives GeV central values with statistical errors and the separate variation across the specified hadron-window, sink, chiral-subset and potential fits:

calibration	channel	m (GeV)	σ_{stat} (GeV)	Δ_{fit} (GeV)
$m_\rho = 0.77526$	π	0.13799628	0	0
	ρ	0.77526	0	0
	N	1.130525	0.030231	0.053203
$\sqrt{\sigma} = 0.44$	π	0.106231	0.001401	0.011761
	ρ	0.596801	0.007873	0.066072
	N	0.870287	0.024191	0.076896

The rho-anchored pion and rho values are fixed calibration identities; their vanishing conditional fit errors do not express physical accuracy. The correlated nucleon ratio is 1.458252 ± 0.038995. Independent diagonal-time fits are retained as controls, including the rho-anchored nucleon value 1.127499 GeV. The inherited lightest nucleon quality flag is also retained.

This calculation treats the time covariance explicitly; the chiral regression uses diagonal mass-error weights and is not a fully correlated errors-in-variables fit. Its errors exclude sea quarks, continuum and finite-volume extrapolation, anchor uncertainty and native-clock matching. The same quenched finite source supplies all results in `model/correlated_hadron_spectrum` [1].

2.27 Open items

Open are the absolute lepton scale, the quark masses and the split between up and down, the quark and lepton mixing matrices, the neutrino masses, the Higgs mass term, which is of cell size in every regulator examined, the declared parts of the coupling chain, and a law that writes the continuous port phases.

3 Native populations and conditional thermal laws

The native population process, a reversible bath extension and a calibrated free particle gas have separate dynamical laws. Their observation counts, costs and temperature parameters are kept distinct. The finite constructions below give complete probability records; their interpretation as an early-Universe history requires a physical state, energy and clock map.

3.1 Stationary native populations and the thermal obstruction

Consider the closed triangular cell with three loads $x_i \in \{0, \dots, 5\}$ and opposite face classes $k, -k \in \mathbb{Z}_6$. Its unchanged native seam repair and register acceptance give 1296 states and 7776 original seam/coin records, each attempted event having weight 1/6. Rejected twists retain their load repair,

and no-operation records count as attempts. The conserved total load is $M = \sum_i x_i$. The native cost

$$E(x,k) = 5\sum_i x_i^2 + 2\big(C(k) + C(-k)\big), \qquad C = (0,5,8,10,8,5),$$

does not increase on any positive transition of the row kernel K.

Proposition 3.1 (Full-support equilibrium obstruction)**.** *In every sector $1 \le M \le 14$, no full-support probability law is stationary for K. In particular, $\exp(-E/T)/Z_M(T)$ is neither stationary nor detailed-balanced for any finite $T > 0$. The endpoint sectors $M = 0, 15$ have identity dynamics on their six flux states.*

Proof. The pointwise drift $d = KE - E$ is nonpositive. Each nontrivial sector has an executed strict-descent edge, so $d(s) < 0$ for some state s. A stationary law would satisfy $\sum_s p(s)d(s) = \langle pK, E\rangle - \langle p, E\rangle = 0$, contradicting full support. A strict-descent edge also has zero reverse probability. The generic finite push/pull argument is kernel-checked in `Lean/Dynamics/StationaryLyapunov.lean`; the source supplies its exact drift premises. □

The positive-edge graph has 46 closed classes: 36 absorbing states (q,q,q,k) and ten three-state classes of balanced load permutations with zero flux. Each three-state kernel is $\frac{1}{2}I_3 + \frac{1}{6}\mathbf{1}\mathbf{1}^T$. Thus all stationary measures form the simplex of arbitrary mixtures of the 46 class-uniform laws. There are 66 recurrent states and 1230 transient states. Gibbs weights restricted to this recurrent set are stationary at every positive temperature, but their inter-class weights are an additional choice. The classification and every original attempt record are reproduced by `model/native_thermal_admission` [1].

3.2 Exact absorption, lifetimes and boundary response

Let τ be the first attempted-event count at which a closed class is reached. For class c, write $h_c(s) = \Pr_s(X_\tau \in c)$, $t(s) = \mathbb{E}_s\tau$, $g(s) = \mathbb{E}_s\tau^2$ and $u_c(s) = \mathbb{E}_s[\tau \mathbf{1}_{X_\tau \in c}]$. Their boundary values on closed states are $h_c = \mathbf{1}_c$ and $t = g = u_c = 0$. On transient states,

$$h_c = Kh_c, \qquad t = 1 + Kt, \qquad g = 2t - 1 + Kg, \qquad u_c = h_c + Ku_c.$$

Ordering by load square and register cost reduces exact rational elimination to native blocks of at most six states. The source survival bound $\Pr_s(\tau > n) \le 25^{-\lfloor n/32 \rfloor}$ proves uniqueness and finiteness; all equations are verified on every original state.

For independent uniform initial loads and initial flux zero, the limiting oriented-face distribution, in class order $0, \dots, 5$, is

$$\left(\frac{12817}{13824}, \frac{499}{27648}, \frac{25}{1728}, \frac{1}{128}, \frac{25}{1728}, \frac{499}{27648}\right).$$

The exact mean lifetime is 5557709/1125000 attempts, approximately 4.94019, and the largest mean over all starting states is 15. The second moment is 177070889/2812500. All 46 terminal probabilities and their conditional lifetimes, including zero-probability classes, are retained. These are relaxation lifetimes, with no particle-decay width assignment.

After absorption the face class is fixed. Coupling each trajectory to its eventual class gives

$$\mathrm{TV}\big(\mathrm{class}(p_n), \mathrm{class}(p_\infty)\big) \le \Pr(\tau > n) \le 25^{-\lfloor n/32 \rfloor}.$$

The three-state recurrent kernels contract their nonconstant modes by 1/2. Splitting at $m = \lfloor n/2 \rfloor$ consequently gives the full state bound $\mathrm{TV}(p_n, p_\infty) \le 25^{-\lfloor m/32 \rfloor} + \frac{2}{3}\,2^{-(n-m)}$. For example, 96

attempts guarantee face-class error at most $1/15625$ for every initial law. This source-count criterion quantifies cessation of native class changes without a cosmic temperature or time.

Absorption is affine in preparation and contracts total variation. The supplied asymmetric input $(0, 1, 5, 1)$ gives terminal oriented contrast $\Pr(k = 1) - \Pr(k = 5) = -91/384$. Averaging the two opposite faces is even for every state, so that contrast is an oriented record and is not baryon number. A separate control resets to this input with probability $1/10$ after eight attempts from $(0, 1, 5, 0)$. Both original reset/no-reset branch vectors, of masses $1/10$ and $9/10$, are propagated independently and sum exactly to every saved marginal. Relaxation counts for this supplied boundary operation start after the reset. The complete distributions and bounds are reproduced by `model/native_freezeout_response` [1].

3.3 A declared reversible bath and its cooling records

A thermal extension can be constructed explicitly on the same states. For $x \neq y$ set $s_{xy} = \max(K_{xy}, K_{yx})$ and $\Lambda = 2 \max_x \sum_{y \neq x} s_{xy} = 12$. The symmetric support adds 3924 reverse edges to 4104 native directed nonidentity edges. It has 26 connected components: the fourteen nontrivial load sectors and twelve endpoint singletons. This reverse support is necessary for full-support detailed balance; its rates are declared choices.

For $0 < q \leq 1$ and $\beta = -\log(q)/2$, propose y with probability s_{xy}/Λ and accept with probability $\min\{1, q^{(E_y - E_x)/2}\}$. Keep rejected proposals and idle records. The 17352 complete records per attempt give

$$\frac{P_{xy}}{P_{yx}} = e^{-\beta(E_y - E_x)}.$$

Hence each component has its Gibbs equilibrium, while component weights are conserved and free. Exact rational checks use $q = 1, 3/4, 1/2, 1/4$. The construction is a bath extension; the obstruction for unchanged native repair is unaffected.

For $p' = pP$, let $Q = \sum_{xy} p_x P_{xy}(E_y - E_x)$ be heat into the system and let S use natural logarithms. Detailed balance gives

$$S(p') - S(p) - \beta Q = \sum_{xy} p_x P_{xy} \log \frac{p_x P_{xy}}{p'_y P_{yx}} \geq 0.$$

The right side is relative entropy between normalized forward and reverse joint laws. Accepted records reverse their endpoints; rejected proposal labels and idle marks are retained. The identity therefore holds for the complete marked law as well.

Two executed schedules use $q = 3/4, 1/2, 1/4, 1/2, 3/4, 1$ with 32 or 256 attempts per setting, retaining full state laws and finite-time departures from Gibbs equilibrium. Exact coefficient-rounding errors and positive sparse-sum bounds give final probability L^1 error below 9.1×10^{-13} and 7.3×10^{-12}, respectively, assuming binary64 products and sums without overflow or underflow, without renormalization. The cost, bath inverse temperature and event count have no physical calibration. The records and entropy identity are reproduced by `model/native_thermal_completion` [1].

3.4 Conditional thermodynamics of the calibrated spectrum

The separate calibrated quadratic menu contains 17 particle types, with 90 fermionic and 28 bosonic degrees per momentum. Antiparticles and both neutrino helicities are counted once; no populated sterile species is added. On its two actual nonzero momenta the fermion partition function has 180 canonical factors. The global boson occupation cap of two gives $1 + 56 + 56 \cdot 57/2 = 1653$ states.

Complete Gibbs enumeration has high-temperature mean total boson number $3248/1653 < 2$. It differs from an uncapped Bose gas even on those same momenta. All energies are excitation energies above the selected quadratic vacuum.

For a declared continuum free-gas extension, put $y = m/T$, $\epsilon = \sqrt{x^2 + y^2}$ and $f = (e^{\epsilon} \pm 1)^{-1}$, with plus for fermions. Per degree,

$$\frac{n}{T^3} = \frac{1}{2\pi^2} \int_0^\infty x^2 f \, dx, \qquad \frac{\rho}{T^4} = \frac{1}{2\pi^2} \int_0^\infty x^2 \epsilon f \, dx, \qquad \frac{P}{T^4} = \frac{1}{6\pi^2} \int_0^\infty \frac{x^4 f}{\epsilon} \, dx.$$

At zero chemical potential, $s = (\rho + P)/T = dP/dT$. Differentiation at fixed mass gives positive heat capacity and $0 < c_s^2 = (dP/dT)/(d\rho/dT) \leq 1/3$; integration by parts gives $\rho - 3P \geq 0$. Independent fugacity series check the numerical quadrature. Its reported errors are numerical estimates, with no certified interval or physical-accuracy claim.

The massless limit gives $g_\rho = g_s = 427/4 = 106.75$ and number count $191/2$. Quark/lepton number fractions are $108/191$ and $27/191$; their energy fractions are $252/427$ and $63/427$. These are conditional ideal-gas counts. With the additional assumptions $a^3 s =$ constant and a flat Friedmann density law, the same curve gives $a(T)/a(T_*) = [s(T_*)/s(T)]^{1/3}$ and $H(T)/H(T_*) = [\rho(T)/\rho(T_*)]^{1/2}$. Neither cosmological equation is derived from these particle counts.

The charged-fermion coefficients are held at the supplied renormalization scale $\mu = 91.1876\,\mathrm{GeV}$; the bosons use a supplied tree background and the neutrinos an optional Weinberg prescription. The calculation supplies no thermal pole matching or temperature running. It omits confinement and hadron replacement, a thermal Higgs background and neutrino decoupling. Compared at the same tabulated temperatures with the interacting equation of state of Saikawa and Shirai [37], the ideal energy count is about 6.52% higher near 10 GeV and 2.44% higher near 1 TeV. Near 10 MeV it is about 4.56 times the tabulated count, exposing the inadmissible low-temperature free coloured gas. The comparison is not a native nucleosynthesis or relic abundance prediction. Its complete finite-source controls, conditional curves and input ancestry are reproduced by `model/particle_thermal_history` [1].

3.5 Complete finite thermal number records

On the same calibrated quadratic source, the simultaneous one-particle energy, charge, colour Casimir, helicity and momentum eigenspaces give 84 fermion groups and 28 boson groups. For a group g, define $N_g = \sum_{j \in g} \beta_j^\dagger \beta_j$ using the source's canonical quasiparticle fields. If $L_g = \{0, \ldots, r_g\}$ for a fermion group of rank r_g, and $L_g = \{0, 1, 2\}$ for capped bosons, then

$$\Pi_{g,n} = \prod_{\ell \in L_g \setminus \{n\}} \frac{N_g - \ell I}{n - \ell}$$

are complete orthogonal projections. The N_g commute, so products of these projections define a joint instrument on every occupation, including every untouched reference. Each group contains whole colour fibers. Its number projections therefore commute with the global colour action and preserve coherence within each colour fiber. They do not measure a many-body total colour Casimir.

The fermion count law has 2^{132} outcomes, represented by 84 complete binomial factors. The boson count law has 435 nonzero-rank outcomes, whose projector ranks sum to 1653. All count vectors with total boson occupation above two have the exact zero effect. To attach the helicity labels, the calculation constructs the actual circular-polarization rotation and its full 1653-dimensional symmetric-Fock lift. This rotation commutes with the excitation energy and charge, hence leaves the finite Gibbs state invariant. Thus the original occupation weights can be pushed forward to joint group counts without measuring colour coordinates or substituting a different thermal state.

At the supplied temperatures $T = 0.1, 1, 10, 100\,\mathrm{GeV}$, complete group and species count laws retain every occupation and its original weight. Count fractions mean $\mathbb{E}[N_g]/\mathbb{E}[N_{\mathrm{total}}]$, with empty trials retained. Before any additional particle probe,

$$\Pr(N_{\mathrm{total}} \le 1) = \frac{1 + \sum_{i\in F} e^{-E_i/T} + \sum_{j\in B} e^{-E_j/T}}{Z_F Z_{B,\le 2}}.$$

The many-excitation probability is about 1.98×10^{-5} at $0.1\,\mathrm{GeV}$; at $1\,\mathrm{GeV}$ the complementary vacuum-or-one-excitation probability is about 2.79×10^{-18}. The complete number instrument therefore handles thermal preparations far outside a one-particle sector. These are finite calibrated count measurements under a declared number-readout service, without a native temperature or cosmological abundance identification. Complete words, factorized laws, logarithms of underflowed probabilities and independent energy, charge and momentum controls are reproduced by `model/thermal_particle_records` [1].

3.6 Certified equilibration within each bath component

The reversible 1296-state bath has a quantitative equilibration bound without assuming that its numerical eigensolver returns exact eigenvectors. On a component let π be its normalized Gibbs law and $S = \mathrm{diag}(\sqrt{\pi})P\,\mathrm{diag}(\pi^{-1/2})$. Its diagonal entries are rational at the four declared q values, and its off-diagonal entries are rational multiples of square roots of rational powers of q. Integer square-root enclosures therefore bound every matrix entry. For a numerical eigenbasis V and diagonal D, let

$$\|V^*V - I\| \le \epsilon \le \tfrac{1}{2}, \qquad \|S - VDV^*\| \le \rho.$$

Both bounds include matrix-entry and floating dot-product rounding. The polar unitary U satisfies $\|V - U\| \le \epsilon$ and $\|V\| + 1 < 3$, so

$$\|S - UDU^*\| \le \rho + 3\|D\|\epsilon.$$

Weyl's inequality encloses every eigenvalue. Laziness gives nonnegative eigenvalues. A certified lower spectral gap $g > 0$ consequently implies

$$\mathrm{TV}(pP^n, \pi) \le \min\left\{1, \frac{1}{2}\sqrt{\pi_{\min}^{-1} - 1}(1-g)^n\right\}$$

for every initial law supported in that component. The global bound is the component-weighted sum; no mixing between the 26 components occurs.

All 56 nontrivial component/temperature cases have positive certified gaps. The minima at $q = 1, 3/4, 1/2, 1/4$ are approximately 1.3663280×10^{-2}, 3.2004108×10^{-3}, 9.2784406×10^{-5} and 1.8385806×10^{-7}. The twelve singleton components require zero equilibration steps. At the final cut of the six-setting cooling schedule, the 32-attempt and 256-attempt protocols have total-variation distance from their component-weighted Gibbs laws in $[0.0965537485972, 0.0965537485982]$ and $[0.0015552032635, 0.0015552032709]$, respectively. These bounds concern terminal state marginals; the complete attempt archive retains its initial-state information. The exact component weights, all original records, gap enclosures and rounding bounds are reproduced by `model/native_bath_mixing` [1]. Its cost temperature and attempt count have no identified GeV or cosmic-time conversion.

3.7 Quantum thermal collisions on the calibrated particle source

A different supplied bath acts directly on the calibrated source with 180 fermionic modes and 56 globally capped bosonic modes. For a whole fermionic colour group g of rank r_g, its Gibbs density is scalar on each occupation subspace. Declare the collision

$$\Phi_g(X) = \tfrac{1}{2}X + \tfrac{1}{2}\tau_g \operatorname{Tr}_g X.$$

The public flag records hold or replacement. A replacement transfers the old group and its reference correlations into a quantum bath register and supplies a Gibbs purification. The complete dilation has Kraus coordinates $\sqrt{\tau_g(a)/2}|a\rangle\langle b|$, in addition to $I/\sqrt{2}$. Keeping a, b coherently in the bath preserves whole-colour covariance; the public pointer does not measure a colour component.

On the actual capped boson space let Π_n project onto total number $n = 0, 1, 2$, and set $q_j = e^{-E_j/T}$, $Q = \sum_j q_j$. With a positive dyadic $\alpha \le 1/(Q+2)$, use

$$K_{n,j,-} = \sqrt{\alpha}\, a_j \Pi_n, \quad K_{n,j,+} = \sqrt{\alpha q_j}\, a_j^\dagger \Pi_n, \quad K_{n,0} = \sqrt{I - \alpha R}\, \Pi_n, \qquad R = N + \mathbf{1}_{N<2} \sum_j q_j (N_j + 1).$$

Since $R \le Q + 2$, the Kraus adjoint products sum to identity on all 1653 states. There are 339 fine quantum-coordinate operators, including 112 exact zeros, and 171 public number/group records. Mode coordinates inside a colour group are retained as quantum bath coordinates. Every occupation edge satisfies Gibbs detailed balance.

Two boson sweeps dominate the completely positive vacuum replacement map with coefficient $\delta = 2\alpha^2$: on number sectors $0, 1, 2$, the retained two-hold, removal/hold and two-removal paths have coefficients $(1 - \alpha Q)^2$, $\alpha(1 - \alpha Q)$ and $2\alpha^2$. All are at least δ. After n fermion sweeps the probability that all 84 groups have been replaced is $(1 - 2^{-n})^{84}$. Thus, for every source/reference state, the unhalved terminal trace distance to $\tau \otimes \rho_R$ is at most

$$\min\{2,\ 2[1 - (1 - 2^{-n})^{84}] + 2(1 - \delta)^{\lfloor n/2 \rfloor}\}.$$

The complete archive retains the transferred information and is not asserted to converge to an input-independent state. Sufficient sweep budgets for error below 10^{-6} at $T = 0.1, 1, 10, 100\,\mathrm{GeV}$ are 82, 4648, 23914 and 42158. These are proved budgets, distinct from the executed 32- and 256-sweep thermal schedules.

The schedules start at supplied 100 GeV Gibbs equilibrium and use $10, 1, 0.1, 1, 10, 100\,\mathrm{GeV}$. They retain every fermion group occupation factor and all 1653 boson populations at each cut. Whole-group replacement can produce non-Gaussian fermion factors; their count laws are not replaced by binomials with the same mean. Rational Boltzmann enclosures and positive sparse recurrences bound the complete source-state numerical L^1 error by 2.12×10^{-11} across the executed cuts. At fixed Hamiltonian, Gibbs preservation gives $\Delta S - \beta Q = D(\rho\|\tau) - D(\Phi\rho\|\tau) \ge 0$, also with an untouched reference. The collision service and bath preparation are supplied; native repair preparation and a physical clock are separate premises. The complete channel and records are reproduced by `model/quantum_thermal_relaxation` [1].

3.8 Pair production under a declared expansion

A supplied scale factor can also change the actual two-momentum fermion Hamiltonian while preserving its mass kernel: replace $p_z = \pm 1$ by $p_z = \pm 1/a$. The exact parent mass-basis transformation splits the 360-dimensional Nambu symbol into 90 physical pairs. Each literal block is $p\sigma_z + sm\sigma_x$, with an exactly checked sign $s = \pm 1$; its particle-hole partner has the opposite sign.

The named positive-energy columns fix their phases. For positive-mass Bogoliubov matrix S and $D_s = \mathrm{diag}(1, s)$, the even Fock unitary is $D_s X S X D_s$, where X exchanges the two coordinates. A fixed exterior-algebra permutation from the original modes to these pairs retains the sign of every occupied inversion. The product of the disjoint even pair unitaries therefore defines the complete source unitary, including arbitrary references.

A sudden change $a = 1 \to 4$ gives vacuum pair probability

$$c_m = \frac{1}{2}\left[1 - \frac{m^2 + 1/4}{\sqrt{m^2+1}\sqrt{m^2+1/16}}\right].$$

The executed smooth-ramp comparisons use $a(s) = 1 + 3(3s^2 - 2s^3)$, durations $\tau = 1, 8$ in the declared GeV^{-1} parameter, and $64, 256, 1024$ exact midpoint constant segments. All twelve fermion types and their canonical conjugate partners are retained. The massless mode has exactly zero production. On initial occupation f, a pair with Bogoliubov coefficients (α, β) has probabilities

$$((1-f)^2 - (1-2f)c,\ f(1-f),\ f(1-f),\ f^2 + (1-2f)c), \qquad c = |\beta|^2,$$

in order $00, 10, 01, 11$, and even coherence $\rho_{00,11} = s(1-2f)\alpha^*\beta^*$. This specifies the full Gaussian output for the supplied vacuum and 1 GeV thermal inputs. The final instantaneous number projectors have physical momentum $\pm 1/4$, with the same charge, colour and helicity group lineage. Their joint law is the product of all 90 two-group count polynomials; all 181 total-number outcomes and 84 group marginals are retained.

Directed integer bounds enclose every pulse. Exact dyadic product centers with accumulated operator errors avoid interval wrapping: if the current error is e, the next pulse-center error is ε and product rounding contributes r, unitarity of the ideal pulse gives $e' \le e + \varepsilon(1+e) + r$. Displayed count probabilities come from consistent pair parameters, with independently bounded interval laws. These floating point diagnostics are neither arbitrary interval midpoints nor renormalized outputs; their comparison tolerance is not a certified rounding bound. The interval laws enclose the exact ideal probabilities. The vacuum mean total count is about 3.28341935 for the quench, 2.60741353 for the $\tau = 1$, 1024-segment protocol and 0.03359484 for the $\tau = 8$, 1024-segment protocol.

The finite pulses also have a controlled continuous-ramp comparison. Put $h = \tau/N$, $M_1 = 9/(2\tau)$ and $M_2 = 117/(2\tau^2)$. A centered interaction-picture Dyson expansion bounds each pair's operator error by

$$\delta_m \le N\left[h^3\left(\frac{M_2}{24} + \frac{2mM_1}{12}\right) + \frac{3}{2}\left(\frac{M_1 h^2}{4}\right)^2\right].$$

The linear midpoint term cancels, the commutator has norm at most $2mM_1|t - t_c|$, and the higher Dyson terms obey this bound when $M_1 h^2/4 < 1$. The full Fock bound sums over all 90 pair multiplicities; twice that sum bounds reference-extended unhalved trace distance, capped at two. A single-pair error is not a full-source error. For mean counts, $0 \le N_{\mathrm{pair}} \le 2$ gives the sharper direct bound $|\Delta\langle N\rangle| \le \sum_{\mathrm{pairs}} \min(2, 2\delta_m)$, without a factor of 180. At 1024 segments the exact continuous-ramp vacuum mean lies in $[0, 0.2282]$ for $\tau = 8$ and in $[2.6040, 2.6109]$ for $\tau = 1$, both strictly below the quench mean. This certifies slower-ramp suppression for the two declared profiles. The scale law and evolution parameter are supplied. No boson production, Einstein backreaction, interacting pole matching or physical cosmological abundance is inferred. Complete states, phase checks and interval count laws are reproduced by `model/expanding_quadratic_particles` [1].

4 Internal quantum numbers

4.1 Complete response and the compact Lie algebra

The carrier is the oriented icosahedral boundary, with twelve ports, thirty seams and twenty faces. Its real port module under the proper rotation group A_5 decomposes as

$$V_{12} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \qquad \dim V_{12}^{A_5} = 1.$$

Assume a faithful real-linear response map $D : V_{12} \to \mathfrak{u}(\mathcal{K})$ whose image is the complete connected response algebra and is closed under commutators, with proper carrier rotations acting through the same connected inner response group up to operationally silent centralizer factors. The gauge paper [7] proves that the compact classification then gives

$$\mathfrak{g} \simeq \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3), \tag{1}$$

because inner rotations fix a centre of dimension at most one and a centre-free $\mathfrak{su}(2)^4$ cannot have a one-dimensional fixed space, which leaves the semisimple split $11 = 3 + 8$. It also constructs an explicit current law on the protected native carrier realizing $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Twelve ports alone do not imply (1), and carrier rechartings are distinct from physical spatial rotations of a relativistic scattering theory. This paper consumes the result and its twelve current matrices.

4.2 The matter module and its family factor

A declared spin lift to five internal coordinates and the nonvacuum even exterior module give

$$M_{15} = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \qquad \dim M_{15} = 10 + 5 = 15,$$

the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching [14]. The source-current lift preserves all brackets and has an exact intertwiner with this module. Its simultaneous primitive-charge and color/weak Casimir readouts give the following all-left convention.

Label	Color	Weak	$6Y$	Components
Q	$\mathbf{3}$	$\mathbf{2}$	1	6
u^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	-4	3
d^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	2	3
L	$\mathbf{1}$	$\mathbf{2}$	-3	2
e^c	$\mathbf{1}$	$\mathbf{1}$	6	1

The labels abbreviate computed finite quantum numbers; their identification with laboratory Weyl fields requires the physical current, spin and kinetic maps. The gauge paper [7] gives the anomaly scan, which selects this module among exterior masks up to conjugation, the $\mathbb{Z}_6$ global form and the conditional family theorem. For the particle construction we explicitly choose $M_{15} \otimes \mathbb{C}^3$, with gauge currents acting trivially on $\mathbb{C}^3$. A two-dimensional radial kernel or a winding number three is no proof of three generations, and equivalent triplet representations on spatial, color and family factors do not identify those factors.

Exchange records have a separate finite route. On the native ordered-cover transport, let $p_h(x)$ be the permutation parity accumulated by branch h. A coherent ordering bit gives

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I \otimes X, \qquad [K_h, Q] = 0.$$

An odd exchange loop acts as X on the bit, and the corresponding colored fermionic lift acts as $-$SWAP. The sectors survive complete record-adaptive native schedules in this specified memory design; dephasing or resetting the bit changes that conclusion. These are exchange-history sectors, distinct from the spatially localized texture charge of Section 7.

4.3 The conditional field action

Choose one complex weak doublet H of hypercharge $1/2$, canonical positive kinetic forms, physical Weyl statistics and a Lorentz-covariant local field interpretation in $3+1$ dimensions. Restrict the polynomial action to operator dimension at most four, modulo integration by parts. The invariant grammar then gives the gauge kinetic terms, scalar kinetic term and potential, fermion kinetic terms and exactly three charged Yukawa lines. Independent gauge couplings, topological theta terms and an additive vacuum-energy constant are also free. In the four-component convention the charged Yukawa part is

$$\mathcal{L}_Y = -\overline{Q}_L \widetilde{H} Y_u u_R - \overline{Q}_L H Y_d d_R - \overline{L}_L H Y_e e_R + \text{h.c.}$$

Each Y_a is a complex 3×3 matrix. The native invariant-line computation solves all twelve current equations for both Higgs components and gives one line for each allowed origin $(6Y_1, 6Y_2) = (-4,1), (1,2), (-3,6)$. Tensoring these lines with the gauge-trivial family factor leaves exactly the three matrix coefficients. Canonical kinetic normalization must precede any reading of their singular values as action masses [7, 31, 30].

For the explicitly chosen broken potential

$$V(H) = \lambda_H (H^\dagger H - v^2/2)^2, \qquad \lambda_H > 0,$$

the vacuum shell has $|H| = v/\sqrt{2}$. Taking $H = (0, v/\sqrt{2})$ gives the following conditional quadratic table.

Field directions	Action mass or coefficient
Three charged leptons	$m_{e,i} = v\, s_i(Y_e)/\sqrt{2}$
Three up and three down flavors	$m_{a,i} = v\, s_i(Y_a)/\sqrt{2}$, each in three colors
Three minimal left neutrino fields	Zero in the dimension-four charged grammar
Charged weak vectors	$m_W = g_2 v/2$
Neutral massive weak vector	$m_Z = v\sqrt{g_2^2 + g_Y^2}/2$
Electromagnetic direction	Zero quadratic mass
Eight color directions	Zero quadratic mass; no free physical gluon claim
One radial scalar	$m_h = \sqrt{2\lambda_H}\, v$

The neutral vector matrix follows from the coefficient of $\frac{1}{2} A^T M^2 A$ in $|DH|^2$; this factor of two matters. The three zero scalar Hessian directions are gauge-orbit directions. Neither the sign of the broken potential nor its quantum vacuum is selected by the finite representation.

For a supplied Weyl principal symbol and the antisymmetric pairing $\epsilon_{\text{spin}} \otimes \overline{K}$, squaring the complete Nambu symbol cancels its momentum–mass cross terms, so

$$E_j(\mathbf{p})^2 = c^2 |\mathbf{p}|^2 + s_j(K)^2.$$

This is a theorem of that supplied relativistic action. The spatial source of Section 7 has additive bands instead, and the formula does not transfer to it by matching K alone.

4.4 Fermion operators, gauge links and the regulator boundary

The internal permutation P_{15} lifts to $P_{90} = I_2 \otimes P_{15} \otimes I_3$ on the configured spin, internal and family factors. Its exterior lift obeys, on every occupation sector,

$$\Gamma(P_{90})a^\dagger(f)\Gamma(P_{90})^* = a^\dagger(P_{90}f), \qquad [d\Gamma(T_A), a^\dagger(f)] = a^\dagger(T_A f).$$

Thus the same twelve currents act on the full odd-CAR field; auxiliary even occupation masks do not supply its statistics. Exact source controls also intertwine the three Weyl axes and all complex Higgs coefficients [1]. For the sine regulator the eight nodes have chiralities $\chi_b = -(-1)^{b_1+b_2+b_3}$. Opposite-winding particles carry the same internal representation, whereas holes carry its conjugate: $\mathrm{Tr}_{15}\, Y^5 = 5/9$ changes sign under conjugation. This fifth trace distinguishes representations; it is not an anomaly coefficient. The complete local free-field addition/removal law assigns weight $1/8$ to each whole valley cell. A valley projector fails to commute with local hypercharge, so selecting such a detector branch does not remove mirrors from a local gauge theory.

There is also a finite interacting gauge attachment. On the specified sixty-state links, the hopping $\sum c_x^\dagger (ic\sigma_j/2)U_{xy}c_y + \text{h.c.}$ commutes with every full-Fock local Gauss generator. The same fifteen-state matrices have zero linear and symmetric cubic anomaly tensors, counting the internal Weyl representation once rather than its two spinor rows. These are exact finite identities. The quantum-link raising operators are nilpotent and cannot be replaced by identity backgrounds to import the free dispersion. A compatible gauge state, continuum scaling and mirror treatment are additional premises for an interacting chiral field interpretation.

4.5 Flavor, neutrinos and conserved defect labels

The canonical family-frame freedom is $\mathrm{U}(3)^5$, and it classifies every same-current, canonical-kinetic family embedding. It preserves the charged singular spectra and basis-invariant mixing data. In particular $V_{\mathrm{CKM}} = U_{uL}^\dagger U_{dL}$ is determined by the same left-handed weak current; relabeling families cannot repair a mass hierarchy or generate physical CP violation. A real symmetric coefficient class has $\mathrm{tr}[Y_u Y_u^\dagger, Y_d Y_d^\dagger]^3 = 0$ [29].

The native current also supplies one LH invariant line, and its symmetric family lift enters the complete action with six free complex Weinberg coefficients. A supplied dimension-five operator $(L\cdot H)^T C_5 (L\cdot H)/(2\Lambda) + \text{h.c.}$ gives $M_\nu = v^2 C_5/(2\Lambda)$ and Takagi singular masses. A calibrated instance has one massless neutrino and two nonzero coefficients. The charged-lepton construction supplies neither C_5 nor its scale.

The source itself conserves $B - L$ exactly (Section 2). The Weinberg operator changes $B - L$ by two units, so the source supplies no C_5 and no Majorana pairing; its neutrinos are Dirac, and their masses and mixing are open. The Weinberg arm below is a calibrated comparison outside the source.

Conserved additive defect labels must respect the same vertices. If the charged-current mixing graph is connected, the relation

$$q(W^-) - q(e_\alpha) + q(\nu_i) = 0$$

for every nonzero entry forces one common neutrino label N and one common charged-lepton label. A massive Majorana field adds $2N = 0$, which in an integer winding group implies $N = 0$.

If a massless neutrino is excluded from defect assignments and has label zero, connected mixing forces the other neutrino labels to zero as well. Distinct protected winding numbers therefore cannot simply be assigned to the three mixed mass eigenstates. Internal excitations, nonadditive invariants and charged compensating vertices require their own source construction.

5 The spectrum of the native source

Before any mass law is chosen, the source has spectra of its own. This section collects them. No coefficient here is fitted to a particle, every energy is in the source's own dimensionless unit, and each result retains the complete state space on which it is stated.

The native source acts on twelve nonnegative integer port loads of total two, which form 78 configurations with an invariant shell of 66 binary ones. For an oriented seam (a, b) with contrast $d = x_a - x_b$ and a fair placement coin c, repair moves $j = \frac{1}{2}[d - (2c-1)(|d| \bmod 2)]$ units from a to b. Instrument C records (c, j) and instrument E also records the seam-total parity; they share one classical update and differ in the coherence they retain [8].

5.1 Transfer modes, compact cores and a phase excitation

Keep all sixty-six binary total-two configurations. Uniform independent seam selection and fair placement give the complete attempt transfer $P = C/60$, with $2I/3 \le P \le I$, from all 3960 endpoint branches. Exact characteristic polynomial factorization, independently checked by sixty-six Newton traces, finds sixteen distinct eigenvalues and a unique constant ground vector. The largest nontrivial eigenvalue is $(55 + \sqrt{5})/60$.

A compact amplitude core means that both loads lie in a selected port set. The largest confined invariant subspace is the kernel of the outgoing block and all its products with powers of the internal block. Exhausting all 4096 subsets, with exact rank certificates, gives no nonzero such subspace on nine or fewer ports. Six ten-port cores support three modes, with transfer eigenvalues 4/5, 5/6, 13/15 and integrated normalized native autocorrelations 5, 6, 15/2 attempts. Admitted changes of seam weights destroy their confinement, and the full recorded instruments do not conserve their transfer sector. A common invariant amplitude subspace under every native seam uses all twelve ports.

The faithful reconstructed reference $\rho = P/\operatorname{Tr} P$, with $\operatorname{Tr} P = 56$, has a different localized excitation. Let $D = (-1)^{n_0}$ and $\rho' = D\rho D$. It has the same entropy, configuration probabilities and full eleven-port exterior as ρ. For the source-derived $H = -\log P$,

$$\frac{1}{1008} \le \operatorname{Tr}(H(\rho' - \rho)) \le \frac{1}{672}.$$

The bounds follow from $\|[P, D]\|_{\mathrm{HS}}^2 = 1/9$ and the positive spectral bounds on P. The native C and E unread channels contract the state contrast by 5/6 and 7/10 per uniform attempt, and complete tape-plus-exterior laws agree when compared with the equally evolved reference. The excitation is the reconstructed functional $A \mapsto \ell(DAD)$ and a decaying finite excitation; no native phase gate or coherent preparation is inferred from it.

Interaction is detectable without a fitted spectrum. Every quadratic operator in this fixed occupation dictionary has diagonal $K_{\{i,j\},\{i,j\}} = c + \epsilon_i + \epsilon_j$, regardless of its hopping terms, so its four-pair contrast on $01, 23, 02, 13$ vanishes. For $I - P$ the same contrast is $-1/15$, which puts it at norm distance at least $1/60$ from every such quadratic. A positive logarithm expansion with an explicit tail bounds the distance for $-\log P$ below by 0.01984562. The test retains every native

state and permits arbitrary quadratic hoppings, on-site potentials and an energy offset. Arbitrary re-encodings lie outside it.

5.2 Reversible seam repair and an observable gauge flux

A declared coherent repair couples a propagating amplitude to the internal $3 + 2$ response code on the same carrier [2]. It changes the committed-record dynamics. Write $|\pm\rangle = (|10\rangle \pm |01\rangle)/\sqrt{2}$ on a one-load seam and exchange $|-\rangle|0\rangle_{\mathrm{mem}}$ with $|+\rangle|1\rangle_{\mathrm{mem}}$, fixing the orthogonal states. The removed seam disagreement is retained in coherent memory. Reversible exchanges also implement $|20\rangle|0\rangle \leftrightarrow |11\rangle|2\rangle$ and $|02\rangle|0\rangle \leftrightarrow |11\rangle|3\rangle$ for double loads. These are unitary finite rules, not classical load averaging or a proof of fermionic exchange statistics.

With sixty oriented seam addresses, the controlled local operation Q and uniform-address embedding A give the walk $W = Q(2AA^* - I)$. On the one-load source and its parity memory the discriminant is

$$A^*QA = \mathrm{diag}\left(I - \frac{L}{60},\, I - \frac{D+B}{60}\right),$$

where B is the carrier adjacency and D its degree matrix. Consequently $A^*W^nA = T_n(A^*QA)$ and $\cos\omega = \lambda$ for each discriminant eigenvalue. The 1440-dimensional program has an exact finite wave response in its own step clock. Its first population update agrees with native repair; its second single-port return is $643/900$, rather than the classical $101/120$. Its reference and apparatus therefore require their own source contract.

For a five-component internal vector h and unitary link $G_{ba} = G^*_{ab}$, use

$$m_{ab}h = \frac{|a\rangle h - |b\rangle G_{ba}h}{\sqrt{2}}, \qquad p_{ab}h = \frac{|a\rangle h + |b\rangle G_{ba}h}{\sqrt{2}}.$$

The seam exchanges $m_{ab}h|0\rangle$ and $p_{ab}h|1\rangle$. Its orientation-reversed partner cancels the parity cross terms, giving magnetic and signless blocks $P_G = I - (D - B_G)/60$ and $I - (D + B_G)/60$. The 7200-dimensional coupled walk conserves $Q_5 = \mathrm{diag}(2I_3, -3I_2)$ for block-unitary links. This is an internal conserved charge; it does not establish relativistic spin, statistics or the physical identities of the three- and two-dimensional blocks.

The link controller generates the binary cover of the actual sixty carrier rotations and retains its central sign in every executed overlap word. The central lift is $Z = \mathrm{diag}(I_3, -I_2)$, so the mixed code is not a projective representation of the sixty-element group itself. The gauge theorem's holonomy-cover premise permits this action. Finite cover-valued links are closed under recharting by that cover; arbitrary continuous frame changes require the corresponding transformed link domain. Complete internal response and full apparatus control are distinct interfaces whose quotient must be specified before applying the twelve-dimensional response theorem.

A single link Z with all other links identity produces two triangular Wilson loops with trace 1, compared with 5 in the flat source. Exact carrier enumeration gives

$$\mathrm{Tr}\, B_G^3 - \mathrm{Tr}\, B_0^3 = -48, \qquad \mathrm{Tr}\, P_G^3 - \mathrm{Tr}\, P_0^3 = -\frac{1}{4500}.$$

The embedded three-step return trace changes by $-1/1125$, and a prepared doublet endpoint return has a nonzero complete probability contrast. This is a gauge-invariant flux response of a propagating source amplitude. Changing that one link back removes both fluxes, so the specified operations do not protect them against arbitrary link repair. A finite spectral gap in this background is not a physical rest mass or a continuum mass-gap theorem.

The same coupled walk prepares its reference through the complete instrument $K_+ = (2I + W)/3$, $K_- = \sqrt{2}(I - W)/3$ on $I/7200$. Since $\operatorname{Tr} W = -6380$, the successful state is

$$\rho_W = \frac{(5/4)I + (W + W^*)/2}{2620} \geq \frac{I}{10480}, \qquad p_+ = \frac{131}{810}.$$

It commutes with W, and independent recorded restarts have failure probability $(679/810)^k$ after k trials. This reference belongs to the reduced walk algebra; conditioning on preparation records changes it. For an implemented local cover frame S, the transformed links execute $W' = SWS^*$ and the filter prepares $\rho' = S\rho S^*$. The apparatus explicitly undoes the frame, reads its owned detector and restores the frame. Every finite adaptive branch consequently obeys $B'_h = SB_hS^*$, including preparation failures and detector complements; complete history laws agree with arbitrary untouched quantum references. A bare unchanged pointer can disagree: the exact source example gives 1/120 instead of 1/60. This finite chart agreement retains its declared apparatus domain. Its modular generator depends on $\cos\omega$ and is not the quasienergy clock. The reflection, coherent storage, code, links and preparation are declared architecture. Dynamical gauge fields, interacting many-particle records, a chiral field limit and measured masses are outside this finite construction.

Joint defect and wave dynamics. A finite coupled program retains ten link words with one regional flux, fixed boundary links and an exterior compensator [3]. An interior edge move T_e flips its bit when exactly one adjacent face has flux and otherwise holds; it is an involution. With Π_e the wave's endpoint projector, its fixed joint clock is

$$B = \bigoplus_g W_g, \qquad C_e = I \otimes (I - \Pi_e) + T_e \otimes \Pi_e, \qquad U = C_4C_3C_2C_1C_0B.$$

This is unitary on 72000 dimensions and preserves regional boundary flux and Q_5. The link register remains coherent during idle updates. Changing its initial defect position while holding one wave preparation fixed changes a complete one-tick wave-pointer probability from 1/120 to 0. Holding the link input fixed and changing the prepared wave changes a face-position probability from 1 to 0. These reciprocal marginal responses distinguish interaction from propagation on an external background. A coherent preparation also gives two equal register Schmidt weights, without assigning a physical tensor factorization to a gauge system.

The same U has trace -47850. Its complete heralded filter on $I/72000$ prepares $\rho_U = [(5/4)I + (U + U^*)/2]/42150 \geq I/168600$, with success probability 281/1080 and all failure flags retained. A simultaneous interior vertex-star link flip and central matter-frame change defines $\Gamma^2 = I$ with $[U, \Gamma] = 0$. The $+1$ Gauss sector has dimension 36000; its normalized faithful reference has denominator 21075. The reciprocal one-tick readouts are gauge invariant and survive this constraint. The fixed boundary, controlled moves and preparation are declared architecture.

Uniform link, port and address amplitudes with blank parity memory and a triplet component define three orthogonal pure fixed backgrounds $U\Omega_j = \Gamma\Omega_j = \Omega_j$. A complete apparatus prepares each with probability 1/10 per fresh trial. Five face measurements, both filters $(I \pm U^2)/2$ and four Fourier bins give forty complete records; the minus-filter outputs are orthogonal to the background and carry total weight 283/7200 conditional on its preparation. These are source-generated finite frequencies, without a rest-mass assignment.

Local Gauss-dressed exchanges connect cells with a strict finite circuit cone. The tested gapless spectator branch is quadratic at small momentum. A distinct shared-link occupation law second-quantizes each W_g and uses endpoint number parity for its link controls. It agrees with U at degree one and retains all occupations. Exact cubic executions resolve eight and sixty-four momenta with

complete 32- and 256-outcome instruments. Their first moments equal $(-29/30)\prod_i[(1-i)/2+(1+i)\cos q_i/2]$; the tick changes a defect face with probability $1/450$. A fixed-link control has the same moments and zero face movement, so moments alone do not identify interacting spectral poles.

The specified positive clock $H = \sum_x[I - \mathrm{Re}\,U_{F,x}] + (\pi/4)\sum_{\langle xy\rangle} d\Gamma(I - S_{xy})$ has constrained empty-wave grounds and $H \le (2+3\pi)N$. Its uniform-link triplet energies are $0, (5-\sqrt{5})/60, 1/10, (5+\sqrt{5})/60$, with localized-port weights $1/12, 1/4, 5/12, 1/4$. The branches $E_m(k) = m + (\pi/2)\sum_i(1-\cos k_i)$ require different leading squared relativistic speeds $\pi m/2$; their diffusive limits leave a quartic mass-shell residual. An exact two-wave contact collision is available on this H, but these energies do not identify a relativistic particle spectrum.

A positive free field of actual source pairs. Adding the flat gauge orbit to the five odd orbits permits local exchange of the whole inside/exterior flux pair. Five face labels describe locations of one pair type. The complete receiver probabilities are $119/240, 1/240, 1/2, 0$, conditional on recorded Gauss success. A two-lane Clifford route retains all 121 cell states and has one-object limit $\omega = \pm\sqrt{|p|^2+4}$, with error $ta(3P^2+6P)$ on $|p_i| \le P$. Its mass coin uses the literal face count $D = 2$ with coefficient one. Positive many-body energy and statistics require an occupation law and reference in addition to this frequency equation.

A separate complete graded source assigns twenty carrier slots per cell, one for each face and Clifford component. Its vacant slot encodes the flat Gauss orbit; its occupied slot encodes the actual paired-flux orbit. Every local occupation is retained. Declare fermionic grading and the local Wilson Hamiltonian with one-body symbol

$$h_a(p) = \sum_i \alpha_i \frac{\sin(ap_i)}{a} + \beta\left[2 + \sum_i \frac{1-\cos(ap_i)}{a}\right],$$
$$E_a(p)^2 = \sum_i \frac{\sin^2(ap_i)}{a^2} + \left[2 + \sum_i \frac{1-\cos(ap_i)}{a}\right]^2.$$

The grading, regulator and face-count coefficient are declared architecture; ordinary qubit realizations require parity strings. This Hamiltonian is not a logarithm identified with the routing tick. Filling the negative bands gives a unique finite Slater ground with $10V$ source pairs and $E_{\mathrm{sea}} = -10\sum_p E_a(p)$. Relative to that same reference,

$$H_a - E_{\mathrm{sea}} = \sum E_a(p)(b^\dagger b + d^\dagger d) \ge 0.$$

The full gap is two and the nonvacuum neutral gap is four. A particle adds one source pair; a hole removes one from the reference. Their opposite relative occupation charges are not electric charges. Native gauge currents vanish on these native-empty slots.

On $|p_i| \le P$, the bound $\|h_a - (\alpha\cdot p + 2\beta)\| \le b = a^2P^3/2 + 3aP^2/2$ controls dynamics by $|t|b$ and the filled-band projector by $b/2$. Wick expansion controls all finite free-field words. The resulting effective field has positive particle/antiparticle energies, spin one half, causal anticommutators and $E^2 - |p|^2 = 4$. The Wilson corner energies $2 + 2n/a$ exclude additional finite-energy valleys. The five face channels remain degenerate neutral copies. This is a controlled free field, without an interacting chiral or Yang–Mills mass-gap claim.

An actual four-mode block supplies a complete energy readout. A fermionic reservoir SWAP, with both initial settings and final flags recorded, gives four preparation alternatives. Nine particle/hole occupation projectors on the entire block give 36 joint records. At rest the six populated weights are $1/4, 1/8, 1/8, 1/4, 1/8, 1/8$. Addition and removal each give energy two; failed preparation has ground and energy-four neutral-pair components. A nine-level work battery completes

this operation to an exactly energy-conserving unitary by extending each total-energy shell, including edge shells. Equality with the driven preparation holds for energy-resolved records; tracing the battery can dephase cross-energy coherence. A nonzero momentum control gives $E = \sqrt{10}$ at $a = 1, p = (0, 0, \pi/2)$. The dimensional rest-energy relation is $mc^2 = 2u_E\,\mathrm{GeV}$, with u_E unspecified.

Native currents and chiral propagation. Bare flux has trivial native gauge charge, and the pure-face commutant of the literal interaction is scalar. Five dark native orbitals instead carry $R = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$, with actual-current weights $(\bar{3}, 1, -2/3) \oplus (3, 2, 1/6) \oplus (1, 1, 1) \oplus (1, 2, -1/2) \oplus (\bar{3}, 1, 1/3)$. The additional central degree current is separate. On this even module $U_F = I$ and $K = I - \mathrm{Re}\,U_F = 0$. A declared cell encoding places each occupied spin–native object in a literal compensated-flux slot. Antisymmetrizing within the cell and taking the graded tensor product between cells embeds every occupation of $\mathcal{F}(\mathbb{C}^2 \otimes R)$, preserving actual currents and the cell support of even operators. Grading, binding and transport are architecture choices; the idle law supplies no binding energy.

A complete antiperiodic cubic momentum menu supports $h_L(p) = -\boldsymbol{\sigma} \cdot p \otimes I_R$. All regulator momenta are retained. The negative-band sea has positive particle and hole excitations with opposite helicities and conjugate charges; no independent mirror family is inserted. Fourier projection gives the free Weyl limit with Sobolev tail control. Its finite position kernel has algebraic long-range hopping, so this proof supplies neither a finite causal cone nor an interacting local gauge limit.

Quantum Higgs coupling and the field dictionary. The inherited symmetric tensors $C_{c,k}$, for up, down and charged-lepton channels, give the spin-antisymmetric pair interaction $\frac{1}{2}c^T(\epsilon \otimes C_{c,k})c\,\Phi_k + \mathrm{h.c.}$, with $\epsilon = i\sigma_2$. Here Φ_k is the scalar field for the up channel and its adjoint for the down and lepton channels. A five-state scalar memory retains its vacuum, two particle states and two antiparticle states. The quantum field $\Phi_k = a_k + b_k^\dagger$ satisfies the exact finite gauge Ward identities; replacing it by a number is a different background calculation. In the full 320-state active lepton–scalar block with unit lepton Yukawa, unit scalar-number energy and empty quark spectators, the ground energy is -2 with multiplicity two and the spectral gap is $(5 - \sqrt{21})/2$. Conditioned on ground preparation, the reservoir and energy instrument has 56 records. Ground preparation succeeds with probability $1/160$; the full preparation/readout law has 69 records, including thirteen preparation failures. A retained compensator balances the reservoir charge; the source plus apparatus obey all local Gauss constraints. Within the prepared-ground experiment, each reservoir setting is selected with probability $1/2$. Successful addition or removal has joint weights $9/40$ and $1/40$ at excitation energies 1 and 3, respectively. The lower global gap has zero weight for this probe. Failed preparations are retained. These are finite onsite energies, without a relativistic mass assignment.

For the separately specified neutral background $H = (0, v)$, set $M = v\sum_c y_c C_{c,1}$ and $\Delta = \epsilon \otimes M$. The Nambu matrix is

$$B(p) = \begin{pmatrix} h_L(p) & -\Delta^* \\ \Delta & -h_L(-p)^T \end{pmatrix}.$$

For each tensor pair (l, r) with sign s_{lr}, the canonical variables $\Psi_L = c_l$ and $\Psi_R = -s_{lr}\epsilon c_r^\dagger$ reduce this matrix to seven Dirac blocks and one neutral left-Weyl block. All occupations survive this invertible change of variables. The paired blocks obey $E_c^2 = |p|^2 + |vy_c|^2$. Their actual electric charges are $2/3, -1/3, -1$; color acts on both quark chiralities and weak currents only on the left. This determines conditional up-type, down-type and charged-lepton roles for one family. It selects no generation, Yukawa coefficient, physical Higgs vacuum or measured mass. The finite antiperiodic menu excludes $p = 0$; $|vy_c|$ is the rest parameter of the background continuum formula.

Complete constrained two-cell spectroscopy. A uniform open-edge member retains twelve fermion modes, one six-state quantum link and two five-state quantum scalars: $2^{12}6\cdot5^2 = 614400$ active states. Both local charged-lepton Yukawa coefficients, scalar number energy and link-excitation energy equal one; quark Yukawa coefficients vanish. Its hopping is $ic_0^\dagger\sigma_3U_{01}c_1/2$ plus its adjoint. Exact projection onto every local Gauss generator gives 1053 neutral physical states. Complete rational eigenbasis certificates across every lepton-number and spin-projection sector place the unique ground in $L = 2$, $2S_z = 0$, with

$$-4.554826639 < E_0 < -4.554826631, \qquad E_1 - E_0 > 0.394804412.$$

These intervals bound the full constrained Hamiltonian, including states outside the empty-input sector. A binary filter on the maximally mixed physical state prepares this ground with probability $1/1053$ and retains failure with probability $1052/1053$, together with the compensating preparation reference.

An actual fermionic swap with a reservoir and compensator preserves the local Gauss charges and total lepton number. Its four branches are $(a^\dagger, -n, -a, I-n)/\sqrt{2}$ on this even ground. Joint conserved-sector and energy-window measurements give 460 conditional records, or 461 records including failed preparation. For the right-cell spin-up probe, the first addition and removal transfers are respectively $1.052043848 \pm 8\times10^{-9}$ and $0.909798632 \pm 8\times10^{-9}$ source units; their conditional record probabilities exceed 0.21838924 and 0.23089293. The right-positive sector's absolute ground has $L = 4$ and is dark to this single removal, which reaches $L = 3$. Charge conjugation reflects the boundary: it pairs right addition with left removal, rather than asserting equality of the two right-cell thresholds. Odd sectors have exact Kramers doublets, while the open source has no translation dispersion or full rotation symmetry. A separate two-link periodic construction has an exact translation and complete phase measurement. Joint source/reference translation-phase records translate a retained compensator with the source. At two cells, odd-sector momenta are $\pm\pi/(2a)$, with equal squared momentum, so their energies cannot separately determine both a mass and a propagation speed. Appending a blank return link does not transfer the open dynamics: an actual physical pair preparation has full intertwining residual squared $3/16$, entirely outside that embedding. An independent complete calculation on the periodic source gives 3424 neutral and 2456 states in each compensated charged Gauss sector. The unique neutral ground has $(L, 2S_z, T) = (2, 0, +1)$, energy $-4.33817023295 \pm 1.5\times10^{-8}$ and gap greater than 0.25770063535. Translation composed with conjugation pairs addition and removal at the same boundary. Both first transfers lie at $0.97009612853 \pm 3\times10^{-8}$ source units, with original-swap probability in $[0.2224969640, 0.2224969715]$. The complete adaptive energy and joint reference-phase instrument has 2627 original-attempt records, retaining ground-preparation success $1/3424$ and its failure. The equal opposite phase weights at each energy follow from the retained reference position; they supply only one squared momentum.

The quantum Higgs field provides an exact operator connection. With its actual finite scalar matrices, the charged contraction $J_e = \phi_0^\dagger c_{L,10} + \phi_1^\dagger c_{L,11}$ has the gauge currents of the conjugate-singlet creator. The up and down contractions obey their native tensor identities on every local fermion occupation. For a charged creator T, the same-H equation is $H_tT - TH_0 = [H_{\rm hop}, T] + [H_Y, T]$ on the complete physical coordinate blocks. The charged Higgs-composite residue in the first periodic line is rigorously in $[0.4230841174, 0.4230841250]$. Its interference with hopping is retained; no scalar background or free-field equation is substituted. The gauge-neutral odd contraction $\phi_0c_{L,11} - \phi_1c_{L,10}$ has nonzero interacting evolution. Conserved local matter/link parities force its separated bare correlator to vanish in a symmetry-invariant reference; a transporting dressed field requires its own construction.

A full periodic cubic family retains all thirty native modes per cell, quantum links, both scalar conjugates and three commuting translations. Its exact operator identities do not transfer the reduced lepton ground. A separate large-link-penalty member admits a complete invariant low-energy graph: on an actual fifteen-state Gauss block at penalty 128, a six-state dressed generator has full evolution error below $10^{-12}|t|$. A controlled connected energy is strictly positive at penalty 65536. These finite interaction estimates supply no nonzero interacting continuum. Raw raising-link paths have norm amplification $d^{r/2}$ on a blank reference, so a classical-link comparison needs additional control. Contractive link matrices are a declared alternative with their own reference and spectrum. The computed finite spectral records identify neither relativistic masses nor observed generations (`model/coherent_defect_wave/chiral_joint/periodic_spectrum`, `model/coherent_defect_wave/chiral_joint/higgs_composites`, `model/coherent_defect_wave/chiral_joint/field_limit`, [1]).

The same periodic ground supports complete neutral and bosonic instruments. For $J_\nu = \phi_0 c_{L,11} - \phi_1 c_{L,10}$, the actual success operator $J_\nu/2$ and its adjoint setting retain their positive square-root failures and all 298 target states in each setting. Their first transfer lies at $0.62435981 \pm 3 \times 10^{-8}$ source units. The local-star symmetry pairs the translation phases and forbids separated bare transport. A scalar-number probe and two closed-link traces retain all 502 target states, elastic events and failures. They have positive weights at a common inelastic line $1.70011247 \pm 4 \times 10^{-8}$ source units. Ground-filter failure is retained throughout. Neither line is identified with an observed neutral lepton or boson (`model/coherent_defect_wave/chiral_joint/neutral_composite`, `model/coherent_defect_wave/chiral_joint/periodic_boson_records`, [1]).

Physical local reference and finite quantum transport. All three native Yukawa channels and the quantum scalar act on the full $5\,2^{30}$-dimensional local source. Exact cyclic reductions give 36 positive spectral weights from the empty preparation and 49 from a neutral scalar–lepton preparation. The complete physical ground follows from a separate comparison of all zero-boundary-charge sectors. A fourteen-mode particle–hole transformation gives a collective $\mathfrak{sp}(4) \oplus \mathfrak{sl}(3)$ action. Its full quark character inventory excludes every physical quark module except one from energies below -11.13. The exceptional module couples to all lepton and scalar states in a complete 104-dimensional physical space, including 56 dark states at energy one. Exact root isolation proves a ground doublet near -11.23058529425 and the full physical gap

$$\frac{25207584560067}{281474976710656} \;\le\; \Delta_{\mathrm{loc}} \;\le\; \frac{25207584560069}{281474976710656}.$$

The ground has spin one-half, raw $L = 1$ and zero centered charge $L - I$. Its stationary spin-invariant density is $P_g/2$. Exact charge-conjugation and spin identities fix its mean native occupations to $(3, 6, 1, 2, 3)$, with total fifteen. These are reference occupations; the local gap is not identified with a particle rest mass (`model/coherent_defect_wave/chiral_joint/local_reference`, [1]).

Literal compressed Peter–Weyl multiplication on 119 quantum link states retains all 37 allowed coordinate channels, including 27 transitions between excited sectors. The full internal transporters are contractions, all endpoint currents are preserved, and all 24 proper cubic rotations act on the actual source Hamiltonian. These coordinate transitions break the minimal link law's local-star obstruction. With all native couplings one, the first neighboring neutral scalar–lepton transfer moment is $-151i/32$ at fifth order. Every native matter block contributes. A complete physical preparation and full-H remainder certify positive finite arrival without an added neutral hopping term. Separately, all five native matter representations share a dual-native reference and common observation time $1/1508$. Complete second powers of all thirty source components give a prepared-minus-unprepared local detector contrast at least $1/81866304$. The arriving charged objects retain

their compensating flux and reference (`model/coherent_defect_wave/chiral_joint/peter_weyl_links`, `model/coherent_defect_wave/chiral_joint/family_transport`, [1]).

Four actual hops around a contractible square return the matter excitation and leave a gauge-invariant link trace. The exact fourth moments are $-1/256$ for the singlet and $-1/768$ for the colored up-conjugate block. All chronological orders, native currents and spin signs are included. Full-state bounds give nonzero energy-dependent eliminated loop kernels and complete loop/no-loop records under the original Hamiltonian. A magnetic term is not inserted. An exactly unitary rational clock of the unchanged H has operator error bounded by $|tB|^7/(100800n^6)$ for n substeps, where $\|H\| \le B$. It preserves the certified neutral arrival and all-family detector contrast. A separate exact 72-state rational execution retains both loop outcomes and their complement. These finite results supply no many-cell Lorentz vacuum, physical photon or gluon phase, observed species or GeV mass (`model/coherent_defect_wave/chiral_joint/matter_induced_plaquette`, `model/coherent_defect_wave/chiral_joint/finite_clock`, [1]).

A controlled interacting ground and its detector line. The physical local reference admits a controlled finite interacting extension. Let $H_\lambda = H_0 + \lambda T$ on the side-two periodic source, with all onsite Yukawa, scalar and link coefficients one. The full kinematic dimension is $D = (5\,2^{30})^8 119^{24}$. Exact local charged-sector bounds give the atomic physical gap $g = 144856130473581/3518437208883200$ and $\|T\| \le 360$. Every omitted local occupation, scalar and link state belongs to the complementary-space bound. A canonical graph isometry maps the 256 atomic ground states into the isolated interacting band. Its second-order generator is

$$K_2 = -24A_{\mathrm{edge}}I + 2J_{\mathrm{edge}}UH_{\mathrm{cube}}U^\dagger, \qquad 0.14771920681900 < J_{\mathrm{edge}} < 0.14771920681901.$$

Here H_{cube} is the twelve-edge spin-half Heisenberg cube; U is an explicitly verified local spin change of basis. The exchange follows from the complete charged pair resolvent. All scalar shifts are retained. The actual side-two sublattice parity removes odd orders, and the full remainder is bounded by $25(360\lambda)^4/g^3$. This parity argument does not apply to the side-four source, whose longer hopping paths can join equal sublattices (`model/coherent_defect_wave/chiral_joint/atomic_reference_control`, `model/coherent_defect_wave/chiral_joint/atomic_even_remainder`, `model/coherent_defect_wave/chiral_joint/charged_ground_exchange`, [1]).

Exact diagonalization by polynomial factorization covers all 256 spin states. The cube ground is a unique singlet; its lowest energy is the least root of $\epsilon^3 + 5\epsilon^2 - \epsilon - 9 = 0$. The first excitation is a triplet at energy -4. Actual source translations put its three components at $(\pi,0,0),(0,\pi,0),(0,0,\pi)$, while the ground has zero momentum. At $\lambda = 2^{-29}$, the full physical Hamiltonian has a unique ground and gap in $[6.95741157484, 9.85455023443]10^{-19}$ source units. Full cubic symmetry preserves the threefold first-cluster degeneracy. No continuous spin symmetry is assumed for the full hopping Hamiltonian (`model/coherent_defect_wave/chiral_joint/atomic_spin_geometry`, `model/coherent_defect_wave/chiral_joint/atomic_momentum_response`, [1]).

A stricter coupling $\lambda = 2^{-34}$ resolves a complete detector on that same interacting reference. With the actual full-source spin current, put $O_a = \sum_x (-1)^{x_a} S_x^a$ and $B = \sum_a O_a \otimes |a\rangle/(60\sqrt{3})$. A retained vector reference carries the inverse momentum; a scalar reference records pump failure. Since $\|O_a\| = 60$, the full Halmos dilation of B is unitary and preserves Gauss constraints, fermion parity and the joint cubic symmetries. Exact atomic residues and full-H projector bounds give first-line probability in $[0.00204894419, 0.00211090162]$ and outside-line probability in $[0.00001386725, 0.00001938783]$, conditioned on ground preparation. The excitation energy lies in $[8.20758426381, 8.21034719056]10^{-22}$ source units. Each source momentum retains its

reference and one third of the corresponding total probability. The kinematic preparation from I/D through the physical ground filter succeeds with probability $1/D$; its retained failure need not satisfy source-only Gauss. The physical protocol below preserves Gauss on all branches. Together with pump failure and all line/outside outcomes, the complete probability is one. These spectral filters are declared finite algebraic services, with no efficient native implementation asserted (`model/coherent_defect_wave/chiral_joint/collective_spin_response`, [1]).

On the physical local ground, the charged composite apparatus separately has 44 complete records, 40 positive. Scalar number and Yukawa probes share the exact partner energy $1 - 2e_g \simeq 23.4611705885$ and have 22 complete records, including all zero and elastic outcomes. Their reference and boundary charges are explicit. These local responses and the finite interacting collective line are distinct observables. The nonzero-momentum collective gap supplies neither a zero-momentum rest mass nor a growing relativistic limit, measured species identification or GeV scale (`model/coherent_defect_wave/chiral_joint/ground_composite_response`, `model/coherent_defect_wave/chiral_joint/ground_scalar_response`, [1]).

Physical preparation and odd source response. A finite algebraic controller starts from physical vacua, constructs the actual local ground doublets by polynomial projection, and prepares the native cube trial with retained fermionic spin references. A Householder unitary and a two-copy energy-agreement filter use the actual $H_0 + 2^{-34}T$. The Cayley filter needs no exact interacting ground vector or ground-energy input. With 119772148350245317615208 4669 terms, preparation succeeds with probability at least 0.999999942661984; the conditional source trace distance from its true ground is at most $2.355922487\,10^{-7}$. Every failure branch remains physical. Exact algebraic synthesis, controlled matrix inversion and the finite schedule are declared services; the construction supplies no efficient native circuit (`model/coherent_defect_wave/chiral_joint/interacting_ground_preparation`, [1]).

The separate finite native service compiler executes a two-state singlet code of the actual 320-state lepton/scalar source. Its exact $HV = Vh$ has $h = \left(\begin{smallmatrix} 0 & -2 \\ -2 & 1 \end{smallmatrix}\right)$, so one simultaneous source pulse gives the full continuous code evolution; six pointer pulses supply the complete native C/E readout. A compensated $U(1)$ code also checks 491520 CAR/Gauss columns and all 32768 amplitudes. Rational control-error bounds retain export failures and reset timeouts. Coherent load/export and signed analog port control are explicit services; the passive classical repair run does not implement them. The finite general compiler includes padding and an explicit product-formula cost, without executing the full non-Abelian cube preparation (`model/native_quantum_service_bridge`, [1]).

On the thirty-mode reference vacuum, the fermion transfer is

$$B_r = (2d_r)^{-1/2} \sum_{s,a \in r} r^\dagger_{sa} c_{0sa}, \qquad B_r^\dagger B_r = N_r/(2d_r) \le I.$$

Its full dilation preserves joint Gauss charges and total fermion parity. The reference energy is $4N_{\mathrm{ref}}$; a bound over every reference occupation and source link sector proves that the neutral ground with blank reference is the unique physical ground of this enlarged Hamiltonian. Complete positive local resolvents have nine energies for e^c removal and seventeen for L removal. Residual bounds transfer their weights to bins of the actual full interacting Hamiltonian without assuming that unseen eigenstates are absent from those bins:

native removal	source energy bin	selected probability
e^c	$[0.6906800974, 0.6926332644]$	$[0.4197041521, 0.4201989707]$
L	$[0.0959119144, 0.0978650814]$	$[0.3210643365, 0.3214971263]$.

The complete outside probabilities are enclosed respectively by $[0.0799404942, 0.0801565287]$ and $[0.1785579463, 0.1788807366]$; pump failure is about one half. The exact probabilities sum to one. These ground-reference intervals acquire the stated preparation error under the constructive protocol. Total source/reference excitation energy adds the declared reference cost four. A localized charged response does not identify an isolated propagating pole, a rest mass or an empirical lepton; the static compensator is retained in that distinction (`model/coherent_defect_wave/chiral_joint/interacting_charged_response`, [1]).

The other three fermion-current probe families. Literal u^c, d^c, Q removal uses the actual native quark ground module, not an empty Fock reference. The normalized six-factor tensor embedding maps its 140 states into a verified 224-state CAR image; its target metric contains the required factor $1/6$. All scalar and lepton factors are retained. Exact number-field resolvents give thirteen visible local energies for each probe. The Q target has twenty-five distinct roots, twelve with exactly zero residue. Every complete probe measure sums to $1/2$, including the normalized color/spin contraction with the static quantum reference. The equal-coupling local u^c and d^c measures coincide; exact degeneracy after lattice hopping is not assumed.

Full-source residual bounds give strong excitation bins $[0.7815710013291, 0.7815763657472]$ for both singlets and $[0.7819863054154, 0.7819916698335]$ for Q. Their original-attempt probabilities exceed 0.4953979660650 and 0.4957017902099, respectively. All other bins, preparation failures, pump failures and outside records are retained. The reference adds four on success. This completes the five internal fermion-current probe families, without turning energy lines into species or supplying asymptotic colored states, quark propagation or GeV masses (`model/native_quark_response`, [1]).

The same quark operators also have a quantum-Higgs spectral connection. For every original annihilator $A_i = c_i$, the literal composite $C_i = [H_{\mathrm{local}}, c_i]$ obeys the full gauge, spin and raw-charge covariance identities on all occupations. Its actual target image is $C = H_{\mathrm{target}}A - AH_{\mathrm{source}}$; no scalar expectation is inserted. Contracting all indices with the complete thirty-mode reference gives $\|A\| \le 1$, $\|C\| \le 2$ and $[h, A] = C + [2^{-34}T, A]$. For the same full-source bin P,

$$\mathrm{Re}\langle PAg, PCg\rangle \ge Lw_{\mathrm{lower}} - 720\,2^{-34} = \gamma > 0.$$

The bounds are $\gamma > 0.3871888488$ for u^c, d^c and $\gamma > 0.3876321760$ for Q. Fair-setting pumps $(A \pm C/2)/2$, followed by the complete hit/miss measurement, give original signed hit contrasts above 0.0967969710 and 0.0969078028. Every pump, spectral and preparation failure remains. The actual sixty-mode source/reference commutators and full target images are checked. Positive total cross weight proves a shared contribution in the bin, without isolating a pole, replacing the composite by a canonical fermion or identifying a physical quark mass (`model/native_quark_higgs_join`, [1]).

For the separately selected elementary and composite pumps, put $a = PAg$, $b = PCg$, and let e, d be the bin center and half-width. Then $\|b - ea\| \le r\|a\|$, where $r = d + 720\,2^{-34}/\sqrt{w_{\mathrm{lower}}}$. The original two-source preparation's certified pair fidelity bounds its purification error ϵ. With $s = \sqrt{w_{\mathrm{lower}}}$, the normalized admitted states, including the same retained environment, satisfy

$$D(\rho_A, \rho_C) \le \frac{r + (r + 2 + e)\epsilon/(s - \epsilon)}{e} < 5.193\,10^{-6}.$$

Every subsequent common finite detector history inherits this bound. The different initial setting labels are excluded from the comparison; original pump/bin success rates remain distinct and every failure is retained. No later measurement of the differing setting label is claimed to have close statistics.

Sharper windows use the same full source. With $v = 360\,2^{-34}$ and atomic gap g_{at}, $PTP = 0$ gives $8E_0 - v^2/(g_{\mathrm{at}} - v) \le E_g \le 8E_0$. Orthogonal spectral projection retains more than 0.4197064484 unnormalized probe weight in the source-excitation enclosure $[0.6916539987293, 0.6916593631475]$. A narrower window $[0.6916566390288, 0.6916567228480]$ retains weight above 0.31477359. These bounds retain every unseen charged state. Almost all weight in a nested bin need not place the largest individual residue there: an exact finite perturbation counterexample separates those properties. A refined spectral selector therefore cannot silently replace the wider bin's selected line in another construction (`model/native_charge_resolution`, [1]).

The actual two-copy preparation also supports a complete twenty-bit energy detector without another ground projection. Its conditional source-energy mean lies between 0.6916566792649 and 0.6916566826119, with original bin admission above 0.419706188794. The blank-link identity $P_{\mathrm{blank}} T P_{\mathrm{blank}} = 0$ fixes the selected atomic first moment; the complete second moment bounds its lost tail. Preparation error is multiplied by the admitted bin width, and the rational phase clock contributes less than $1.04\,10^{-10}$. Every pointer label, preparation failure and reference state remains. A known rational neutral-energy baseline replaces an unknown exact ground-energy phase. Bin admission is a declared exact spectral service; the calculated mean is neither a single eigenvalue nor a GeV mass (`model/native_charge_centroid`, [1]).

An elementary probe and a quantum Higgs composite at one energy. For $A_\alpha = c^\dagger_{\alpha,e^c}$ and the actual five-state-scalar composite $B_\alpha = -\epsilon_{\alpha\beta}\phi^\dagger c_{\beta,L}$, the full operator identity is $[h, A_\alpha] = B_\alpha + \lambda[T, A_\alpha]$. Both probes are odd color/weak singlets with $Y = 1$ and raw lepton increment -1. On the actual ground and the charged bin projector P,

$$\mathrm{Re}\langle PA_\alpha g, PB_\alpha g\rangle \ge L w_{\mathrm{lower}} - 720\lambda > 0.28988126.$$

At least one actual energy consequently has nonzero elementary and composite residue. The native cubic symmetry gives a common G_1 doublet there. Complete interference pumps $(A \pm B/2)/2$, with retained charge/spin references and physical preparation, have original setting-hit contrast above 0.07247007595. All sixteen apparatus labels and preparation failures remain. This proves a same-Hamiltonian spectral connection without a classical scalar substitution. The positive-cross-residue selector differs from the maximum elementary residue selector; the composite is not a canonical fermion on every occupation sector, and no physical weak-scattering law follows (`model/native_higgs_spectral_join`, [1]).

The same equation bounds the normalized filtered states. Put $a = PAg$, $b = PBg$, let e, d denote the bin center and half-width, and write $w_{\mathrm{lower}} \le \|a\|^2$. Then

$$\|b - ea\| \le q\|a\|, \qquad q = d + 720\lambda/\sqrt{w_{\mathrm{lower}}}, \qquad D(\widehat{a}, \widehat{b}) \le q/e.$$

Here D is half the trace norm on the corresponding pure densities. A depth-three finite preparation retains eight trials and seven filters. After including its error, the narrow-bin successful elementary and composite states have trace distance below $1.68597220\,10^{-7}$. Every common subsequent complete detector history has the same total-variation bound by channel contractivity. Their original pump/bin success bounds, 0.3147735248783 and 0.0376460324440, are distinct. All sixteen apparatus labels, preparation failures, references and control work remain. Selecting the actual positive-cross-residue doublet in this bin gives a rest-frequency input for the parametric free Dirac construction below, with explicitly redeclared spectral couplers. No interacting field law transfers by changing that selector (`model/native_higgs_filtered_state`, [1]).

The quantum weak current. The canonical particle–hole map preserves all twelve native gauge currents independently of the scalar state. Its Hamiltonian identity with a prescribed scalar background is a separate conditional statement. For the actual five-state operator, $\phi_k^3 = 0$ and $\|\phi_k\| = 1$ imply

$$\|(\phi_k - v)\eta\| \geq \frac{|v|^3}{1 + |v| + |v|^2}$$

on every normalized state, including entangled references. A nonzero exact scalar eigenbackground is excluded at this cap. The explicit product-state average over n copies has variance proportional to $1/n$, so this obstruction does not exclude collective scalar limits.

The same source supplies the physical dressed density $W = \sum_a \mathrm{tr}(\sigma_a \phi \phi^\dagger) j_a$. Its full active spectrum is $\{-1, -1/2, 0, 3/2, 2\}$ and all Gauss commutators vanish. Exact atomic contractions with full-source errors give $\langle W \rangle \in [0.0495974, 0.0506134]$ and $\mathrm{Var}(W) \in [0.1517363, 0.1531928]$ on the actual ground. A current pump followed by the finite two-source Cayley detector has original nonzero-phase probability above 0.0379338, including imperfect preparation. The compensated charged pump, energy admission and subsequent current readout give conditional expectation in $[0.0903329, 0.0928015]$, including both control intervals. The complete five-outcome current law retains all failures. Current measurement can change energy, so this is an ordered instrument rather than a commuting energy/current measure. It establishes a dressed weak response, without a W-boson pole or a relativistic weak cross section (`model/native_weak_current_interface`, `model/native_weak_current_response`, [1]).

The elementary/Higgs addition branch has its own calculated response. The source antiunitary sends $\phi_i \mapsto \phi_i^\dagger$ and $j_a \mapsto -j_a$, so the same original detector cannot inherit the removal-channel mean. Its transformed operator is

$$\Theta W \Theta^{-1} = -\sum_{a,i,j} \overline{\sigma_a[j,i]} \phi_i^\dagger \phi_j \, j_a \neq W.$$

In all three complete charged blocks, exact atomic contractions give mean $-0.0376996702797\ldots$. Full-source projection and evolution bounds, with the actual eight-trial preparation, give $\langle W \rangle_A \in [-0.07004020, -0.00539719]$ and $\langle W \rangle_{B/2} \in [-0.07005609, -0.00538131]$ in the radius-$128v$ bin. Both preparations therefore have a nonzero signed current response on the unchanged interacting source. Their conditional five-label laws differ by less than $3.972\,10^{-6}$; their original success weights remain unequal. Every one of the eighty setting/pump/energy/current labels and all preparation failures is retained. Antiunitarity reverses the source time; the bound covers either sign. This does not infer CP violation from comparing two different transformed observables (`model/native_higgs_current_response`, [1]).

A local alternative retains the first three preparation/pump/bin services and replaces only the last drive. The exact current projectors give

$$f(W) = \frac{4W^4 - 4W^3 - 13W^2 + 25W}{15}, \qquad f(\{-1, -1/2, 0, 3/2, 2\}) = \{-2, -1, 0, 1, 2\}.$$

Couple $f(W)$ to the Hermitian generator of a neutral five-state cyclic pointer. Its rational 40-bit approximation P_{40} has norm below $4\pi/5 + 9.078\,10^{-13}$. All 400 local source/pointer states are retained. For $G = f(W) \otimes P_{40}$ and $\tau = 2^{-20}$, the actual pulse is $\exp[-i(\tau H_{\mathrm{ext}} + G)]$, without subtracting H_{ext} from the drive. Double Duhamel gives

$$\|e^{-i(\tau H_{\mathrm{ext}} + G)} - e^{-i\tau H_{\mathrm{ext}}} e^{-iG}\| \leq \tfrac{\tau}{2} \|[H_{\mathrm{ext}}, G]\|.$$

The full onsite Yukawa terms, including both quark terms, have norm at most 56; all incident hops contribute at most $90\,2^{-34}$. Thus $\|[H_{\mathrm{ext}}, f(W)]\| \leq 224 + 360\,2^{-34}$. Including pointer quantization,

complete-history total variation is below 0.000268446640. Both decoded current means remain below -0.0043 at total clock $3 + 2^{-20}$, with every earlier failure retained. The polynomial control preserves Gauss constraints throughout. Its final drive norm is below 5,270,718 and gross switching work below 10,541,436. It is a declared local apparatus primitive; the earlier full-matrix preparation and spectral services are unchanged (`model/native_local_current_readout`, [1]).

Gauge-invariant composite spectroscopy is also used in the independent lattice study of Maas, Martins and Wieland. Its scalar–fermion correlation matrices provide a useful comparison method. That study uses vectorial fermions without Yukawa couplings on a different prescribed gauge–scalar action. Its numerical spectra therefore supply neither a chiral realization nor missing premises for the native source above.

Charge transport and its energy cost. The same prepared source and elementary removal have six orthogonal neighbor-arrival routes. At source time 2^{-24} their unconditional union probabilities exceed $3.8580\,10^{-37}$ for e^c and $6.2900\,10^{-37}$ for L. The amplitude estimate includes imperfect preparation before squaring and retains every complementary outcome. It does not identify those arrivals with the selected low energy bins.

For that distinction, let Π_y require gauge-singlet matter at every cell except y, retaining all scalar, fermion, link and reference states. Gauss' law forces a nonblank link path from the origin reference to y. With ℓ its graph distance, the complete local sector bounds imply

$$\Pi_y H_0 \Pi_y \geq (8E_0 + \ell + \delta_r)\Pi_y, \qquad \Pi_y T \Pi_y = 0,$$

where $\delta_{e^c} > 0.10058529$ and $\delta_L > 0.09688849$. If P_I is a selected full-Hamiltonian energy bin, with upper energy $b < a_y = 8E_0 + \ell + \delta_r + 4$, the separated-spectrum Sylvester identity gives

$$\|\Pi_y P_I\| \leq \frac{360\,2^{-34}}{a_y - b}.$$

Thus at every source time the conditional probability of any of the seven remote single-charge locations is below $8.656\,10^{-15}$ for e^c and $1.699\,10^{-15}$ for L. The eighth location and the multi-charge cloud complete the location measurement; preparation, pump and energy-selection failures give thirteen sequential records. The cloud can contain screening. An explicit positive-norm state has a locally compensated reference and a remote charge pair without a flux path back to the origin. The bound therefore concerns the unscreened single-charge domain, rather than unrestricted confinement (`model/coherent_defect_wave/chiral_joint/interacting_charged_transport`, `model/coherent_defect_wave/chiral_joint/interacting_charge_localization`, [1]).

The charged-particle criterion also depends on the intended field phase. Under the relativistic Gauss-law and asymptotic-field hypotheses of Buchholz's infraparticle theorem, a charged Coulomb sector has no sharp mass eigenstate. Its physical mass target is a dressed or inclusive energy–momentum threshold with specified detector resolution. The finite matrix bins above establish neither that Coulomb phase nor a relativistic threshold. Demanding an isolated charged pole in every continuum phase would impose an inappropriate acceptance criterion.

A complete charged field and its native attachment. A declared electric/magnetic architecture on the thirteen-vertex icosahedral cone retains thirteen fermionic modes, a charge reference, longitudinal Gauss constraints and 5456 transverse states at total occupation at most three. Exact reduction gives electric energy $E_C = \rho^T (D^T D)^+ \rho/2$. Literal finite Peierls hopping of strength $1/24$ preserves the constraints and couples charge motion to the field. A full-Hamiltonian residual certificate, including arithmetic error, locates the unique ground of the complete 70928-state one-charge

sector:

$$E_1 \in [1.8738097549651658, 1.8738097549671659].$$

The reference contributes one unit. Transverse dressing has probability between 0.00007224653680 and 0.00007224655239, and lowers the energy by about 0.0001750058 relative to the zero-photon compression. All thirteen position probabilities are retained; these spatial levels are not thirteen particle species.

A compensated fermionic rotation prepares the specified packet with probability 1/4. A twenty-qubit phase pointer measures the actual H, retaining every number and phase outcome. Its neutral-energy reference phase and rational controlled-evolution gates are explicit. Including preparation, acceptance lies in $[0.2499183323303489, 0.2499819397665590]$ and the accepted ground fidelity exceeds 0.99999999439512. The source evolves throughout the apparatus schedule. Exact coefficient and linear-solve services are declared; the finite specification is not an efficient local circuit (`model/affine_gauss_source`, `model/affine_gauss_mobile`, `model/affine_gauss_full_spectrum`, `model/affine_gauss_charged_clock`, [1]).

The subsequent complete position/field instrument keeps all 13×2 outcomes and uses the actual accepted state. Its conditional transverse excitation probability lies between $7.09795005\,10^{-5}$ and $7.35247970\,10^{-5}$. For ground fidelity at least f and ground projector probability p, the lower record bound is $[\sqrt{fp} - \sqrt{(1-f)(1-p)}]^2$ when $p > 1 - f$, and zero otherwise. This fidelity bound avoids substituting a perfect ground projection. A finite-duration interaction-picture apparatus records position at the start of its readout interval while the source keeps evolving (`model/affine_gauss_clock_observables`, [1]).

The native attachment places a complete interacting eight-cell block at each vertex. Its local reference adds a conjugate of the thirty-mode reference, sharing the vacuum, and two charge ports. This enlargement is essential: particle-hole conjugation sends the original $4N$ to $120 - 4N$, rather than preserving its energy. Every native H and Gauss sector is retained. A selected actual e^c energy E_* in its certified bin supplies odd cubic-spin G_1 matrix units; the selected moving sector obeys

$$H_{\mathrm{join}} - (13E_g + E_{\mathrm{vac}}) = [H_{\mathrm{affine},1} + (E_* + 2)I] \otimes I_{G_1}.$$

Its reference costs four. This selected hard-core channel is not a full spinor Fock algebra or the unpenalized Hamiltonian's full charged ground. A separate declared port penalty has its own sector-exclusion proof.

The whole-bin alternative couples normalized $P_J c_s g$ and keeps the actual native H on all of J. The bin half-width gives a finite-time transfer error $\epsilon|t|$. At source time 1/4, unconditional radial and boundary arrival each exceed $3.06\,10^{-5}$; arrival together with nonvacuum transverse field exceeds $1.67\,10^{-6}$ and $2.26\,10^{-6}$ respectively. Six records retain preparation and exact ground-admission failure, pump failure, outside-bin weight and both arrival outcomes. Ground admission is an additional spectral service. A unitary field-parity map relates opposite-charge positive-time records; the native antiunitary alone reverses time. These results establish a finite native-current/field coupling. Cubic G_1 symmetry, a boundary energy and finite arrival do not identify a relativistic electron (`model/native_abelian_field_join`, `model/coherent_defect_wave/chiral_joint/interacting_charge_conjugation`, [1]).

Successive position and energy measurements retain their backaction on that same native attachment. A separated spectral cluster controls the actual whole-bin Hamiltonian directly; its long energy clock is not approximated by a short-time transfer estimate. At time 1/4, the original-attempt probability of arrival followed by acceptance at the admitted charged ground exceeds $4.50\,10^{-6}$ at the apex and $1.13\,10^{-6}$ at an adjacent boundary site. The accepted infidelities are

below 0.001041 and 0.004354, respectively. A coherent spatial pump gives accepted infidelity below $2.14\,10^{-8}$. Preparation, ground admission, pump, spatial complement and all phase labels remain in the complete law. The ground belongs to the admitted native bin; the full unpenalized charged-sector ground is a different question (`model/native_abelian_energy_records`, [1]).

A complete charged occupation code and free Dirac records. Two complete native constituents encode a spin doublet's vacuum, two one-particle states and its occupied spin singlet. Their actual energies above the product ground are $0, E_*, E_*, 2E_*$. An independent conjugate pair supplies the antiparticle modes. The sixteen occupations form a reducing CAR code inside the full native carrier; every complement is retained. With $m = E_*$, declare on a periodic cubic macro lattice

$$H_a = \sum_{x,j}(h_{xj} + H_{\mathrm{ref},xj} - E_g) + \sum_{q,s,x,y} a^\dagger_{xqs}[\sqrt{m^2 I + L_a} - mI]_{xy} a_{yqs}.$$

The kinetic term is positive on the entire carrier, so the unique native product ground remains the actual ground. The code has exactly $H_a = d\Gamma(\Omega_a)$, where $\Omega_a(p)^2 = m^2 + 4a^{-2}\sum_i \sin^2(ap_i/2)$. The square-root couplers have full spatial range; they are declared architecture, without a microscopic finite-range light cone.

Normalized positive and negative Dirac frames $U(p), V(p)$ give the same-clock field $\Psi_a = Ua\, e^{-i\Omega_a t} + Vb^\dagger(-p)e^{i\Omega_a t}$. Both terms have the same boundary charge. The source excitations have positive energy, and the opposite phase follows from creation. On $|p| \le P$, $0 \le \sqrt{m^2 + |p|^2} - \Omega_a \le a^2 P^4/(24m)$. Explicit momentum, volume and tail bounds transfer complete bounded apparatus laws to the free Dirac field, including its induced spin-half Lorentz action and causal anticommutator. A finite momentum projection is not strictly local.

Charge-conserving swaps into retained fermionic apparatus doublets give a literal sixteen-state active circuit. At $p/m = 3/4$, two swaps of angle $\pi/4$ separated by $T = \pi/(2\Omega_a)$ have probabilities

$$(p_{00}, p_{01}, p_{10}, p_{11}) = \left(\frac{313}{500} + \frac{12\sqrt{2}}{125}, \frac{137}{500} - \frac{12\sqrt{2}}{125}, \frac{1}{10}, 0\right).$$

The other twelve spin-partner records are retained. Finite native preparation with six retained attempts bounds its complete record error below 10^{-9}, using about $2.40\,10^9$ nats of displayed capacity. An explicit programmable matrix drive implements preparation and finite-duration detector pulses under the common raw Hamiltonian; additional synthesis workspace and passive local compilation are separate resources. This free vectorlike e^c pair supplies neither the electron's weak chirality nor the affine field's interaction, and no measured mass or GeV scale selects m (`model/native_block_dirac`, [1]).

One finite spinor current coupled to its transverse field. A declared open-cube Wilson–Maxwell member uses thirty-two complete native constituents. The positive elementary/Higgs source selector fixes its rest-frequency parameter m, and the specified spacing is $a = 1/m$. The open Wilson matrix is $H_W = A + \beta M$, where $M \ge mI$ and $\{A, \beta\} = 0$. Thus $\mathrm{Re}\langle \beta v, H_W v\rangle \ge m\|v\|^2$ proves $|H_W| \ge m$ without assuming that A and M commute. Its actual positive/negative frames give $\Psi = Ua + Vb^\dagger$, including pair terms in the local charge density. The Fock unitary uses the explicit grouped-mode to site-coordinate permutation. All native complements remain, with a declared global spectral-code compression of the new kinetic and current couplers.

The same cube cochains supply five transverse modes of frequencies $(2, 2, 2, \sqrt{6}, \sqrt{6})/a$, retaining all fifty-six states of total occupation at most three. The positive top-boundary completion is part of H_{ph}. Finite coordinate commutators at that boundary remain; no unrestricted Weyl identity

is imported. Seven longitudinal labels in $[-4,4]$ and a compensator in $[-32,32]$ enforce the full source Gauss constraints by an exact isometry. The common Hamiltonian uses the same coupling in e^{igQ_e} and $g^2\rho_{\mathrm{ref}}^T G\rho_{\mathrm{ref}}/(2a)$.

For its free matter/photon input the complete first pair-plus-photon coefficient has two independent certificates. An exact contraction gives

$$\frac{\sigma^2}{m^2}=\frac{965}{1456}+\frac{5\sqrt{6}}{36}-\frac{\sqrt{14}}{48}+\frac{\sqrt{21}}{336}+\frac{\sqrt{91}}{546}+\frac{7\sqrt{546}}{1872}>1.$$

With declared $g=1/64$ and $mT=1/64$, the complete nonlinear Peierls, Coulomb and all-order Duhamel bounds give original pair-plus-photon probability above $4.75055\,10^{-8}$ and a matter/field connected response above $4.71131\,10^{-8}$. The deterministic native trial has an exact reducing-code decomposition: its code component is the free vacuum, with weight at least p_0^{32}, and all remaining weight is an explicit native-complement outcome. There is no initial ground projection or discarded preparation. The pointer retains 1160 physical counts and a distinct outer-Gauss failure label. Its displayed source/reference/pointer capacity is below 10479 nats; coefficient synthesis workspace is separate.

The input is nonstationary under the actual coupled Hamiltonian, and programmable preparation and readout pulses use its same clock. This is a controlled finite current/field interaction, not emission from an interacting vacuum or an asymptotic scattering cross section. The global spectral compression requires a spatial locality/refinement proof before an interacting Lorentz limit or empirical electromagnetic identification follows (`model/native_wilson_maxwell`, [1]).

The full interacting ground and a positive local completion. A comparison over the retained native complement is necessary. Let $E_{\mathrm{base}}=32E_g+m(3+\sqrt{6})$. Exact filled-band contractions give

$$C_0=\frac{4055}{4368}-\frac{\sqrt{21}}{14}-\frac{3\sqrt{39}}{208}-\frac{27\sqrt{91}}{728},\qquad h=-\sqrt{3}+\frac{6\sqrt{7}}{7}+\frac{17\sqrt{13}}{13},$$

where $C_0=\langle\rho^T G\rho\rangle/2$ and $h=-\langle H_{\mathrm{hop}}(0)\rangle/m$ in the free filled-band vacuum. A weighted Schur bound, retaining the full Coulomb operator, proves $P(H-E_{\mathrm{base}})P\geq(m/50000)P$. The full code complement instead contains a single neutral native triplet in any of the thirty-two constituents, at energy $E_{\mathrm{base}}+\Delta_{\mathrm{native}}$ with $\Delta_{\mathrm{native}}\in[8.20758,8.21035]10^{-22}$. Its active lowest eigenspace has rank 96 and lies below the entire interacting code. Zero-energy preparation spectators multiply that rank.

The explicitly changed source

$$H_+=H+mc(g)\sum_x(I-P_x),\quad c(g)=g^2 2^{-48}\left\lceil 2^{48}(C_0+hv/2)\right\rceil,\quad v=\frac{1}{16}+\frac{1}{12\sqrt{6}},$$

uses each macrocell's complete occupation-code projector P_x. This positive completion makes no scalar energy subtraction. Its chosen coefficient uses only the source contractions, leaves all code energies unchanged and vanishes at $g=0$. A q-defect complement costs an additional $qmc(g)$. Every P_x reduces the full native Hamiltonian, Gauss dressing and count effects, so complete pattern-preserving count histories are invariant under the change. Cross-pattern phases and energy differences change.

The completed source has a unique active code ground, internal code gap above $1.7170018m$ and full physical gap above $2.52078609\,10^{-10}$. Four actual trials and three finite Cayley energy-agreement filters prepare it with success above 0.9995943286660 and trace error below

$1.417923\,10^{-7}$. No ground projector or unknown ground-energy phase is used. All partners and filter labels remain. The registers require less than 42006.105 nats before the final 1161-label count pointer; synthesis workspace is separate. The original-attempt $(1,1,1)$ count probability exceeds $1.41692895\,10^{-6}$. This is stationary occupation in the free quasiparticle/photon frame, not emission from the ground. The completion is an architecture choice. Its macro-local penalty does not remove the kinetic/current terms' global code guard or establish a full-source spatial propagation cone (`model/native_wilson_maxwell_ground`, [1]).

A controlled photon cutoff on the same cube. Let $P_K = 1_{N_{\mathrm{ph}} \le K}$. The actual finite link is $\exp(igP_KQ_eP_K)$; it is not replaced by $P_K \exp(igQ_e)P_K$. The completed free photon Hamiltonian is exactly the compression of $m[d\Gamma(\nu) + \operatorname{Tr}\nu/2]$. Each of the twenty-four oriented matter coefficients has norm at most $2m$. In the interaction picture its field factor is a displacement of radius $r = |g|\sqrt{1/16 + 1/(12\sqrt{6})}$. A product of n displacements from vacuum has coherent radius at most $R = nr$. Projected Duhamel and the Poisson factorial moment give

$$b_K(R) = \min\left\{2, \frac{R^{K+1}}{\sqrt{(K+1)!}}(1 + \sqrt{K+1}R)\right\}.$$

For $x = 48mT < J + 2$, the complete embedded-state error obeys

$$\delta_K \le \sum_{n=0}^{J} \frac{x^n}{n!} b_K(nr) + \frac{2x^{J+1}}{(J+1)!\,[1 - x/(J+2)]}.$$

The factorial tail supplies a constructive $K \to \infty$ comparison at fixed cube, coupling and time. At $K = 3$, $g = mT = 1/64$, $\delta_3 < 1.863\,10^{-9}$. The actual native preparation has comparison pair-plus-field probability above $4.7504776747\,10^{-8}$ and connected response above $4.7112356073\,10^{-8}$. Overflow probability is at most δ_3^2. Complete record laws include the overflow label and every native complement; no rare-event normalization enters these estimates. The unbounded oscillator is a mathematical comparison, not a physical infinite register.

On the reducing code, let $E = P_T + gDG\rho_{\mathrm{ref}}$ denote the total electric flux and $B = CQ$ its magnetic curl. The transverse projector Π_T and longitudinal projector satisfy $\Pi_T + DGD^T = I$. For the dressed oriented hop F_e, the two electric commutator contributions therefore give

$$[E_e, F_f] = g\delta_{ef}F_f, \qquad [B_f, E_e] = iC_{fe}, \qquad [\rho_x, F_e] = D_{ex}F_e.$$

These relations yield the finite-graph Maxwell/current equations and a local incidence algebra. Bounded electric/magnetic Hadamard tests have complete-history cutoff bounds, including disturbance and every control outcome. The finite cap has nonvanishing boundary commutators; no uniform operator-norm or spatial-cone conclusion is inferred. The complete ground and its explicit positive completion are treated above; growing-volume and ultraviolet limits require separate estimates (`model/native_wilson_maxwell_refinement`, [1]).

The interacting ground under photon refinement. The same positive completion works at every photon cap $K \ge 3$. On the complete code, in units of m, let $H_0 \ge 2N$ be the free operator and $A = 96g\sqrt{v}$. The literal Peierls links give

$$\|V_{\mathrm{hop}}\psi\| \le A\|\sqrt{N+1}\psi\|, \qquad V_{\mathrm{hop}} \ge -\epsilon H_0 - \left(\frac{A^2}{8\epsilon} + 2\epsilon\right) I.$$

At $\epsilon = 1/10$, the same vacuum contractions and a weighted Schur bound give a unique code ground with gap above $1.3284353m$. The completed full physical gap stays above $2.52078609\,10^{-10}$. The uncompleted source retains its lower native-complement ground.

Recorded code admission avoids resolving the much smaller complement gap. Keep all sixteen admission flags of four actual trials and use

$$h = (H - E_{\mathrm{base}})/m, \qquad C = (I - ih)(I + ih)^{-1}$$

only on the admitted code, extending by identity on failure. The known native/free baseline is not an unknown interacting ground-energy input. The ground-to-excited Cayley chord exceeds $3/2$. Three complete 4096-label energy-agreement filters give original success above 0.9993248562, conditional purification error below $8.235\,10^{-8}$ and original preparation-plus-$(1,1,1)$ probability above $4.5122\,10^{-7}$. A two-copy, one-filter 64-label setting gives a positive original record above $6.4978\,10^{-8}$.

A uniform resolvent estimate, with the full perturbation norm at most $48m$, controls the literal cap-dependent Cayley services. At $K = 2^{64}$ the complete four-copy preparation/count error is below $1.254\,10^{-11}$ and its two-field-history error below $7.304\,10^{-10}$. All partners, flags and overflow remain. The elapsed clocks are respectively six and four duration-one service slots for the two settings; control pulses explicitly cancel the same raw background. Their source/reference/pointer budgets are below 42817.043 and 21425.219 nats, excluding controller workspace. These matrices are bounded analytically, not enumerated. The source Hamiltonian and its positive completion are unchanged (`model/field_record_addenda/native_code_ground_preparation`, `model/native_wilson_maxwell_ground_refinement`, [1]).

Ordered quark-type arrival and energy records. On the original native charged source, every one of six flux routes obeys $P_0 B^\dagger T P_e T B P_0 = P_0/128$. At an autonomous wait of one, a complete arrival projection followed by an active-source-plus-reference spectral measurement has original one-route/bin probability above $2.3275\,10^{-23}$ for u^c, d^c and $2.3348\,10^{-23}$ for Q. The corresponding source excitation windows lie in $[1.7815710013, 1.7815763658]$ and $[1.7819863054, 1.7819916699]$, after the fixed reference cost four and actual neutral ground are subtracted. All 88 or 160 coarse records, dark local lines, failed preparations and outside-bin outcomes remain. The energy operator acts as identity on the retained preparation partner. Disjoint energy projectors without the intervening arrival measurement instead satisfy $\Pi_2 e^{-itH_{\mathrm{total}}}\Pi_1 = 0$ exactly. Thus the positive ordered event includes measurement backaction; it is not an autonomous energy gain or a rest-mass determination (`model/field_record_addenda/native_quark_spatial_spectrum`, [1]).

Spatial records and an interaction witness. Growing open boxes have $V = N^3$ vertices and retain four complete native constituents per vertex. Full-box preparation banks, all local code flags and first-success routing bound failure by $V(1 - p_0^4)^R$. The transverse photon minimum frequency is $2\sqrt{2}\sin[\pi/(2N)]$, so the cube's photon gap is not uniform in volume. Heat bounds for the actual Hodge operator, finite-range sign polynomials for the free sea, longitudinal dipole estimates and bounded-hopping Dyson tails give complete positive-time local-record convergence at fixed spacing. The preparation is a specified nonstationary sea, not an assumed coupled vacuum. Literal finite field caps and every failure outcome have separate error allowances.

A fair recorded electric kick or identity, actual coupled wait $mt = 2^{-32}$ and complete sine-Hadamard current readout give a strict interaction witness. With $J_e = i(F_e - F_e^\dagger)$ and $u = 2^{-16}$, $\Pr(+) = \frac{1}{2}[1 - \langle \sin(uJ_e)\rangle]$. A source-only integration-by-parts identity gives $-\mathrm{Re}\langle F_e\rangle > 1/345$ in an explicit interior finite box. Uniform field-moment bounds control the full nonlinear short-time

drift. The complete finite law differs from the matched zero-coupling law by more than $1.7183\,10^{-10}$ after all failure and cutoff allowances. The conservative sufficient box has 8200 cells per axis and roughly $5.054\,10^{15}$ nats; no such matrix is enumerated. This is an interacting local-record limit at fixed spacing, not a relativistic ultraviolet or scattering theorem (`model/field_record_addenda/native_wilson_maxwell_spatial_family`, `model/field_record_addenda/native_wilson_maxwell_interaction_witness`, [1]).

A revised regulator and its distinct ground-sector requirement. The original free Wilson symbol has $E_F^2 - m^2 = (1+ma)\Omega_\gamma^2 + 2a^{-2}\sum_{i<j} w_i w_j$, where $w_i = 1 - \cos(ap_i)$. At $a = 1/m$ its axial squared coefficient is two. The declared regulator $\Gamma = \alpha_1\alpha_2\alpha_3\beta$ gives

$$h_\perp = m\beta + a^{-1}\alpha\cdot\sin(ap) + a^{-1}\Gamma\sum_i w_i, \qquad E_\perp^2 - m^2 = \Omega_\gamma^2 + 2a^{-2}\sum_{i<j} w_i w_j.$$

It retains the native field and charge frame and a vectorlike four-component carrier; ordinary parity and time reversal fail at finite spacing. The revised cube, with its old scalar offset retained and declared complement coefficient 21/2, has full physical gap above 0.3711801 source units. Its new vacuum and prepared interaction are recomputed. This free-cone correction does not itself establish an interacting mass shell (`model/field_record_addenda/native_orthogonal_wilson_bridge`, [1]).

The revised regulator has a separate growing-box proof in the original field and Gauss frame. With W the graph Laplacian divided by two and $A_\theta = K + (\cos\theta\,\beta + \sin\theta\,\Gamma)W$,

$$h_\theta^2/m^2 = I + 2\cos\theta\,W + A_\theta^2 \geq I.$$

This identity does not assume that the derivative and Laplacian commute at an open boundary. The equal-rank endpoint projectors define a finite algebraic sea-rotation service. Charge conjugation gives zero initial current and mean density; fresh sign-projector, heat and dipole bounds then establish complete positive-time local records at fixed spacing. A fresh hopping integration by parts strengthens the complete finite interaction-law separation to more than $3.4360\,10^{-10}$.

The ground-sector extension requires another explicit source choice. The old raw vacuum offset is extensive: the cube coefficient 21/2 fails to select the code ground on sufficiently large boxes, at zero coupling. A neutral onsite two-occupation trial with photon vacuum has energy at most $E_{\mathrm{base}} + 18mV$, for every finite cap and any coupling. Every noncode state is bounded below by $E_{\mathrm{base}} + mc$. Thus the source-defined choice $c(V) = 18V + 1$ places the global ground in the code with at least m separation from the native complement. Code-admitted record laws are unchanged. This variational sector result does not determine the coupled ground vector, its internal gap or a relativistic charged threshold (`model/field_record_addenda/native_orthogonal_wilson_spatial_join`, [1]).

Charged clusters and complete joint energy records. The finite orthogonal cube at photon cap three admits a sharper fixed-charge calculation. With the same field, Gauss map and old raw sea offset, declare only the complement coefficient $c = 23/2$. Exact free positive levels start at $\sqrt{5}/2$, $\sqrt{17 - 4\sqrt{3}}/2$, $\sqrt{13}/2$ and $\sqrt{33 - 4\sqrt{7}}/2$, in units of m, with multiplicities $2, 4, 2, 2$. The uniform interaction estimate alone does not certify the first two states. It does isolate the first eight in each active charge sector $Q = \pm 1$: the separation from higher states exceeds $0.01652645m$, and the margin to the native complement exceeds $0.21532398m$. These are fixed-sector ranks, excluding zero-energy spectators, not counts of particle species. Charge conjugation pairs the two

code spectra. The central 2π action is odd fermion parity, but the fixed corner reference breaks full cubic rotation symmetry; a continuous spin representation is not inferred.

Two-copy code admission and a 64-label energy-agreement filter precede an even two-mode fermionic pointer pump. It preserves $Q + R$, the distinct source/pointer charge $Q - n_{f+} + n_{f-}$, and joint parity. The actual reference $R = -Q$ retains cost four. Charged and neutral energy clocks have 2^{24} and 2^{18} labels. Five complete driven slots retain all 2^{53} records, including failed admissions, empty-pointer outcomes and every rejected energy result. The original joint acceptance exceeds 0.56935668; conditional fidelity to the charged cluster tensored with the actual neutral ground exceeds 0.99991834. Source, reference and pointer capacity is about 20980.4 nats, excluding coefficient workspace. The energy comparison obeys

$$0.57009819m < E_{\text{charged}} - E_{\text{ground}} - 4 < 2.35071144m.$$

It contains Coulomb and boundary-reference effects. Exact finite matrix services are declared and their background cancellation and work are bounded; the full huge matrix is not enumerated. This is a controlled charged energy record, not a relativistic mass or a species assignment (`model/field_record_addenda/native_orthogonal_wilson_charged_band`, [1]).

Charged preparations with a receding reference. The growing orthogonal family also admits the compensated local probes $D_+ = Yc_f^\dagger Y^\dagger$ and $D_- = Yc_f Y^\dagger$, with a fixed positive or negative β spinor at an interior site. Its free band bound $m \le |h| \le 10m$ gives conditional pump success at least 11/20. All assembly flags and nonvanishing pump failures remain. As every boundary and the static reference recede, complete local gauge-invariant records converge at fixed spacing and bounded source time. The compared alphabet is the local observer's record; remote fine flags remain in each finite source. Reference phases vanish by the dipole bound, while a three-dimensional heat estimate gives $G(x,x) \le 33$ for the point Green function needed by the energy comparison.

For the normalized charged Slater packet $P_q = P + quu^\dagger$, exact Wick contraction cancels the rank-one classical mean-density term against its exchange covariance. Relative to the neutral prepared sea, the actual-H expectation, after subtracting the displayed reference cost $4 + (mg^2/2)G(r,r)$, has a bulk limit in $[8081/8192, 81953/8192]m$ at $g = 1/64$. No Hamiltonian counterterm is introduced. A complete local density readout has limiting signed outcome imbalance at least 121/1600 immediately after preparation; this bound is not asserted at arbitrary positive time. The input is nonstationary: its positive prepared energy is not a charged threshold above the interacting vacuum or a rest mass (`model/field_record_addenda/native_orthogonal_wilson_charged_spatial_records`, [1]).

Neutral propagation and a controlled field limit. The actual eight-cell ground g and isolated triplet e_α supply dressed native-current matrix units $A_\alpha = P_g O_\alpha P_t/(60\sqrt{r}) = |g\rangle\langle e_\alpha|$, where $r > 0$ is the computed current residue and $\Delta \in [8.20758, 8.21035]10^{-22}$ is the source gap. A declared number-conserving exchange between complete blocks gives an isolated moving band $E(k) = \Delta - 2J\sum_i \cos k_i$ and a uniform positive gap. Its controlled spacing limit is Schrödinger evolution.

A different, explicitly paired architecture places N such blocks at each cubic site. Write $A^{(N)}_{x\alpha} = N^{-1/2}\sum_j A_{xj\alpha}$ and $X = A^{(N)} + (A^{(N)})^\dagger$. On the full physical carrier declare

$$H_{a,N} = \frac{1}{a}\sum_{x,j}(h_{xj} - E_g) - t_a \sum_{\langle xy\rangle,\alpha} X_{x\alpha}X_{y\alpha}, \qquad t_a = \frac{\Delta(1-a^2)}{12a}, \quad 0 < a \le 1.$$

Its clock is $e^{-itH_{a,N}}$. On the exact permutation-symmetric sector, $(A^{(N)}_{x\alpha})^\dagger = b^\dagger_{x\alpha}\sqrt{1 - n_x/N}$, including the capacity boundary. Every higher native sector is retained. A variational argument places a

full-source ground in this sector. Occupation-tail and square-root bounds compare its energy and vector with a specified Gaussian ground. In particular, $0 \le E_N - E_{\mathrm{free}} \le \epsilon_E$ implies squared ground overlap at least $1 - \epsilon_E/\Delta$.

The quadratic comparison has

$$\omega_a(p)^2 = \Delta^2 \left[1 + \frac{1-a^2}{3a^2} \sum_i (1 - \cos(ap_i)) \right] \longrightarrow \Delta^2 + \frac{\Delta^2}{6}|p|^2.$$

Complete bounded field instruments retain their defect flags and a finite orientation reference. Explicit capacity choices bound a two-instrument, three-evolution record law by 19/20000, with separate spacing and volume errors. This is a controlled free Klein–Gordon description with three internal scalar components. The component labels do not furnish spatial spin one. Replication, spectral dressing and the spacing-dependent clock are declared architecture; no empirical mass or GeV conversion selects their coefficients (`model/interacting_block_propagation`, `model/interacting_block_quadrature`, [1]).

A vector readout of the same neutral source. An explicitly different field instrument on those same blocks uses $\tau = \Delta t$, $\xi = \sqrt{6}x$, $\Phi_\alpha = \sqrt{\Delta} Q_\alpha$ and $\Pi_\alpha = P_\alpha/\sqrt{\Delta}$. With limiting mass $m = 1$ and $\Omega_0(p) = \sqrt{m^2 + |p|^2}$, declare

$$A_0(p) = \frac{i p_\alpha \Pi_\alpha(p)}{m \Omega_a(p)}, \qquad A_i(p) = \left(\delta_{i\alpha} + \frac{p_i p_\alpha}{m(\Omega_0 + m)} \right) \Phi_\alpha(p).$$

The momentum and factor i in A_0 are required for Hermiticity and the Lorenz identity. The induced Lorentz action has polarization sum $-\eta_{\mu\nu} + p_\mu p_\nu / m^2$, three positive polarizations and the Proca equations. Its commutator is a differentiated Klein–Gordon commutator and vanishes at spacelike separation. The finite source has no exact boost symmetry; the controlled record limit carries that action.

Two specified finite members retain sixteen complete instrument outcomes and have total-variation error at most $9.5\,10^{-10}$ against the free vector comparison. Capacity, state, evolution and filter errors are separate. The rest frequency is the native Δ in the original clock; no W/Z identity or GeV scale is selected. This is a free massive spin-one readout of one declared source, not a deduction of spatial spin from the dimension of an unfiltered internal triplet (`model/interacting_block_proca`, [1]).

A finite collision distinguishable from free evolution. The same paired Hamiltonian has a finite-N channel forbidden by its free quadratic comparison. Prepare two different internal components in Fourier packets with disjoint input and output momenta but equal total momentum. The free evolution preserves each component's momentum; the capacity square root produces a connected quartic vertex. For four momenta on one frequency shell,

$$\langle k', l' | V_1 | k, l \rangle = \frac{t_a A(k)}{V} \left[\left(\frac{D}{\omega_k} \right)^2 + 1 \right] > 0, \quad A(k) = 2 \sum_i \cos k_i, \quad D = \Delta/a,$$

when $A(k) > 0$, with $H_{a,N} = H_{\mathrm{free}} + V_1/N + R_N$. Global number-weighted bounds control R_N, the second Born term, packet admission and finite preparation before squaring the signal. The experiment retains its hit, complement and preparation failures. It uses explicitly prepared nonstationary states; an approximate interacting ground is not substituted in a rare-event bound.

An independently interval-certified diagnostic on a five-block cycle evolves the complete 1024-state reducing carrier, retaining all three components. With $D/\Delta = 2$, $t/\Delta = 1/8$ and $\Delta T = 1$, its preparation-inclusive final hit probability is enclosed by $[0.00296630042443, 0.00308409427253]$. The associated occupation records are quantum marginals, rather than classical particle paths. At $N = 1$, $H_N = P_N H_{\mathrm{free}} P_N$; the admitted packet projection changes the separate momentum supports. Consequently the diagnostic's nonzero response alone does not certify a channel forbidden to the free evolution of those same admitted states. An exact Gaussian-kernel calculation of that matched free evolution, without an intermediate oscillator cutoff, gives the preparation-inclusive interval $[0.00218695122151, 0.00228825603677]$. The native-minus-free contrast is therefore at least 0.00067804438766. Both comparisons retain the same admitted input, output projector and quantum reference; the complete native complement has an explicit common-carrier embedding. The large-N theorem separately bounds that admission error below its connected interaction signal. Three-dimensional finite members also have strictly positive controlled collision channels. Their interaction vanishes as $N \to \infty$ in the established refinement. Neither the cycle nor those finite records proves a nontrivial interacting relativistic ultraviolet limit, an asymptotic scattering matrix or a Standard Model species assignment (`model/interacting_block_scattering`, [1]).

A family effect and an energy clock. For the declared cyclic family register, native repair columns give the one-tick amplitude $M = \sqrt{2}\, C(\theta)/3$, where $C = I/\sqrt{6} + (e^{i\theta} R + e^{-i\theta} R^{-1})/\sqrt{12}$ and $R^3 = I$. Positive eigenvalues of C have squared values with Koide ratio $2/3$. The literal two-tick amplitude is instead $\sqrt{3} I/9$. An unconditional collision extension realizes $A = M^\dagger M$ as a generator: N fresh retained ancillas over time t give marginal-channel diamond error at most $\min(2, 4t^2/N)$ from e^{-itA}. The projector phase gate, inverse tick and schedule are explicit architecture inputs. The XOR probability $2/9$ does not alone choose a Hamiltonian phase.

The separate scalar-memory construction contains three CAR families and $5\,2^{18}$ states. On its stationary vacuum-generated component, put $S = 4\sum_f y_f^2$, $D_0 = \sqrt{\omega^2 + 4S}$ and $D_f = \sqrt{\omega^2 + 4S - 8y_f^2}$. Its lower charged line is $m_f = 4y_f^2/(D_0 + D_f)$; a second line, nine dark states and the no-operation record are retained. For $y_h > y_l > 0$, $m_h/m_l > (y_h/y_l)^2$. Consequently this response cannot remove the $+9.83354$ ppm excess in the muon/electron comparison of the $\theta = 2/9$ squared-amplitude reading. Three independently occupiable families, a common three-state register and a neutral classical Higgs background are distinct source constructions; these finite identities do not transfer their energies to the spatial interacting source (`model/finite_family_clock`, [1]).

The retained finite candidate producers separate these inputs further. The optional native quarter-turn implements the oriented record phase; the family shift, joint analyzer and definite tail preparation are additional operations. Independent-object readout cannot realize that particular Schmidt-rank-two analyzer. In the tested transport dressing, retaining each virtual flag's clock energy gives the common equation $d = w/(d - \omega)$ and common residue, leaving family ratios unchanged. Omitting that energy defines a different model; its successful muon-ratio weight is fitted. A two-seam junction does generate nonzero CP violation, with one native assignment giving $(|V_{us}|, |V_{cb}|, |V_{ub}|, J) \simeq (0.07994105, 0.02311042, 0.01860003, -0.00003424005)$. None of its tested native, conditional or fitted assignments passes the registered four-observable comparison. The single-site first-order hierarchy failure does not exclude all junction strengths. These finite results are reproduced from retained original producers; no absent phase is silently replaced by zero (`model/particle_construction_scope`, [1]).

A defect class and a particle representation. The six-class register preserves colour triality and weak centre parity: $k = 3n_2 - 2n_3 \pmod 6$. Its 32 exterior states have class multiplicities $(4, 6, 6, 4, 6, 6)$, while the independently computed fifteen-state current representation gives the internal weights of one Standard Model fermion family. This comparison requires the full currents. For example, the centre acts trivially on both summands of $\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$, whereas the full quadratic Casimir is $3(I - P_1)$. Two quarks also contain a $(\overline{\mathbf{3}}, \mathbf{1})_{1/3}$ channel with the gauge charges of d^c, but their even fermion parity differs from a single odd fermion. A class-conserving fusion therefore needs its complete physical reference and exchange law before receiving a particle interpretation. Auxiliary exterior degree alone is not the physical fermion grading (`model/particle_catalogue_correspondence`, [1]).

Common interacting state and its scope. Finite quantum-link transport, the quantum Higgs pair terms and invariant scalar and electric energies can be placed in one Hermitian Hamiltonian with exact local Gauss conservation. On its nonzero physical subspace, $\rho_\beta = P_G e^{-\beta H} P_G / \operatorname{Tr}(P_G e^{-\beta H})$ is a specified faithful stationary reference, with $\beta > 0$ a declared input. For $\|H\| \le B$, the filter $L = P_G e^{-\beta(H+BI)/2}$ on a maximally mixed physical input prepares it; $F = \sqrt{I - L^\dagger L}$ retains the failure branch. A two-cell quantum creation/transport control on 122880 active states gives $\|RH^2\Omega\|^2 = 3$, with $R\Omega = RH\Omega = 0$ and a strictly positive, error-bounded arrival probability at a specified finite time. Its empty input is distinct from the thermal reference. The free filled sea is not substituted for this interacting state. The separate shared-link collision source has a positive finite encounter response, but its density vertex fails a scalar boost test and its bounded contact vanishes in the stated one-flux/one-native refinement window. A nonzero connected interacting limit, controlled physical reference and mass renormalization do not follow from either finite interaction. An energy conversion supplies units only: the background relation is $m_c c^2 = u_E |v y_c|$ GeV, without a source selection of v, y_c, u_E [3].

5.3 The charged spectrum of the interacting cube

The open eight-site Weyl/contact cube is a separate interacting source with a complete charged spectrum. In its exact hidden-spin frame the source is

$$K = \mathrm{d}\Gamma(A \otimes I_2)/16 + \sum_x n_{x\uparrow} n_{x\downarrow} - N/2, \qquad A = A^T, \quad A^2 = 3I,$$

at $c = 1/8$, $U = 1$. All source matrices come from the full 65,536-state integer Hamiltonian. The reference is the unique half-filled ground, with hidden spin zero, eta zero and energy in $[-4.12405873337830, -4.12405873337828]$. Its uniqueness uses the theorem of Lieb on the half-filled Hubbard model [15] together with a full-number-sector bound. Hidden spin and eta are internal algebras, and neither carries a Lorentz-spin assignment.

Lines visible to single-fermion probes. Every local one-spin addition or removal measure has exactly 228 distinct positive energies, each with rigorously positive weight. The signed retarded measure has 456 atoms, each charge branch has total weight 1/2, and every eight-site residue matrix has visible rank four per hidden spin. The smallest weight is near 5.63×10^{-21}, and no threshold discards a line.

Lines invisible to them. Combine a local fermion removal with a neighboring hidden-spin flip,

$$C_x = \Big(\sum_{y \sim x} S_y^+\Big) c_{x\downarrow}, \qquad S_y^+ = c_{y\uparrow}^\dagger c_{y\downarrow}, \qquad \|C_x\| \le 3.$$

On the singlet reference the output has total hidden spin 3/2 and eta 1/2. The linear probes have hidden spin 1/2 and K commutes with hidden spin, so the complete dynamical cyclic spaces of the two responses are orthogonal. Exact integer residuals certify all 138 spectral lines in the composite probe's reducing block, each with strictly positive weight, and direct calculations of $C^\dagger C$ and $C^\dagger[K, C]$ on the unreduced source independently enclose the total weight and energy moment.

Table 7: Certified charged lines of the centered cube, in source-clock energy units. The first six of each spectrum are shown; the complete tables retain every line and weight.

Line	Linear probe	Relative to first	Composite probe	Relative to first
0	0.361682711806	1.000000000	0.402468515770	1.000000000
1	0.402293044056	1.112281652	0.422089756328	1.048752237
2	0.415174263333	1.147896346	0.438990742922	1.090745551
3	0.438196496809	1.211549467	0.479938217500	1.192486365
4	0.462192207579	1.277894111	0.485785522474	1.207014967
5	0.474429777956	1.311729210	0.494035650369	1.227513783
Largest	3.886434754951		2.974756508168	
Conditional mean	0.49955163275		0.52537938729	

Each composite residue matrix is twice its local weight times one of the source's two kinetic-band projectors, of exact rank four, with 69 lines on each band. Spin and eta ladders give 2,208 visible composite states per charge sector. Their orthogonal union with the linear probes covers 4,032 of the 11,440 states of that number sector, and the block therefore does not exhaust every charged composite. Independent source equations and all 192 spatial energy moments retain the contact contribution and the kinetic/contact interference. The finite spatial projectors supply no continuous momentum dispersion.

The composite response can be read. A compiled pulse instrument implements the success Kraus operator $C_0/3$ in 1,008 source pulses on the authenticated hidden-frame ground preparation, retaining all 7,776 raw selector records, native aborts and coherent field-reference amplitudes. Its success probability is approximately 0.08261623248916, one ninth of the certified composite spectral weight. It detects the total response; no clock resolving the 138 energies is executed.

These lines give one clean exclusion. The visible energy spread is below eleven, so no three distinct lines with one positive unit reproduce the charged-lepton mass ratios. That excludes this finite gap dictionary, and it leaves other sources, refinements and unseen composites untouched.

5.4 Complete conserved quadratics

The labels of these spectra come from a complete classification.

Theorem 5.1 (Complete quadratic conservation)**.** *On a finite connected open rectangular box with V sites and two fermion modes per site, let*

$$H_\mu = \sum_{\langle x,y=x+e_j\rangle} c_x^\dagger(\tfrac{ic}{2}\sigma_j)c_y + \text{h.c.} + U\sum_x n_{x\uparrow}n_{x\downarrow} - \mu N, \qquad c, U > 0.$$

Modulo scalars, the real space of self-adjoint even conserved operators of degree at most two is spanned by $N - V$ and S_1, S_2, S_3 when $\mu \neq U/2$, and by these together with η_x, η_y when $\mu = U/2$, where

$$S_a = \tfrac{1}{2}\sum_x (-1)^{\sum_{j\neq a} x_j} c_x^\dagger \sigma_a c_x, \qquad \eta_+ = \sum_x (-1)^{x_1+x_2+x_3} c_{x\uparrow}^\dagger c_{x\downarrow}^\dagger.$$

The traceless conserved Lie algebra is hidden $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ *for* H_0 *and hidden* $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ *for the centered generator* $K = H_{U/2}$.

Proof. Commutators with H_μ preserve the number change of each term, so the number-preserving and anomalous parts separate. The one-particle restriction gives $[h, q] = 0$ for a one-body matrix q, hence kinetic commutation on the whole Fock algebra and commutation with the doublon count. An input with one spectator and its doublon output isolate each intersite coefficient and force it to vanish. The kinetic condition $q_y = \sigma_j q_x \sigma_j$ then transports one Hermitian root matrix over the connected box, which gives the four stated coefficients. For a pair creator, comparing the vacuum-to-pair entry with its spectator entry gives $U\Delta_{pq} = 0$ on distinct sites. Onsite terms obey $(U - 2\mu)f_x = 0$, and at $\mu = U/2$ the kinetic condition forces the alternating profile, whose real and imaginary parts are η_x and η_y [16]. □

Exact full-CAR coefficient ranks and independent occupation-matrix checks verify the finite instances, and centering changes the conserved pairing algebra as well as the charged energies. Higher-degree currents, Lorentz stress and an interacting chiral continuum are outside the theorem. The radial defect source has its own classification: every conserved sum of onsite even operators is eta-pairing SU(2), and on the mass-64 source every center Pauli current has leakage norm above 1/64 outside the low quartet. The complete native color, weak and hypercharge action therefore cannot be inherited from that onsite-even algebra.

5.5 One-body massless and massive sectors

The same source has exact one-body sectors. On the periodic cube of eight sites per axis a complete spectral detector resolves all 1,024 one-particle modes, including sixteen zeros and six pairs of signed shells, with every pointer and abort outcome retained. A stationary one-particle state cannot occupy only six sites, and the infinite lattice has no square-summable stationary one-particle state.

With centered clock $H-(U/2)N$ and $c = v/a$, the continuum is the free massless field in all eight Weyl valleys, $E = v|\mathbf{p}|$, throughout $0 \le U$, $U = o(c^{2/7})$ [8]. Supplied quadratic mass coefficients give two Dirac copies at each rest energy $|m_0 \pm |\mathbf{m}||$, with continuum dispersion $E^2 = v^2|\mathbf{p}|^2 + M^2$ and a proved finite-spacing error. This uniform family has at most two distinct positive rest energies, so a direct assignment cannot reproduce three distinct charged-lepton masses. The ordered-spin continuum has nonzero positive-energy bands at every small momentum. Complete energy–symmetry readouts bound the original response near a specified dispersion, with aliases, clock leakage and all failures retained. Selection-field and zero-field histories have a same-input comparison bound; a finite prepared state is not identified with the infinite ordered vacuum. Relaxed inverse susceptibility determines an infinitesimal-response cost, not a particle mass. A uniform nonvanishing shrinking-tube bound, Lorentz covariance and surviving continuum interactions are unproved. Matching a light speed fixes length unit $= cT$; determining T requires an independently identified same-generator clock.

6 The charged mass operator

6.1 Pixel capacity and the mass law

The pixel variable is the dimensionless capacity ratio $P = a_{\mathrm{cell}}/\ell_P^2$, distinct from a projector P_j in the memory controller. The declared self-reference relation reads

$$P = \varphi + \sqrt{\pi}\,\alpha, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

In the heat/running construction the inner response supplying α depends on P through a unified scale, a heat-sum equation, low-energy couplings and charged masses, so the loop has the form

$$P \longrightarrow (\alpha_U, v, \text{field census}) \longrightarrow (M_f, J_f) \longrightarrow A_{\text{response}} \longrightarrow \varphi + \frac{\sqrt{\pi}}{A_{\text{response}}}.$$

The same masses and current normalization appear on both sides. A fixed point of a declared response proves consistency of that map, and its identification with the physical electromagnetic response is a further claim.

The controller is a native family memory. It stores one of twelve port addresses, each with a rank-one family projector P_j. A C event rotates the port label, a joint event then applies the rotation of E, and other records reset the address to port two. The fair-coin rates of the C and E events are $p = 1/12$ and $q = 1/6$, and all 128 raw record labels are retained. Declared current probes lower these rates to the values that follow.

The mass law used here is a chosen response architecture with icosahedral registers and determinant normalization. Its charged shape uses a three-phase Koide-type parametrization [9, 13]. The architecture has target-informed provenance: its numerical comparisons are conditional postdictions even when its execution reads no measured masses. The quark exponents, scalar quartic, neutrino operator and physical flavor mixing are not determined from the premises A1–A3 of the registry [35]. Independent validation requires consequences not used to choose it. The operator realization below applies to the displayed $\beta = 4$ branch and to the normalized coefficients of the explicit $\beta = 7$ branch.

6.2 A pixel mass law in native memory dynamics

A chosen simulator specification can realize the conditional pixel-dependent charged mass law on the native gauge-family carrier. The construction uses the twelve-port reset controller and the complete native current instruments. Its design inputs are independent left and right probe durations, a rule enabling feedback on a successful C event, and a common feedback amplitude. The conditional mass-law architecture is target-informed; evaluating that architecture without a charged-mass input does not establish a prospective prediction.

The selected response architecture gives

$$\kappa = \frac{\alpha_U/50}{1 + \alpha_U/31 + \alpha_U^2/310}, \qquad \chi = \frac{\alpha_U^2/512}{1 - \alpha_U/77}, \tag{2}$$

$$\zeta = \frac{(P-\varphi)^2/(21\pi)}{1 - \alpha_U/27 - \alpha_U^2/135}, \qquad r_j = 1 + \sqrt{2}e^{-\chi}\cos\left(\frac{2}{9} + \zeta + \frac{2\pi j}{3}\right), \tag{3}$$

where $\varphi = (1+\sqrt{5})/2$ and $j = 0, 1, 2$. Its charged coefficients, ordered increasingly, are

$$t_i = \frac{m_i}{v} = \frac{e^{\kappa} r_i^2}{\left(2\,6^{14}\prod_{j=0}^{2} r_j^2\right)^{1/3}}, \qquad \frac{v}{E_*} = P^{-1/2}\exp\left(-\frac{2\pi}{4\alpha_U}\right). \tag{4}$$

The register dimensions and determinant exponent in these formulas are part of the selected architecture. The scale E_* is a source coordinate; assigning it an energy in laboratory units requires a clock calibration.

Let $t_i > 0$ be these three charged coefficients and $\lambda_i = t_i/\sum_j t_j$. For the native projectors P_2, P_3, P_{10}, direct multiplication gives $\operatorname{tr}(P_i P_j) = 1/5$ for distinct indices and the spanning-vector

Gram determinant $d = 2/5 - 2\sqrt{5}/25$. Consequently the matrix $Y = uP_2 + v_1P_3 + wP_{10}$ has elementary eigenvalue invariants

$$\operatorname{tr} Y = u + v_1 + w, \qquad e_2(Y) = \frac{4}{5}(uv_1 + uw + v_1w), \qquad \det Y = duv_1w. \tag{5}$$

Choose its three positive weights as the roots of

$$z^3 - z^2 + \frac{5}{4}e_2(\lambda)z - \frac{e_3(\lambda)}{d} = 0, \qquad w < u < v_1. \tag{6}$$

It follows that $\operatorname{spec} Y = \{\lambda_i\}$.

The complete C-enabled memory coefficient, with $a = p(1-q)$, $b = pq$ and $D = 1 - a^2 + ab$, is

$$Z_C = \frac{(1-a)P_2 + a(1-a)P_3 + abP_{10}}{D} - (1-p)P_2. \tag{7}$$

It has trace p; disabled records have not been removed or normalized away. Put $A = v_1(v_1 + w)$ and $B = v_1(1 - v_1 + w) + (v_1 + w)w$. The inverse equations

$$-Ap^2 + Bp - w = 0, \qquad q = 1 - \frac{v_1}{1 - p(v_1 + w)} \tag{8}$$

give the exact identity $Z_C = pY$. This follows by substituting the second equation and reducing each of the three coefficient differences by the quadratic in the first equation.

At each of the supplied forward and measured-endpoint pixel coordinates, rational Taylor enclosures and polynomial endpoint signs give exactly three roots of (6), and exactly two solutions of (8) satisfying $0 < p < 1/12$, $0 < q < 1/6$. For the forward coordinate the first solution is approximately $p = 0.01915526454$, $q = 0.05474497127$. The native charged-lepton support Casimirs are $9/8$ and 3, so the complete probes implement these rates with the declared durations

$$\theta_L = \frac{\arcsin\sqrt{12p}}{\sqrt{9/8}}, \qquad \theta_R = \frac{\arcsin\sqrt{6q}}{\sqrt{3}}. \tag{9}$$

Both probes retain all thirteen outcomes. Their product retains all 169×128 current and repair labels per old memory input. The covariant binary grouping retains 512 labels per input and all twelve old inputs.

Choose $g = \sum_i t_i/p$. Then gZ_C has exactly the three coefficients t_i. Its diagonalizing unitary acts on the gauge-trivial family factor, commutes with every native gauge current, and preserves the canonical kinetic form. Thus this is an operator realization of the conditional mass law, up to the allowed family frame. It does not assert an additional alternating-group intertwiner. The source-coordinate enclosures also give $\sum_i t_i < 1$ by bounding the cube of that sum.

The coefficient is the generator of a controlled limit of the full correlated memory. Apply $\exp(-ihgX \otimes P_{\text{new}})$ on enabled records and identity on disabled records. Let Ψ_h be this complete classical/quantum channel. Its classical transition satisfies $T = (1-p)R + pS$, where R resets to port two and S is stochastic. For every normalized initial classical/quantum state z_0, including an arbitrary untouched finite reference, the reduced state after N steps of size $h = \tau/N$ obeys

$$\left\| \sum_{\text{memory}} \Psi_{\tau/N}^N(z_0) - \mathcal{U}_\tau\left(\sum_{\text{memory}} z_0\right) \right\|_1 \le \frac{4g\tau}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{g^2\tau^2}{N}, \tag{10}$$

where $\mathcal{U}_\tau$ is generated by $gX \otimes Z_C$. For a stationary product preparation the first term vanishes.

To prove the bound, let $\Pi z = \pi \sum z$ and $e_k = (I - \Pi) z_k$. The sum of trace norms on memory blocks gives $\|Te\|_1 \le p\|e\|_1$ on zero-sum blocks and $\|\Psi_h - T\| \le 2gh$. Hence $\|e_k\|_1 \le 2p^k + 4gh/(1-p)$. The reduced stationary step and the mean unitary step have the same first derivative; their two Taylor remainders sum to at most $4g^2h^2$. The reduced error due to e_k is at most $2gh\|e_k\|_1$. Summing these errors proves (10).

This limit is in dimensionless simulator time. A finite unread channel need not equal its limiting unitary. The feedback pulse service is an explicit design choice; compilation into a smaller native gate alphabet is a separate requirement. The forward pixel coordinate has no complete physical endpoint proof, and the measured-endpoint coordinate inherits measured-alpha calibration. No physical pole-mass identification, renormalization map, operational energy scale, or complete quark and neutrino spectrum follows from this finite realization.

6.3 Memory correlations beyond the mean unitary

The native stationary memory has three distinct projector rays, because $P_9 = P_{10}$. Summing their complete temporal correlations gives a controlled correction to the mean evolution. Let T be the reduced column-stochastic transition, E its enabled-pulse part, π its stationary column, $\Pi = \pi \mathbf{1}^T$ and $w = E\pi$. The reset identity contracts zero-sum memory vectors by $p < 1$, so the group inverse $\mathcal{R} = (I - T + \Pi)^{-1} - \Pi$ sums every lag. The one-sided covariance and its positive symmetric part are

$$K = \tfrac{1}{2}(\operatorname{diag} w - ww^T) + E\mathcal{R} \operatorname{diag} w, \qquad D = K + K^T \succeq 0,$$

where positivity follows from the limiting covariance of accumulated pulse counts. For $H_i = gX \otimes P_i$ and $L_i = -i[H_i, \cdot]$, the reduced generator is $G_0 + hG_1$, with $G_0 = \sum_i w_i L_i$, $G_1 = \sum_{ij} K_{ij} L_i L_j$. Its antisymmetric coefficient gives the coherent ordering potential $H_{\mathrm{ord}} = -(i/2) \sum_{ij} K_{ij} [H_i, H_j]$; the symmetric part gives the dissipator $-\frac{1}{2} \sum_{ij} D_{ij} [H_i, [H_j, \cdot]]$. Both belong to the same correction, and $H_{\mathrm{ord}} = I_2 \otimes A$ is a family potential with a different auxiliary grading from a scalar mass.

For stationary product preparation, any untouched reference, $h = t/N$ and fixed t, the corrected completely positive semigroup has an explicit rational $O(h^2)$ finite-time diamond-norm bound on the unread channel. Complete record tapes and correlated initial memory need their own terms.

6.4 Complete records, capacity and recovery

The native twelve-address family memory admits an exact smaller encoding. Every raw branch preserves the recurrent set $S = \{2, 3, 9, 10\}$. Let E_S embed four memory addresses into these source addresses. For all 128 raw labels, all twelve original old-memory labels and every real pulse size, the corresponding reduced branch B_y obeys

$$K_y (E_S \otimes I) = (E_S \otimes I) B_y.$$

Multiplication proves equality of all joint records and terminal work states for any initially encoded state, including entanglement with an arbitrary reference. Reset to address 2 and stationary memory qualify. Address 0 admits one fully retained startup event, landing at 2 or 9, followed by exact recurrent encoding. These maps encode one complete law, and physical independence of capacity would need more.

The native matching code obeys the ideal recovery identity $R_y K_y E = \sqrt{w_y} E$, with $\sum_y w_y = 1$, for every retained branch, on both matching codes, both repair laws and all thirty seams. Specified logical evolution between corrected repairs is therefore preserved even conditional on the records, and faithful recovery cannot shift its poles.

Changing feedback duty changes the evolution. The native always-duty and C-duty generators have ratios approximately $1 : 195.117 : 3579.826$ and $1 : 53.588 : 271.403$. Their normalized quadratic traces differ exactly, so neither a family relabeling nor a common clock rescales one into the other. Dividing by event probability changes only the common scale. The specified nine-transition preparation and two-clock inverse have stationary-kernel error at most $1/2579890176$, which controls a first-order kernel approximation.

6.5 Flavor from current-gated memory

Reading a specified memory's angle, reset rule, duty policy and feedback law determines its stationary coefficients, and these are source-derived for that specification. The declared native current gates the complete repair record differently in the up, down and charged-lepton sectors, and the static mean-action service has certified full-action spectra and nontrivial real quark mixing. For $\theta = \pi/5$, duty "always" and unit Higgs norm, the dimensionless coefficients are as follows.

Channel	Three ordered coefficients		
Up	0.00015764215	0.04279127122	0.95705108663
Down	0.00014014156	0.04369077654	0.95616908190
Charged lepton	0.00003179436	0.02148604675	0.97848215888

Every coefficient in the retained real-projector class has $\mathrm{Tr}[H_u, H_d]^3 = 0$, and complex family basis changes cannot create charged-current CP violation. All four default duty policies fail direct identification of the charged-lepton ratios: under a common positive normalization and relative corrections bounded by δ, the unconditional and C-duty cases require $\delta \geq 0.79167$ and $\delta \geq 0.59931$, and E and CE duties have an exact zero coefficient. Other controllers and source angles lie outside this comparison.

The correlated memory has a controlled realization of these same mean coefficients. For duty $D \in \{1, C, E, CE\}$, retain every path and apply $e^{-ih(H_0+\chi_D(r)V_{F(m,r)})}$, with $\|V_m\| \leq L$ and $h = t/N$. The reset probability $1 - p$ gives the reference-safe reduced bound

$$\delta_N \leq \frac{4pLt + 4(1+p)L^2t^2}{N(1-p)}.$$

The complete archive differs by at most $\sqrt{2\delta_N}$ in unhalved trace distance from the *same* record law with the mean unitary on the target. Initial correlations and all fine outcomes are retained. The interaction-picture estimate contains no $\|H_0\|$ factor. The physical CAR pairing lift has norm at most 14 per cell at unit neutral Higgs amplitude, using directed tensor ranks $3, 3, 1$. Its record-conditioned controller is specified wiring. The finite-volume bound does not identify target pulse time with auxiliary gate overhead, prepare a vacuum, or give a volume-uniform limit.

The four charged-lepton coefficients survive this limit. Their action ratios $\sum m_i/(\sum \sqrt{m_i})^2$ lie near $0.76758, 0.57975, 0.95024, 0.77544$, respectively, and exclude $2/3$. A separate coherent source construction supplies a different action. For the regular three-cycle R, its fully retained analyzer has block $C/\sqrt{3}$, where

$$C = \sqrt{10/72}\, I + \sqrt{5/72}(e^{i\phi}R + e^{-i\phi}R^2), \qquad Y = C^2/3.$$

In the positive chamber $|\phi| \leq \pi/12$, the action singular values of Y satisfy Koide exactly. A projection kick with fresh ancillary flags realizes the logical mean $X \otimes Y$; keeping all eight outcomes

at every step gives complete-output error at most $9t^2/(4N)$. The full exterior lift of the analyzer acts on charged ancillary CAR copies. Vacuum compression of its conjugated charged-Higgs kick is exactly the physical pairing with coefficient Y on all ninety-mode occupations. Committing classical vacuum/nonvacuum flags and retaining the entire charged quantum archive gives unhalved trace error at most $108|v_H|^2t^2/N$. The fresh modes, analyzer and pulse clock are specified apparatus; their overhead is retained. This coherent controller is distinct from the four memory duties and leaves ϕ explicit. Ordinary Yukawa masses are singular values of Y; the square in $Y = C^2/3$ belongs to this particular source construction.

An additional neutral loss flag gives a complete attenuation controller. Before the Fourier analyzer it multiplies both rotation-branch amplitudes by $r \in [0,1]$ and sends their remaining amplitudes to retained loss records. The sixteen-outcome instrument has

$$C_r = \sqrt{10/72}I + r\sqrt{5/72}(e^{i\phi}R + e^{-i\phi}R^2), \qquad Y_r = C_r^2/3, \qquad p_{\text{loss}} = 5(1-r^2)/36.$$

The full-CAR compression theorem applies to this enlarged apparatus. In the positive chamber its quotient is $Q = (1+r^2)/3$; the exact controls $r = 1, 3/5, 0$ give $2/3, 34/75, 1/3$. All outcomes are retained without renormalizing the selected effect. Thus $r = e^{-\chi}$ is constructible, but the unchanged primitive records do not select that control. Its native logical analyzer and complete weak round are executed; a full many-fermion native tape is not claimed (`model/particle_candidate_atlas/BARE_EFFECTIVE_CONTROLLER.md`).

With the same native Weyl kinetic symbol and a chosen filled-negative-energy Gaussian reference, the quadratic action has three charged Dirac family branches

$$E_j(\mathbf{p})^2 = c^2|\mathbf{p}|^2 + |v_H|^2\sigma_j(Y)^2.$$

The lattice sine symbol converges uniformly on each compact momentum set, with Hamiltonian error at most $ca^2P^3/2$ for $|p_j| \le P$; a positive charged mass floor controls the reference-projector error. All eight valleys and their mirror currents remain. These are family-resolved free branches, not three lower thresholds of the full electric-charge sector. The complete phase score reads their unordered coefficients, while the ordinary measured repair maps are phase-blind. The full-CAR and free-field proofs are in `model/charged_lepton_response/`.

The same Clifford proof accepts every positive Hermitian family matrix. Writing $V(Y)$ for the native charged pairing, real linearity gives $V(gZ_C) = g\sum_r w_r V(P_r)$ and $\|V(P_r)\| = 2$ for a rank-one family projector. Complete C-duty feedback therefore supplies the entire pixel-dependent coefficient to this full-Fock free-field map. For a finite region of V cells the correlated-history theorem uses $L = 2Vg|v_H|$ and $p_{\text{joint}} = 1-(1-p)^V$, with all original records. This establishes conditional free masses for the deformed coefficient without assuming exact Koide balance. The charged-only free action also has massless spectators: three zero-momentum down-type color modes form a global color singlet of electric charge -1 and zero energy. This countercontrol imposes no local Gauss constraint and identifies no hadron; it excludes reading the three family masses as the three lowest unrestricted electric-sector thresholds (`model/particle_candidate_atlas/CHARGED_FIELD_ADMISSION.md`).

Including all three native Higgs tensors defines a separate complete charged action. With the explicit common choice $Y_u = Y_d = Y_e = \text{diag}(e_k)$, its directed native matrix has rank seven and satisfies $D^\dagger D + DD^\dagger = P_{\text{charged}}$ of rank fourteen. Every remaining zero mode is neutral. The complete quasiparticle Fock law then obeys $H - E_G \ge e_{\min}|Q|$, attained in the light charged-lepton rest branch. This removes the charged zero-mode obstruction on that action while retaining all eight valleys. The common family choice gives equal up, down and lepton spectra, so it does not fit unequal observed quark masses. The exact tensor, full-charge and retained-neutral certificates are in `model/openwave_full_higgs_floor/`.

A distinct confined native Clifford and quantized Maxwell member joins localized position, ground-relative energy, current and field angular momentum. Complete oscillator shells and a positive operator-valued CAR lift isolate a sixteen-dimensional charged ground band containing eight identical doublets with exact $J^2 = 3/4$. Preparation selects one spin-up line in one native copy, including a nonzero Maxwell angular share. The global charge -1, trap and coupling are declared. One electron energy anchor calibrates every compared photon cap; its cap-two static magnetic coefficient lies in $[1.998275353408213, 1.9982753545060563]$. This is a finite source result, without a measured anomalous-moment fit, three-family transfer or continuum claim (`model/openwave_quantum_rotational/`).

The literal antiperiodic side-four native source also permits transfer of a prepared neutral $e^c LL$ spin-half packet, with full-source third moment $-3i\sqrt{2}/64$. Separate declared neutral-family and weak-contact members supply complete three-flavour oscillation records and a nonzero heavy-to-light charged transition with neutral ejecta. They do not supply measured PMNS parameters, a physical neutrino mass or a muon lifetime. Their complete records and source assumptions are given in `model/openwave_neutral_transfer/`, `model/openwave_neutrino_oscillation/` and `model/openwave_weak_decay/`.

6.6 One hyperfine spectrum, charge profiles and complete gauge records

One native hyperfine action. A positive current interaction $h[\frac{3}{2}N_q N_{\bar{q}} + 2\mathbf{S}_q \cdot \mathbf{S}_{\bar{q}}]$ is the sum of quark–antiquark pair triplet squares. It vanishes on every pure-quark state and shifts a spin-one one-pair meson by $2h$, leaving its spin-zero partner unchanged. Together with a calibrated coefficient of the existing local spin square, one declared action gives a neutral vector energy 0.77526 GeV and an actual $S = I = 3/2$, charge-one baryon energy 1.232 GeV, while preserving the five specified proton, neutron, pion, axial and nuclear-binding calibration agreements. These are seven empirical inputs. The two added mass conventions are Breit–Wigner numbers, not derived complex resonance poles or widths. The complete 120, 64 and 2520 baryon, meson and nuclear singlet spectra, their coloured preparation failures and all magnetic partners are retained. Exact ground-line bounds test transfer of the native current and charge records, without identifying states from fitted labels alone. For the complete local 2^{24} matter Fock space, exact SU(3) fusion gives $\sum_x C_x^2 \le (22/7)\sum_e E_e^2$ on the physical zero-boundary-flux Gauss carrier. An independent quark/antiquark-count bound also retains the positive pair and baryon-density terms. Both apply to the same conditional quadrupole/string-length attachment; neither is a four-dimensional pure-gauge mass-gap assertion. The complete source and instruments are specified in `model/openwave_common_hyperfine_spectrum`.

Native charge profiles and their independent radius test. On the same seven-input strong source, the literal native isospin raising operator transports the complete neutron ground-ray certificate to the proton ray. Their mass difference is 0.00129333251 GeV. The intrinsic neutron charge is positive at graph radius 1/3 and negative at 2/3. A maximally mixed input on all 120 physical colour singlets gives a ground-ray filter probability 1/120; all 2024 occupation effects and their failure records are retained through a compressed positive-operator measurement. Its nonselective mean source work on that input is zero.

With the shared quadrupole/string length near 0.7559087287 fm, the orientation-averaged charge records give a proton root-mean-square radius near 0.3345681563 fm and neutron charge second moment near -0.03731193543 fm^2. The common length cancels in $-\langle r_n^2\rangle/\langle r_p^2\rangle$. Its certified interval is disjoint from the comparison box formed by the CODATA2022 proton radius and PDG2025 neutron mean-square radius. No single length therefore fits both radii on this selected source. The graph orientation average is specified; no relativistic Sachs-current or continuum

nucleon-pole identification follows from these records (`model/openwave_baryon_form_factors`). The complete source and instruments are specified in `model/openwave_baryon_form_factors`.

Complete strong-source reduction and a linear potential. On the same seven-input action, the complete native baryon singlet carriers have dimensions $120, 364, 816, 1540$ on two through five cells. Eliminating the full long-span complement yields the exact Schur action $H_{\mathrm{eff}}(E) = A + V(E - D)^{-1}V^{\dagger}$, together with its normalization metric and native current vertex $R_f^{\dagger} J R_i$. Rational complement bounds control their variation across a stated energy window; full finite-time records retain the complement and all failures. Graph-dependent masses and axial responses are diagnostics, without a continuum QCD running-coupling claim. For a pinned quark and diquark at nonadjacent interior sites, all 40 internal singlet states obey $H_{\mathrm{pin}}(r+1) - H_{\mathrm{pin}}(r) = (4\kappa/3)I$. This exact total-source identity includes local, spin, isospin, hopping and Gauss-electric terms. Its common conditional length gives a constrained inter-charge slope $1\,\mathrm{GeV/fm}$, explicitly used as an attachment target. The result concerns prescribed positions and does not identify the unpinned ground energy or a four-dimensional pure-gauge mass gap. The complete source and instruments are specified in `model/openwave_strong_step_scaling`.

Calibrated interacting source poles. The complete native three-quark Gauss source has five compact cluster fields per translation cell, with matrix spectral measure of zeroth moment I_5. Exact bounds retain every relative configuration outside this frame; the squared compact-to-complement coupling norm is $5/64$. Two declared counterterms vanish on the entire calibrated two-cell operator. With the inherited neutron and axial inputs, the neutron rest pole lies in $[0.939565387072, 0.939565456806]$ GeV, and the native axial magnitude in $[1.274757, 1.275843]$; native isospin gives the proton shift and vector amplitude one. The fixed-coefficient control has disjoint axial interval $[1.291676, 1.292715]$, so preserving two-cell fits alone does not preserve spatial refinement.

The pole residue is rank one in each tested momentum and spin/isospin fibre. If q bounds complementary probability, s bounds the compact-ground angle and τ the actual rational preparation's angle, its pole weight obeys

$$Z \geq (1 - q)\left[1 - (s + \tau)^2\right] > 0.9999997$$

at rest. The complete five-coordinate filter succeeds with probability $1/5$; its $4/5$ failure, other spectral outcomes and work are retained. This certifies observable poles of an interacting one-dimensional source, without identifying physical four-dimensional Lorentz particles (`model/openwave_hadron_refinement_flow`, `model/openwave_baryon_poles`).

Propagation with the same native rest operator. A positive correlated kinetic operator covers every colour-singlet triquark shape of occupied span at most one. Its common rule is $J = \zeta M^{-1}$ for the complete compact rest matrix M and $\zeta = (\hbar c/a)^2/(2u^2)$; the action vanishes on the complete three-quark rest fibre. Exact five-band and infinite-complement bounds give width $[0.1450650, 0.1450654]$ GeV and neutron inertial mass $[0.91037, 0.97069]$ GeV. The electron unit, strong length and physical speed are supplied. Full Fock positivity supplies neither higher-number rest cancellation nor exact Lorentz dispersion (`model/openwave_baryon_dispersion`).

Uniform native particle bands. The complete interacting baryon source has an isolated simple ground pole at every momentum on its Brillouin circle. An exact finite shortcut splits its five

compact triquark coordinates into three nearby light coordinates and two high coordinates, retaining the latter with the complete infinite relative-position tail. Subtracting the common scalar part of inverse-rest propagation reduces the remaining momentum derivative bound to less than 0.178 source units. A rational interval cover then certifies a ground isolation gap above 0.127 source units and high-complement probability below 2.7×10^{-6} for both charges. These are unchanged calibrated one-dimensional source poles, with all spectral complements retained. They supply a uniform input for normalizable packets; no physical four-dimensional particle identification follows. Exact source records are in `model/openwave_uniform_pole_bands`.

Normalizable source particles and complete records. The explicit momentum amplitude $f(k) = \sqrt{15/(16\kappa)}[1 - (k/\kappa)^2]$ on $|k| \le \kappa = 1/512$, zero outside, has unit norm on the infinite chain and finite position variance $5/(2\kappa^2) = 655360$ cell^2. Uniform full-source projector bounds give neutron and proton band probabilities above 0.9999614. After conditioning on the band outcome, more than 98.4025% lies within 8192 source cells of the origin. The same native weak-current packet response has vector amplitude above 0.983876558018 and axial interval $[1.226387005970, 1.324212994030]$. Preparation, band, current and position outcomes retain every failure, reference and work term. These are actual normalizable states, rather than fixed-momentum plane waves. The exact band filter remains a declared nonlocal spectral service; finite local compilation and physical four-dimensional particle identification are separate obligations. Exact records are in `model/openwave_normalizable_particle_records`.

Native weak response on the same propagating poles. Exact CAR contraction of the complete compact triquark source gives $V_P = I_5$ and $A_P = 5/3 \oplus \left(\begin{smallmatrix} 1 & 2/3 \\ 2/3 & 1 \end{smallmatrix}\right)^{\oplus 2}$. The full fixed-charge axial square satisfies $(M - I)(M - 4I) = 0$, retaining current-produced spin and isospin channels outside the pole frame. Complement and actual rational-preparation bounds transfer these currents to the isolated interacting source poles at six momenta. The rest vector amplitude is exactly one; the rest axial interval is $[1.274757331914, 1.275842668086]$, with its strong axial calibration retained. At lattice phase π the vector amplitude remains above 0.994488237318. A complete charge-conserving current instrument records success, failure, every complementary spectral outcome and bounded source work. These source responses do not establish autonomous beta decay or a physical lifetime. Exact records are in `model/openwave_pole_weak_current`.

Local strange families and an energy-conserving shell. The complete two-cell native source uses two positive number/pair coefficients to calibrate charged kaon and Lambda energies to 0.493677 and 1.115683 GeV. Complete 816- and 256-state colour-singlet carriers retain the failed Sigma, Xi and neutral-kaon comparisons. The charge-conjugate strong action and nonzero native weak vertex are explicit. Its global number correction prevents separated-cluster additivity, and its full two-cell outgoing channel is closed even after the weak norm bound (`model/openwave_strange_composites`).

A finite-range extension of the complete native strange action restores additivity of separated colour-singlet clusters. With the same positive inverse-rest-operator propagation rule, full relative-position bounds give $m_\Lambda \in [1.11568462, 1.11568488]$ GeV and $m_{\pi^-} \in [0.13959620, 0.13959623]$ GeV in the supplied electron unit. These inherited-calibration bulk energies retain their shifts from the two-cell fits. The proton and pion ground-band energy sum crosses the Lambda rest interval between $\tan(k/2) = 1/6$ and $1/5$, proving an open strong-sector energy shell. The unfiltered weak current requires an interacting scattering-state transfer; no strange lifetime follows from the shell

crossing alone. The complete construction and records are specified in `model/openwave_cluster_bandwidth`.

A native current identity for the bulk strange source. The complete compact *uds* frame has 56 shapes and its $S = 1/2$, $I = 0$ Lambda sector has seven coordinates with Gram matrix G. Exact CAR contraction of the inherited strangeness-changing weak current into all 1,656 nonzero five-body occupation outputs gives

$$6c^2G \;\le\; W_P^\dagger W_P \;\le\; 7c^2G, \qquad c = \sin\theta\cos\theta.$$

Thus every compact Lambda state has nonzero response, independently of a fitted eigenvector. The current preserves occupied spatial span; a complete source-pole complementary probability bound q therefore gives $\|P_{\mathrm{out}} W \psi_\Lambda\|^2 \ge 6c^2(1-q) > 0$. At fixed total momentum, the calibrated bulk Lambda source pole lies in $[1.115684628607, 1.115684878081]$ GeV. All current outputs and probe failure are retained. A separately enclosed proton–pion energy shell is present in the same source, but its interacting five-body scattering projector has not been identified with this compact current response. No physical Lambda lifetime follows. Exact records and the remaining transfer premise are specified in `model/openwave_strange_shell_current`.

Native proton–pion fields and complete strong evolution. Literal CAR contraction projects the complete native weak output onto 23 independent antisymmetrized proton–pion fields. Relative to the seven-coordinate Λ Gram matrix, the projected-current square has eigenvalues $6/5$, $241/150$ and $(277 \pm \sqrt{54409})/200$, with multiplicities $1, 2, 2, 2$, before the inherited Cabibbo factor. The complete autonomous five-body Hamiltonian includes both correlated inverse-rest triquark and meson terms. Its exact separated-field response derivative has squared norm $0.02761018661589379\ldots$ times the Cabibbo factor squared for a specified localized Λ coordinate. Direct separated contraction vanishes; the full derivative is nonzero. A spectral-theorem remainder on the complete carrier certifies positive finite-time response, retaining every orthogonal output and probe failure (`model/openwave_full_strong_current`).

Normalizable strange-particle records. The native Λ trial has normalized momentum amplitude proportional to $1 - (k/\kappa)^2$ for $|k| \le \kappa = 1/512$. Its leftmost-cell variance is exactly 655360 and its interacting ground-band probability exceeds 0.9997. The seven-field residue theorem and a bounded-momentum perturbation estimate suffice; isolation over the whole Brillouin circle is unnecessary for this trial. All 72 native weak CAR words preserve the leftmost occupied cell. The complete current and 23-field channel therefore act as the identity on the spatial packet times their exact internal maps. Actual trial probabilities are Gram contractions, with all preparation, other-channel and probe failures retained. Every nonzero internal branch keeps the same packet. This principal experiment uses no exact band filter; optional full-pole conditioning is an explicitly nonlocal service. Source rest-pole GeV intervals retain their calibrations, and the packet mean is a distinct energy observable (`model/openwave_lambda_wavepacket`).

The native breakup spectrum. The proton–pion energy shell belongs to the essential spectrum of the complete interacting five-body strong Hamiltonian. Exact CAR/Gauss factorization on separated singlet supports includes both correlated cluster operators. The finite Schur theorem gives factorially controlled relative tails; normalized spatial boxes then produce a singular Weyl sequence at the native Λ rest energy. Its coefficients use the isolated full-source spectral projections, and the executable certifies quantitative residual bounds, without executing a finite observer preparation of

these Weyl vectors. An outgoing wave operator and nonzero on-shell weak spectral weight are not established, so neither a physical lifetime nor a $3+1$-dimensional particle dictionary follows. The source is one dimensional and retains its inherited GeV calibrations (`model/openwave_cluster_weyl_sequence`).

Nuclear isospin and native charge response. An occupancy-gated three-cell spin interaction vanishes on every sector with at most six quarks, preserving all seven two-cell calibration domains. One triton mass input fixes its coefficient. Exact colour exterior characters reduce complete 572572- and 176729280-singlet carriers to 64 and 256 low coordinates with full complement bounds. The triton-assigned interval $[2.8089192030, 2.8089230704]$ GeV contains its 2.80892113668 GeV input (`model/openwave_light_nuclear_curve`).

The native isospin intertwiner fixes the strong triton–helion separation at 1.29333251 MeV; the supplied nuclear comparison is 0.52952556 MeV. Thus an isospin-scalar nuclear interaction cannot repair both masses. A charge-breaking contribution of 0.76380695 MeV is required. A declared native-charge Coulomb pair kernel, using the inherited electromagnetic coupling and strong length with no nuclear fit, gives full-source helion and alpha intervals $[2.80884329, 2.80884820]$ and $[3.74016768, 3.74019421]$ GeV. Both held-out comparisons fail. The finite collinear pair kernel carries no asserted dynamical Gauss-field or physical halo transfer. The complete construction and records are specified in `model/openwave_nuclear_isospin`.

The complete spin multiplets, preparation failures and high spectral complements are retained. The Coulomb comparison and the positive correlated-propagation source are distinct members: their higher-number spectra have no asserted transfer.

Covariant matter and compact Maxwell refinement. The declared compact-field matter source uses the actual native charged spin coordinates and the positive kinetic operator $\sqrt{m^2 + (\boldsymbol{\sigma} \cdot \boldsymbol{\Pi})^2} - m$. Its Pauli magnetic commutator is retained. Every integer electric flux, compensating charge reference and native measurement complement remains in the source. Complete position–flux records have rational tail and finite arithmetic bounds, with preparation and absorption work included. A positive Wilson energy lifts all seven extra free lattice corners. The free matter and quadratic Maxwell symbols share $c^2 = 1/128$ and have controlled spatial refinement. The construction preserves excitation number; no vacuum pair current, signed-Dirac transfer or interacting Lorentz theorem is supplied. Charged mass ratios and the GeV anchor retain their input ancestry. The complete source and instruments are specified in `model/openwave_dirac_maxwell_refinement`.

The dynamical field connection. A field-dependent signed-band transformation also transforms the quantum field momentum. For $E = -i\partial_\theta$ and a smooth periodic unitary $U(\theta)$, the exact identity is

$$U^* E U = E + \mathcal{A}, \qquad \mathcal{A} = -iU^* \partial_\theta U, \qquad U^* E^2 U = (E + \mathcal{A})^2.$$

A complete compact-rotor diagnostic executes both bands and every integer flux. Exact blocks of dimension two certify the full ground and show nonzero field-induced band conversion. Omitting the connection changes the spectrum and measurement work even though the fixed-field matter square is scalar. The native charged mass parameters are inherited inputs to this diagnostic. The result identifies a required transfer term for a signed-Dirac completion; it supplies no equivalence to the full native fermion theory or interacting Lorentz-covariant continuum. The construction and complete records are specified in `model/openwave_dirac_field_connection`.

Resolved charge profile and native effective coupling. A declared Wilson coefficient $1/2$ resolves the charged core on the positive Pauli–Maxwell member. A retained scan over $1/16, 1/4, 1/2, 1, 2$ selects this coefficient by finite spatial resolution, without empirical radius or running data. Full integer-flux ground and resolvent certificates give a radius of approximately 0.523 cells for each charged family and prove a half-suppression crossing between $3\pi/4$ and $7\pi/8$ per cell. The literal native scalar-charge intervention gives a certified ground-energy derivative equal to the same form factor after separate reference subtraction. Executed positive and negative probes independently cross half coupling inside the spatial Brillouin interval. The same current satisfies the Ward identity and has four strictly ordered imaginary-frequency responses, each enclosed to absolute error 10^{-9}. Both spin branches, preparation failure, the true field-vacuum subtraction and complete pointer/readout work are retained. The resulting GeV sector energies are reference dependent and are not observed particle poles; no physical running-coupling curve or interacting continuum is inferred. The construction and complete records are specified in `model/openwave_resolved_core`.

Native clocks and the phase resource. The phase boundary admits a branchwise proof. An energy-invariant joint source and battery state, an energy-conserving unitary, and a battery readout commuting with battery energy give an energy-invariant conditional source state for every outcome. Exact execution retains all 4×81 native input and battery-word records. A two-copy native exchange creates relative coherence in a degenerate energy block, whose free frequency is zero; its interaction-driven population frequency is 2λ, with λ a declared coefficient. An energy-mixing battery readout creates the missing phase and has explicit mean work $m(p-1/2)$. These results exclude the stated finite phase-free collision route without excluding other declared observer architectures. The construction and complete records are specified in `model/openwave_clock_phase_resource`.

Continuous quotient winding and fractional charge. The native fractional charges admit a continuous quotient-group lift. For $G = (SU(3) \times U(1))/\mathbb{Z}_3$, the universal-cover kernel is $\{(e^{2\pi ik/3}I, 2\pi k/3) : k \in \mathbb{Z}\}$, so $\pi_1(G) = \mathbb{Z}$. An explicit generator has electromagnetic phase lifts $2\pi(2/3)$ and $2\pi(-1/3)$ on the actual quark coordinates, compensated by colour. Complete 120- and 364-state Gauss carriers satisfy the same endpoint identity. Their electric eigenstates have uniform conjugate-angle density; the continuous group loop does not establish a localized classical texture or a quark rest mass. The construction and complete records are specified in `model/openwave_quotient_homotopy`.

Full-Haar gauge records and spatial blocking. An actual spatial cube retains all six plaquettes and every Haar representation. Complete native contractions and whole-complement spectral bounds give a positive source gap and strictly positive adjacent-face Wilson covariances at the declared coupling. All fifteen face pairs, all 64 joint outcomes and the apparatus-work bound remain. Separately, the exact bounded-Wilson identity for the shared-link source controls its coupling derivative without differentiating an unbounded truncated matrix. A fixed physical gap supplies only a necessary source-clock tangent, not a measured beta function.

Subdividing the cube gives an isometric coarse Haar/Gauss embedding. Complete parity enumeration and independent Haar contraction certify a positive coarse Wilson coefficient and a uniform remainder for the blocked magnetic factor. Matching the electric term in the unchanged clock requires $g_f^2 = g_c^2/4$ at half spacing; magnetic mismatch and induced operators remain. These are finite-source and spatial-blocking results, without a full-Hamiltonian renormalization, GeV glueball or four-dimensional continuum identification (`model/openwave_cubic_haar_source`,

`model/openwave_haar_coupling_flow`, `model/openwave_gauge_refinement_clock`).

Interior-resolved full-Haar refinement. The actual refined cube has 27 vertices, 54 links and 36 faces. Arbitrary-valence Clebsch–Gordan fusion trees retain every Gauss intertwiner, including the two and five singlets of four and six fundamental legs. The vacuum, 36 four-cycles and 164 six-cycles give the complete 201-mode frame through electric energy 9/2; every omitted state has electric energy at least 5. All 7,236 face columns and their exterior fusion multiplicities enter the certificates. The coarse $(a, g^2) = (1, 64)$ and fine $(1/2, 16)$ sources have the same electric clock under edge subdivision. Their full-complement gap and Wilson-variance bounds retain interior modes and the eight-link boundary field. The exact first spectral moment for a loop of length L is

$$m_1 = \frac{\alpha L}{4}(1 - \langle W^2 \rangle), \qquad v e^{-t m_1/v} \le C(t) \le v e^{-t\gamma}.$$

Here v is the connected variance and γ a certified full-source gap. These are finite spatial-refinement comparisons, without an inferred running coupling or a physical continuum identification (`model/openwave_refined_haar_source`, `model/openwave_refined_haar_spectrum`).

Controlled memory of omitted gauge modes. For this complete frame, $QHQ \ge 80Q$. Full Wilson action computes the leading positive Schur correction

$$G = PHQ(16E_Q + 9)^{-1}QHP, \qquad \|K(t)\| \le \|QHP\|^2 e^{-80t}.$$

Its 3,260 nonzero entries include every output intertwiner. A Neumann ratio 9/89 bounds the infinite-channel zero-energy self-energy error in source-energy units by $9\|QHP\|^2/(89 \cdot 80) \le 0.00175460$. A separate exact control rejects autonomous boundary-only compression. At $t = 1/1024$, retaining interior modes improves the Haar-vacuum semigroup-defect bound by a factor greater than 225. This controls one specified probe and finite memory. Exact autonomy and a uniform continuum refinement are not inferred.

The full calculation is specified in `model/openwave_haar_memory_closure`.

6.7 Positive energy response and common gravitational records

An autonomous two-state geometry register couples to both complete native eight-level factors through their positive ground-referenced energies. For $H_g = \omega(I - \sigma_x)/2$, $Q = q\sigma_z$, and the positive three-dimensional harmonic profile F, the source $H = H_g + e^Q K_s + e^{FQ} K_t$ has an explicit square-root ground energy. Its mixed source/probe energy derivative is strictly negative. Rational bounds certify attraction for all 49 excited native pairs at three separations, with both self energies subtracted. Complete autonomous records conserve combined energy and retain measurement work. The positive energy coupling is a declared construction; the reciprocal scalar source's opposite-charge control is not transferred to it. Its dimensionless Newton coefficient follows from the mode stiffness, while physical strength and Einstein dynamics require separate attachment. The complete construction is `model/openwave_universal_energy_geometry`.

Native lapse and finite curvature. For the complete two-state geometry source, the weak-source derivative of the actual reference-normalized clock lapse is $-2\sinh(q)\tanh(qF)/\omega$. Its linear-profile error is bounded by $2\sinh(q)(qF)^3/(3\omega)$. Commuting lattice derivatives give exact linearized Bianchi identities on the declared metric variation, including every boundary halo. Their curvature Laplacian differs from the nearest-neighbour Poisson Laplacian by $(a^2/4)\sum_i \Delta_i^2$; that correction

is retained. This identifies a finite response and error budget, without deriving a physical metric assignment or nonlinear Einstein limit. The source and complete records are specified in `model/openwave_source_ward`.

Joint transverse light and native clocks. A complete three-dimensional Maxwell complex has 882 directed edges and 540 transverse coordinates, with Gauss law imposed at every vertex. The declared positive-frequency field, two native clocks and the quantum geometry share one autonomous Hamiltonian. Complete finite records give arrival delay 0.006354 ± 0.000509, transverse displacement -0.003216 ± 0.000050, and Fourier-direction change -0.004595 ± 0.000057, in source units. The joint native clock phase ratios are approximately 0.992837 and 0.992829. Photon vacuum, no-arrival, polarization complements and apparatus work are retained. These are finite constitutive Maxwell records, with no canonical metric quantum-field or Einstein-continuum transfer asserted. A separate attachment of the electron energy and strong length fails both the observed Newton strength and light speed; it supplies no fitted or derived physical G. The complete source and instruments are specified in `model/openwave_transverse_gravity`.

Common-unit compatibility. For the positive native-energy geometry, write u for the electron-anchored energy unit, a for the edge length, and $g = 1/(4\pi\kappa)$ for its weak-source coefficient. The two unit equations imply

$$\frac{G_{\mathrm{model}}}{c_{\mathrm{model}}/c} = \frac{g}{u^2}.$$

Thus a change of length alone cannot match both physical constants. The speed-matched length is about 386.159 fm. Exact rational stiffness and arctangent enclosures then give $G_{\mathrm{model}} \simeq 4.05075 \times 10^4\,\mathrm{GeV}^{-2}$. Retaining the electron anchor would require a stiffness factor about 6.04×10^{42}; that adjustment would be a calibration to G. This rejects the specified unit attachment, without excluding other declared geometry sources. Six Lean identities establish this compatibility condition, the source scaling relations and the inherited capacity law. The construction and complete records are specified in `model/openwave_gravity_unit_constraint`.

Calibrated collective gravitational strength. A distinct member fixes each register frequency, $H_g = (\omega/2)\sum_i (I - X_i)$, while retaining $Q_N = (q/N)\sum_i Z_i$. Its weak response decreases as $1/N$ and its free register gap stays ω. With the declared $q = 1/8$, $\omega = 1$ and $g_0 = Sq^2/(2\pi\omega)$,

$$\frac{g_0}{N}\left(1 - \frac{q^2}{3N^2}\right) \le g_N \le \frac{g_0/N}{1 - q^2/(2N)}.$$

Choosing the integer $N \simeq 6.04 \times 10^{42}$ using the observed Newton coefficient makes weak gravity and the free-symbol light-speed unit equations compatible with the unchanged electron anchor. This is a calibration of the declared hardware to G, without an independent prediction of its value. Full coordinate readout costs $N\omega/2$ source units. No common strong/geometry spatial theory or dressed-photon identification is proved (`model/openwave_collective_gravity_strength`).

6.8 Native calibrated states, complete currents and common mass records

A family of declared native-current constructions supplies calibrated composite states, massive neutral families and complete observer records. Exact representation identities, finite spectral instruments and Gaussian contractions reduce the full admitted sources. The positive spin-gradient strong member fits five stated data and supports an actual three-dimensional beta-current transfer.

The weak decay member has controlled massive-neutral survival intervals and an unused tau-width-ratio comparison. Source members, reference states and empirical inputs are stated separately; the boundaries of each transfer are part of the result.

Positive native spin gradients. The positive interaction $\lambda \sum_a (S_0^a - S_1^a)^2$, with the declared rational coefficient $\lambda = 5/2$, changes the actual native action while preserving its colour, spin, isospin and cell-reflection symmetries. Five disclosed inputs calibrate the proton, neutron and charged-pion energies, the axial magnitude and the deuteron binding. Exact colour and angular-momentum reductions retain all 2,520 six-quark singlet states. The even spin-one, isospin-zero energy is approximately 1.87561294497 GeV, with its binding fixed by the input. The odd spin-zero, isospin-zero and odd spin-one, isospin-one channels lie approximately 0.33047 and 0.16430 MeV above their compatible two-nucleon thresholds. The rational coefficient is a retrospective architecture choice; the complete comparison menu and failed controls are retained. A spin-three, isospin-zero bound channel relative to its two-Delta threshold is explicitly unassigned. Positive native-colour bounds and complete baryon spectra control every independent-copy quark-fragment partition of six. Integer preparation amplitudes, complementary failures and a retained charge reference supply a complete finite instrument. These are finite mass and binding records with stated calibrations, without a continuum nuclear or orbital-parity identification.

The native axial-state mechanism. The four-input positive action without a spin-gradient term supplies a separate exact mechanism control. Its native vector current is one and its calibrated axial coefficient is enclosed in $[1.27529999995, 1.27530000005]$. Only about 3.58% of the baryon state is co-located. Split-site correlations evade the local multiplicity-one scalar-action value $g_A = 5/3$ without multiplying the weak current by an effective form factor. The positive colour square vanishes on every original two-cell singlet, including a separated quark–diquark state; that control supplies no spatial-binding theorem.

Operational rest-energy spectroscopy. The five-input spin-gradient strong action can be coupled to the full native light-family contact and its filled massless-neutral reference before measuring energy. The charge-zero and charge-one sectors retain 1,400 and 360 states. Exact spin and reflection reductions give complete 199- and 51-dimensional prepared blocks. The neutral preparation has substantial lines near 0.939547847, 0.939570731 and 0.939580102 GeV, with probabilities 0.53512, 0.33840 and 0.11574. The charged preparation has a dominant line near 0.938279942 GeV with probability 0.999978. These contact outputs use the fixed strong coefficients without refitting. A finite Fourier register supplies a complete energy instrument, including all pointer bins, preparation failures, posterior-state operators and the idle charge reference. Registers of dimensions 1024, 4096 and 16384 have controlled finite-source approximation bounds. The finest bin spacing is approximately 7.9844 keV. This is apparatus resolution, without a physical decay-width or infinite-volume pole interpretation. The electron unit is calibrated, and the massless-neutral finite reference is distinct from the fitted three-family massive Dirac source. The filled-neutral spectroscopy reference and the massive Gaussian reference of the angular calculation are specified separately. Their contact energy shifts are not transferred by matching particle labels.

Common strong-force controls. For the four-input control in the complete native three-quark singlet sector, the colour and baryon-number costs reduce to $A + (8U/3 - 4A/9)P_{\mathrm{split}}$. Preserving the coefficient of P_{split} preserves the calibrated axial state. A common positive total-spin square changes the Delta–proton splitting by exactly $3k_s$, while min–max bounds the rho–pion splitting

between $2k_s$ and $2k_s + 2J_s$. The required coefficients are incompatible: a Delta fit gives a rho candidate near 0.33530 GeV, and a rho fit gives a Delta candidate near 1.89194 GeV. A positive colourful electric-current square can instead calibrate a neutral scalar energy at 0.1349768 GeV with a disclosed fifth datum; its physical neutral-pion wavefunction is not identified. A local colour-spin alternative has the positive completion $h\sum_x[4N_x^2 - \sum_{a,k} d\Gamma(T_a \otimes \sigma_k)_x^2]$, proved by Cauchy on every particle-number sector. Complete native-current reductions and feasible four-input controls are retained, together with failed positive-coefficient searches. These results constrain the stated architectures without claiming a universal strong-interaction exclusion.

Nuclear and spatial controls. The exact native Casimir and cell-swap reductions resolve all 2520 six-quark singlets, or 240 at fixed zero spin and isospin projections. The four-input action without a positive spin gradient has its lowest even $(S, I) = (1, 0)$ and odd $(0, 0)$ lines about 0.19121 MeV above two nucleons. Fitting only its edge weak-current coefficient binds both channels, including an unassigned odd state. These are distinct controls of the five-input spin-gradient member described above. For three quarks the native colour isometry gives $C_X^2 = \frac{2}{3}n_X(3 - n_X)$. On a four-cycle the positive edge-colour law costs $8\kappa/3$ for an adjacent quark–diquark pair and $16\kappa/3$ for an opposite pair. Complete 120-, 364- and 816-state singlet spectra retain finite-range spatial attraction and geometry-dependent energies, without a continuum-confinement claim.

Colour strings and fixed-source spatial records. For an open chain with zero external colour flux, Gauss elimination gives $E_x = \sum_{y \le x} C_y$ and a positive electric action $\kappa \sum_x E_x^2$. The native three-quark singlet identity $C_X^2 = 2n_X(3 - n_X)/3$ implies the exact string cost $4\kappa(x_{\max} - x_{\min})/3$. On two cells, replacing U by $U - \kappa/2$ preserves the entire colour-singlet strong and colour-preserving contact action. Integer checks retain all 120 baryon and 2,520 nuclear singlets. With the same coefficients on chains of two through five cells, the complete singlet dimensions are 120, 364, 816 and 1,540. The neutral lowest energies are approximately 0.93956542194, 0.9396642972, 0.939664299718 and 0.939664299718 GeV. The two-cell mass is calibrated; the larger-chain values are fixed-source outputs. Their axial response is approximately 1.6658, so the fitted two-cell axial value does not transfer under this spatial extension. A positive source bound controls relative-separation tails uniformly in chain length. For one shared physical length, the exact relation $\sigma^2 Q = (4\kappa m_e/3)^2 Q_{\text{graph}}$ tests the string tension and nuclear quadrupole together. Using the stated quadrupole comparison and the illustrative 1 GeV/fm tension benchmark, every $\kappa \le 2U$ preserving a nonnegative local colour cost fails the simultaneous match. The resulting upper tension is below 0.429 GeV/fm. This is a finite-source comparison, without a continuum confinement or string-breaking claim. Using $Q_d = 0.285699\,\text{fm}^2$ as the quadrupole comparison[1] under the stated graph-to-quadrupole attachment, the declared $\kappa = 4$ sets the tension to approximately 0.00360537 GeV/fm. The termwise nonnegative-colour restriction gives at most 0.4286771 GeV/fm under that same length attachment. It is sufficient rather than necessary for full-source positivity. The exact local Fock bound $C_x^2 \le 3N_x$ and Gauss incidence inequality allow the number term to control a negative local colour coefficient. On the five-input source this admits a shared length near 0.755908729 fm and the illustrative tension 1 GeV/fm. Both comparison values select the attachment; no independently predicted dimensional scale follows (`model/openwave_string_positive_cone`).

Three-dimensional native beta angular records. The complete native baryon current, on the calibrated strong action, gives $a = (1-g^2)/(1+3g^2) = -0.1065439613$, $A = 2g(1-g)/(1+3g^2) =$

[1] M. Puchalski, J. Komasa and K. Pachucki, Phys. Rev. Lett. 125, 253001 (2020), arXiv:2010.06888v2.

-0.1194352515, and $B = 2g(1+g)/(1+3g^2) = 0.9871087098$. These coefficients follow from an explicit Pauli contraction checked against all 120 native baryon coordinates. The axial magnitude is a calibration input; comparisons with neutron angular data share that input's experimental ancestry. An explicit quartet and current recontraction transfers these results to the positive spin-gradient strong action with its five disclosed calibrations. The finite three-dimensional Weyl sources retain all valleys, helicities, calibrated neutral families and the two-contact form factor. On cubes of sides four and eight, a negative momentum-parity record survives the entire Fock-space Duhamel remainder $\delta = (54Gt)^2/2$, with probability error at most $(8Gt + \delta)\delta$. Preparation failure, complementary quantum records and measurement work are retained. The point-contact spin coefficients do not identify the finite anisotropic source with a physical continuum angular distribution. The displayed GeV energies use the electron anchor; the strong free-channel endpoint is distinct from an interacting pole endpoint. The reproducible source and proof are in `model/openwave_beta_angular`. The executed five-input strong source has a certified negative positive-minus-negative electron momentum probability below -2.6887×10^{-14} at $t = 10^{-4}$ and $G = 1/64$, conditional on its spectral preparation. The scalar-kinetic control has exact zero parity contrast; a separate one-dimensional chiral-ring control also has a negative contrast. Momentum sign and group velocity are distinct instruments on the finite regulator. The PDG 2025 zero-recoil comparisons are $a_0 = -0.1044 \pm 0.0007$, $A = -0.11958 \pm 0.00021$ and $B = 0.9807 \pm 0.0030$.[2] The axial input shares experimental ancestry with these data. The deviations in a_0 and B are retained; the finite calculation does not supply an independent simultaneous precision fit.

6.9 A calibrated neutral family tensor and its native oscillation records

A common neutral multiplier of the charged hierarchy fails a scale-independent test. The ratio of the solar to atmospheric squared-mass gaps lies below the normal-ordering range of the NuFIT 6.0 comparison, while the inverted assignment gives its complement, close to one. Exact analyzer intervals exclude both assignments for every positive common multiplier.

An explicit Dirac family tensor uses the six normal-ordering oscillation parameters in table 1 of NuFIT 6.0[3], including IceCube24 and Super-Kamiokande atmospheric data, as calibration inputs. With the declared lightest mass 0.001 eV, its masses are

$$(m_1, m_2, m_3) = (0.001,\ 0.008712060606,\ 0.050139804547)\,\text{eV}.$$

The same conditional charged analyzer and electron anchor fix the energy conversion. Gauge-neutral singlet coordinates complete the Dirac pairs. They contribute no gauge anomaly, and their sterile coordinates share the masses of the active coordinates. Conserved Dirac lepton number excludes a Majorana term.

The full 288-mode source retains the native weak-contact tensor and its filled reference. Direct native contractions produce an isometry into all 144 outgoing channels. Complete free-flight records retain active and sterile-coordinate outputs, the production complement and reference work. Over atmospheric phase magnitude at most 64π, the exact complete record law differs from the calibrated relativistic vacuum oscillation law by total variation below 4.1×10^{-9}. This estimate follows from the positive source dispersion and complete current/readout isometries. Independent contraction of the charged-antiparent current gives the conjugate flavour protocol. At atmospheric phase π, the two complete protocols give electron-flavour neutral and antineutral probabilities 0.04794190585 and 0.03465026894, respectively. Their contrast follows from the calibrated CP phase. The six

[2] Particle Data Group, Neutron listing, 2025 edition.

[3] I. Esteban et al., JHEP 12 (2024) 216, arXiv:2410.05380.

empirical oscillation parameters are fitted inputs; physical beam preparation is outside this free-flight comparison. The matter and decay constructions below state their own sources and transfer domains.

A native matter potential and complete medium records. Direct and fermionic-exchange contractions of the literal weak current on a neutral and rest-electron port give a twelve-state operator with characteristic polynomial $x^9(x-1)(x+1)^2$. Its unpolarized electron mean is $(P_e/2 - I/4) \otimes I_{\text{spin}}$. The calibrated weak unit therefore gives $V_{\text{CC}} = \sqrt{2}G_F n_e$, with its common neutral-current term retained. Compressing by the actual massive active weights retains all three neutral masses, both helicities and sterile outputs. For N fresh electron-spin ports, complete Kraus-word composition gives an exact finite instrument. Its neutral-record distance from the coherent mean is at most $2B^2e^{2B/N}/N$, where $B = 5|G_{\text{Weyl}}n_eT|/4$. A declared 2.5 GeV, 1300 km medium gives $P(\nu_\mu \to \nu_e) = 0.0697011$, versus 0.0491046 in vacuum; the antineutral values are 0.0197144 and 0.0351399. These are scenario outputs from the calibrated tensor and a forward-current port architecture, without a claim of equivalence to unfiltered scattering or a measured beam-event fit. All preparation, electron-spin and sterile records and apparatus work are retained. The source and proof are in `model/openwave_neutrino_matter`.

Native leptonic widths and their physical units. The actual family current and three-body phase space give $\Gamma = G^2M^5f(m/M)/(1536\pi^3)$. With the conventional weak normalization, $G = 2\sqrt{2}G_F$. All nine calibrated massive-neutral channels are retained for each charged transition; a rational bound places their total width within 1.7×10^{-8} relative to the massless-neutral value. The widths are $3.00876260 \times 10^{-19}$ GeV for the muon-like electron channel and $4.04898950 \times 10^{-13}$ GeV, $3.93790076 \times 10^{-13}$ GeV for the two tau-like leptonic channels. Their ratio is 0.972563836, compared with the unused PDG 2025 fit 0.9730 ± 0.0022. The weak coefficient is a calibration input with muon-lifetime ancestry; the muon result is a normalization check and lacks radiative corrections. The positive-frequency source has an exact parent-plus-all-daughters invariant sector. Rational Volterra and finite-bin survival bounds transfer to its calibrated massive-neutral member: exact angular coarea gives $\|\sqrt{\rho_m} - \sqrt{\rho_0}\|_2 \le 2\mu_{\max}^2$, so the inclusive probability error is at most $4Gt\mu_{\max}^2$. This controls a finite weak-lifetime interval with all angular dark channels retained. Hadronic tau channels and a total physical tau lifetime are not derived. The source and proof are in `model/openwave_lepton_widths`. For the supplied weak normalization, the exact massive muon-like survival probability at 2.1876500221 μs differs from e^{-1} by less than 1.223×10^{-7}; at four such intervals the bound is 4.892×10^{-7}. The finite energy-bin error is included. The measured muon lifetime is 2.1969811 μs, so this radiatively uncorrected normalization check has a -0.4247% discrepancy. The independent on-shell tree relation using $\alpha(0), M_W, M_Z$ instead gives 2.35418 μs and fails that precision comparison. The tau ratio is compared with the unused HFLAV fit in the PDG 2025 tau listing; the weak coefficient and unit constants use the named physical-constants table. Each prepared-parent decay member retains its full admitted sector. The tau member contains both leptonic continua; a subsequent muon-daughter decay and the hadronic tau vertices are outside that reducing-sector construction.

A controlled outgoing meson interval. For the complete native bound-meson current, the six parent coordinates couple to every admitted emitted channel through $C^\dagger C = 24I_6$. Each of the nine family blocks is one $4L^2$ operator plus its exact mass offset. With $\lambda = \sqrt{24}/64$, the reconstructed first-emission vector satisfies

$$\|b(t) - b_1(t)\| \le \lambda^3 t^3/6.$$

This follows from unitarity and two substitutions in the exact memory equation; it retains every return and subsequent emission. A 128-site calculation at source time $1/10$ certifies a strictly positive infinite-line light-channel probability above $7.8898788951 \times 10^{-8}$ (and original-input probability above $1.8935709348 \times 10^{-8}$) after local-coefficient, binary64, Taylor-tail, cubic-feedback and finite-volume errors are included. The original preparation succeeds with probability greater than $6/25$ and its failure is retained. Full-source energy is conserved; pair/no-pair readout work has magnitude at most $2\lambda^2 t$. The fixed resonance-selected box and current are unchanged. The volume-sensitive two-site peak supplies no irreversible rate or physical strange-particle identification. The source, explicit error bounds and complete records are in `model/openwave_meson_memory_window`.

One positive semileptonic mass source. A declared two-cell, three-family extension retains all 192 native fermion modes, the four-input strong action, the charged analyzer tensor and the oscillation-calibrated Dirac neutral tensor. Its actual semileptonic current has coefficient $G = 1/64$. Its lepton spatial kinetic coefficient is zero; the massive Dirac branches belong to the full on-site family operator. The complete one-quark/one-family local current has exact norm 6, giving $|V| \le 18N_Q$ and $H \ge (m_{\min} - 18|G|)N_{\text{colourful}} + H_L$. The retained coefficients obey $m_{\min} - 18|G| > 1$. Consequently this finite matter source has a unique massive Dirac reference and full charge ± 1 threshold exactly the electron unit. The three charged and three neutral Dirac branches coexist on its quark-vacuum sector; purely leptonic contacts and photons are absent.

The pure lepton reference has active covariance $I/2$ and zero active–active anomalous covariance. Exact Wick contraction gives $\langle V \rangle = 0$ and $\langle V^2 \rangle = 3 \sum_r \langle J_{Q,r}^2 \rangle$. For the actual integer native baryon trial, the complete interacting energy variance is therefore its retained strong residual squared plus $G^2\kappa$, with $\kappa \simeq 36.091699949922$. The spectral theorem gives at least 0.964754199267 conditional probability in a half-width $0.255499475345\,\text{MeV}$ window about each calibrated nucleon mean. The corresponding original native-occupation-seed probability exceeds 0.042440240225; preparation failure and the outside-window outcome are retained. These exact rational bounds avoid enumeration of 2^{48} lepton states without truncating the Hamiltonian. They identify finite energy records, not physical pole masses or widths. The full operator polynomial, native intertwiner and complete instrument are in `model/openwave_common_mass_source`.

Longitudinal charge on the common matter source. The complete 192-mode native matter action admits an exact compact-rotor Gauss completion. On two cells, the integer flux and reference constraints are $E_x - 6Q_x - R = 0$ and $E_0 + E_1 = 0$. Every matter occupation has a unique flux/reference image, giving a unitary map onto the complete physical constraint kernel. All charged hopping and cross-cell weak-current terms are dressed by their literal charge shifts. The full native charge commutator vanishes, including the charged Dirac reference transformation.

With declared stiffness $\kappa = 1/8$, the longitudinal Green energy is $\kappa(Q_0 - Q_1)^2/8$. A specified diagonal self-energy subtraction $-(\kappa/8)\sum_a q_a^2 n_a$ preserves every single-lepton analyzer mass. Its coefficient is fixed by the graph Green function; it is not selected by a mass refit. The bare control raises each unit-charge rest energy by $7.98435860453125 \times 10^{-6}$ GeV. The full physical charge-one threshold of the normalized source is exactly the electron anchor: the colourful floor exceeds it and the two-charged-lepton floor is $63/32$ electron units.

Exact native baryon trials give rational changed energy means and full interacting energy variances, using the pure Dirac reference current covariance $3I/4$. Complete spectral-window records retain the preparation failure and the outside-window outcome. These means are not nucleon pole masses. The source includes a quantum compensating background and longitudinal energy, without independent transverse photon modes, a calculation of the physical electromagnetic coupling,

or a continuum gauge theorem. The source, integer Gauss controls and moment certificates are implemented in `model/openwave_common_electromagnetism`.

Moving records on the longitudinal matter source. Replicate the complete 192-mode longitudinal matter block and add the positive Dirac excitation kinetic energy $\sqrt{m_j^2 + L_a} - m_j$, with $L_a(p) = 4\sum_i \sin^2(ap_i/2)/a^2$. The entire operator is dressed by the product of the native Gauss isometries, including the quantum reference shifts between blocks. Full-Fock positivity and the exact one-electron threshold are preserved. In the exact colourful-vacuum, one-lepton reducing sector the dispersion is $E_a(p) = \sqrt{m^2 + L_a(p)}$, with group speed at most one and one minimum per massive branch.

A standalone Taichi kernel executes all 48 positive source coordinates and 96 native particle/hole probe rows on 16, 32 and 64-site periodic probes. Every position outcome and the one-half preparation failure are retained. On a common momentum ball $|p| \le P$, the complete record variation from the same-input continuum positive-energy source is at most $\min\{1, s|t|a^2P^4/(24m)\}$. The finite electron packet variations are approximately 0.0069404, 0.0017534 and 0.0004396 at the three spacings. An electron continuum four-vector $(E, p) = (5/4, 3/4)$ supplies an explicit boost-to-rest comparison with a bounded regulator defect.

The source retains a preferred frame and its square-root spatial kinetic is nonlocal. These results concern complete records of an exact reducing sector; they do not establish a unitary finite-grid Lorentz symmetry, interacting-object contraction, transverse photons or a common matter/light continuum. The source and portable executor are in `model/openwave_boosted_records`.

Three charged minima of a finite-range native source. The full native Higgs tensor admits a declared nearest-neighbour regulator with charged-branch dispersion

$$E_k(p)^2 = \frac{\sum_j \sin^2(ap_j)}{a^2} + \left[m_k + \frac{\sum_j (1 - \cos(ap_j))}{a}\right]^2 .$$

For $x_j = ap_j$, differentiation gives $\partial_{x_j} E_k^2 = 2\sin x_j [am_k + 1 + \sum_{i \neq j}(1 - \cos x_i)]/a^2$. Since $m_k > 0$, each charged branch has exactly one local and global minimum on the Brillouin torus, at zero momentum. The native colour and charge operators identify three colourless charged Dirac branches, with electron-anchored energies 0.00051099895069, 0.1056594144959 and 1.7769849708134 GeV. The complete side-four execution retains all 11 520 doubled rows, all colourful and neutral modes, and complete creation/arrival records. The charged corner states are retained eigenstates with higher energy and negative curvature in at least one direction. The neutral sine valleys and the degenerate neutral reference are retained. The regulator coefficient is declared, and the analyzer phase 2/9 has retrospective ancestry. No interacting particle identification or chiral gauge continuum is inferred. On momenta $\|p\| \le P$, the operator error from the continuum Dirac branch is at most $aP^2/2 + a^2P^3/6$; complete same-input record variation is at most $\min\{1, |t|(aP^2/2 + a^2P^3/6)\}$.

6.10 Two native clocks and a controlled common-clock limit

The actual eight-level native energy admits an exact Jordan–Wigner representation as three normal fermion occupations. Two occupations therefore supply independently prepared clock bits within the same source cell. Their joint Ramsey instrument retains all settings, both clock outputs and the spectator complement. They share one fixed quantum geometry and scalar-wave source of dimension 80000.

Complete records resolve a finite response difference. At event 16, the difference between the two background-induced phases after normalization by their respective bare gaps lies in

$$[3.06881,\ 3.21525] \times 10^{-7},$$

corresponding to 31.37–32.87 parts per million relative to the second clock response. An exact dyadic partial-transpose witness in the vacuum-field controls also certifies correlations that entangle the clocks through their common geometry. Mean wave-arrival delays measured by both clocks are positive and use the same receiver law. These finite source effects are distinguished from a measured equivalence-principle violation.

There is nevertheless a controlled common-clock limit. For the declared probe family $H_\lambda = H_{\mathrm{env}} + \lambda B(an_a + cn_c)$, with $B = \exp(Q_{\mathrm{right}})$, both normalized clock phases converge to the same integral of the Heisenberg-evolved B. For $0 < \lambda \le 10^{-3}$ and source parameter at most 8, their disagreement is bounded by $1500\lambda/(1 - 600\lambda^2)$, uniformly over arbitrary environment states and passive references initially independent of the two prepared $|+\rangle$ clocks. This follows from the exact finite Duhamel remainder and the complete joint instrument. No physical time or gravitational constant calibration is assumed.

A complete native kinetic kernel in OpenWave. The periodic native kinetic and analyzer-pairing source admits a portable Taichi implementation on all ninety canonical fermion modes per site. Its complete 180-row doubled symbol has $H(p) = M + \sum_j \Gamma_j \sin p_j$, with the native Clifford relations and $\|H\| < 2$. The fourth-order Taylor step is contractive for $2t/n \le 2$, and its exact-arithmetic evolution error is at most $2^5 t^5/(120n^4)$. Complete charged-addition records include preparation failure $1/2$ and every source-coordinate outcome. Cubes of sides four and six retain 11520 and 38880 doubled coordinates. A dense native stencil and an independent Fourier solution test the Taichi action; a reversed kinetic sign is detected. At twenty steps to $t = 1$, the largest recorded arrival-law discrepancy is below 6.85×10^{-8}, while the analytic operator bound is 1.667×10^{-6}. Floating norm drift is retained separately. The engine executes in the OpenWave namespace with its Taichi dependency; no upstream model award, physical clock or interacting continuum follows from this port. The complete replay is `model/openwave_platform_kernel`.

6.11 Native electric charge of a dressed quantum state

Fill all seven native weak-singlet coordinates in both spin states and leave the doublets empty. Every original gauge current annihilates this reference; its electric charge is exactly zero. For each actual signed Higgs pair define $d_L = c_L$ and $d_R = s\epsilon c_R^\dagger$. The canonical CAR transformation preserves both spin components and gives $Q_{\mathrm{pair}} = q(N_d - 2)$. The three up, three down and one lepton charges sum to zero, so the full native current obeys

$$Q_{\mathrm{native}} = \sum qN_d$$

without a reference-charge offset. The mass pairing becomes a Dirac bilinear. A constant chirality rotation identifies its positive absolute kernel with the corresponding quantum rotational block, including its actual current derivative and field coupling. This attaches one charged block, rather than identifying all sixteen Clifford coordinates with one native family.

The declared extension retains all native pairs, families and neutral coordinates, together with a charge-conjugate native copy. Its neutral confined Weyl kernel has energy at least $1/2$, as follows from $T^2 = (5/2)T$ and the exact positive decomposition of $N_{\mathrm{spatial}} + T/2 - I/2$. The ordered positive CAR lift gives $H \ge 16N_{\mathrm{charged}} + N_{\mathrm{neutral}}/2 + N_{\mathrm{photon}}$. The light charged ground energy

lies below 16.5, every additional neutral costs at least 1/2, and every two-charged sector costs at least 32. Thus each charge-one sign has a four-dimensional lowest band containing two spin-half doublets. The complete preparation selects one line with actual native charge of magnitude one and dressed $J^2 = 3/4$. The absolute kernel, trap, conjugate copy, fixed Higgs background and empirical electron unit are explicit; local longitudinal gauge completion and a physical particle dictionary do not follow from this attachment.

Infinite-photon confined spin-half band. On the complete finite matter shells, the transverse photon oscillator can be extended to its full Fock space. The positive form $H = \sqrt{16^2 + B_\infty^2} + N_{\mathrm{m}} + N_\gamma$ has two reduced levels below 17 and all other levels at least 17. Projection to the matter/photon vacuum identifies the low space equivariantly with spin 1/2; its exact Casimir is 3/4 and its energy is a scalar on that space. For a trial with photon number at most C, fourteen additional photon levels evaluate $B_\infty^{14}\phi$ exactly. The global estimate

$$0 \le \sqrt{16^2 + x} - p_6(x) \le \frac{33x^7}{2048\,16^{13}}, \qquad x \ge 0,$$

then gives a vector residual and Temple enclosure without an operator-norm bound on the infinite field. Three independent finite witnesses enclose the same infinite-source energy and certify photon tails. The spatial trap, field coupling and electron calibration are retained inputs; the result removes the photon cap and does not establish a spatial continuum or an observed-electron magnetic moment.

6.12 Quantum magnetic response

Same-source quantum Larmor dynamics. For the complete cap-two source, a uniform probe gives $H(b) = \sqrt{16^2 + (B + bE_z)^2} + N_{\mathrm{m}} + N_\gamma$. The exact static derivative satisfies $KX + XK = BE_z + E_zB$ and restricts to $PXP = \mu\sigma_z$ on the dressed spin-half band. Exact angular-sector interval witnesses enclose both finite-field branches. Their precession coefficient $g_{\mathrm{pre}}(b) = 2E_0|E_+(b) - E_-(b)|/|b|$ converges to the same static value $4E_0|\mu|$. A source-derived propagator estimate bounds all joint matter, photon and spectator-reference records over finite time; the complete executable instruments retain filter failures and photon outcomes. This result retains the finite regulator and declared field and mass parameters, without assigning a physical electron error budget.

6.13 Neutral production and inclusive weak conversion

Production, neutral flight and recoil on one source. On an odd periodic ring, retain the complete native charged and neutral Higgs kernel, its filled reference and every spectator. The actual charged-current frame is C, and its positive-band left weights are $w_i(n) = (E_i(n) + \sin(2\pi n/L))/(2E_i(n))$. A charged recoil register makes the source-current contraction

$$K_{ink';amk} = \frac{C^*_{ai}\sqrt{w_i(n)w_a(m)}}{L}\delta_{k',k+m-n \pmod L}.$$

Opposite momenta give $\sum_n w_i(n) = L/2$, hence $(K^\dagger K)^2 = K^\dagger K/4$ and $\sqrt{I - K^\dagger K} = I + (2\sqrt{3} - 4)K^\dagger K$. A rest-family input succeeds with probability $1/(4L)$ for every normalized recoil input. The same quadratic Hamiltonian then propagates every neutral momentum, family and recoil coordinate. The receiver uses the actual reverse current and its complete square-root failure operator. Source failure, detector failure and all received family/momentum/recoil records sum to one. Each detector

aperture is a separately chosen complete experiment; a uniform aperture choice retains that classical label on every outcome.

A sharp source recoil momentum makes the spatial response uniform, while a localized recoil register produces moving interference profiles. A separate retained mass record removes the time dependence of the spatially summed family-interference signal; it is distinct from a momentum recoil record. At side five the native untagged family contrast between source times zero and nine is approximately $1.17942\,10^{-4}$. Source and detector charge references each cost $1/4$ in the declared units, and their preparation and readout work are retained. These are current/spectral apparatus services on one free source, without a measured neutrino fit, an autonomous decay claim or an identification with the separate gauge-singlet composite.

Inclusive charged-current decay with a finite survival bound. Consider the positive-frequency part of the native charged-current contact interaction, its Hermitian reverse, and every spin and momentum channel of a middle charged particle, a lighter charged particle and two distinct massless neutral species. This is a declared contact member, with continuum relativistic dispersion and free complementary native fields. The three conserved combinations $N_p + N_j = 1$, one for each daughter species, restrict the admitted full Fock evolution exactly to one parent or three daughters. The native weak generators give unit cross-family coefficient. With $u^\dagger u = 2E$, the parent-spin average and complete daughter-spin sum give

$$\overline{|\mathcal{M}|^2} = 8G^2(P \cdot p_{\bar{\nu}})(p_e \cdot p_\nu).$$

The neutrals are distinct and contribute no identical-particle factor. The two-body phase-space volume is $\lambda^{1/2}(S, a, b)/(8\pi S)$, so its three-body factorization and the parent factor $1/(2M)$ give

$$\Gamma = \frac{G^2 M^5}{1536\pi^3} f(r), \qquad f(r) = 1 - 8r^2 + 8r^6 - r^8 - 12r^4 \log r^2, \quad r = m/M.$$

Here G has dimension energy to power minus two, and r is the actual native analyzer ratio. The numerical coefficient follows from the complete current and phase-space contraction.

In units $M = 1$, choose the ultraviolet amplitude $u(E) = (2 - E)^2$ for $m \le E \le 2$. The inclusive spectral density is $\rho(E) = E^5 f(m/E)(2 - E)^4/(3072\pi^4)$. Its exact Volterra survival equation has kernel $C(t) = \int \rho(E) e^{-i(E-1)t}\, dE$. Rational derivative bounds give $\int_0^\infty |C| < 1/2000$ and $\int_0^\infty t|C| < 1/250$. If $\alpha = \int_0^\infty C$, unitarity and the Volterra equation imply

$$|a(t) - e^{-G^2\alpha t}| \le G^2 B_1(1 + G^2 B_0 t), \qquad \Gamma = 2G^2 \operatorname{Re}\alpha.$$

At $G = 1/64$, survival at one and four nominal lifetimes differs from e^{-1} and e^{-4} by at most 0.000050782 and 0.000197266. Rounding the entire reservoir energy to bins of width h adds at most th to any survival probability; $h = 2^{-60}$ gives an additional error below $7.11\,10^{-10}$ through four lifetimes. This large finite-carrier statement is an analytical bound. Executed smaller carriers retain every emission bin, the survival complement, and the measurement energy work $W = -2G \operatorname{Re}(a^* \langle \sqrt{\rho}, b \rangle)$.

The finite realization retains its orthogonal source complement and arbitrary external references. The cutoff and contact strength are declared inputs; there is no transfer to the unfiltered contact source, measured weak lifetime or exact irreversible finite-volume decay. Electron anchoring supplies a conditional conversion of the calculated width to GeV.

6.14 Bound native composites and a common energy scale

Bound native colour composites. On a two-vertex graph retain the full fifteen-coordinate native matter module, two spin coordinates and all sixty fermionic modes. With the actual native colour currents T_x^a and the spin current S_x on the entire colourful subspace, declare

$$H = mN + t \sum_{\langle xy\rangle,r} (c_{xr} - c_{yr})^\dagger (c_{xr} - c_{yr}) + U \sum_{x,a} (T_x^a)^2 + J \sum_x S_x^2, \qquad (m,t,U,J) = (1, 1/8, 1, 1/4).$$

Every term after mN is positive, so $H \geq mN$ controls the full Fock complement and the empty matter vacuum. The action preserves global native colour, weak and hypercharge symmetries. Its graph kinetic term, number energies and positive interaction strengths are declared architecture; no local gauge-link dynamics or equivalence to a Weyl/Higgs action is asserted. Idle opposite-charge or conjugate-colour apparatus references have declared zero energy and are retained in every preparation outcome.

The complete three-quark colour singlet is $\mathrm{Sym}^3(\mathbb{C}^8)$, dimension 120, inside the 2024-dimensional three-quark sector. The quark/conjugate-quark colour singlet has dimension 64 inside a 576-dimensional sector. Explicit exterior-colour contractions give isometric reducing embeddings into these full occupation sectors. On a graph of maximum degree z, a local spin-half three-quark trial has energy at most $3m + 3J/4 + 3zt$, whereas every two-plus-one fragment threshold is at least $3m + 8U/3 + 3J/4$. A local spin-zero meson trial has energy at most $2m + 2zt$, whereas its two-fragment threshold is $2m + 8U/3 + 3J/2$. Thus the executed binding margins are at least $55/24$ and $67/24$, respectively, without enumerating the 2^{60}-dimensional carrier.

The lowest internal-spin-half baryons of charges zero and one are degenerate. For the spin-zero mesons, all three charges $-1, 0, 1$ share the same energy:

$$E_{\mathrm{baryon}} = 3.544903181009, \qquad E_{\mathrm{meson}} = 2.229589021818.$$

The probability that all constituents occupy the same cell exceeds 0.993. Complete spectral preparation includes its failure outcome and every original occupation record. Spin-one mesons, the neutral meson singlet and spin-three-halves baryons are retained. Spin is the internal rotation representation on this finite graph. These finite bound eigenstates establish neither relativistic hadron identity nor physical lifetimes or continuum confinement. The source, exact reductions and complete records are in `model/openwave_native_composites/`.

A common-scale obstruction for the symmetric control. Three conserved family coordinates with the native analyzer number energies give exact colourless constant-mode energies m_f. The empirical electron unit 0.00051099895069 GeV then preserves the conditional charged-lepton values 0.105659414496 and 1.776984970813 GeV. It puts the light bound meson below 0.00115 GeV and baryon below 0.00183 GeV, excluding the charged-pion and proton assignments on this action. Weak symmetry also forces the charge-zero and charge-one baryons to be degenerate for every U, J, t. For families a, b, $H_{a,b} = H_{1,1} + (m_a + m_b - 2m_1)I$, so the common binding correction cancels from family differences. Its light/second-family difference 0.105148416 GeV cannot equal the kaon-minus-pion comparison 0.35410661 GeV by changing that correction. These exact controls explain the need for distinct hadron coefficients in the calibrated construction. Complete source spectra and comparisons are retained in `model/openwave_composite_scale`.

6.15 Autonomous geometry and complete wave deflection

Autonomous finite quantum geometry. The native eight-level excitation operator K admits the declared source $H = H_g + e^{Q_0} K_0 + e^{Q_1} K_1$ on all $64(C+1)^2$ states, with two product oscillator

caps and graph Laplacian $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Both native excitation populations and total source energy are conserved; geometry energy responds autonomously. Exact dyadic eigenframe residuals enclose every native conditional block. At $C = 4$ and source time 10^4, a rational partial-transpose witness for two initially product native clocks lies in $[-0.054402654, -0.054401892]$. All nine product Pauli instruments retain every geometry and complementary native outcome. All 49 positive-energy source pairs have negative finite mutual ground interaction energy. Deleting the spatial edge factorizes the source and prevents this clock entanglement. The graph, stiffness 10^4, cap and clock unit are declared; physical G, irreversible relaxation, continuum gravity and a transfer to the static wave profile are not inferred. The construction and certificates are in `model/openwave_quantum_geometry`.

Complete finite wave bending. On the central plane of the declared native 31^3 lapse profile, the scalar source $H = \sqrt{SD^*\overline{AD}S}$ retains all 961 wave modes and the vacuum, where $S = e^{(1+3\gamma)q/2}$ and $A_e = e^{(1-\gamma)q_e}$. Complete position and bounded velocity instruments admit arbitrary references. At source time 16 and $\gamma = 1$, a finite packet has transverse displacement $-1.9417333 \times 10^{-5}$ and velocity change $-3.6701444 \times 10^{-6}$ relative to the identical flat preparation. An independent local second-order wave evolution agrees to numerical roundoff. A side 47 control retains the sign; reflection reverses it. The principal ray dispersion and complete wave records are explicitly compared: $\gamma = -1$ has zero principal-ray deflection and a nonzero finite-wave residual. The scalar planar extension, static source, γ and units are declared. No physical G, three-dimensional photon limit, observed lens angle or transfer to the autonomous quantum-geometry source is inferred. The finite law, analytic polynomial tail bound and separate arithmetic residuals are in `model/openwave_wave_deflection`.

6.16 Connections to observed particle and field structure

The finite constructions reproduce several characteristic mathematical structures used in descriptions of observed particles and fields. Each comparison below states the generated record, its calibration and its physical domain. These results use declared source members; combining their conclusions into one interacting realization requires the corresponding source-transfer statements. A structural agreement, an empirically anchored energy and a held-out numerical prediction are different tests.

Quantized electric charge. The native Fock charge operator has an exact discrete spectrum, including the charged-lepton unit and quark charges $2/3$ and $-1/3$. Actual charged arrivals carry the corresponding central current and regional Gauss flux, with complete failure and reference outcomes. Charge cannot be continuously divided within a fixed native representation. This is a finite compact-current realization of charge quantization. It does not equate texture winding with electric current or by itself identify the observed electromagnetic field. The complete source calculations are in `model/openwave_charge_protection` and `model/openwave_native_charge_flux`.

Charged rest energy. The exact native CAR and local Gauss isometry attach a positive energy to a dressed charged state, including the root-reference field. Equal-depth tree dressing adds its explicit electric energy without changing the matter current on its finite-orbital member. Separately, the full spatial/photon charged block, without tree dressing, has a trap-release threshold 16, conditionally 0.000509529040252 GeV under its retained electron conversion. The empirical electron anchor and its source conventions are explicit; this energy is not an independent precision electron-mass prediction. The threshold has no normalizable minimizing state. The distinction between a

free threshold and a localized trapped state is proved in `model/openwave_spatial_completion`. Its enlarged spatial domain does not inherit the finite-orbital tree energy or neutral-sector bounds.

Particle persistence. The complete finite charged packet has a proved persistence domain. For the spatial/photon source $H_\Omega = \sqrt{16^2 + B^2} + \Omega N_{\mathrm{m}} + N_\gamma$, positivity and oscillator coercivity isolate a spin-half band when $\min(\Omega, 1) > 3/64 + (3/8192)(5/7)^{5/2}$. Shell convergence controls the full source in this trap domain. This is a quantitative stable-band construction. The result supplies neither a self-bound localized electron after trap removal nor a nonlinear classical soliton.

Magnetic response near two. The actual differentiated current, including the photon field and its complement, gives a static coefficient in $[1.99827530667, 1.99827540121]$. Global divided-difference estimates and finite moment witnesses control the infinite-photon source; finite-time precession uses the same current. The proximity to two is a quantitative structural comparison with a spin-half magnetic response. Its deviation does not reproduce the observed electron anomaly. Field normalization, trap and regulator are declared inputs, as specified in `model/openwave_infinite_current`.

Spin one half. The finite-orbital dressed band carries $J^2 = 3/4$ and the two spin projections $\pm 1/2$ in units of $\hbar$. The native CAR charge map and longitudinal Gauss isometry preserve this representation on that member. Independently, the spatial/photon charged block includes complete matter shells and photon angular momentum, with a spin-half band and full-space convergence in its stated trap domain. The tree dressing and magnetic response bounds are not transferred to this enlarged spatial source. These exact representations agree with the spin structure of a fermionic particle; physical species identification is an additional comparison.

The double-cover rotation law. Literal proper cube rotations act on the kinetic/pairing source and its filled reference. An odd state changes sign under 2π relative to an idle reference and returns under 4π. Complete oriented spin-frame records return under 2π, while a declared two-copy apolar tensor record returns under π. These distinct record periods are computed in `model/openwave_spin_frame_records`. They realize the spinorial double-cover structure and do not prove an exchange-statistics theorem or a fundamental nematic field.

Particle–antiparticle conversion to radiation. The actual native CAR current converts a neutral particle–hole pair into two transverse quantum modes. A positive full-Hamiltonian probability bound, complete observations and conservation of total energy, charge and momentum establish this finite conversion mechanism. The source uses a declared resonance, reference quench and field cap; measurement work is retained. The calculation in `model/openwave_radiative_pair` supplies no physical positronium branching ratio or infinite-volume annihilation rate.

A stable neutral candidate. The periodic link-penalty-two source contains an isolated stationary neutral band. Its finite polynomial/polar preparation has success above 0.99999999999639 and stationarity error below 4.000002×10^{-16}. This supplies a stable neutral sector with complete failure records. Exact location conservation makes its momentum band flat, so these properties do not identify a moving relativistic dark-matter particle or a cosmological abundance. The preparation and obstruction are both retained in `model/openwave_neutral_preparation` and `model/openwave_neutral_dispersion`.

Electrostatic spatial dependence. The declared Gauss Green operator gives the Coulomb $1/r$ potential and the associated static inverse-square force in its controlled spatial domain. A separate autonomous eighteen-state native-current experiment supplies actual force and displacement records. This connects a constraint law to finite dynamics; its coupling is supplied and is not a prediction of the observed fine-structure constant. The source and reference domains in `model/openwave_pinned_coulomb` and `model/openwave_autonomous_coulomb` are retained.

Two transverse Maxwell waves. A finite cochain field source has two transverse Maxwell modes, exact finite-occupation dynamics and independently computed real-space propagation. Discrete constraints remove longitudinal radiation in the stated sector. This reproduces the transverse wave structure of electromagnetism on the declared lattice. A shared physical particle, coupling and elapsed-time calibration is required for a common electromagnetic interpretation; see `model/openwave_field_observables`.

A massive second-order wave law. For the full native kinetic/pairing source, charged frames obey

$$h_k^2 = \left(\sum_i \frac{\sin^2(ap_i)}{a^2} + e_k^2 \right) I.$$

Complete spatial wave records and a controlled low-momentum limit give the massive Klein–Gordon form. Every finite-derivative valley and spectator is retained. This is a source-derived wave identity with stated momentum/spacing error, rather than an identification of all interacting particle masses or an elimination of lattice multiplicities.

Discrete atomic levels. The declared Gauss atom has negative standing-wave levels under independent volume and spacing variation. In its atomic units, the first excitation gap approaches $3/8$, and the tested second-shell splitting decreases to 0.000410646363. This is the hydrogen-like orbital pattern generated by the specified kinetic and Gauss source. Mass and coupling are supplied. Precision physical hydrogen, fine structure and radiative shifts are outside this calculation, whose refinement evidence is in `model/openwave_atomic_refinement`.

Charged-lepton numerical comparison. The separate analyzer-family comparison gives electron-anchored values 0.00051099895069, 0.1056594144959 and 1.7769849708134 GeV. The muon residual is approximately +9.83 parts per million. The phase $2/9$ has retrospective selection ancestry, and the electron fixes the common unit. These small residuals are useful quantitative comparisons, with those empirical dependencies retained; they are not three independent parameter-free mass predictions. The full physical particle dictionary and interacting mass-shell identification require further same-source tests.

6.17 Source domains, complete records and physical identification

The target is one executable OPH construction whose native observers recover all the physical observations of our Universe, including its effective fields, gravity and particle spectrum. Hardware and software may be reverse-engineered; neither architectural uniqueness nor an a priori derivation of the design is required. Architectural choices define a legitimate candidate. Its complete outputs must be derived and compared with observation, with empirical calibration distinguished from independent prediction. The starting premise is a self-referential mathematical structure, without an independently existing background spacetime or external global clock. Computational

events have a partial order; physical spacetime and mutually calibrated clocks must be recovered from compatible records. An operational observer has readback, record, feedback, prediction and continuation interfaces. A minimal controller can qualify when it implements those interfaces; no consciousness premise is imposed. Auxiliary generator times in the constructions below are calculation parameters whose identification with native observer clocks must be stated and verified. Exact invariant sectors and positivity bounds avoid enumerating entire Fock carriers while retaining their full complements. Each construction supplies its own complete preparation and observation law. Separate source results require an implementation or transfer theorem before they constitute a common interacting particle and gravitational model.

Native compact Gauss completion. The actual native charge identity $Q_i = \sum_a q_a N_{ia}$ permits a compact rotor tree with local constraints $E_i = 3Q_i$ and a retained root charge $R = -\sum_i E_i$. The exact physical isometry dresses every native Fock word with these fluxes. Wilson-linked positive kernels retain the full ordered CAR floor. Electric energy reduces to $H_E = (\kappa/2)\sum_i d_i Q_i^2$ for branch depths d_i. For an equal-depth star, the entire one-electron block differs from the quantum source by the constant $\kappa/2$; its current, spin and complete observer laws transfer through the isometry. This energy includes the charged reference's longitudinal field. The declared orbital graph does not establish a physical Euclidean Coulomb law or an isolated-electron rest mass.

Infinite-photon magnetic current and precession. For the fixed complete matter shells, the photon-coordinate Weyl operator $B(x)$ is unbounded globally but is a finite Hermitian matrix at each x. Divided differences give the uniform derivative bound $\|K'(x)\| \le \|E_z\|_{\mathrm{HS}} = \sqrt{5}$. A global differentiated Taylor remainder is controlled by a finite trial's twenty-sixth radial field moment. Fourteen padded photon degrees evaluate its current and energy witnesses against the full photon source. The resulting static coefficient is enclosed in $[1.99827530667, 1.99827540121]$ in the unchanged source conventions. The same uniform derivative and isolated-band gap give finite-time Larmor bounds with arbitrary references and every field outcome retained. Physical field calibration, electron anomaly and spatial trap removal are separate identifications.

Spatial completion and trap-release threshold. On the full spatial and photon carrier, the declared positive source $H_\Omega = \sqrt{16^2 + B^2} + \Omega N_{\mathrm{m}} + N_\gamma$ has an exact isolated spin-half doublet whenever $\min(\Omega, 1) > v_0 = 3/64 + (3/8192)(5/7)^{5/2} < 1/16$. Complete-shell core and form convergence, together with oscillator coercivity, give convergence of fixed low energies without equating a truncated square root with a compressed infinite operator. Gaussian width-L trials give

$$E_\Omega \le 16 + \frac{3}{64L^2} + \frac{3L^2}{8192(1 + 2L^2/5)^{5/2}} + \frac{3\Omega}{4}(L^2 + L^{-2} - 2).$$

Hence trap release has threshold 16, with no normalizable state attaining it. This excludes a demonstrated self-bound localized ground, not an ordinary free-particle mass threshold. The retained electron unit assigns that threshold approximately $0.000509529040252\,\mathrm{GeV}$, conditionally.

Native bound-meson charged-current conversion. Retain the positive two-cell native colour/spin action, all three analyzer family energies and the complete 180-mode fermionic carrier. A declared native port word gives the quark family current $C = F^\dagger V F$. The lepton reference fills all twelve neutral coordinates. The autonomous semileptonic source uses every local spin and family current,

$$H = H_{\mathrm{hadron}} + H_{\mathrm{lepton}} + GV, \qquad V = \sum_{x,\mu} \eta_\mu (Q_+^\mu L_-^\mu + Q_-^\mu L_+^\mu + 2Q_3^\mu L_3^\mu).$$

Conserved charge forces at most one emitted charged lepton and neutral hole on the prepared one-quark/one-antiquark sector. Exact colour reduction, total spin and cell parity reduce its complete 888-state block to 150 states without removing a reachable outcome.

The heavy-family and light-family spin-zero mesons have identical orbital wavefunctions. Their vector-current form factor is one, and the complete bound-to-bound pair matrix element is $C_{01}/\sqrt{2}$. The native current expectations are -3 and $5/2$. With $G = 1/64$, a declared outgoing frequency $q = (m_2 - 2)/2 - 11G/4$ equates their interacting mean energies. Full evolution at source time 720 gives conditional pair probability approximately 0.994907948; the spectral-preparation probability is approximately 0.2483446023, and its failure is retained. Independent complete-block propagation checks the reduction. The finite evolution also returns toward the parent, so this is a coherent finite-box conversion.

Separate complete position and momentum instruments retain 2664 active colour/occupation records, the full source complement and preparation failure. Their different energy work is calculated from the full native occupation Hamiltonian. The port word, contact strength, filled reference and resonance are adjustable architectural inputs. No empirical hadron mass, lifetime or mixing datum enters, but the result is neither a parameter-free physical strange-decay prediction nor a repair of the rejected physical mass assignment. Numerical probabilities have independent residual checks without a certified all-arithmetic enclosure.

The exact resonant orbit of the native weak current. The unfiltered 270-mode contact has 31464 quartic words. Its complete free-mode transformation has 55575 ordered terms and 299 Bohr-frequency labels. Exact normal ordering and a positive nonzero-frequency gap give an invariant two-state resonant orbit whose current is

$$\overline{V} = \begin{pmatrix} 173/2 & 1 \\ 1 & 85 \end{pmatrix}, \qquad p_{\mathrm{receiver}}(t) = \frac{16}{25} \sin^2 \frac{5Gt}{4}.$$

The $3G/2$ detuning follows from the actual contact; no selected transition operator or occupation penalty defines the source. Prepared-domain bounds below control its approximation by the full Hamiltonian, retaining every nonresonant outcome and arbitrary references. The finite momentum box, weak coupling and source parameter have no physical lifetime calibration. The source orbit is reproduced by `model/openwave_current_averaging`.

Native-charge mixing. One conjugate and one same-representation copy of the actual colourful weak-singlet module form an anomaly-cancelling vectorlike extension. The gauge-invariant native-hypercharge mixing gives $M_q = \left(\begin{smallmatrix} e_f & 0 \\ GY_R & M \end{smallmatrix}\right)$. For $G, M > 0$, its light singular values obey $0 < s_{u,-} < s_{d,-} < e_f$, while all heavy partners remain. The declared example $G = M = 1$ retains13 charged Dirac channels per family, the neutral Weyl coordinate and all finite-derivative valleys. Exact Ward identities, cubic anomaly cancellation and a rational charged energy-floor certificate accompany complete256-outcome active CAR addition records, success and failure, and the opposite-charge reference. The unchanged electron calibration supplies conditional GeV branch gaps. These are consequences of supplied extra matter and mixing coefficients; physical quark masses, heavy-particle identities and an interacting pole interpretation are not established. See `model/openwave_vectorlike_flavour`.

Balanced-current baryon control. The equal-number-energy action with positive colour, spin and weak squares has a complete 120-dimensional colour-singlet baryon sector, with a spin-half/isospin-half ground band near 3.73359579. The exact integer and rational instruments

retain all 2024 original three-quark occupation outcomes and failure. A number-centred graph charge readout gives positive inner and negative outer neutral charge, about ± 0.000575083 in graph units. Using the free vectorlike gaps as number coefficients gives $0 < E_n - E_p = m_d - m_u < m_e$. The conditional graph energies 0.001709996503 and 0.001769595845 GeV fail the nucleon mass comparisons and forbid ground-state beta conversion. These are specific failures of the balanced-current/free-gap attachment, not of the separately calibrated hadron construction. The exact source and complete instruments are `model/openwave_baryon_ordering`.

Dynamic geometry and its source law. A declared seven-by-seven scalar grid retains 64 native matter sectors, two harmonic geometry coordinates and all 49 wave modes plus vacuum. Harmonic extension induces geometry stiffness and inertia from the same graph. The positive source $H = H_g + e^{Q_0}K_0 + e^{Q_1}K_1 + h(FQ)$ includes wave backreaction. Complete wave, geometry and native-clock records are computed using its exact reducing blocks. For comparison of occupation caps, the fixed-master compression below keeps the action fixed; rebuilding e^Q after compression is a different source. `model/openwave_dynamic_wave` gives the dynamic action, and the fixed-master experiment states its own complete-record bounds and event convention.

Source stress and the test-clock domain. The autonomous wave construction determines its field force through $hG_a + G_a h = 2(D_a K_q + K_q D_a)$, with $h^2 = K_q$ and $D_a = \mathrm{diag}(F_a)$. The complete finite Ehrenfest equation retains the oscillator boundary term; geometry, native and field energy rates sum to zero. Full stress and momentum instruments retain all complementary outcomes. Interval certificates for all64 native vacuum sectors show a frequency-ratio spread of [87.8991138, 87.8991463]ppm among the seven dressed native clocks. A gap/resolvent theorem gives $|(f(s)-f(0))/s - \langle e^{Q_1}\rangle_0| \le [(17/250)^2/(9/50)]s$ on the stated probe domain. Thus the common test-clock response has a quantitative backreaction bound, while exact finite-strength universality fails in this construction. The source choices and separate clock preparations are retained; physical G, laboratory clocks and an Einstein stress tensor are not inferred. See `model/openwave_dynamic_stress`.

Complete native six-quark binding and graph quadrupole. A declared spin/weak-balanced current action on the full native sixty-CAR carrier has an exact $N = 6$, colour-zero, $S = 1$, $I = 0$ reduction from11700 occupation words to27 states. Exact integer Casimir projectors and a positive Gram factorization certify completeness. The six-quark Rayleigh energy is approximately 7.22070970417 and lies more than 0.24648187228 below two baryons of the same balanced source; all other fragment partitions have threshold at least $6+8/3$. The member without weak-current cost is certifiably unbound in this target channel. For the number-centred graph charge moment, $\langle Q_{zz}\rangle = \langle N_0 N_1\rangle/18$ on the isospin singlet. Binding forces it strictly positive; the certified ground comparison gives approximately 0.00263818496 in squared graph-separation units. The equal spin/weak costs also give a retained degenerate $S = 0$, $I = 1$ partner. Complete preparation/failure records and their stationary-ground error are included. The coefficient equality is architectural; physical deuteron mass, length, shape and species identification are not inferred.

The vectorlike mass comparison. For the declared $G = M = 1$ mixing, the light up/down gaps in the three analyzer families are approximately $(0.00042518, 0.00048478)$, $(0.0879117, 0.1002362)$ and $(1.46748, 1.68031)$ GeV on the empirical electron unit. Every heavy branch is retained, with gaps between 7.14 and 8.21 GeV. These are source branch energies, not identified quark or extra-particle masses. The observed quark spectrum is not reproduced, and family labels alone do not

supply a physical assignment. Complete intervals and current records are in `model/openwave_vectorlike_flavour`.

Reuse of finite instrument theorems. For every exact Hermitian projector P, the outcome maps $\rho \mapsto P\rho P$ and $\rho \mapsto (I-P)\rho(I-P)$ form a completely positive, trace-preserving summed instrument. This general finite theorem is formalized in the finite event-algebra library by the theorem `luedersChannel_isCPTP`. Its repeatability and complement-exclusivity results use the same algebraic premises. The source is `Lean/EventAlgebra/LuedersPhaseInstrument.lean`. The baryon and six-quark rank-one filters satisfy those premises by their stored exact integer vectors. Thus their failure channels and quantum references use the general theorem, with the concrete matrix identities checked by the executable certificates. Those concrete instances are not additional Lean proofs. The same formal library also proves that an effect table alone does not determine its state-update instrument. A declared preparation service cannot therefore be inferred solely from matching its outcome probabilities.

6.18 Mass-only controls and finite nuclear reductions

The mass-only member of the positive native-current action fits $m_p = 0.93827208943$, $m_n = 0.93956542194$ and $m_{\pi^\pm} = 0.13957039\,\mathrm{GeV}$ with three empirical inputs. An exact min–max slope bound of $2/9$ proves uniqueness within its declared coefficient family. Its complete three-quark and meson singlet sectors have dimensions 120 and 64. It is distinct from the four-input axial state described in Section 6.8.

The axial discriminator. The local colour-singlet representation is $\mathrm{Sym}^3(\mathbb{C}^2 \otimes \mathbb{C}^2) = (\frac{1}{2},\frac{1}{2}) \oplus (\frac{3}{2},\frac{3}{2})$, with multiplicity one. An integer colour contraction gives $g_V = 1$, local $g_A = 5/3$ and squared excited-current leakage $8/9$. Schur's lemma prevents a local scalar preserving both symmetries from changing this coefficient. On the mass-only source the positive separated-colour floor

$$\gamma = \frac{8U}{3} - \frac{4A}{9} - \frac{3}{4}(J_s + J_w) - 3t$$

and $\|I_- J_+^z\| \le 6$ give $|g_A - 5/3| \le 36t/\gamma$, hence $g_A \in [1.665734201267, 1.667599132066]$. This excludes the neutron comparison 1.2753 ± 0.0013 on that source. The split-site four-input construction changes the state rather than the native current, so the local multiplicity-one restriction does not apply to its full ground state. The exact controls are in `model/openwave_baryon_weak_formfactors` and `model/openwave_axial_state`.

Nuclear spectra of the mass-only member. Exact integer projectors give 27 levels at fixed spin projection $+1$ in the six-quark colour-singlet $(S, I) = (1, 0)$ sector; the full spin multiplet has dimension 81. Its binding exceeds 1.4999965731 electron units, about 0.766497 MeV, below every same-source fragment threshold. A positive edge current shifts the spin-zero isotriplet by 2η; its three components are above their respective two-baryon thresholds on the two-cell source. The excluded deuteron binding comparison 2.22456640 MeV fails. These masses, binding signs and partner exclusions are specific to the mass-only action, rather than the axial-calibrated action.

For an L-cell graph, exact local three- and six-quark colour-singlet dimensions 20 and 50 give the full locally singlet six-quark dimension $50L + 400\binom{L}{2}$. If the coloured complement is above C, $a = \max(PHP) < C$ and off-block hopping has norm $b \le 6tz$, every low level lies within $b^2/(C-a)$ below its compression; the next level is at least C. This controls paths and cycles with two to four cells, retaining excited bands, failures and an additional bound $(S, I) = (0, 0)$ channel. The two-cell

isotriplet exclusion does not transfer to all these graphs. See `model/openwave_nuclear_exchange` and `model/openwave_nuclear_refinement`.

Finite conversion of the mass-only member. The native semileptonic contact $G = 1/64$ retains all 70 four-occupied states of its eight-mode lepton sector. With the baryon sector this gives 8400 states, 1400 at charge zero and 398 at fixed twice-spin projection one. The identity $N_e = N_{\mathrm{up}} - 1$ retains virtual two-electron states. At source parameter 512 the complete floating propagation gives conditional one-electron probability 0.96855747 and original-input probability about 0.32285249, retaining preparation failure and alternative complete position and momentum instruments. The 0.78233355931 MeV strong/free-channel endpoint does not include the filled-reference shifts ∓ 7.984 keV. Recurrence and the failed axial comparison preclude a physical neutron lifetime interpretation. The full weak-dressed energy instrument and the axial source's outgoing momentum record are separate calculations, given in Section 6.8; the finite conversion control is `model/openwave_baryon_beta`.

Prepared-domain control of the unfiltered native current. Exact normal ordering of all 31464 native contact words retains the scalar 84, quadratic contractions and every quartic channel. Complete current columns from the derived parent/receiver orbit have nonresonant gaps above 0.00481278. The CAR identity $\sum_{|J|=b} \|a_J \psi_n\|^2 = \binom{n}{b}$ bounds coefficient matrices on particle-number sectors without truncating the source evolution. For its two-dimensional prepared domain, integration by parts gives $\|(U_I(t) - e^{-iG\overline{V}t})P\| \le GA + G^2 t(K + 5A/4)$. The first-peak coefficient is below 3367930.443. Thus $G = 2^{-30}$ certifies a receiver probability within 0.00628 of 16/25, while $G = 2^{-44}$ retains probability error below4×10^{-7}. All full-source complements and arbitrary references are included. The $G = 1/64$ control is uncertified; no observed weak lifetime or irreversibility is inferred. See `model/openwave_reachable_averaging`.

Outgoing volume and a native meson memory kernel. Keep the two-site bound meson, native family word, full semileptonic current and contact coupling $G = 1/64$, and extend only the outgoing lepton ring to L sites. The original fermionic carrier is retained with $12(L-2)$ added coordinates. The filled neutral reference and charge identity $Q = -w - N_e = -1$ restrict this preparation to zero or one emitted electron–hole pair without a particle cutoff. The exact charge and spin sector has $24{+}216L^2$ states; spin zero and reflection reduce the prepared invariant subspace to $6 + 36L^2$ states. In its parent/emission decomposition,

$$H = \begin{pmatrix} A & GC^\dagger \\ GC & D \end{pmatrix}, \qquad C^\dagger C = 24 I_6, \qquad P_{\mathrm{pair}}(t) = \frac{3}{512} t^2 + O(t^3).$$

The coefficient follows from the native Pauli contraction, three emitted families and family unitarity. Eliminating the complete emitted sector gives

$$\dot{a}(t) = -iAa(t) - G^2 \int_0^t C^\dagger e^{-iD(t-s)} C a(s)\, ds.$$

The operator D retains its neutral-current interaction. This identity supplies a memory kernel rather than an assumed decay rate.

With the same resonance-selected box held fixed, the pair probabilities at source time 720 are 0.994908, 0.244041, 0.063225 and 0.023513 for $L = 2, 4, 6, 8$. The selected plane-wave form factor is $2|C_{01}|/(\sqrt{2}L)$ and its contact detuning is $G(1/2 - 1/L)$. Consequently the two-site peak does not

define a volume-stable irreversible lifetime. A same-source infinite-ring comparison uses outgoing hopping norm below 103 and first differing hop order $R = (L-2)/2$:

$$\|(U_L(t) - U_\infty(t))P_{\text{parent}}\| \le 2\sum_{k\ge R}\frac{(103|t|)^k}{k!}.$$

For $L = 32, t = 0.01$ this is below 2.546754×10^{-12}. The separate 36,870-state floating propagation gives pair probability approximately 2.183×10^{-7}; its roundoff is outside this volume bound. Complete position and alternative momentum records retain preparation failure, all matching-colour outcomes and full-source measurement work. The source clock is not identified with seconds, and neither a measured strange-meson lifetime nor physical particle identification is asserted.

A rigid neutral-hierarchy control. On the same 288-mode Dirac carrier, the declared tensor $\epsilon R\,\mathrm{diag}(m, 1, M)$ with $\epsilon = 2^{-32}$ gives three neutral masses proportional to the charged analyzer. It retains both helicities, the sterile coordinates, the filled reference and the native weak contact. The exact complete current isometry has 144 outgoing channels and a production complement; it is a declared apparatus, distinct from autonomous decay. The rigid hierarchy fails the scale-independent squared-gap test in Section 6.9 for every positive common multiplier. Its conditional absolute masses therefore carry no measured neutrino identification. The calibrated neutral tensor uses the same current identities with explicitly different family coefficients; `model/openwave_dirac_neutral_current` retains the rigid control.

6.19 A fixed quantum geometry source and controlled compression

On a seven-by-seven scalar grid, two harmonic geometry coordinates use exact induced stiffness and inertia matrices. Retaining occupations zero through four in each coordinate gives a fixed source with 64 native matter sectors, 25 geometry states and 50 field states including vacuum. Its total dimension is 80000. The source contains the native energy–lapse coupling and the positive scalar-wave square root. Its nested comparisons compress this fixed action; exponentiating a smaller compressed coordinate defines a different action.

Complete arrival-history isometries certify the comparison for every input and untouched reference on the span of native levels 00 and 10, geometry vacuum, and field vacuum or the specified packet. The cap-three compression has complete-record total-variation error below 0.000358 relative to the cap-four source. A finite Schur bound and an exact rational trial give the native 10 one-wave ground interval

$$1.35873578062 < E < 1.35873578145.$$

All 64 native vacuum spectra are certified. The same master source gives transverse displacement in $[-0.001776299, -0.001775262]$ grid sites after eight evolution events, and conditional arrival delay in $[0.0105311, 0.0105598]$ in the event-parameter convention. Every receiver failure and field vacuum outcome is retained. These finite bounds prove neither an infinite geometry-cutoff limit nor physical gravitational units. The evolution word defines an experimental event order, without asserting a fundamental global clock.

6.20 A native clock read at wave arrival

The same 80000-dimensional matter–geometry–wave source prepares the right native cell in an equal coherent superposition of levels 0 and 1. A retained setting bit selects either the X or Y clock instrument when the wave arrives. All native complements, geometry outcomes and timeouts

are included. The resulting four active setting/sign frequencies determine the clock coherence without assigning simultaneous sharp X and Y values. Independently prepared vacuum-clock controls provide a strictly increasing phase calibration for the sixteen event labels.

Using this measured event-to-cycle dictionary, the source-on minus source-off mean arrival difference lies in

$$[0.0013506,\ 0.0013547] \quad \text{native clock cycles.}$$

The phase of the actual aggregate arrival-clock coherence gives a separate positive response in $[0.0004679, 0.0004687]$ cycles. It is a circular phase statistic, rather than the mean arrival delay. Complete finite numerical bounds control both quantities. A finite-repetition theorem also certifies the sign of the phase response with retained failure records; its large repetition count is an analytic resource bound.

Within a terminal event, clock, field and geometry effects act on distinct tensor factors. Their products commute, and every permitted ordering has the same full joint record. Copying these orthogonal records supplies two observer readouts without copying an arbitrary quantum state. This closes an operational clock–wave comparison on one specified construction. No physical-second calibration, Einstein proper-time identification or fundamental global-time variable is assumed.

Finite theorem interfaces. The exact native colour projectors and their representation meanings are reused across the three- and six-quark sources. Their actual current intertwiners are checked before any sector reduction. The finite complete instrument theorem in `Lean/EventAlgebra/LuedersPhaseInstrument.lean` applies once the concrete projector identities hold, retaining failures and arbitrary quantum references. Positive-root identities and uniqueness are formalized in `Lean/SourceLaw/MatrixLimits.lean`; the geometry calculation adds its stated executable residual and operator-norm estimates. The numerical instances and quantitative bounds are not additional Lean theorem counts.

6.21 Complete source transport and quantitative field records

Exact Fock and field transport. The complete 192-coordinate native CAR carrier admits 24 even pair implementers on all 48 lepton coordinates, with identity on the 144 colourful coordinates. Their exact inverse transforms give every canonical coefficient of the 114-term source and all 24 weak currents. Pair terms are retained. Charge, Gauss constraints, preparation, arbitrary untouched references and measurement work transform through the same unitary. For a smooth periodic field-dependent implementer, the full electric operator transforms as

$$U^*EU = E + \mathcal{A}, \qquad \mathcal{A} = -iU^*\partial_\theta U, \qquad U^*E^2U = E^2 + E\mathcal{A} + \mathcal{A}E + \mathcal{A}^2.$$

The connection and derivative term are necessary even when a fixed-field matter square is scalar. A complete four-state Fock example retains both odd occupations and every integer flux, with exact nonzero measurement work. The native longitudinal Gauss carrier has dimension 2^{192}; adding an independent rotor gives an infinite-dimensional carrier. There is no onto unitary equivalence between those different architectures. The exact algebraic transport therefore does not identify a common interacting continuum or additional physical rest masses (`model/openwave_fock_field_transport`).

Finite references and the complete clock experiment. A finite energy-conserving exchange gives source coherence $i\sin\vartheta\, e^{imt}/2$ and energy $m\sin^2\vartheta/2$, retaining every success and failure. Coevolution of both finite phase references, followed by energy-compatible readout, cancels the free

source phase in every terminal probability. A fresh terminal phase is a supplied timing resource. A distinct static controller executes the actual calibrated electron, muon and tau masses at common scale $L = 4096$, common times and coupling $g = 2/3$. The exact reducing carrier has dimension $256 \cdot 4 \cdot 2 \cdot 2 = 4096$; it retains every occupation record. The full source has the reducing projector $R = (P_{\mathrm{vac}} + P_j) \otimes I_{\mathrm{memory}}$, with an explicit $R^{\perp}$ terminal outcome and lower bound $H_{\mathrm{total}} \geq -L \max(|g|, 2)$. Its source-pair phase obeys the exact local error bound

$$\left| \frac{d\langle A \rangle}{dt} - im\langle A \rangle \right| \leq L|g|P_{\mathrm{prep}} + 2LP_{\mathrm{read}}.$$

Complete component-energy and terminal-measurement work accompany the finite trajectories. Numerical propagation has convergence diagnostics, without an outward arithmetic certificate. The moving controller packet, phaseful references and final occupation instrument are declared resources; no autonomous detector compilation, self-starting limit cycle or physical seconds follows (`model/openwave_finite_phase_clock`).

Gauge-invariant localized quark records. The full $L^2(SU(3) \times U(1))$ link carrier and thirty native source and reference CAR coordinates support an open-string operator B with $B^2 = 0$ and $B^*B = N_{\mathrm{family}} P_{\mathrm{ref,vac}}/3$. Its controlled seed/path superposition has fractional source charge and a compensating quantum reference. The contractions $X = B + B^*$, $Y = i(B - B^*)$ give the complete physical instrument

$$\left\{ \frac{1}{4}G(I+X)G, \frac{1}{4}G(I-X)G, \frac{1}{4}G(I+Y)G, \frac{1}{4}G(I-Y)G, I-G \right\}.$$

Seed preparation on Gauss success has outcomes P_{seed} and $G - P_{\mathrm{seed}}$, with failure absorbing. The path energy is $2 + \ell(q^2/32 + 1/12)$ for integer hypercharge labels $q = 2, -1$. The displayed complex quadrature mean is $\langle B^* \rangle$. Complete charge and flux profiles are alternative terminal measurements, with their distinct work. The electric idle Hamiltonian and coherent preparation are supplied. This establishes finite localized charge records, without autonomous hopping, a spontaneously wound texture, confinement or a physical quark mass (`model/openwave_localized_quark_record`).

Literal weak-probe energy and the Lambda spectral measure. The normalized native Lambda packet uses $\kappa = 1/512$ and a trial preparation of probability $1/7$. All 72 native weak words, 522 nonzero current outputs and 19140 full-Hamiltonian successors enter exact packet moments, including every translated-packet cross term. The successful probe's energy mean and variance are enclosed by

$$\langle H_5 \rangle \in [2.418046697462, 2.418046697463] \text{ GeV},$$
$$\mathrm{Var}(H_5) \in [0.085780468809, 0.085780468810] \text{ GeV}^2.$$

The mean energy increase is $[1.302361802725, 1.302361802726]$ GeV; this is probe-induced energy, not the calibrated Lambda rest energy. Nonselective work retains probe failure. For $D = (H_5 - E)^2 + \eta^2$ and every full-source trial x,

$$\langle u, \eta^2 D^{-1} u \rangle \geq \frac{\eta^2 |\langle u, x \rangle|^2}{\|(H_5 - E)x\|^2 + \eta^2 \|x\|^2}.$$

All 23 native proton–pion columns and the literal current supply such witnesses. Their complete effect and complement certify a broad-window probability above one half; the tested narrow windows have zero certified lower bound. Zero lower bounds do not establish zero spectral weight. A Weyl

sequence at the rest energy establishes spectral membership, without proving current overlap or a physical decay density. A controlled shrinking-window limit and outgoing-current identification are required (`model/openwave_lambda_spectral_measure`).

Dynamical electromagnetic nuclear bounds. A compact $U(1)$ cycle extends the actual nuclear source with electric coefficient $a = 1/16$ and magnetic coefficient $b = 1/8$. The exact Gauss isometry retains all matter occupations and every integer cycle flux. Its unique positive field vacuum is gauge invariant; hence $H_{\text{field}} - E_\gamma \geq 0$ on the full carrier. An $n = 0$ trial gives an upper bound using the actual native compression plus longitudinal V_0, without deleting field polarization. The complete GeV intervals are

triton	$[2.808919203020, 2.809127749492]$
helion	$[2.807625870510, 2.808017057294]$
alpha	$[3.739123982204, 3.739741842803]$

The triton input lies in its interval; both held-out comparisons fail. The positive-addition class cannot lower the excessive alpha mass. For general prepared matter state ρ, the field preparation energy is $\mathrm{Tr}(\rho V_0) + b - E_\gamma$, in $[\mathrm{Tr}(\rho V_0), \mathrm{Tr}(\rho V_0) + b]$. Uniform conditional quartet states give longitudinal means $9/32$, $3/4$ and $39/32$ in source units. The interval $[0, b]$ applies only to the zero-charge vacuum service. These bounds preserve all spin multiplets, failures, thresholds and inherited calibration; they are not a fitted Coulomb potential or successful nuclear-spectrum prediction (`model/openwave_nuclear_em_transfer`).

Energy-dependent full-complement spectral certificates. For the actual refined Haar source, let P be the complete 201-mode frame, $T = QHP$ and $C = QHQ = D + B$, where $D = 16E_Q + 9$, $C \geq 80Q$ and $\|B\| \leq 9$. Every omitted representation and intertwiner remains. For real $z < 80$, define

$$\Sigma(z) = T^*(C - z)^{-1}T, \quad G(z) = T^*(D - z)^{-1}T, \quad J(z) = T^*(D - z)^{-2}T.$$

Factoring the resolvent around $D - z$ gives

$$\|\Sigma(z) - G(z)\| \leq \frac{9\|J(z)\|}{1 - 9/(89 - z)}.$$

All 8143 off-block rows evaluate these shifted Gram matrices; exact finite-frame inertia transfers to the full source. In the common electric clock, the coarse and refined gap intervals are

$$[96.00000060968, 96.00000069820], \qquad [47.99403921427, 48.00529277681],$$

with widths reduced by factors above 361 and 5.36. These are finite-source energies, with no GeV or continuum identification. A general finite-word Neumann remainder is bounded by $b_T^2 M^{n+1}/[(d + M - z)^{n+1}(d - z)]$ when $\|T\|^2 \leq b_T^2$; only the zeroth matrix term is executed here. Conversely, plaquettes vertex-disjoint from a fixed probe give the exact volume obstruction

$$\|QHP\|^2 \geq \frac{\beta^2 N_{\text{far}}}{4}, \qquad N_{\text{far}} \geq N_{\text{faces}} - 48.$$

Orthogonal Haar output supports prevent cancellation. Local incidence and interaction-picture propagation estimates replace the unavailable global volume-uniform norm bound; they do not establish a continuum mass gap (`model/openwave_energy_schur`).

Finite-strength control on the full geometry register. For the declared $H_g = (\omega/2)\sum_i (I - X_i)$ and $Q = (q/N)\sum_i Z_i$, Perron positivity orders all total-spin blocks. The complete geometry-sector spectral gap is determined by the two highest-spin Jacobi matrices, with exact rational Sturm inertia. This is not a gap across different matter sectors. For finite nonnegative energies, write $H = H_g + E_s e^Q + E_p e^{FQ}$. Convexity gives a product-comparison Hamiltonian and a positive remainder. For $M \geq E_s + E_p$, its ground gap is at least $d = \omega - Me^q\mu_2/2$ when $d > 0$, and the ground-ray angle is at most $Me^q\sqrt{\mu_4}/(2d)$, where μ_2, μ_4 bound the second and fourth Q moments of the comparison product ground uniformly over $E_s + E_p \leq M$. Centered observable bounds control the finite lapse; Hellmann integration controls mutual force for all 49 positive native-energy pairs. A full Duhamel comparison bounds the complete quench records. Independently prepared contexts at times zero and 355/113 recover the static normalized lapse coefficient with the stated finite error. They do not identify a single continuously operating observer or physical seconds (`model/openwave_collective_finite_strength`).

Fluctuation–response normalization. The same complete collective geometry gives $g_N = S\,\mathrm{Cov}(Q, e^Q)/(2\pi\omega)$ and $W\,\mathrm{Var}(Q) = q^2\omega u/2$, with $\mathrm{Var}(Q) = q^2/N$ and coordinate-readout work $W = N\omega u/2$. Thus an independently determined native covariance would fix the weak gravitational coefficient without fitting it again. For the continuous source parameter in $H_g + E_s e^Q$, at $E_s = 0$, the complete coordinate measurement attains the geometry ground state's quantum Fisher information. Exact spectral sums imply $4v(1-q^2/(3N^2))^2/\omega^2 \leq I \leq 4v/[\omega^2(1-v)^2]$, where $v = q^2/N$. The zero-source quench information is at most $4I$ at every time. These bounds quantify the measurement resources of this specified geometry family; they do not bound all observer architectures. The inherited register count uses measured G, so its predicted fluctuations and response retain that calibration ancestry.

For $q = 1/8$ and $\omega = 1$, the upper information bound satisfies $1/I_{\mathrm{upper}} = 16N - 1/2 + 1/(256N)$. The Cramér–Rao inequality therefore requires at least $16N$ independently prepared geometry copies for a locally unbiased estimate of the continuous source parameter with unit native variance. The finite ground-state preparation, coordinate readout cost and measured-G ancestry are retained; this is a bound for the specified geometry family, not for every observer protocol (`model/openwave_gravity_noise_normalization`).

Conditional cosmological normalization. For the declared Einstein–dust–radiation–scalar comparison, the regular adiabatic mode reduces exactly to $T = (3/2)(H/a)\int_0^a da'/H(a')$ and $\Phi = 3/2 - T$. The full evolution equations and Hamiltonian constraint vanish under this substitution. With $t = -\log a$ and $F = e^{-t}/(H/H_0)$, the background gives $-2 \leq h \leq 0$ and $0 \leq h_N \leq 33/8$, hence $|F_{tt}| \leq 9F$. Outward background intervals, trapezoidal remainders and the complete improper-integral tail enclose the present normalized metric coefficient in $[0.734179279095, 0.734257299186]$, compared with $[0.709108311479, 0.709181069590]$ for the same-density constant-Λ control. These are conditional regular long-wavelength coefficients; the strict zero-wavenumber mode has zero comoving density and no measured growth or microwave-background comparison is implied. Exact rescalings $H_0 \mapsto \lambda H_0$, $\rho_{i0} \mapsto \lambda^2\rho_{i0}$, $L \mapsto L/\lambda$ and $k \mapsto \lambda k$ preserve the dimensionless equations. Their homogeneous perturbations also admit arbitrary amplitude. An absolute cosmic density and primordial normalization therefore require additional native source relations.

`model/openwave_capacity_normalization_transfer`.

6.22 Complete source acquisition and local refinement

A common matter and compact-field carrier. The full 192-CAR native source keeps all 114 source terms and 24 even pair intertwiners. Two parallel $U(1)$ links and an integer quantum reference solve the Gauss constraints exactly as $E_1 = 3Q_1 - n$, $E_2 = n$, $R = -3(Q_0 + Q_1)$, $n \in \mathbb{Z}$. The entire physical carrier is $\mathcal{F}_{192} \otimes \ell^2(\mathbb{Z})$. For the inherited source H_0, define the declared positive extension

$$H = H_0 + \frac{(3Q_1 - n)^2 + n^2}{32} + \frac{1}{64}\sum_{j=1}^{96}(d_{0j} - d_{1j})^*(d_{0j} - d_{1j}) + \frac{1}{64}\left(I - \frac{S + S^*}{2}\right), \quad S|n\rangle = |n+1\rangle.$$

It is self-adjoint on $D(n^2)$, positive, and has compact resolvent on this finite spatial graph. All native cross-cell terms carry their exact integer dressing. Charge, local current, reference and complete-instrument work identities hold on every retained matter/flux sector. The constant pair maps have zero connection; any field-dependent replacement retains $E + \mathcal{A}$ and every term of its square.

The full one-lepton reducing operators have second eigenvalue bounded from below by $1/16$ after subtracting their inherited rest coefficient. A rational trial with exact full integer-tail residual gives $\mu - v/(1/16 - \mu) \le E_0 \le \mu$ by Temple's inequality. The inherited coloured and charged occupation floors exceed the cycle vacuum bound $1/64$, while neutral-only states share its field operator. This proves that the cycle vacuum is the full-source ground. Subtracting this vacuum yields a common charged gap increase approximately 7.535487×10^{-6} GeV with the supplied electron unit. Neutral one-lepton shifts cancel exactly. This is a calculated architecture dependence, not a physical rest-mass prediction. For every native preparation at $n = 0$, the added mean is $9\alpha\langle Q_1^2\rangle + t\langle K\rangle + \beta$; the operator difference is unbounded on the full flux space. A native-process compiler and an independent colour gauge field are not supplied by this $U(1)$ extension (`model/openwave_matter_flux_source`).

A separate optional normalization uses the exact computed spectral number $\delta = E_{\text{charged}} - E_v$, with $0 < \delta < 1/64$, and defines $H_\delta = H - \delta N_{\text{ch}}$. The inherited inequality $H_0 \ge cN_{\text{colour}} + N_{\text{ch}} + H_{\text{neutral}}$, $c > 1$, gives $H_\delta \ge cN_{\text{colour}} + (1-\delta)N_{\text{ch}} + H_{\text{neutral}}$. Every coloured or charged sector remains above the cycle vacuum, and the neutral-only sector retains exactly H_v plus positive terms. Thus the global vacuum is unchanged and the same coefficient restores all inherited charged one-lepton rest coefficients; neutral ones are unchanged. This is an optional source convention with its own generator, not a native-process derivation or a new physical mass prediction. Weak transitions can change N_{ch}, so current and work formulas must be evaluated for H_δ.

Repeated finite mass-clock records. A finite phase reference supports two native mass-clock cycles when source reset is implemented as an energy-conserving exchange with an explicit empty battery. For preparation coordinates $c^2 + s^2 = 1$, the two isolated source coherences are $ise^{imt_1}/2$ and $isce^{imt_2}/2$; depletion reduces contrast without changing the ideal frequency. Complete source and resource occupations give exact gap-switching work $m(1 - c^4/2)$, with battery energy and correlations retained. A static 512-position controller executes the two cycles on a 32768-dimensional reducing carrier for all three charged rest energies and two preparation strengths. A rational Taylor remainder and explicit arithmetic bounds enclose propagation, reset failure, record overwrite and both flight phase increments. The second flight has a resolved finite controller bias; exact preparation-strength independence applies to the isolated collision identity, not to this complete finite detector. The supplied phase references, clock packet, terminal measurement work and orthogonal native outcomes are retained. Sustained replenishment, physical clock units and native twelve-port execution of this protocol remain separate requirements.

Three-refinement local gauge memory. On open full-Haar $SU(2)$ cubes subdivided by $L = 1, 2, 3$, take $\alpha = 32/L$, $\beta = L^3/32$ and a fixed physical unit-square perimeter of $4L$ links. The Haar probe is supplied, rather than an interacting vacuum. Fundamental and spin-one Wilson characters have variances $1/4$ and $1/9$ and exact electric frequencies 96 and 256 at every subdivision. Electric evolution preserves support; a local Duhamel comparison gives

$$\delta_L = \min\{2, 2\beta N_{\mathrm{inc}}|t|\}, \qquad N_{\mathrm{inc}} = 5 \ (L = 1), \quad N_{\mathrm{inc}} = 8L - 4 \ (L \geq 2).$$

Normalized real correlation errors are at most $2\delta_L$ and $3\delta_L$. Both correlations exceed 0.97 at common time $t = 1/4096$, without a global low-energy projector or representation truncation. The probe is nonstationary, so its normalized correlation is not confined to $[-1, 1]$ by a ground-state argument.

For a Hermitian contraction A, the complete instrument $K_\pm = (I \pm \eta A)/2$, $K_0 = \sqrt{(I - \eta^2 A^2)/2}$, $\eta = 1/2$, has all nine two-time records. Its weighted moment is exactly $\mathbb{E}[s_0 s_t] = \eta^2 \operatorname{Re} \operatorname{Tr}(\rho A(t) A)$, including backaction and arbitrary idle references. The Haar derivative identity bounds the two-readout work by $32/3$ and $512/27$ source units. Probe preparation and intervention times remain explicit resources. At fixed time the local error grows as L^4; controlled windows of order L^{-4} supply no fixed-time continuum or Clay theorem (`model/openwave_local_gauge_memory`).

Quantum acquisition of the finite tensor stencil. All 60 columns of the native coordinate Wilson source carry ten source observables: energy, three momenta and six coframe action derivatives. Three compact declared metric profiles give a full 34-site staggered stress/curvature stencil. Ten same-state mean records suffice for every component. Complete three-outcome instruments retain thirty exact probabilities, the zero events and arbitrary quantum references. Their nonselective trace-norm disturbance is at most $3/8$, and source work is at most $3\|H\|/8$ per acquisition. Independent preparations prevent incompatible sequential readouts from changing the target state. A two-state control changes $\langle X \rangle$ from 1 to $3/4$ after a prior Z acquisition, exposing that distinction.

If C is the maximum coefficient norm and each native mean has error ϵ, every null contraction has error at most $8C\epsilon$. The sharp inverse $2 + \sqrt{3}$ and measured variation budget yield

$$\delta = (2 + \sqrt{3})(\widehat{q} + 8C\epsilon) + \widehat{\tau} + 2C\epsilon, \qquad |\Delta\lambda| \leq \widehat{w} + 8C\epsilon + 8\delta$$

at unit spacing. The complete sufficient Hoeffding budget is 3,239,348,495 independently prepared copies for $\epsilon = 10^{-3}$ and joint confidence 0.99. Every support point and zero halo is retained. This closes the finite acquisition interface for the stated carrier and readout assignment; its actual nonzero Einstein residual and physical metric/stress identification remain explicit (`model/openwave_stress_curvature_instruments`).

Narrow Lambda weight and autonomous quark transport. For the actual native weak current applied to a normalizable Lambda packet, an approximate inverse of the complete interacting Hamiltonian gives $\|1_{[E-w,E+w]}(H)u\| \leq \|u - (H - E)x\| + w\|x\|$. The rational witness includes native proton and pion ground trials at eight relative separations. Every occupation successor and translation return contributes to the residual. The conditional weight in the window of half-width 1 MeV is below 0.026622, compared with the moment upper bound 0.048210. These are complete finite-resolution spectral bounds, with the inherited electron and strong calibration. They imply neither an on-shell density nor a physical decay rate.

The complete source and records are in `model/openwave_lambda_narrow_weight`.

Gauge-covariant native hopping transports a fractional-charge endpoint with its complete reference-supported string. A full-carrier intertwiner reduces the specified preparation to

$$H|x\rangle = (2 + \sigma x)|x\rangle - \kappa(|x-1\rangle + |x+1\rangle).$$

where $\kappa = 1/8$ and $\sigma = q^2/32 + 1/12$. Rational exponential remainders enclose all charge, flux and pointer records on five- and nine-vertex carriers. The entire fermion and rotor complements remain in the source. Energy conservation gives the all-time conditional bound $\langle x\rangle - 1 \le 2\kappa/\sigma$, equal to $6/5$ and $24/11$ for native charges $2/3$ and $-1/3$. This autonomous source-unit transport theorem supplies no physical quark rest-mass or continuum identification.

The source and complete records are in `model/openwave_quark_string_transport`.

Stable nuclear attraction and full-input work. A neutral harmonic field coupled to a gauge-invariant four-site occupation projector supplies a signed nuclear interaction while preserving every sector with fewer than twelve quarks. Completing the full operator square gives $H \ge H_{\mathrm{EM}} - g^2 \sum_b O_b/(2\omega)$, an extensive lower bound. Exact coherent-state trials provide the corresponding complete-source variational upper bounds; the oscillator is not truncated. With declared $g = 1/8$ and $\omega = 1/16$, the helion and alpha comparisons fail. No held-out mass selects either coefficient. For the original full singlet input, an independent Casimir trace gives electric preparation energies $8955/9152$ and $2949/1360$ source units for three and four cells, respectively, including all charge sectors.

The alpha interval is $[3.739060107335, 3.739677967934]$ GeV; helion remains in $[2.807625870510, 2.808017057294]$ GeV. The necessary $|g| \ge 1.694976066305$ at fixed oscillator frequency is a diagnostic obstruction, not a chosen coefficient or sufficient fit. All oscillator and integer-flux complements remain (`model/openwave_nuclear_field_attraction`).

Complete covariance measurements and normalization. For $Q = (q/N)\sum_i Z_i$ in the product X vacuum, two complete coordinate records x, y give the unbiased statistic $D = (x-y)(e^x - e^y)/2$ for $C = \mathrm{Cov}(Q, e^Q)$. With $v = q^2/N$, its second moment is bounded by $3e^{2q}v^2$ and $C \ge e^{-q}v$. A median of independent block means therefore has a certified relative-error sample bound independent of N. Its measurement work per pair is $N\omega u$, and its coordinate resolution is a supplied resource. The complete source gives the weak rest slope $M(1) = \langle e^Q\rangle$ and the photon slope $M(f_\gamma)$. Using one clock and length yields $G_{\mathrm{weak}}/(c_\gamma/c) = SC/[2\pi\omega u^2 M(1)M(f_\gamma)]$. For a far photon reference, $M(0) = 1$ and this reduces to $Sq\tanh(q/N)/(2\pi\omega u^2)$. The register count requires independent observational ancestry; the calibrated count does not predict Newton strength. Finite-source physical identification and the native-process implementation are not established by this covariance protocol.

The same pair data also estimate $M(1)$, and a union bound propagates both relative-error intervals through the complete unit map. No independence between the two estimators is assumed. Geometry-copy counts for covariance estimation and local-unbiased source-energy estimation concern distinct parameters. The former can be independent of N while its detector resolution and work remain N dependent (`model/openwave_geometry_covariance_protocol`).

Finite-wavelength cosmological transfer. For the supplied Einstein–dust–radiation–capacity-scalar model, put $U = U_m$, $v = U_r - U_m$, $R = \Phi + U$ and $w = y - sU$. These comoving variables remove the exact adiabatic zero mode. At $N = \log a = -1$, the declared correction $\Delta_m = -2Q\Phi/(3\Omega_m)$ enforces the complete Hamiltonian constraint. An eight-coordinate affine

interval equation transports the k^2 response to $N = 0$; the exact variation-of-constants equation retains the complete k^4 remainder. The certified metric and comoving-dust coefficients lie in

$$\Phi_1 \in [-0.008281806997, -0.005759370982], \qquad \Delta_{m,1} \in [-1.455879943233, -1.454618725225].$$

Full finite-wavelength records are enclosed for $k/H_0 = 1/100, 1/50$, with overlapping independent 1024- and 2048-step certificates. The supplied late preparation is distinct from a primordial continuation. Gravity, density fractions, horizon choice and the common physical unit remain input premises; no early-Universe abundance or CMB match follows (`model/openwave_capacity_finite_wavelength`).

Native-process admission. OpenWave lists OPH as M10 with all 31 criteria untested. Finite theorems, calibrated extensions and implementation replays are supporting results. They do not substitute for an in-platform native carrier process generating the criterion's observable without installing the target field law. The admission gate preserves this distinction (`model/openwave_m10_review_gate`).

6.23 Exact repair response and complete observer records

One literal finite repair grammar fixes the twelve-port carrier, fair C/E law, endpoint coin and record-memory rule. Every experiment declares its schedule and input: the response and event-clock tests use independent uniform seam selection; causal-order and interaction controls use fixed literal seam words. Its local instrument retains all 25 raw slots, including zero hold and register padding. On the port-pair federations, three-event experiments retain all 15 625 raw histories. Disjoint events commute after relabelling their recorded event addresses; events sharing a port retain their causal order. The execution ordinal is an addressing convention rather than a fundamental global clock. Source preparation, randomness, blank memory and the coherent apparatus services are distinct declared resources (`model/openwave_native_process`).

The complete finite response. In the full global one-load sector, with graph Laplacian L, adjacency A, port degrees d_i and M seams, the exact unread-source channel is

$$\operatorname{diag}(\Phi\rho) = T \operatorname{diag}\rho, \quad T = I - \frac{L}{2M}, \qquad (\Phi\rho)_{ij} = \left[1 - \frac{3(d_i + d_j) - 2A_{ij}}{4M}\right]\rho_{ij} \quad (i \neq j).$$

Every endpoint matrix unit and every global one-load column is covered on 240, 960 and 3840 ports. The identity tensors with an arbitrary idle reference. The complete observer history is retained by the underlying instrument; the displayed marginal is sufficient only for the fixed independent seam schedule. It is also the exact linear response at the maximally mixed stationary input.

Let C sum each cell's twelve populations and S lift a cell population uniformly. The natural cell projection has

$$\|CT - (CTS)C\|_{1\to 1} = \frac{11}{12M}, \qquad Q = I - SC.$$

The complete complement gives the exact memory correction $CTQ(QTQ)^{\tau}QTS$. A memoryless cell process therefore fails on explicit basis columns. This response has no supplied translation representation or wave-time intertwiner identifying it with the separately declared twelve-direction cosine dispersion (`model/openwave_native_response`).

Fast propagation with complete native arrivals. Since the maximum port degree is six, $P = I - L/6$ is stochastic and

$$T^t = \sum_{k=0}^{t} \binom{t}{k} q^k (1-q)^{t-k} P^k, \qquad q = \frac{6}{2M}.$$

Positive uniformization certifies up to 80 640 original attempts using 128 local powers and an outward rational binomial tail. No source carrier or virtual-power remainder is discarded. Three graph-ball widths, including a predeclared radius-two control, use the common event-intensity coordinate $\tau = t/(2M)$ on all three graphs. Native crossing-current records determine first arrival at graph distance four; an observer flag retains arrival and nonarrival while the physical source continues. For the radius-two preparation on 240 ports at $\tau = 4$, the certified first-arrival probability is approximately 0.547397035737486. All corresponding coherent offdiagonals decay at their initial port addresses, so occupation transport is not a coherent translating packet. The calculations supply neither a physical speed nor a continuum limit (`model/openwave_native_packet_propagation`).

Conservation and interacting native records. Every actual current obeys $n'_a - n_a = -j$ and $n'_b - n_b = j$; complete histories therefore give the exact continuity equation and all cut-current identities. With a supplied known initial occupation, records reconstruct the final positions and retain explicit invalid-record outcomes. On a connected fixed-load carrier the native branch star algebra is the full matrix algebra: E-effect signatures distinguish configurations, and occupied-vacant edge exchanges connect them. Its source-only commutant is scalar. Total load is conserved, but a modulo-six accumulated current record is not a protected source charge or an identified Wilson flux (`model/openwave_native_defect_records`).

Two actual native loads give a nonfactorizing interaction observable. Two overlapping repairs produce connected record-current covariance 1/8. After one and two two-event blocks, the complete native pair densities are 1/2 and 3/8; independent labelled walkers with the same one-point means differ by 1/4 and 9/64. A complete additional E repair acquires a zero-imbalance statistic equal to that pair density in the native two-load sector, retaining its post-measurement state. The equality does not extend to the comparison process's collision sectors. All local operator columns and the full spectator complement remain. This is an exclusion interaction of the executed source, not a compiled unitary matter field or QFT scattering law (`model/openwave_native_interaction`).

Event modes, phase readout and energy attachment. For a graph eigenvector v, the native current record supplies $\Delta(v \cdot n) = (v_b - v_a)j$. On the elementary H66 carrier, four-event record means recover three exact first-moment multipliers

$$\lambda_\pm = \frac{55 \pm \sqrt{5}}{60}, \qquad \lambda_0 = \frac{9}{10}.$$

The complete instrument is contracted for the requested moments; its expanded four-event tree is not claimed to have run. Stationary current covariances separately satisfy $\mathrm{Cov}(\Delta f_0, \Delta f_n) = -\mathrm{Var}(f)(1-\lambda)^2\lambda^{n-1}$ for $n \geq 1$. These are three decay sectors, without a particle-family or physical frequency assignment (`model/openwave_native_event_clock`).

Every repair-only history map is a scalar times a partial injection, so its effect is diagonal in occupation. Phase twins with the same diagonal have identical complete record laws. A constructive extension loads a two-dimensional H66 code into the declared six-level response, executes its actual coherent H pulse and follows it with a native E measurement. The complete instrument retains the orthogonal 64-dimensional source outcome and all four processor complement outputs; all

4356 source matrix-unit trace identities are checked. This gives a phase-sensitive finite readout conditional on the supplied coherent service. It neither generates that service by repair alone nor compiles the full stress and curvature stencil (`model/openwave_native_geometry_readout`).

The unread repair channel has positive real decay eigenvalues and zero event phase. Its pure-port output differs from any unitary output by trace distance at least $d_i/(2M)$. Identifying one-load coherence decay rates with rest masses is an additional prescription; its full rate ratio is at most 30/23 on these graphs and fails the inherited electron-anchored charged-lepton hierarchy. No GeV rest-mass interval is obtained from this member. A Euclidean transfer or coherent dilation remains a distinct possible reconstruction with its own algebra, clock and readout proof (`model/openwave_native_rest_energy`).

Capacity and physical normalization. For export-oldest memory and a controller reading only the last committed record, concatenating archive and FIFO is an exact complete-instrument intertwiner across FIFO capacities. Every exported record and terminal outcome remains. A halt-on-full controller gives a different process. Raw storage, classical record entropy and source information capacity are distinct; none is an identified horizon area. Native event laws are also invariant under a rescaling of an assigned physical clock, while H_0 and Λ scale inversely and inversely quadratically. A homogeneous perturbation has an independent amplitude freedom. The finite-wavelength cosmology attachment therefore requires a physical clock, horizon map and preparation normalization (`model/openwave_native_capacity_history`).

An author-run headless endpoint calculation in the pinned OpenWave M10 layout reproduces every local operator column with an independent rational oracle and rejects an altered conservation rule. This authenticates the finite source calculation and its declared platform interface. It is not a maintainer criterion award: the official M10 column has zero validated criteria and all 31 cells untested (`model/openwave_native_platform_gate`).

6.24 Quantum scalar completion and calibrated mass conventions

A finite interacting extension keeps the scalar quantum. Its carrier is a vacuum plus the conjugate Higgs doublet. The actual charged-Higgs tensor defines a CAR-even vertex that commutes with all twelve local Gauss generators. Exact occupation arithmetic on its complete 192-state active scalar/fermion carrier gives rank-one vertex norm 2. The same complete feedback theorem therefore applies with the common quantum gauge/Higgs Hamiltonian. This is a finite gauge-invariant interacting action limit; vacuum selection and the interacting chiral continuum do not follow from it (`model/particle_candidate_atlas/INTERACTING_CHIRAL_JOIN.md`).

The zero-phase program has ratios $1 : 1 : (34 + 24\sqrt{2})$. A separate calibrated test uses only $m_\mu/m_e = 206.7682827$ to set $\tan(\phi/2) = 55785269584552807/500000000000000000$. The source then gives $m_\tau/m_e \simeq 3477.4416363$, within one quoted standard uncertainty of the held-out CODATA 2022 value 3477.23 ± 0.23. With the additional electron unit 0.51099895069 MeV, the conditional value is 1776.9690272 MeV, compared with 1776.86 ± 0.12 MeV. These correlated comparisons are not a joint significance test or exact central agreement. The phase is fitted and the default is unchanged. The same conditional value is 1.7769690272 GeV; the separate PDG2026 tau average is 1.77693 ± 0.00009 GeV. Agreement with these individual intervals does not require equality to their central estimates.

A distinct three-input test fits all three charged central masses with the complete attenuator. Put $x = (\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu})$, $A = \sum_j x_j/3$ and $B = \sum_j x_j e^{-2\pi i j/3}/3$. Choosing $r = \sqrt{2}|B|/A$, $z = B/|B|$ and an overall energy factor $s = 3A^2/a^2$, with $a = \sqrt{10/72}$, gives $s\,\mathrm{spec}(C_r^2/3) = \{m_e, m_\mu, m_\tau\}$ by inverse Fourier transformation. The quoted central values

admit $r \simeq 0.99999669509849$, $\arg z \simeq 0.22222476189459$ and $s \simeq 6.7791801525\,\mathrm{GeV}$, with all sixteen records retained. Here $r^2 = 3Q - 1 < 1$, so attenuation suffices. The exact radical construction and its rational controller approximation have an output enclosure error below $10^{-36}\,\mathrm{GeV}$. All three measured masses select this controller; that numerical bound is neither a physical error budget nor a held-out prediction. The factor s is no identification of the physical Higgs expectation value, and interacting threshold or resonance matching is an additional premise (`model/particle_candidate_atlas/CALIBRATED_TRIPLET.md`).

The separate pixel-dependent response law also has a complete C-duty realization: two admissible controllers obey $\mathrm{spec}(gZ_C) = \{m_j/v\}$, including every disabled record. The mass modes diagonalize the entire coefficient, not its nonorthogonal port-projector constituents. Its retained capacity coordinates give the rounded ratios

Conditional source branch	m_2/m_1	m_3/m_1
Forward coordinates	206.768300595181	3477.365536596023
Measured-α endpoint	206.768282739123	3477.365262268937
Supersymmetric proxy, $\beta = 4$	206.768297922262	3477.365495531309
Minimal census, $\beta = 7$	206.782621310399	3477.644439943646

Only the measured-α endpoint lies inside both quoted individual CODATA2022 ratio intervals. It inherits measured-α input; the architecture is target-informed. The two closure roots depend on their specified census, scale and response kernel. Their amplitude changes with the phase, giving $Q = (1+e^{-2\chi})/3 < 2/3$. Transferring only their phase to the balanced analyzer changes the action. Pixel capacity is a declared response coordinate, without an identity to total cell count or hardware memory. The interval calculations and ancestry are in `model/charged_lepton_response/P_DEPENDENCE.md`.

The two shapes can describe bare and effective coefficients. A physical common-source identification requires a derived family-dependent correction. The response architecture stipulates $\chi = (u^2/512)/(1 - u/77)$. At a common phase, set $B = (I + T)^2$ and $D = (I + e^{-\chi}T)^2$, where $T = (e^{i\delta}R + e^{-i\delta}R^2)/\sqrt{2}$. When both root matrices are positive, the supplied kinetic choice $Z_L = Z_R = BD^{-1}$ whitens B to D exactly. This solves an inverse matching problem; it does not calculate a source self-energy. Passive field changes preserve canonically normalized masses. A common-source identification requires independent kinetic and mass-response data and must distinguish $\delta(P, u)$ from the fitted phase (`model/particle_candidate_atlas/COMPATIBILITY.md`).

Reversing the native half-turn order preserves the unordered masses and conjugates the coherent phase. The exact all-lag ordering potential has the form $I \otimes i\kappa_{\mathrm{ord}}(R - R^{-1})$, with coefficient $\kappa_{\mathrm{ord}} = 603(1 + \sqrt{5})/6666724$ and an accompanying dissipative correction. It commutes with the coherent coefficient, so its conjugation cannot select that phase. An exact inverse memory realization establishes a specified action, without selecting its control law or physical clock. The actual default all-lag diffusion also leaves the circulant algebra; it is not the displayed scalar attenuation. At finite pulse angle h, disabled and enabled records alone give full-CAR vacuum purity loss at least $2w_0w_3(1 - \cos^4 h)$ in the unread working channel. For $h \notin \pi\mathbb{Z}$, this excludes an exact unitary mass-only replacement on that register. It neither discards the full archive nor excludes a physical interaction correction in a different source model (`model/particle_candidate_atlas/NATIVE_RESPONSE.md`). The eighteen-channel candidate dictionary retains the internal source weights, these distinct charged-action families and the separate eleven protected objects. It assigns no physical spin, mass, width or defect-to-species map (`model/charged_lepton_response/CANDIDATE_MAP.md`).

A conditional Standard Model completion makes the physical target explicit. Supply a local Lorentzian chiral gauge action on three copies of the fifteen-state module, a complex Higgs

doublet, positive kinetic forms and a symmetry-breaking vacuum. Canonical normalization then gives the fermion action masses from the singular values of the normalized Yukawa matrices. For $\langle H\rangle = (0, v/\sqrt{2})$, the tree relations are $m_f = v\sigma_f(Y_f)/\sqrt{2}$, $m_W = g_2 v/2$, $m_Z = v\sqrt{g_2^2 + g_1^2}/2$ and $m_h^2 = 2\lambda v^2$. The unbroken electromagnetic and color gauge fields have zero gauge-boson mass parameters. Supplying fermionic CAR and the chiral spacetime action assigns spin 1/2 and Fermi statistics to matter, spin 1 to gauge fields and spin 0 to the Higgs. Conjugate representations specify antiparticles; these assignments require more than the internal weights.

The calibrated comparison keeps two explicit action arms. The inherited common-scale running coefficients give tree values 80.848, 92.2264 and 131.280 GeV for W, Z and the radial Higgs mode. These are not measured poles, and its three massless neutrinos fail the nonzero oscillation splittings. The separate Weinberg arm retains the same charged action and verifies both supplied neutrino mass-squared splittings using its invariant Higgs tensor. Its normal ordering, lightest mass and Majorana phases retain their calibration or declared-choice ancestry; transport to the common scale has no certified error. A dimension-five action requires an explicit effective-theory window and an ultraviolet completion for an all-energy claim. The minimal zero entries are not overwritten.

Independent Yukawa and mixing inputs provide a coherent conditional action without imposing exact Koide on measured masses. Retaining the coherent Koide coefficient instead requires an independently derived interaction correction. Compatibility with all physical reference targets is conditional on radiative, infrared, resonance and neutrino matching: no complete numerical fit or interacting continuum is supplied. The ninety-coordinate chiral extension also requires an explicit map from the sixty-mode selected source, including its state and clock.

For an interacting comparison, independent kinetic forms Z_L, Z_R and a mass correction E require bounds in the same state and clock. If $\|Z_{L,R} - I\| \le \epsilon_{L,R} < 1$ and $\|E\| \le \eta\sigma_{\min}(M_0)$, then each singular value of $Z_L^{-1/2}(M_0+E)Z_R^{-1/2}$ lies between its original value times $(1-\eta)/\sqrt{(1+\epsilon_L)(1+\epsilon_R)}$ and $(1+\eta)/\sqrt{(1-\epsilon_L)(1-\epsilon_R)}$. Exact allowed corrections to this source coefficient violate Koide. No interacting correction is computed here. A stable charged infraparticle may have a lower spectral boundary without a mass-shell atom; unstable leptons require a resonance prescription. The interacting gauge vacuum, mirror treatment and observable matching are separate from the free theorem.

The numerical discrepancy has a sharp conditional matching floor. If $r_0 = 1$ and $[L_i, U_i]$ are the reference ratio intervals, the smallest required multiplicative spread is

$$D_{\min} = \max\left(1, \frac{\max_i(L_i/r_i)}{\min_j(U_j/r_j)}\right).$$

The corresponding additive singular-value optimization has minimum $\eta = \max(0, \max_{ij}(L_j r_i - U_i r_j)/(L_j + U_i))$ after optimizing one common unit. Exact diagonal controls attain these bounds. Assigning the entire change to componentwise observable matching, the forward pixel branch requires a relative correction of at least 3.21×10^{-8} for the stated ratio windows, far above its numerical enclosure width. This is a necessary correction size, not a calculated physical correction or a rejection of every interacting completion (`model/particle_candidate_atlas/MEASURED_MASS_JOIN.md`).

Capacity self-consistency does not select the attenuation law. Holding the inner heat, scale and coupling equations fixed, the supplied change $\chi(u) \mapsto \chi(u) + 10^{-14}$ preserves a root in each of two certified closure intervals. The original strict self-map margins exceed the map perturbation bound 4×10^{-16}. The two attenuation and Koide-quotient ranges are disjoint, and the complete C-duty inverse admits both. This compares different specified laws; uniqueness for any fixed original law is unaffected (`model/particle_candidate_atlas/CAPACITY_NON_SELECTION.md`).

6.25 Pixel feedback with one charged-fermion mass prescription

The pixel feedback must use the same mass coefficients as the declared field and memory operators. Let $u = \alpha_U$ and retain

$$M_U/E_* = e^{-2\pi}P^{1/6}, \qquad v/E_* = P^{-1/2}e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi}\log(M_U/m_Z).$$

Here E_* is a declared energy unit. The same conditional three-lepton coefficients are inserted into the charged-fermion integral, together with all six declared quark coefficients, including the finite top contribution. With the prescribed factor $1 - 3\alpha_3/\pi$ on the quark terms, the resulting inverse response $A(P)$ defines $P = \phi + \sqrt{\pi}/A(P)$. This kernel supplies no charged-vector or charged-scalar terms, confining hadronic spectral measure, threshold matching, or derivation of the native current's physical Thomson limit.

There is an exact obstruction to substituting the minimal Standard Model census while retaining $\beta = 4$. Assume $P \geq 1$, $M_U \geq m_Z$, positive couplings and threshold-free one-loop coefficients $b = (41/10, -19/6, -7)$. Set $x = \alpha_2^{-1}$ and $y = \alpha_3^{-1}$. Eliminating the unified coupling and scale ratio gives $0 < y \leq x$ and

$$\frac{\pi(18x + 5y)}{46} = 2\pi - \frac{2}{3}\log P + \frac{1}{2}\log\left[\pi\left(\frac{1}{x} + \frac{69}{333x - 218y}\right)\right].$$

The logarithm's argument is at most $8\pi/(5x)$. The function $9\pi x/23 - 2\pi - \frac{1}{2}\log[8\pi/(5x)]$ is increasing and exceeds one at $x = 6$, so $x < 6$ and both nonabelian couplings exceed $1/6$. Their heat-sum denominators are at least one; direct outward summation with infinite-tail bounds places their combined numerators below $1/80$. The required pixel entropy is $P/4 \geq 1/4$, a contradiction. This excludes the stated scale and census combination, independently of the readout kernel.

Two other declared maps have certified local fixed points. The supersymmetric running proxy with $\beta = 4$ gives $P \simeq 1.630971514159316$ and $A \simeq 137.000994766342$. The minimal census with the explicit scale choice $\beta = |b_3| = 7$ gives $P \simeq 1.628533369611833$ and $A \simeq 168.815082928453$. The first requires extra supersymmetric fields absent from the native quadratic carrier. The second supplies a scale choice, without identifying the Higgs scale physically with color transmutation. For each map, verified implicit-root signs, infinite heat tails and interval automatic differentiation give a strict self-map and $|g'| < 0.00734$ on the reported coupling interval. Uniqueness is local to that interval.

Rational interval inversion over both complete root enclosures admits both native memory solutions, with $0 < p < 1/12$ and $0 < q < 1/6$. The gain $g = \sum_j (m_j/v)/p$ preserves the unconditional duty probability, so the same full-record memory coefficient has the required three singular energies. Independent simultaneous-root and quadrature checks accompany the proof. These are conditional consistent feedback calculations; neither is a complete physical particle or fine-structure prediction.

The beta coefficients in the feedback calculation are obtained from the actual internal-current traces with the intended chiral field multiplicities. This assumes that those are the surviving continuum fields, and it does not count or remove spatial regulator valleys or doublers. The minimal-census root and the supersymmetric proxy are different scale designs, not two calibrations of one proved physical theory. The absence of a positive solution for the minimal census with $\beta = 4$ is a mathematical inconsistency of that specified combination; it is not an exclusion of other admitted OPH designs.

The three increasing coefficients returned by these closed conditional maps are dimensionless singular coefficients m_i/v.

Scale design	m_1/v	m_2/v	m_3/v
Supersymmetric proxy, $\beta = 4$	2.07203042×10^{-6}	4.28430204×10^{-4}	7.20520710×10^{-3}
Minimal census, $\beta = 7$	2.07103435×10^{-6}	4.28253912×10^{-4}	7.20232110×10^{-3}

The interval certificate encloses these values and both admissible memory inverses over each complete root box, so the table and the controller refer to the same conditional source coordinates.

7 Localizing the same current and mass operators

7.1 Topology, total energy and the radial source

The seed of the defect hypothesis is a precise finite frustration statement. For a finite connected graph, abelian edge offsets b_e and positive weights w_e, define

$$\Phi_b(x) = \sum_{e:u\to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e].$$

If a cycle has nonzero signed offset sum, then $\min_x \Phi_b(x) > 0$: the possible costs form a finite set of positive-weight subset sums, and a zero cost would solve every edge equation and force zero cycle sums [5]. This is frustration at fixed compatibility data. Protection of b against an update that changes its class, localization of its energy and a rest mass all need more.

For a one-cochain b on the full carrier complex, face curvature is $F = \delta b$ and $b(\partial c) = F(c)$ for every two-chain c. The icosahedral boundary has trivial first homology, and the native load difference $b_{uv} = n_v - n_u$ is exact, so $F = 0$ and all loop sums vanish. An actual defect invariant therefore requires additional transport or field data.

Consider the two-spin cubic hopping operator with forward link $i\sigma_j/2$ and its Hermitian reverse. A declared radial onsite texture is

$$B_x = M \sum_{j=1}^{3} (-1)^{x_j} \frac{x_j}{|x|} \sigma_j, \qquad B_0 = 0.$$

On the odd radius-one cube at $M = 64$, this literal source has exactly two zero modes and a complementary absolute-energy gap at least 62. The normalized kernel frame W has scalar local density $W_x^\dagger W_x = \rho_x I_2$, with $\sum_x \rho_x = 1$. All 28 retained winding matrices have exact spectral enclosures with every multiplicity and source-matrix rounding error controlled. For every retained odd-degree cube, inversion and complete spectral exhaustion prove exactly two zero modes, including weak-mass cases. Degree-three six-level windows hold two zeros and four nonzero levels, which is no count of generations. A complete-Fock contact theorem admits 34 of 84 supplied contact cases, with a ground pair and unique neighboring charged states.

A radial mass profile also needs a total energy law. In the selected ordered-spin rotation orbit, each admissible charged shell costs at least $\pi J\kappa|q|$ with $\kappa > 0$. A hedgehog retaining nonzero degree on arbitrarily many disjoint shells therefore has infinite excess energy in that ungauged identification, and a fermion bound-mode gap omits this background cost. Screening, compact opposite-charge textures or another specified finite-energy structure are substantive dynamical choices. The compact pair below provides the second kind.

7.2 One local carrier for defect states, currents and mass coefficients

The radial defect operator admits a direct realization of the native matter currents and a prescribed family mass operator on the same localized modes. The construction is a choice of simulator carrier

and coherent pairing service. Its finite operator identity is exact; the choice is not claimed to follow uniquely from overlap consistency.

Let h be the radial operator on $\ell^2(\Lambda)\otimes\mathbb{C}^2$, with its normalized kernel frame $W:\mathbb{C}^2\to\ker h$. Use the strong texture domain, in which the kernel has dimension two and the complementary spectrum has absolute energy at least $d>0$. With

$$J=i\sigma_2,\qquad D_x=(-1)^{x_1+x_2+x_3},\qquad C=DJ,$$

the source identities are

$$C^T=-C,\quad C^\dagger C=I,\quad hC=-C\overline{h},\qquad hW=0,\quad W^\dagger W=I_2,\quad C\overline{W}=WJ.$$

The last identity fixes the quaternion frame through evaluation at the origin and its scalar normalization Gram matrix.

Choose the canonical anticommutation relation algebra on $\ell^2(\Lambda)\otimes\mathbb{C}^2\otimes M_{15}\otimes\mathbb{C}^3$. Here M_{15} is the native matter representation, transported by its actual source intertwiner; the three family copies are declared inputs. For a symmetric all-left Higgs kernel $K\in M_{45}(\mathbb{C})$, define

$$\Delta=C\otimes\overline{K},\qquad B=\begin{pmatrix}h\otimes I & \Delta\\ \Delta^\dagger & -\overline{h}\otimes I\end{pmatrix},\qquad E=\operatorname{diag}(W\otimes I,\overline{W}\otimes I).$$

The antisymmetry of Δ makes it a valid local creation pairing. Multiplication gives the exact isometric intertwiner

$$BE=EB_0,\qquad B_0=\begin{pmatrix}0 & J\otimes\overline{K}\\ (J\otimes\overline{K})^\dagger & 0\end{pmatrix}. \tag{11}$$

Every singular value of K therefore appears with both signs and twice the multiplicity in the localized Nambu operator, including zero values. Nambu coordinates duplicate annihilation and creation operators; they introduce no physical mirror carrier.

For every actual anti-Hermitian native current G, the map $W\otimes I$ intertwines $I\otimes G$ with $I_2\otimes G$. In Nambu coordinates the generator is $\operatorname{diag}(G,\overline{G})$; Hermitian charges use $\operatorname{diag}(Q,-\overline{Q})$. The Higgs background transforms with matter under the uniform internal gauge action. A fixed broken vacuum does not make every gauge generator commute with its mass operator. The resulting map of canonical anticommutation relation algebras intertwines quadratic Heisenberg evolution. On even states, tensoring with the normalized trace on the orthogonal complement gives an exact state inclusion with partial-trace recovery. This complement state is not a vacuum.

Use the positive family matrix $M(P)$ supplied by the memory construction, whose eigenvalues are the retained charged-lepton coefficients $m_i(P)/v(P)$. Choosing $H=(0,v(P)/\sqrt{2})$ and $Y_e=\sqrt{2}M(P)$ realizes those coefficients in $K/v(P)$ itself. The coherent pairing service implements the stationary-mean memory coefficient times its declared gain. Equation (11) does not identify the unread finite memory channel with this Hamiltonian. The calibration provenance of the retained family law is unchanged.

An explicit record service realizes the same Hamiltonian in a controlled weak-step limit. Apply the native current probes to fresh auxiliary legs in their invariant preparations, so their complete twelve-state memory transition is independent of the spatial field. For the updated memory port j choose the field pulse generated by

$$H_r=H_0+\mathbf{1}_C(r)\,gL(P_j),$$

where H_0 contains the radial kinetic term and the chosen quark pairing, L is the local lepton pairing map, and g is the source gain. Every H_r has the same core intertwiner. Linearity gives stationary

mean $\overline{H} = H_0 + L(gZ_C)$, precisely the Hamiltonian above. If $\|H_r\| \le M$ on the fixed finite Fock space, the reset contraction of the complete memory law and a second-order unitary expansion give

$$\left\|\rho_N - e^{-iT\overline{H}}\rho_0 e^{iT\overline{H}}\right\|_1 \le \frac{4MT}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M^2T^2}{N}.$$

The bound includes arbitrary finite untouched references and initial memory correlations; stationary product memory removes the first term. For the displayed cube, $M = 2430\cdot 66 + 1215(1+g)$ suffices. This chosen coherent service uses every record. Its finite unread channel need not be unitary, and its bound is not uniform in volume.

For a complete finite witness, choose the retained quark design $Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1)$ and $Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2})$, and set the three minimal neutrino coefficients to zero. Their exponents are design inputs, not topological index conclusions. On the radius-one cube there are 27 sites, 2430 physical annihilator modes and 4860 Nambu rows. The localized sector has 90 annihilators and 180 Nambu rows: 84 positive, 84 negative and 12 zero rows. These numbers count operator degrees of freedom, not observed species. In particular the zero rows do not establish a relativistic massless neutrino sector.

The full complement is retained. Indeed, with $T = \mathrm{diag}(I, C^\dagger \otimes I)$,

$$T^\dagger B T = \begin{pmatrix} h \otimes I & I \otimes \overline{K} \\ I \otimes K & h \otimes I \end{pmatrix}.$$

Its energies are the sums $\lambda(h) \pm s_j(K)$, not a relativistic square-root dispersion. If $\|K\| \le M$ and $2M < d$, the localized band lies in $[-M, M]$ and the complement has absolute energy at least $d - M$. This is single-quasiparticle isolation, not isolation of all many-body energy sums. In the displayed design, choosing $v(P)$ as energy unit, radial hopping one and texture coefficient 64 admits $d = 62$ and $M = 1$.

The common spatial density follows from $W_x^\dagger W_x = \rho_x I_2$; inversion gives mean position zero. A Fourier transform supplies a momentum distribution, without making the localized state a momentum eigenstate. Occupation of these modes does not create the supplied texture winding. Original site-resolved repair can leak out of the kernel, and interactions can mix the two factors. Thus the exact common finite operator supplies neither defect formation, physical rest masses nor an interacting relativistic continuum.

The certificate retains all twelve native current operators, and its complete finite controls include an unstaggered-pairing counterexample.

7.3 Compact opposite-degree textures and their isolated modes

A finite-energy texture can carry opposite local winding numbers while approaching a constant exterior. On the two-spin cubic hopping carrier, choose an integer $a \ge 1$, write $R = a + 1$, and define

$$f(x,y,z) = \begin{cases} (x, y, z^2 - a^2), & \max(|x|,|y|,|z|) \le R, \\ (0,0,1), & \text{otherwise.} \end{cases}$$

Set $n = f/|f|$ away from $c_\pm = (0, 0, \pm a)$ and $n(c_\pm) = 0$. The Hamiltonian is $h = T + Mn\cdot\sigma$, where the nearest-neighbor hopping has $\|T\| \le \sqrt{3} < 2$ and $M \ge 8$. The texture and its amplitude zeros are explicit design inputs. The complete unit cubic shells about c_+ and c_- have degrees $+1$ and -1 under their normalized barycentric interpolation; the exterior shell is constant. Exact rational regular-value counts check every oriented triangle. This degree classifies the interpolated texture and is not a fermionic superselection charge.

For the selected ordered-spin reference, fill the two core directions by the exterior direction. Each rotated-ground source bond has excess energy in $[0, J]$. With $s = 2a + 3$, precisely $3s^2(s + 1)$ bonds touch the support cube, so

$$0 \le \Delta E_{\text{spin}} \le 3s^2(s + 1)J.$$

The two zero fermion masses and the spin-core fill are distinct declared parts of the construction. The second-quantized Hamiltonian differs from that of the uniform texture by an operator of norm at most $2M(s^3 - 1)$. Its vacuum-energy difference has the same bound, independently of exterior volume. These are total texture-energy bounds; they do not assign a constituent rest mass to a localized mode.

Projection onto the two core sites gives $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$. The exterior block has no spectrum in $(-M + 2, M - 2)$, and $\|B\|^2 = b^2$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. There are exactly four low eigenvalues. Their Feshbach equations give

$$|E| \le r := \frac{b^2}{M - 4}, \qquad \frac{\|\psi_{\text{exterior}}\|^2}{\|\psi_{\text{cores}}\|^2} \le \frac{b^2}{(M - 2 - r)^2}.$$

These bounds are uniform in volume. For a cube of radius $L \ge R + 1$, resolvent path expansion and monotone Feshbach eigenvalue crossings give

$$|E_j(L) - E_j(\infty)| \le \frac{2b^2}{M - r - 2} \left(\frac{2}{M - r}\right)^{2(L+1-R)}.$$

Thus the infinite lattice also has precisely four isolated modes in $[-2, 2]$. At $a = 1$, $M = 64$, $L = 3$, complete rational interval certification of all 686 eigenvalues excludes zero from all four low levels and from their infinite-volume limits. The numerical levels are approximately ± 0.003905899328 and ± 0.003906437151. The finite-volume filled-sea energy difference is approximately 127.9700 in the same hopping units.

Let W span this four-dimensional band, $A = W^\dagger h W$, and $C = D_{\text{stagger}} i\sigma_2$. Particle-hole symmetry gives $C\overline{W} = WS$ with $S = W^\dagger C \overline{W}$ antisymmetric and unitary. The actual twelve native current operators intertwine through $W \otimes I_{45}$. The same local pairing $\Delta = C \otimes \overline{K}$ has the exact core

$$\begin{pmatrix} A \otimes I & S \otimes \overline{K} \\ (S \otimes \overline{K})^\dagger & -\overline{A} \otimes I \end{pmatrix}.$$

Its energies are those of A plus or minus the singular values of K. Consequently the compact texture shifts the retained family coefficients; dropping A gives an incorrect operator inclusion. A local Pauli perturbation at the two cores preserves every shell datum while changing these energies. Winding alone therefore does not pin a zero mode for this carrier. The construction supplies a compact background and controlled localized spectrum, without autonomous formation, stability, a relativistic mass shell or physical particle identification.

7.4 Protected texture records and controlled band transport

A declared local repair law protects the compact pair's two winding charges. Let f_x be its unnormalized reference texture and fix the zero cores and constant exterior. At every noncore site retain a rational vector v_x with $\|v_x - f_x\| \le \eta \|f_x\|$, $0 < \eta < 1/4$, using $v_x/\|v_x\|$ in the actual Pauli potential. A one-vertex proposal is accepted only if it satisfies this budget and has nonnegative scalar product with the old vector and every other vertex of every incident shell triangle. Shared-face data are checked on both shells. Rejected proposals remain recorded identity operations.

Proposition 7.1 (Guarded winding and uniform isolated band)**.** *Every admitted finite history preserves the two shell degrees. Its spatial Hamiltonian has exactly four isolated modes with* $|E| \le r = b^2/(M-4)$ *and complementary spectrum outside* $(-g, g)$*,* $g = M-2$*, where* $b^2 = 7/4$ *for adjacent unit shells and* $b^2 = 3/2$ *for separated shells. For two admitted band projectors* Π_v, Π_w*,*

$$\|\Pi_w - \Pi_v\| \le \beta = \frac{4M\eta c}{d^2}, \qquad c = \frac{g+r}{2}, \quad d = \frac{g-r}{2}.$$

The statements are uniform in the exterior volume.

Proof. During an accepted replacement, every interpolated shell vector is a convex combination of nonzero vectors with pairwise nonnegative scalar products. Its squared norm is at least $\sum_i t_i^2 \|v_i\|^2 > 0$, so normalization gives a shell homotopy. Every noncore onsite eigenvalue stays $\pm M$ and the four core coordinates stay zero; the compact-source Schur proof therefore gives the same band count and localization bounds. Normalization gives $\|H_w - H_v\| \le 4M\eta$. On $|z| = c$ both resolvents have norm at most $1/d$. Their identity and the contour formula for the two projectors give the displayed bound. □

For $\beta < 1$, define

$$S = \Pi_w\Pi_v + (1-\Pi_w)(1-\Pi_v), \qquad U = S(S^*S)^{-1/2}, \qquad S^*S = 1 - (\Pi_w - \Pi_v)^2.$$

Since $S\Pi_v = \Pi_w S$ and the positive factor commutes with Π_v, U is unitary and $U\Pi_v U^* = \Pi_w$. The literal particle-hole relation $H_v C = -C\overline{H_v}$ implies $\Pi_v C = C\overline{\Pi_v}$, and the polar formula gives $UC = C\overline{U}$, hence $UCU^T = C$. Thus $U \otimes I_{45}$ transports all twelve uniform native currents and the same pairing tensor $C \otimes \overline{K}$. For $W' = UW$, the pairing matrix $W'^* C \overline{W'}$ is unchanged, while $A_w = W'^* H_w W'$ retains the actual changed spatial energies. Second quantization also transports $N_v = d\Gamma(\Pi_v \otimes I_{45})$ and $\lambda(N_v - 90)^2$.

The explicit source uses $M = 64$, $\eta = 1/512$ and four equiprobable proposals at each record: two opposite transverse changes at the shared-face center, one forbidden zero and one forbidden core change. Its three texture states have transition rows $(1/2, 1/4, 1/4)$, $(0, 3/4, 1/4)$ and $(0, 1/4, 3/4)$. All twelve state/proposal records are retained. Each branch applies its texture transition and $\sqrt{q_r}\,\Gamma(U_r)$, with identity control for a repeated texture, so the complete instrument preserves trace. The band is transported exactly at record endpoints. Omitting U_r instead gives leakage probability at most $\min(1, n^2\beta^2)$ after n quenches and intervening source evolution; the numerical controls exhibit nonzero leakage.

A nearest-neighbor snake through the $N = 2(2L+1)^3$ spatial-spin coordinates compiles each U by adjacent Givens rotations: at most $N(N-1)/2$ two-mode gates and N onsite phases. The complete $N = 686$ controls replay 234,955 two-mode gates per nontrivial texture pair. Addressed quadratic pulses, their timing and drift cancellation are supplied services; this is not a compilation into unchanged C/E repair. Intermediate pulses need not preserve the band, and the controller supplies the endpoint energy change, bounded by $2r$ on the one-body band. Erasing the guards permits zero-degree textures with the same four-mode count. The result establishes controlled protection and transport on this law, without autonomous core motion, local Gauss dressing, physical species identification or relativistic dispersion.

For this same compact pair, the two core sites have equal bipartite parity. Odd Schur hopping terms therefore cancel. At texture amplitude 64 the static four-mode band obeys $|E| \le 1/256 + \eta$, with $\eta = 2365097/60283957504 < 1/25000$. Combining it with the zero-phase charged coefficient $m_{\min} = (15 - 10\sqrt{2})/216$ gives a paired static gap greater than $1/40000$. Its complete positive charged response lies in four separated interval groups of multiplicities $8, 8, 4, 4$; each core probe

on the paired Gaussian reference has combined band weight greater than 99/100. The exact band operator retains $A = W^\dagger h W$. Its paired energies are $E_j \pm \sigma(K)$, not relativistic mass quadrature. Neither the four interval groups nor their multiplicities count physical species. The fixed texture breaks translations, and its two-component spinor space admits no nonzero scalar mass matrix anticommuting with all three kinetic Pauli matrices. Moving defects and a source-selected paired reference require independent constructions (`model/particle_candidate_atlas/DEFECT_FIELD_JOIN.md`).

7.5 Gauge-covariant paired compact bands

The compact source extends to supplied classical unitary internal links on its 343-site radius-three lattice. The gauge paper [7] proves that with onsite texture amplitude 64 and pairing of norm at most one, the complete Bogoliubov–de Gennes operator keeps a separated band whose particle grading defines 180 canonical annihilation modes and an integer occupation N_r. The quartic $\lambda(N_r - 90)^2$ of Section 7.16 is therefore well-defined on every link background and equals the source compact interaction at trivial links. Transforming link and pairing together preserves band and interaction covariantly, and declared polar controllers transport the grading between records.

The same paper gives the separate native quantum-link design: a 60-state link on the five matter blocks with exact local non-Abelian Gauss constraints on the full Fock space, and a conditional rotor completion that leaves five link-energy differences unselected. A literal native two-site raising-link example has a noninteger naive occupation even on a Gauss singlet, so the canonical band above does not transfer to operator-valued links by substitution. A dynamical quantum-link realization of the interacting compact band is not established here.

7.6 Native record response

The fixed biased binary source also supplies an observable response without an attached field action. Give ports $0, 1$ weight 25 and the other ten weight 8, select each of the thirty seams uniformly, and tag current and contrast repairs with equal probability. Let P be its column-stochastic occupation transition and $\pi(x) \propto \prod_a \theta_a^{n_a(x)}$ its stationary law. On every complete record define $Y_e = 60\,\mathbf{1}[\text{contrast law, seam } e](n_a - n_b)$ for $e = (a, b)$, with zero score on reserved addresses. This uses the reported incoming contrast. If B_e is the transition weighted by that score, then

$$\mathbf{1}^T B_e = d_e^T, \qquad B_e \pi = \operatorname{diag}(\pi) g_e, \qquad g_e = \frac{\theta_a - \theta_b}{\theta_a + \theta_b} \mathbf{1}[n_a + n_b = 1].$$

The second identity includes the measuring repair's backaction. With rows d_e^T and g_e^T collected in D, G, and $m = D\pi$, the ordered stationary covariance and its discrete-attempt resolvent are

$$C(k) = D P^{k-1} A \quad (k \geq 1), \qquad A = \operatorname{diag}(\pi) G^T - \pi m^T,$$
$$\sum_{k \geq 1} z^{-k} C(k) = D(zI - P)^{-1} A.$$

All 22 equal-weight seams have zero columns in $C(k)$, despite nonzero same-time shot noise. A passive occupation correlator omits this backaction. Exact ranks 66 for $[\mathbf{1}^T; D; \ldots; DP^{12}]$ and 65 for $[A, PA, \ldots, P^{10}A]$, together with the squarefree characteristic polynomial, show that all 65 nonstationary population relaxation poles appear. They are positive real decay factors, without a particle-mass interpretation. The biased occupation means require pair correlations: adjoining one-step images raises their span from 12 to 19.

The calculation retains all 721 interpreter slots and arbitrary untouched references. Complete record probabilities observe only diagonal source blocks, so the rank statement is population observability, not quantum-state tomography. After 1600 unread preparation attempts, the reduced source/reference half trace distance is at most $\varepsilon = 542159/10^{12}$; any fixed future r-fold score moment has error at most $2\,60^r\varepsilon$. The preparation archive can remain correlated. Exact population observability alone supplies no sampling or stable pole-inversion bound (`model/native_record_response`, [1]).

Positive transfer from complete records. Write $\Pi = \mathrm{diag}(\pi)$. Detailed balance gives the symmetric transfer $S = \Pi^{-1/2} P \Pi^{1/2}$, and the actual minimum diagonal $\min_x P_{xx} = 791/990$ gives

$$\frac{296}{495} I \le S \le I, \qquad H_{\mathrm{tr}} = -\log S \ge 0.$$

For each raw flag the native reversal satisfies $T_{\bar{h}} = \Pi T_h^T \Pi^{-1}$. It reverses word order and exchanges incoming and outgoing contrast using the retained current. If F is any scalar future-record cylinder and f_F its conditional expectation on the initial configuration, its reflected form across n unread attempts is $\langle f_F, (P^T)^n f_F \rangle_\pi \ge 0$. The preceding observability rank makes the reflection quotient exactly 66-dimensional. Centered contrasts therefore have the positive reflected spectral representation

$$D P^n \Pi D^T - m m^T = \sum_{j=1}^{65} \lambda_j^n v_j v_j^\dagger, \qquad v_j \ne 0.$$

The ordinary oriented covariance $C(1)$ is different: the centered score $Y_{(0,5)} - Y_{(0,1)}$ has the naive unreflected quadratic form $-83481968/1858725825$. Reflection must use the actual record dictionary.

Complete single-record conditional probabilities span dimension 42. Two records suffice: the constant, eleven contrast functions and fifty-four ordered contrast products give an invertible feature matrix X. With $G = X^T \Pi X$ and $B_h = X^T \Pi T_h^T X$, correctly reflected moments of at most five attempts determine

$$A_h = G^{-1} B_h = X^{-1} T_h^T X, \qquad G \succeq \frac{55756800}{2129251819009} I.$$

The first feature is constant, so every finite chronological word has law $e_0^T G A_{h_1} \cdots A_{h_m} e_0$. All 721 raw labels, including zero slots, enter this reconstruction. The finite realization and the depth-two rank certificate supply its domain; a dimension bound or known source specification is required to infer arbitrary histories from bounded-depth data. The Gram bound controls inverse perturbations. Uniform native mixing followed by each probe gives conditional score-bias bounds without independent resets, provided the preparation archive is inaccessible to later controls. Martingale concentration then gives finite-error intervals; the conservative sample budget exceeds the interpreter's admitted horizon and is unexecuted. This finite population result does not determine quantum coherences or a physical clock (`model/native_record_response/SHORT_RECORD_RECONSTRUCTION.md`, [1]).

This is a finite Euclidean reconstruction from scalar records. Spectral calculus defines a unitary continuation but supplies no native protocol executing it. Its pure ground direction $\sqrt{\pi}$ is not the native faithful working state, and H_{tr} differs from that state's modular generator $K = -\log \Pi$. For $x = \{0,1\}$ and $y = \{1,5\}$ the exact element $[S,K]_{yx} = \sqrt{2}\log(8/25)/99$ is nonzero; hence $[H_{\mathrm{tr}}, K] \ne 0$. The two clocks cannot be scalar rescalings and shifts of one another. Physical time, spatial locality, operator-valued history positivity and particle masses require additional maps (`model/native_record_response/TRANSFER.md`, [1]).

Operational clock comparison. The distinction has a finite operational witness. For the accessible Y probe on seam $(1,7)$, the working state Π has zero mean under every modular-clock time. The proposed identity identification with $e^{-itH_{\mathrm{tr}}}$ instead gives

$$f(t) = \mathrm{Tr}(Y e^{-itH_{\mathrm{tr}}} \Pi e^{itH_{\mathrm{tr}}}), \qquad f'(0) > \frac{7}{1000}, \qquad f(1/10) > \frac{1}{2000}.$$

A rational degree-ten logarithm enclosure proves the derivative bound. Reality makes f odd, and $\|H_{\mathrm{tr}}\| < 103/200$ controls the cubic remainder. The native preparation and complete analyzer bounds leave an output half-trace separation greater than $1/5000$ if both proposed clock services have half-diamond error at most 10^{-5}. This rejects their identification on the same working reference, including reference-preserving unitary changes of basis. It neither executes the transfer unitary nor excludes a different state/readout map. Bare repair records, with the final carrier traced, are insensitive to any interleaved diagonal phases; the declared coherent analyzer supplies the phase-sensitive readout (`model/native_clock_comparison`, [1]).

Complete records for disjoint repairs. For two fixed native commands with disjoint endpoints, every pair of actual raw Kraus operators commutes on the full 66- or 78-state carrier. Both executions must finish. The comparison permutes command-instance slots with their exported and retained records, and relabels the known program identifiers; their literal digests differ. The resulting complete classical-quantum channels are identical for every initial state and untouched reference. All 25^2 raw-slot pairs, including holds and zero addresses, are retained. Adjacent swaps extend the identity to fixed schedules related by such disjoint interchanges. The literal chronological tapes need not agree, and the shared queue's implementation dependencies are unchanged. Overlapping endpoints, selectors using earlier records and memory timeouts give explicit failures. This is a finite equivalence of observation protocols; it supplies no physical event quotient, spatial dimension or light cone (`model/native_record_causality`, [1]).

Canonical source restriction. The fair connected federation has $V_\ell = 240\,4^\ell$ ports and a declared global occupation m. Its canonical state $\rho_{V,m} = \Pi_m / \binom{V}{m}$, where Π_m projects onto total occupation m, is preserved by each fixed fair native seam's unread source channel. Tracing to any v ports gives the full number mixture

$$\rho_W = \sum_{k=0}^{v} h(k)\rho_{v,k}, \qquad h(k) = \frac{\binom{v}{k}\binom{V-v}{m-k}}{\binom{V}{m}},$$

with impossible binomial coefficients zero. These restrictions compose exactly for nested regions. Orthogonal number sectors give the half trace distance $D(\rho_W, \rho_{v,q}) = 1 - h(q)$; imposing a sharp local number is a different state. For $p = m/V$ and $\beta_p = \mathrm{diag}(1-p, p)$, sampling with and without replacement gives

$$D(\rho_W, \beta_p^{\otimes v}) \le \min\{1, v(v-1)/(2V)\}, \qquad D(\rho_W, |0\rangle\langle 0|^{\otimes v}) \le \min\{1, vm/V\}.$$

A common complete local instrument contracts these bounds, including its records; a correlated external reference needs a joint input bound. The number law requires $O(v)$ rational operations. All thirteen outcomes of a twelve-port restriction at $\ell = 200$ are evaluated exactly with the 408-bit population integer, without allocating the global state. These are static state comparisons. They select no physical capacity map, occupation trajectory $m(V)$, finite preparation, clock or dynamical restriction. The fair federation is distinct from the biased local member used above (`model/native_capacity_restriction`, [1]).

The existing independent-cell grammar supplies a distinct dynamic extension. Append unused 66- or 78-state factors to any admitted native repair/hold program, preserving its parameters, active cell addresses and record controller. Every raw branch is $K_h \otimes I_B$. Consequently partial trace over B intertwines the complete adaptive instrument for arbitrary correlated inputs and untouched references. The comparison retains chronological records, FIFO exports and matched stopping outcomes, and checks the exact lifted program identifier before relabeling it. These restrictions also compose. Declared allocation and specification queries can distinguish the programs. The host evaluator stops before executing products above 64 cells, so equality of executed histories requires a common admitted and evaluated domain. Compiled coherent source-word services have a separate grammar and are not lifted by this theorem. This proves a spectator-factor extension without connected geometry or interactions. A direct single-attempt comparison with the fair federation fails: the same coin record has probabilities 25/33 and 1/2, giving total variation at least 17/66 (`model/native_capacity_restriction/PRODUCT.md`, [1]).

Connected weighted references. The same endpoint rule has a consistent weighted extension on the retained connected federation graph. For positive vertex weights w_i, write $e_m(w)$ for the elementary symmetric polynomial and set $\pi_m(S) = \prod_{i\in S} w_i/e_m(w)$ on $|S| = m$. Every seam preserves this reference by conditional heat-bath detailed balance. Its full restriction to A has number weights

$$h_A(k) = \frac{e_k(w_A)e_{m-k}(w_{A^c})}{e_m(w)}.$$

These restrictions compose. Repeating the local weight word gives the H66 reference conditional on two loads in a cell; cross-cell repair changes that local number. The faithful product family with occupations $p_i = zw_i/(1+zw_i)$ also restricts exactly for fixed fugacity z and inherited weights. The modular transfer $j\log(w_a/w_b)$ is independent of total occupation and graph size. It is not a particle mass.

For a fixed independent seam selector, replacing crossing repairs by a complete hold control changes the t-attempt channel by at most $1-(1-\beta)^t$ in half diamond norm, where β is its boundary-selection probability. Each cell has thirty internal and three crossing seams. Under the uniform seam selector, through the kth internal repair, the no-crossing probability is $(10/11)^k$ at every federation level; increasing exterior capacity does not establish vanishing error in that local clock. Symmetrizing the same population law gives local positive terms

$$h_{ab} = \frac{1}{w_a+w_b}\begin{pmatrix} w_b & -\sqrt{w_a w_b} \\ -\sqrt{w_a w_b} & w_a \end{pmatrix}$$

on $|10\rangle, |01\rangle$, zero on $|00\rangle, |11\rangle$. Their weighted sum annihilates $\sqrt{\pi_m}$. This is a relaxation representation. The weight assignment and fugacity are declared, and the native program admission rule rejects weighted connected federations; no source selection, prepared coherent ground or relativistic limit is inferred (`model/weighted_source_family`, [1]).

Local weighted state limits. The repeated native weight word admits a controlled local limit without enumerating its global carrier. Let B be the number of twelve-port cells, declare total occupation rB with $r \in \{2,4\}$, and choose the positive fugacity solving

$$1000z^2 + 32z - 1 = 0 \quad (r=2), \qquad 800z^2 - z - 2 = 0 \quad (r=4).$$

The product reference $Q_B = Q_1^{\otimes B}$ then has mean rB, and its conditioning on that total is exactly the canonical reference P_B. A binomial-mode bound gives $Q_B(S = rB) \geq 1/(240B)$. Cell

exchangeability and the finite entropy chain rule distribute this conditioning cost: for every set of k cells, including k that do not divide B,

$$D(P_k\|Q_1^{\otimes k}) \le \frac{k\log(240B)}{B}, \qquad \tfrac{1}{2}\|P_k - Q_1^{\otimes k}\|_1 \le \min\left\{1, \sqrt{\frac{k\log(240B)}{2B}}\right\}.$$

Here D is classical relative entropy, the second inequality is finite Pinsker, and every local number sector is retained. A common trace-preserving local instrument contracts the bound; a numerical map with bounded gain requires that gain in the comparison. The instrument must act on every retained local number sector: a two-load coherent service does not automatically extend to this mixture. Correlated references require a joint input-state bound. For $B = 20\,4^{200}$ the one-cell bound is below $2\,10^{-60}$, computed with integer and rational enclosures. Matching mean occupation two does not confine a cell to its two-load sector. The theorem selects neither density nor physical capacity and supplies no preparation time or connected dynamical limit (`model/weighted_local_limit`, [1]).

Local records and finite-volume response. For a fixed repair schedule, a local observation consists of a final port region and selected complete local records at their original event positions. An untouched reference is retained; remote record values and the final exterior are traced. Reverse scanning the schedule adds the endpoints of each retained read and each unread command touching the accumulated support. Every disjoint unread complete channel cancels under partial trace. Thus the complete observed instrument factors exactly through the initial state on this dependency cone and its reference, including initial coherences and correlations. State-independent random selectors can be averaged afterward. Record-adaptive selectors and intrinsic queue halts require separate analysis; the full global archive is a different observable.

Compare with a finite port region whose crossing and exterior commands act as holds, retaining the original selector and clock. Coupling the selector tapes bounds half the diamond distance of the two local output channels by the probability that the dependency cone exits the region. An initial joint state error adds to this bound. For independent seam choices, root set S, maximum degree d, maximum seam probability q_* and graph-distance margin r, a union over ordered paths gives the sufficient bound

$$p_{\mathrm{exit}} \le \min\{1, |S|\tbinom{T}{r+1}(dq_*)^{r+1}\}.$$

The roots include both endpoints of every retained state-dependent read. This conservative cone can exceed the support obtained from the precise record times.

For independent uniform seam selection and repair probability one, on the actual fair connected source at levels zero through two, each cell has thirty internal seams and three distinct exterior neighbors, each with five seams leaving the one-neighbor halo. For a whole-cell root with retained internal-cell records, this reduces cone exit to four transient states $n = 0, 1, 2, 3$, with n counting the halo ports reached. With $M = 630\,4^{\ell}$ seams, the only nonzero entries are

$$Q_{nn} = 1 - \frac{3+4n}{M}, \qquad Q_{n+1,n} = \frac{3-n}{M}, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top}Q^{T}e_0 \le \min\{1, 15\tbinom{T}{2}/M^2\}.$$

Here Q^T denotes the Tth power. At level zero and $T = 30$ this probability is below 0.015. The combinatorial attachment below proves the halo geometry for every returned source routing. Stopping immediately after the kth internal repair instead gives

$$R = 30(D - B_0)^{-1}, \qquad D_{nn} = 33 + 4n, \quad (B_0)_{n+1,n} = 3 - n, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top}R^{k}e_0.$$

The elapsed global time is retained without conditioning. The exit probability is independent of M, equals $677/50061$ at $k = 1$, and is at most $\min\{1, k(k+1)/120\}$. For bounded scores $|f| \le 1$ and an analysis parameter $0 \le q < 1$, the normalized Abel response error is at most

$$2\left[1 - (1-q)\mathbf{1}^{\top}(I - qR)^{-1}e_0\right] \le \min\{2, q/[30(1-q)^2]\}.$$

At $q = 1/2$ the sharper value is $151/3432$. These are complete local-record response bounds, with no pole-energy interpretation. The weighted connected extension has the same conditional locality construction but no native admission; its reference is not assigned to the fair source. The proof and bounded interpreter checks supply neither a physical clock nor a relativistic limit (`model/native_local_response`, [1]).

Occupation-sector readout. On any connected binary source with V ports, the conserved total m and complete records of $V - 1$ fixed-seam repairs along a spanning tree determine the initial configuration. If s_i is the accumulated recorded displacement, the incoming contrast on (a, b) gives the initial difference $d - s_a + s_b$. Propagate these differences to offsets o_i; the common offset is $(m - \sum_i o_i)/V$. Replaying the currents also determines the terminal configuration. Every nonzero complete word therefore has the map $\sqrt{w_h}|y_h\rangle\langle x_h|$, including arbitrary untouched references, and coarse graining by x_h gives the configuration projectors. Without a supplied total, the empty and full configurations share one rank-two branch that preserves their coherence. All other supported words decode a unique configuration. This ambiguity cannot be removed by record-adaptive repair selectors.

The actual connected members at levels zero through two admit trees with 239, 959 and 3839 commands. An exact lazy query of a complete raw word uses the original continuation kernel and retains its archive; it does not turn the expanded evaluator's resource stop into a completed execution. A final tree read permits reverse-current reconstruction of a preceding complete repair history. Its terminal basis state is outcome-labelled. For any controller of at most T repairs, positive placement probabilities retain a no-motion event from every basis input with probability at least $(1 - r + rp_{\min})^T$, where r is the repair probability. On a nontrivial fixed-number sector, this obstructs exact deterministic reset of every input to one target by bounded repair-only control. It permits approximate or heralded preparation and the separately declared coherent services (`model/native_sector_readout`, [1]).

Native correlation sectors. For the fair connected source with M seams, the population transfer is $P_q = I - \sum_{ab}(I - S_{ab})/(2M)$ on the q-occupation sector. Let U add a site to a subset, $D = U^*$ and $W_r = \ker D_r$. The standard harmonic decomposition of the Boolean slice [32] follows from $DU - UD = (V - 2r)I$ on degree r. Its source-preserved degree-r subspace has dimension $\binom{V}{r} - \binom{V}{r-1}$, for $r \le \min(q, V-q)$, and is unitarily equivalent to $P_r|_{W_r}$. This is a decomposition of the unread transfer, with no assertion that each recorded quantum branch preserves these subspaces.

For two disjoint native seams $(a, b), (c, d)$, put $h = e_{ac} - e_{ad} - e_{bc} + e_{bd}$ and $\phi = (n_a - n_b)(n_c - n_d)$. Then $D_2 h = 0$, and in the uniform fixed-number reference

$$v_q = \mathbb{E}\phi^2 = 4\frac{\binom{V-4}{q-2}}{\binom{V}{q}}, \qquad \frac{\mathbb{E}[\phi(X_0)\phi(X_t)]}{v_q} = \frac{h^* P_2^t h}{4}, \qquad 2 \le q \le V - 2.$$

An actual four-read experiment realizes this curve: read both seams, score their outgoing contrasts $(d_1 - 2j_1)(d_2 - 2j_2)$, run t uniform attempts, then read both seams and score the incoming contrasts $d_3 d_4$. The complete fair instruments are unital; the first scored output is $\phi\tau_q$ and the last scored

effect is ϕ. Thus their joint score equals the displayed covariance with measurement backaction included. Every raw flag is retained. Using incoming contrasts at both ends changes the experiment and fails this identity. A preparation error ϵ in half trace distance gives normalized error at most $2\epsilon/v_q$; v_q decreases as V^{-2} at fixed q. The normalized poles are independent of occupation on a fixed graph, not of graph size or clock. Different correlation degrees may share poles. Neither degree nor pole is assigned a particle identity (`model/native_correlation_sectors`, [1]).

Preparing native correlation records. The stationary response can be prepared at the level of its complete signed score from an arbitrary quantum input in the same fixed-number sector. For possibly different past and future seam contrasts ϕ_i, ϕ_j, let $H_e = (I+S_e)/2$ on occupation functions. Pulling the entire four-read experiment back to its initial cut gives the diagonal effect

$$A_{ij,t} = H_{e_i} H_{f_i}[\phi_i P_q^t \phi_j], \qquad \|A_{ij,t} - \mathbb{E}_{\tau_q} A_{ij,t}\|_2 \le \sqrt{v_q}.$$

The two past seam averages include measurement backaction. This effect has squarefree degree at most four. Put $s = \min(4, q, V-q)$ and $N_s = \binom{V}{s} - 1$. The mean-zero polynomial space is invariant under all site permutations, even when the graph is not vertex-transitive. Its orthogonal projector therefore has constant diagonal N_s in the uniform L^2 normalization. Cauchy–Schwarz and the exclusion gap $\gamma = \lambda_1(L)/(2M)$ give, after n independent fair warmup repairs,

$$\frac{\|Z_R - C_{ij}(t)\rho_R\|_1}{v_q} \le \sqrt{N_s/v_q}\,(1-\gamma)^n.$$

Here Z_R is the terminal reference operator weighted by the complete score, and R is any untouched reference initially correlated with the source. The diagonal input blocks on R are positive and have total trace one, which proves this extension. Warmup records are summed without conditioning; their full joint law is not asserted to be stationary.

Integer square roots and the bound $(1-\gamma_0)^{\lceil 1/\gamma_0 \rceil} \le 1/2$ give sufficient attempt budgets from a rational gap lower bound. On the fair 66-state source, 352 warmup repairs followed by four reads and two lag attempts give normalized error at most 33/65536; exact propagation of all input columns gives a tighter rational enclosure. At fixed density the prefactor grows polynomially rather than with the full sector dimension. The gap and sample cost retain their capacity dependence. All raw records and host stops are retained; assigning score zero to stopped runs adds p_{stop}/v_q to the bound. Single-program budgets retain the original 2^{32}-attempt limit. The same argument applies to any separately implemented degree-four scored effect using its exact centered norm. It supplies neither full-density preparation nor a physical vacuum (`model/native_correlation_preparation`, [1]).

Native spatial motion and clocks. Midpoint subdivision preserves a simple closed triangular complex; each cell has three distinct neighbors. The source assigns their crossing edges to distinct ports, so each of the three halo ports has five seams leaving the halo and none joining another halo port. This proves the four-state halo premise at every combinatorial level for every returned routing. It does not assert successful floating-point construction or host execution at arbitrary levels.

In the existing auxiliary affine cell-centroid dictionary, normalized to unit circumsphere, all ports of a cell share one coordinate. Writing $n = 2^\ell$, there are $30(n^2-n)$ within-face and $30n$ original-face hinge crossings. Their exact total squared length is

$$E_\ell = 20 - 4\sqrt{5} + \frac{-40 + 16\sqrt{5}}{3\,2^\ell} < 12.$$

For a stationary uniform single-unit source, $P = I - L/(2M)$ with $M = 630\,4^\ell$ seams and $V = 240\,4^\ell$ ports. Positivity gives $I - P^T \le T(I - P)$ and therefore

$$\mathbb{E}\|x(X_T) - x(X_0)\|^2 \le \frac{TE_\ell}{MV} \le \frac{T}{12600\,16^\ell}.$$

Markov's inequality gives a displacement-probability bound by dividing by the squared threshold. At $T = \lfloor 1260\,8^\ell t \rfloor$, this mean square is at most $t/(10\,2^\ell)$. It is a finite stationary path bound in comparison coordinates, with no physical light-speed assignment.

For a fixed set of d tick seams, a completed tick consists of a geometric number of remote repairs followed by one incident repair. Both conditional transfers preserve the uniform state. The squared triangle inequality and stationarity bound displacement after k completed ticks by $k^2/(10d\,4^\ell)$. Substituting the mean elapsed time into the deterministic-time bound would not justify this result. A finite attempt cap retains timeout as a separate outcome. Along a shortest path of D edges to a receiver, one positive native history of weight $(2M)^{-D}$ arrives at its first incident tick; that tick has no strict port-distance support bound uniform in capacity. Global attempt locality is unchanged. A final tree detector and the body currents reconstruct the recorded path; the prepared law, auxiliary coordinate dictionary and complete archive access are explicit inputs (`model/native_spatial_scaling`, [1]).

Spatial pair responses. Use one declared $2 \le q \le V - 2$ and its reference τ_q, the same fair graph and the same attempt clock for a menu of disjoint-seam contrast pairs. The complete reflected probes give

$$C_{ij}(T) = \frac{v_q}{4} h_i^* P_2^T h_j, \qquad B_{ij}(T) = C_{ij}(T)/v_q.$$

The matrix $B(T)$ is positive semidefinite; $B(0)$ is the probe Gram matrix, not necessarily the identity. The two-subset coefficients implement the exact intertwiner without changing the working occupation to two. For a radius-r local comparison containing every probe endpoint, maximum degree Δ gives normalized response error at most $2\min\{1, |S|\binom{T}{r+1}(\Delta/M)^{r+1}\}/v_q$. The compared window retains all local number sectors, and remote raw values are traced at this observation boundary. A global archive-dependent controller is outside that locality bound.

The source also fixes a finite exclusion correction. Compressing the independent two-walker generator to distinct symmetric pairs gives $G_{\mathrm{ind}} = G_2 + 2A$, where A projects onto adjacent pairs. Its collision-retaining control has surviving transfer $R = P_2 - A/M$ and an absorbing collision flag. The identity

$$P_2^T - R^T = \sum_{k=0}^{T-1} P_2^{T-1-k}(A/M)R^k$$

is entrywise nonnegative. Its column sum is the complete collision probability, bounding the normalized probe-response difference under the initial coefficient law $|h_j|/4$. No survivor renormalization occurs. On a four-port path, the one-attempt discrepancy is $1/12$; matching the one-body transfer therefore does not determine the pair response.

For bounded macroscopic profiles, subtracting the constant and one-body components of a product of native density fields gives an exact primitive degree-two field on the same q slice. With $V_\ell = 240\,4^\ell$, $q_\ell = V_\ell/3$ and $\lfloor 1260\,16^\ell \tau \rfloor$ attempts, along the existing native density subsequence, these fields converge to $W_\tau(f,g) = {:}Y_\tau(f)Y_\tau(g){:}$, with

$$\mathrm{Cov}(W_\tau(f,g), W_u(h,k)) = K_{\tau,u}(f,h)K_{\tau,u}(g,k) + K_{\tau,u}(f,k)K_{\tau,u}(g,h).$$

Here Y is the inherited Gaussian density field and K its heat covariance. Canonical exponential-moment bounds give uniform eighth moments and justify the product and covariance limits. A final native spanning-tree detector and reverse-current readback access these occupation snapshots; this global observation is distinct from the local four-read menu. The result is the quadratic sector of a Gaussian diffusive field. The finite exclusion correction does not by itself establish an interacting relativistic continuum (`model/native_pair_response`, [1]).

Native attempt time and the local exchange action. The same source fixes $G_q = \sum_e (I - S_e)$ and $P_q = I - G_q/(2M)$. Its bound $10I/21 \le P_q \le I$ gives, by scalar spectral calculus,

$$\left\| P_q^n - e^{-nG_q/(2M)} \right\| \le \frac{3}{5n}, \qquad n \ge 1.$$

Indeed, for $p \ge 10/21$ and $\delta = 1 - p$, $0 \le -\log p - \delta \le 21\delta^2/20$; maximizing $n\delta^2 e^{-n\delta}$ proves the bound. It holds on every occupation sector, independent of its dimension, and bounds each entry of the normalized stationary response matrix. The rescaled time is the exact count $t_n = n/(2M)$, without a replacement Poisson scheduler or a physical time unit.

Generator convergence has a narrower domain. On correlation degrees at most r, $\|G_q\| \le 2r\Delta$ and, for $M > r\Delta$,

$$0 \le -2M \log P_q - G_q \le \frac{r^2\Delta^2}{M - r\Delta} I.$$

At half occupation the full-sector generator difference is at least $M/16$, although its endpoint semigroups obey the dimension-independent bound above. Exact sparse moments give rational enclosures of both response functions. Preparation and local-window errors add with their stated normalizations. The Poisson-count representation of this exponential has a different complete history-length law; the endpoint estimate identifies neither that law nor a native unitary implementation. These results fix a finite local exchange response, with no chiral matter, Lorentzian action or physical mass assignment (`model/native_exchange_clock`, [1]).

7.7 An observable native contact mode

The fair twelve-port action at working reference $I_{66}/66$ has an observable correlated mode. On unordered distinct pairs write $G_2 = \sum_e (I - S_e)$ and let $W_2 = \ker D$, where D sums pair coefficients at each port. Its dimension is 54. If Q projects onto W_2 and A onto adjacent pairs, compression of the independent one-body sum gives $G_2 + 2A$. On W_2 this yields

$$H = G_2|_{W_2}, \qquad H_0 = H + 2QAQ \ge 2(5 - \sqrt{5})I.$$

The coefficient two follows from the literal collision loss; it is not fitted. For the adjacent-coordinate insertion E_A, put $U = QE_A$ and $R_0(z) = (H_0 - z)^{-1}$. Direct multiplication proves

$$(H - z)^{-1} = R_0 + 2R_0U(I - 2U^*R_0U)^{-1}U^*R_0.$$

Below the stated independent-pair threshold, the middle kernel detects eigenvalues with multiplicity. This is a finite comparison threshold, without an infinite-volume scattering interpretation.

Exact integer characteristic polynomials give a fivefold native eigenvalue E at the lowest root of

$$E^3 - 30E^2 + 280E - 776 = 0, \qquad 5 < E < 51/10.$$

It lies more than $2/5$ below the comparison threshold. The actual seam contrast $(n_0 - n_1)(n_2 - n_4)$ has a twelve-dimensional cyclic primitive space. Its exact scalar resolvent has coprime numerator

and denominator, so all twelve distinct primitive poles have positive weight. At this lowest line the normalized weight obeys $1/100 < w_E < 1/50$; decimals are $E \simeq 5.01827276928$ and $w_E \simeq 0.01943798293$. Complete reflected four-E records have raw weight $(2/33)w_E$ and attempt transfer factor $1 - E/60$. All raw branches and measurement disturbance are retained.

The isolated-cell mode does not extend unchanged to the connected source. For any normalized vector in the fifteen-dimensional sum of the cubic eigenspaces, every three-port boundary attachment satisfies

$$\|(I - P_{\mathrm{cell}})G_{\mathrm{federation}}J_{\mathrm{cell}}f\|^2 > 1/100.$$

Exact integer principal minors verify this for all 220 possible boundary triples. Thus its unchanged embedding has residual greater than $1/10$, independent of capacity. A dressed candidate needs a separate residual, spectral isolation, localization or propagation, and accessible-residue bound. At fixed total occupation two the raw probe variance is $8/[V(V-1)]$, so normalized spectral weight alone does not ensure feasible sampling. These are native correlation poles, with no physical particle or mass assignment (`model/native_contact_response`, [1]).

7.8 Connected contact response

The connected comparison has a different threshold from the isolated cell. Let $V = 240\,4^{\ell}$, $M = 21V/8$, and let Q_2 project pair coefficients onto the primitive degree-two space. The actual comparison is $H_0 = Q_2(G_2 + 2A_{\mathrm{adj}})Q_2$ on that space, where A_{adj} marks adjacent occupied ports. Give every port in a cell the same auxiliary affine triangle-centroid coordinate f. Exact icosahedral symmetries give $\sum_i f_i = 0$, $|f_i| \le 1$, $V/5 < A = \sum_i f_i^2 \le V/3$, and $\sum_{ij\in E}(f_i - f_j)^2 < 4$. These are comparison coordinates. The primitive trial vector is

$$w_{ij} = f_i f_j - c_i - c_j, \qquad c_i = \frac{-f_i^2 + A/[2(V-1)]}{V-2}.$$

Its squared norm is at least $V^2/75$, its exchange energy at most $4V/3$, and its contact energy less than $6V$. Rayleigh's principle therefore gives

$$0 < \lambda_{\min}(H_0) \le \frac{550}{V} = \frac{55}{24\,4^{\ell}}.$$

The bound uses the exact cell adjacency and the source routing postconditions; it does not assert that the floating constructor returns at every level. The isolated pole $E > 5$ lies above this connected comparison bottom. Its isolated threshold is therefore insufficient to certify global binding. Bound bands at fixed momentum and resonances require separate tests.

The source also supplies a controlled exterior correction. Retain the 66 pair configurations wholly inside the chosen cell and write $G_2 = \left(\begin{smallmatrix} A & B^* \\ B & D \end{smallmatrix}\right)$. The literal boundary gives $A = G_{\mathrm{cell}} + N_\partial$, $B^*B = N_\partial \le 2I$ and $0 \le D \le 24I$. For $s > 0$ set

$$F(s) = A + sI - B^*(D + sI)^{-1}B, \quad F_J(s) = A + sI - \sum_{k=0}^{J} \frac{B^*(24I - D)^k B}{(24+s)^{k+1}}.$$

The core resolvent is $F(s)^{-1}$. Positive inverse ordering and the geometric remainder give

$$0 \le F^{-1} - F_J^{-1}, \qquad \|F^{-1} - F_J^{-1}\| \le \frac{2}{s^3}\left(\frac{24}{24+s}\right)^{J+1}.$$

Only finite exterior paths enter each coefficient. No projection back to the cell's primitive space is inserted during an excursion. For the normalized four-E probe $u = h/2$, $s = 24$ and $J = 2$,

the connected response $s\langle u,(G_2+sI)^{-1}u\rangle$ is enclosed between 0.660081931825 and 0.660515959604 on the evaluated cell-zero windows at levels zero through two. The isolated value exceeds the connected enclosure by more than 1/120.

For the canonical total-two reference, this response is accessible through complete native lag experiments: with $q_s = 2M/(2M+s)$,

$$s\langle u,(G_2+sI)^{-1}u\rangle = (1-q_s)\sum_{n\geq 0} q_s^n C(n)/v_V, \qquad v_V = \frac{8}{V(V-1)}.$$

Truncation at N costs at most q_s^{N+1}. Preparation errors and independent sample costs have explicit allowances; host stops remain distinct outputs. The discount is an analysis parameter. It adds no native random scheduler or physical clock. The negative-spectral-parameter resolvent bound does not continue a pole through the exterior spectrum at positive energy (`model/native_connected_contact`, [1]).

7.9 Positive-energy correlations from open records

On the finite configuration transfer space, the constant unit vector Ω is the unique ground of $H_M = -2M\log P_q$. It differs from the native working density $I/\binom{V}{q}$. For normalized primitive pair fields, the complete reflected record statistics determine, using the isometric primitive two-occupation representative of H_M,

$$W_{ij}(t) = \langle h_i, e^{-itH_M} h_j\rangle/4.$$

This matrix has a positive spectral measure on nonnegative energies. It is a reconstructed ground correlation, without a claimed unitary implementation on the working carrier.

The continuation has a controlled capacity dependence. On primitive degrees at most r, choose $b = \lfloor M/(2r\Delta)\rfloor \geq 1$, $Q = P^b \geq I/2$ and $u = 2Mt/b$. With $c_0 = 1$, $c_{k+1} = c_k(k-iu)/(k+1)$ and $N = \max(1, \lceil |u| \rceil)$,

$$V_m(Q) = \sum_{k=0}^{m} c_k (I-Q)^k = \sum_{j=0}^{m} a_j Q^j, \quad \|V_m(Q) - Q^{iu}\| \leq 2^N - \sum_{k=0}^{m} \binom{N+k-1}{k} 2^{-k}.$$

Substituting the actual lag-bj four-E statistics gives an explicit signed complex reconstruction. Its systematic error adds $A_m\eta$, where $A_m = \sum_j (|\Re a_j| + |\Im a_j|)$ and η bounds each normalized prepared score. For $m+1$ settings, variance-bounded independent repetitions give sampling error at most s with failure probability p when each setting has at least $(m+1)A_m^2/(pv_q^2 s^2)$ shots. This bound can be impractical; preparation, sampling and host-stop costs are not removed by analytic reconstruction. The displayed $t = 1/64$, order-12 example requires 116424316 attempts at level zero under its sufficient warmup bound. The corresponding levels one and two exceed the unchanged 2^{32} attempt ceiling, and the conservative 99% sampling certificate requires more than 10^{17} total repetitions. These are sufficient budgets, without an optimality claim.

The reference distinction has an operational witness. Bare native record effects are diagonal, so an ideal diagonal phase on the working carrier changes no complete record law, even with arbitrary initial reference correlations. In the transfer ground the same field instead has susceptibility $-2\Im W_{ii}(t)$, with a strictly positive finite-kick margin certified by native moments and a Taylor remainder. Identifying these interventions and readouts without an additional map would assign two different responses to one experiment. A physical dictionary must preserve the intervention statistics as well as the reconstructed correlations (`model/native_vacuum_response`, [1]).

Complete finite energy windows. The same blocked transfer defines positive effects $B_k = \binom{n}{k} Q^k (I-Q)^{n-k}$, $0 \le k \le n$, whose sum is the identity. Their probabilities are exact linear combinations of the complete native lag scores at $0, b, \dots, nb$. Assign $k > 0$ the counted-energy label $-(2M/b)\log(k/n)$ and retain $k = 0$ as overflow. At a transfer eigenvalue $x \ge 1/2$, the output is binomial, so

$$\Pr(|k/n - x| \ge \delta) \le \frac{1}{4n\delta^2}, \qquad |E_k - H(x)| \le \frac{2M\delta}{b(1/2-\delta)}$$

outside that exceptional event, for $0 < \delta < 1/2$. The total coefficient amplification of the measured, nonzero-lag moments is $A = 3^n - 1$. Normalized moment error η gives histogram half-ℓ^1 error at most $A\eta/2$. Sampling error s with failure probability p has the sufficient budget $nA^2/(4pv_q^2 s^2)$ independent repetitions per setting. Native local preparation on the fair 66-state source makes the $n = 16$ full-vector preparation error less than 4×10^{-11}, with at most 3871 native attempts per setting. The sufficient one-percent sampling budget exceeds 3.2×10^{25} total repetitions. This is a reconstructed spectral statistic; the effects are not thereby an executed energy instrument. Physical mass, a laboratory energy unit and an autonomous relativistic action require separate identifications (`model/native_energy_identification`, [1]).

A complete controlled energy instrument. The fair 66-state carrier also admits a destructive energy measurement using its declared coherent services. An exact rational eigenframe certificate gives a nearby generator $\widetilde{H}$ with $\|H - \widetilde{H}\| \le 27 \times 10^{-8}$ in the counted convention $H = -60 \log P$, without assuming separated eigenvalues. For each certified normalized ray u, let C be the existing source encoding and p a padding address. The phase word on $z = (Cu + |p\rangle)/\sqrt{2}$ implements $U = I - 2|z\rangle\langle z|$, whose source-success block is $I - |u\rangle\langle u|$ and padding row is $-\langle u|$. Thus a padding flag detects that ray destructively; the surviving complement continues to the next filter. The sixty-six energy effects and final residual sum to the identity by telescoping. This removes the second phase word of the general weak filter, which acts trivially on this surviving complement.

All sixty-six words are separately admitted, totaling 3,853,872 pulses. Exact primitive certificates and the eigenframe residual bound give complete half-diamond error below $1/12000$, including every processor failure, with stored-map gain below $1001/1000$. The ideal instrument is normalized; stored outputs have that bounded gain. This is a sequential composition of declared source services, with no monolithic repair-program admission or generic-host expansion asserted. Every stored success prefix is evaluated with a retained reference control.

The record-prepared noncentral total-two reference described below feeds this same instrument, with exact acceptance $7771/65536$ and explicit preparation rejection. Exact rational matrix products give the entire accepted energy law, including the residual. No successful branch is renormalized in that full output. Repair attempts and service pulses have distinct counters; physical duration and a rest-mass identification are separate maps (`model/native_energy_measurement`, [1]).

The existing source phase experiment supplies a second ideal reference $\rho = P/56 \ge I/84$ on the same configurations. Its exact modular generator is $-\log \rho = H/60 + \log(56) I$, identifying the controlled operator families under a known factor of sixty. The preparation retains its selector timeout and numerical-density admission scope; unread repair is a different evolution. This reference and the diagonal tree reference are operationally distinct: their norm-one occupation-exchange expectations differ by $1/168$, so their half trace distance is at least $1/336$. Their separate regional-modular and energy-clock properties cannot be combined as properties of one state.

A fair ground, local energy and complete readout. For the connected fair binary source put $G = \sum_e (I - S_e)$, $P = I - G/(2M)$ and $H = -2M \log P$. A triangle-cover argument gives

$P \geq \alpha I$, $\alpha = 10/21$, uniformly on the growing federation. The fixed-number uniform vector Ω is the unique ground on each connected occupation sector. On the sixty-six-state source a declared coherent preparation, following eleven native tree attempts, has complete half-diamond error below 1/4,000,000. A local occupation phase creates an excitation with $\langle G \rangle = 50/33$ and $H \geq G$. The matched preparation, control and complete energy detector use sixty-eight words and 3,877,451 service pulses. Scoring all failures as zero gives an unnormalized mean-score contrast greater than 3/50; the excitation is detected without a successful-branch normalization. This is a finite excitation-energy observation, with no stable-particle or rest-mass assignment (`model/native_fair_ground_readout`, [1]).

The same source admits an explicit comparison between its logarithmic clock and local exchange dynamics. Write $\tau = I/D$, $\rho = P/\operatorname{Tr} P$, $c = \operatorname{Tr}(\tau P)$, $\mu = \operatorname{Tr}(\rho G)$ and $a = (1 - \mu/(2M))^{-1}$. For the weighted Hilbert–Schmidt norm,

$$\|[H, X] - a[G, X]\|_{2,\rho} \leq \frac{\|[G, X]\| \sqrt{\operatorname{Var}_\tau(G)/c} + \frac{1}{2}\|[G, [G, X]]\|}{2M\alpha^2}.$$

Exact occupation moments bound the variance by $O(M)$ on the bounded-degree graph. Fixed local observables, times and complete classical intervention histories therefore have error $O(M^{-1/2})$. All failures and zero slots are retained. This is a state-weighted comparison, not an arbitrary-input channel-norm claim. On $A\Omega$, positivity instead gives $\|(e^{-itH} - e^{-itG})A\Omega\| \leq |t|\|[G, [G, A]]\|/(4M\alpha)$, with a corresponding fixed-word bound. A growing ground preparation is an additional premise.

Writing $h_e = I - S_e$ gives antisymmetric local currents $J_{e \to f} = i[h_e, h_f]$, supported on adjacent seams, and exact conservation of G. Native unread repair is a different operation: on the faithful sixty-six-state reference it increases the mean G by 37/840. For its unital channel the mean increase in H is at least as large. The complete energy instrument resolves a score increase greater than 0.034 for the ideal reference, with separately counted preparation error. Source-plus-apparatus conservation requires an apparatus-energy observable. None of these graph currents supplies a symmetric Lorentz stress tensor (`model/native_local_energy_clock`, [1]).

The dilute ground imposes a separate constraint on its observable limit. For a bounded number-preserving operator A on s ports,

$$\operatorname{Var}_{\Omega_{V,q}}(A) \leq 4\|A\|^2 \left(1 - \frac{\binom{V-s}{q}}{\binom{V}{q}}\right) \leq 4\|A\|^2 \frac{sq}{V}.$$

Thus fixed-support bounded connected correlations vanish when $q/V \to 0$, uniformly in controlled time. For a fixed sequence of complete local instruments on supports s_j whose fine Kraus operators each preserve local occupation, the terminal source and archive have half trace distance at most $\min\{1, 2\sum_j \sqrt{s_j q/V}\}$ from the scalar empty-sector protocol. The comparison retains every branch. A clock rescaling cannot restore a vanishing pole residue. Finite density, growing supports, rescaled fields or charged operators require separate limits; this bound does not exclude them (`model/native_dilute_ground_limit`, [1]).

7.10 Controlled transfer state and time

The weighted 66-state source admits a distinct operational construction using its declared coherent services. Its scalar record transfer has

$$S = \Pi^{-1/2} P \Pi^{1/2}, \quad \frac{296}{495} I \leq S \leq I, \quad \Omega_x = \sqrt{\pi_x}, \quad H = -\log S.$$

An eleven-E configuration detector has fine maps $\sqrt{w_h}|y_h\rangle\langle x_h|$. Retained-record routing followed by the native amplitude-tree word with rational weights π_x changes these to $\sqrt{w_h}|\Omega\rangle\langle x_h|$. Thus the ideal prepared source factors from the retained archive and any untouched reference, even for an initially correlated input. The archive can remain correlated with that reference.

Transfer evolution has a finite residual certificate. Rational proposed columns V and diagonal $(296/495)I \le \Lambda \le I$ satisfy $\|V^*V - I\| \le g = 10^{-9}$ and $\|SV - V\Lambda\| \le r = 10^{-9}$, checked by exact rational arithmetic and directed square-root intervals. The polar comparison $W = V(V^*V)^{-1/2}$ is unitary and obeys $\|W - V\| \le g$. Consequently

$$\|S - W\Lambda W^*\| \le r + 2g, \qquad \|\log S - W \log \Lambda W^*\| \le \frac{r + 2g}{296/495}.$$

The logarithm bound follows by integrating the resolvent identity. The actual compiler uses normalized rational columns of V and their existing rank-one phase words; it does not execute the polar map. Telescoping the 66 phases bounds their ideal unitary error by $|t|(r + 2g)/(296/495) + 528g$. Scalar logarithm intervals and each stored primitive's certificate supply the additional numerical allowance.

One complete experiment prepares Ω, phases configuration $a = (0, 1)$ by i, applies the controlled transfer evolution at $t = 1/10$, and decodes the initial configuration of a second eleven-E detector. The unphased control has probability $p = \pi_a = 625/7505$. Writing $u(t) = \langle a|e^{-itH}|a\rangle$, the intervention has

$$p_a(t) = p\,|1 + (i - 1)u(t)|^2.$$

Here $H_{aa} \ge 32/495$ and $\|H\| < 103/200$. A spectral Taylor remainder proves $p_a(1/10) - p > 1/1000$. The complete protocol has 68 separately admitted coherent words, at most 3,877,451 pulses and 22 native E commands. Every first processor failure and padding abort is retained and scored zero. Passing primitive gates bound complete half-diamond error below $1/11000$ per experiment and gain below $1001/1000$. Charging the full trace-norm error for both rounded, potentially unequal-trace outputs leaves a stored-map score increment above $1/2000$. These rounded scores are approximations to the ideal probability law.

This supplies a controlled finite preparation, intervention and readout for the source-derived transfer Hamiltonian. The nine-bit direct compiler and lazy detector retain the generic evaluator's separate resource stops. The parameter t is dimensionless transfer time, not elapsed repair time or laboratory seconds. Autonomous unitary repair, a growing coherent embedding, a physical vacuum and a relativistic particle spectrum do not follow from this controlled experiment (`model/native_transfer_control`, [1]).

7.11 Local preparation and readout on one attempt clock

The fair federation admits a controlled preparation on each twelve-port cell, including every local occupation sector and an arbitrary untouched reference. Run its eleven-edge E tree, then $0 \le b \le 68$ repetitions of two ordered sweeps through the thirty internal seams. Each sixty-attempt population matrix is exactly $A_q/2^{60}$ on sector q. Integer enumeration for all $0 \le q \le 12$ proves double stochasticity and $\sum_y \min_x A_q(y, x)/2^{60} \ge 1/2$. Minorization therefore contracts half-ℓ^1 distance by at least a factor $1/2$ per block.

This population argument has a complete quantum attachment. Each nonconstant tree record identifies a rank-one branch $2^{-11/2}|y\rangle\langle x|$; constant branches preserve the entire empty/full corner P_c. Retain the tree records and trace only the subsequent mixing-record values in the comparison.

The resulting channel is within half-diamond distance 2^{-b} of the tree followed by

$$\mathcal{R}(\rho_{AR}) = P_c \rho_{AR} P_c + \sum_{q=1}^{11} \tau_q \otimes \sum_{|x|=q} \langle x|\rho_{AR}|x\rangle.$$

Here τ_q is the uniform local sector state. Its weights are inherited from the input. All physical records remain in the archive; replacing the source by τ_q while retaining every mixing record has error $1 - 1/\binom{12}{q}$ on a known basis input, regardless of mixing time. Thus this preparation does not erase archive correlations or select a global density or pure vacuum (`model/native_fair_local_preparation`, [1]).

Complete occupation readout uses the same native records. For a connected b_0-port window, track all incident currents, including boundary repairs. If s_i is accumulated displacement, the first occurrence of an internal tree edge yields its initial contrast $d - s_a + s_b$. Tree integration fixes every nonconstant initial bit pattern. Global occupation q is public source specification: empty and full windows are feasible precisely when $q \le V - b_0$ and $q \ge b_0$. A constant pattern is therefore resolved whenever only one is feasible. In particular, for $0 < q < 12$ a fresh eleven-E intracell tree gives exact occupation readout across every feasible local sector, with its terminal disturbance and arbitrary reference.

For a general window and $0 < q < V$, T uniform E attempts cover its tree except with probability $p_{\mathrm{cov}} \le (b_0 - 1)2^{-\lfloor T/M \rfloor}$. Keep coverage failures and unresolved constants as erasures. The full labelled output has half-trace distance exactly $p_{\mathrm{cov}} + (1 - p_{\mathrm{cov}})\eta_{\mathrm{amb}}$ from an initial occupation measurement followed by the same native instrument. The unresolved constant weight is η_{amb}; for the canonical fixed-q marginal it is bounded by $(1 - q/V)^{b_0} + (q/V)^{b_0}$ when both constants are feasible, and vanishes otherwise. This oracle is a comparison channel, not an extra implemented service. Worst-case input error need not be small (`model/native_fair_local_readout`, [1]).

Preparation and readout share the actual attempt counter. With $b = 64$, preparation followed by a fresh intracell tree takes 3862 attempts and literal nodes; the sparse-sector comparison error is at most 2^{-64}. This two-tree program permits $b \le 67$ under the fixed node ceiling. For a 348-port certified window at level six and canonical $q = V/3$, preparation followed by $64M$ uniform attempts uses 3852 nodes and 165154571 attempts, with combined error below 10^{-16}. These are admitted finite instruments, with expanded host stops retained. In a separate short-time comparison with binned Poisson clocks, reserving L fixed native operations costs at most KL/M for K window seams, in addition to the existing cone, selector and joint-state errors. The long readout estimate is not identified with that short-time limit (`model/native_fair_local_clock`, [1]).

The occupation chart also binds an explicitly supplied proper-code load/export service to the actual twelve local source bits and exterior, intertwining every native C/E raw branch. After a fresh E tree, its decoded nonconstant branch selects an occupied/unoccupied seam. A beam, optional phase, inverse beam and complete compiled E detector use 305 pulses. Their ideal unconditional score increment equals the actual nonconstant branch weight a; certified stored primitives give score increment at least $a\,9999999/10000000$, retaining constant outcomes, all first failures and rounded number leakage. The rounded register stays in the apparatus. The supplied quantum coupling is an operational premise; pulses and native attempts have distinct counters without a derived physical time conversion. Sector-uniform mixed states alone are invariant under these number-preserving controls. No autonomous field action or particle mass follows from this conditional experiment.

A faithful noncentral reference from the same records. One sixty-E mixing block after the tree has exact integer transition counts $A_q[x, y]$ with denominator 2^{60}. Every column is faithful and nonuniform on each recognized local sector $1 \le q \le 11$. Choose the explicit observer protocol

$a_i = 1 + \mathrm{depth}(i)$ on the native tree and $W(x) = \prod_{i \in x} a_i$. Define $L_q = \min_{x,y} \lfloor A_q[x,y]/W(x) \rfloor$. Accepting the first $L_q W(x)$ lexicographic coin paths to endpoint x prepares $\sigma_q = \mathrm{diag}(W)/Z_q$ exactly, independently of the recognized tree terminal state y. Exact backward path counts compute the flag from the retained tape. Conditional acceptance is at least $1/12$, and $\sigma_q \geq I/8192$. Rejects and coherent constant-code outcomes remain explicit; unconditional acceptance can vanish on a constant input. The complete source word has seventy-one attempts. The reference observation retains the tree and flag and traces only the sixty mixing values. Since $-\log \sigma_q = (\log Z_q)I - \sum_i (\log a_i) n_i$, natural number-preserving regional modular actions agree on overlaps. This observer-selected protocol supplies no unique vacuum or geometric modular clock (`model/native_noncentral_reference`, [1]).

Occupation current and complete local histories. For a rational profile ϕ on the retained ports, extend ϕ by zero outside and set $A_\phi = \sum_v \phi_v n_v$. Each actual seam record gives $n'_a = n_a - j$, $n'_b = n_b + j$. Thus its weighted occupation increment is $(\phi_b - \phi_a)j$, including crossing seams. A fine history is a scaled partial permutation K_h; if Δ_h is the summed increment, then

$$K_h^* A_\phi K_h = K_h^* K_h A_\phi + \Delta_h K_h^* K_h.$$

This proves the discrete continuity identity for arbitrary input and untouched reference. The decoded initial mask and retained currents determine every local snapshot on the same attempt clock. Keeping that initial label preserves the complete instrument comparison, so its error has no factor for the number of snapshots. Erasures receive no invented absolute trajectory. In the 3862-attempt preparation/readout join this history bound concerns the final eleven detector attempts, with preparation mixing values traced as specified. Occupation conservation supplies no physical gauge charge or stress-energy tensor by itself (`model/native_local_charge_history`, [1]).

7.12 Complete local records at growing capacity

The capacity comparison also admits complete local quantum instruments. Fix observed seams and final ports with endpoints in a finite set S, and a radius-r neighbourhood B with v ports and K internal seams. Retain every observed native tag, coin, raw value and current, its order, and the final quantum state with an untouched reference. Remote records and exact global tick addresses are traced by this observation, while remaining available in the native archive. Choose a common partition $0 = t_0 < \cdots < t_b = t$ with Mt_j integral and run $T = Mt$ native attempts. Each observed event is assigned its partition bin; its order within the bin is retained. Here t is per-seam attempt time, twice the exchange time above, with no physical calibration.

The complete instrument differs from its restriction to B by at most

$$a_N = \min\left\{1, |S| \binom{T}{r+1} (\Delta/M)^{r+1}\right\}$$

in half-diamond norm. Conditional on the input-independent selector word, the backward cone seeded at terminal S contains every observed record root. Leaving B requires a time-ordered path of $r+1$ seams, which proves the bound for arbitrary coherent inputs and references.

Compare the restricted process to independent rate-one Poisson attempt clocks on its K seams, applying the same native instrument at each event. In one interval of length $1/M$, the native window selector is empty with probability $1 - \lambda$, or has each seam with probability $1/M$, where $\lambda = K/M$. The corresponding Poisson probabilities are $e^{-\lambda}$ and $e^{-\lambda}/M$, with every multiple-event

word retained. Their exact total variation is $\lambda(1-e^{-\lambda}) \le \lambda^2$. Coupling all T intervals and applying the identical conditional quantum maps therefore gives

$$\tfrac{1}{2}\|\Phi_B - \Phi_{\mathrm{Pois},B}\|_\diamond \le d_N := \min\{1, tK^2/M\}.$$

Aligned boundaries give the same bin label on matched events. Exact continuous timestamps have a singular comparison with native grid times, and the full global record-count laws differ, so neither observation can replace the declared local binned one.

If initial joint window/reference states differ by half-trace distance ϵ, their complete observed outputs differ by at most $\min\{1, \epsilon + a_N + d_N\}$. For a canonical fixed-$q$ source and an independent common reference, comparison to a declared Bernoulli density p gives

$$\epsilon \le \min\{1, v(v-1)/(2V) + v|q/V - p|\}.$$

Matching rooted windows thus give an explicit comparison between source capacities. If all fixed-radius windows stabilize and their state restrictions converge, the additional Poisson cone bound $|S|(\Delta t)^{r+1}/(r+1)!$ constructs a limiting local instrument. The evaluated source levels zero through two have identical cell-zero radius-one windows with 15 ports and 33 seams, with a verified orientation-preserving record map. This establishes neither a unique all-radius routed refinement nor the physical density or spatial dimension. The proof and exact rational controls retain every overflow and distinguish kernel analysis from bounded native execution (`model/native_local_history_limit`, [1]).

7.13 Rooted source refinement

The exact oriented midpoint stencil supplies a constructive rooted cell topology. Choose the cell that follows child zero at every subdivision around one original vertex. Inserting an additional leading child zero identifies its radius-r dual ball, including incident edges, whenever $2^\ell > r+2$. In each of the five incident original faces, a dual step increases the triangular grid's radial coordinate by at most one. This proves stabilization of every finite corner ball and consistency under restriction. The resulting countable graph retains the five-face corner; no assertion about an arbitrary root sequence follows.

A recursive local evaluator also computes the literal midpoint coordinates, face orientation and port scores without allocating the sphere. It reproduces coordinates bit for bit and routed neighbor rows at the independently evaluated levels zero through three. Exact binary32 score intervals certify first-maximum decisions, including stable ties. Collision repairs require a unique certified assignment. Uncertain decisions are rejected. The finite address domain extends through level fourteen, where every source edge key fits its signed integer representation. These are finite numerical attachments to the oriented topology.

The certified tables give matching windows of cell radii one, two, four and eight at specified levels through fourteen. Keeping every port of a radius-r cell ball contains the required radius-r port ball. Canonical names preserve source endpoint order and therefore intertwine the complete native instruments, including signed currents and zero slots. For declared density p, the canonical totals $q_\ell = \lfloor p\, 240\, 4^\ell \rfloor$ have the local state bound above; the Bernoulli comparison states restrict consistently. The radius-four, level-fourteen window has 348 ports. The observation retains records on the thirty root-internal seams and the final state on the twelve root ports; other window records are traced. At $p = 1/3$ and $t = 1/630$, its complete local-output comparison bound is below 10^{-6}. The corresponding 268435456-attempt description is admitted. The expanded host retains its unevaluated-carrier stop; this is a finite kernel bound, without a claimed execution of the global experiment.

An infinite routed process requires a specified arithmetic continuation. Finite-width level tags and coordinate collapse in bounded local tests prevent treating the literal numerical constructor as a total such specification. Exact coordinates, arbitrary precision and combinatorial port assignment are different continuations. The construction chooses none of them and establishes no unique all-radius routed limit, physical density or Lorentzian field (`model/native_rooted_refinement`, [1]).

7.14 Native clock observables

The declared coherent clock services also give operational noncommuting measurements on the same biased 66-state source. On the existing seam $(1,7)$, pair $|1,k\rangle$ with $|7,k\rangle$ for each of the ten spectators. Let X, Y, Z be the direct sum of the usual Pauli matrices on these pairs, zero on the 46-dimensional complement $P_0 = I - Q$. The existing balanced cap and quadrature words give analyzers U_A with $U_A^\dagger Z U_A = A$. One native contrast-reading repair after the analyzer therefore has effects

$$P_{A,\pm} = \frac{Q \pm A}{2}, \qquad P_{A,0} = P_0, \qquad A \in \{X, Y, Z\}.$$

All 25 raw addresses are retained, including 19 zero maps. The six live maps have coin factors $8/33$ and $25/33$. The zero-contrast maps preserve all coherences within P_0, including coherence between empty and doubly occupied endpoints. The raw update is destructive. Its retained displacement record permits an explicit correction: apply two existing balanced cap rotations when the repair moved, then undo the analyzer. Every fine corrected Kraus operator is a scalar sign times $\sqrt{p_{\mathrm{coin}}} P_{A,\delta}$ on all 66 states. This gives the three-outcome Lüders instrument, with its coin refinement and arbitrary untouched references.

For the full normalized stationary density, no odd-sector conditioning is needed:

$$\langle X\rangle = \langle Y\rangle = 0, \quad \langle Z\rangle = \frac{1649}{7505}, \quad \operatorname{Var} X = \operatorname{Var} Y = \frac{3201}{7505}, \quad \frac{|\langle [X,Y]\rangle|^2}{4} = \left(\frac{1649}{7505}\right)^2.$$

These actual observables and state supply the finite Robertson theorem's inputs. With $\Delta = \log(25/8)$, the working modular clock acts by

$$U_s = \rho_*^{-is}, \qquad U_s^\dagger X U_s = \cos(s\Delta) X - \sin(s\Delta) Y.$$

The literal nine-qubit analyzer words have full-address certificates, retained export failures and a separate multiplication-rounding allowance. These bounds concern the stored numerical primitive matrices, with no hardware or unrounded analytic exponential error claim. The coherent services and their feedback composition are declared observer operations; they are outside the bare repair/hold grammar. This finite measurement menu supplies neither universal implementation independence for every complex effect nor a physical position–momentum relation (`model/native_clock_observables`, [1]).

Configuration readout and repeatable preparation. There is a bounded contrast-record detector on this same sector. Perform one native E repair on each edge of a fixed spanning tree of the twelve ports. Let u_i be the accumulated recorded change of occupation at port i. An incoming contrast d on (a,b) gives the initial difference $n_a(0) - n_b(0) = d - u_a + u_b$; the retained current updates u after that read. The eleven differences and $\sum_i n_i(0) = 2$ determine the initial configuration. Thus every nonzero fine history has a rank-one map $K_h = \sqrt{p_h}\,|x_h\rangle\langle x_0|$, and coarse graining by the decoded x_0 gives the full computational projective measurement. All 25^{11} formal raw words are

retained in an exact lazy representation; $66\,2^{11} = 135168$ words have nonzero maps. One queue slot and ten exports suffice for the eleven-step record protocol. Streamed branch continuation uses the existing native transition kernel and authenticates its records. The stock expanded evaluator's resource stop after three steps is retained as a separate return; it is not equated with the lazy evaluator. The nine-bit coherent words use the direct source compiler; the generic word evaluator has a six-qubit host ceiling.

The terminal configuration is also known from the currents. Existing coherent bit routing maps its code either back to x_0 or to a fixed target code. The ideal corrected maps therefore give the basis Lüders instrument or the replacement channel $\rho \mapsto |x_*\rangle\langle x_*| \operatorname{Tr} \rho$, including untouched references. This supplies repeatable preparation to the existing pair-analyzer architecture: its 4291 settings span all 4356 Hermitian coordinates. Complete compiler bounds and export outcomes accompany these compositions. The rounded maps have bounded normalization error; their positive record functionals retain full rank without output renormalization. Coherent service semantics are declared, and finite tomographic coverage does not prove universal coexistent-effect realization or a growing field-observable family (`model/native_effect_coverage`, [1]).

Rank-one measurement synthesis. The supplied coherent services implement more than the fixed tomographic menu. For any nonzero Gaussian-rational vector v on the 66 source states and rational $0 \le \lambda \le 1$, with each input numerator and denominator of at most 64 bits, consider $E = \lambda|v\rangle\langle v|/\langle v, v\rangle$. In the existing 512-address register, append the amplitude $\sqrt{1-\lambda}$ at one padding address. A binary amplitude tree and diagonal phase word prepare the resulting normalized vector from address zero. Executing the inverse word and routing zero to a genuine source code before complete export and tree readout gives E as the ideal selected effect. All other source outcomes, padding aborts and processor failures are retained. No extra register or conditioning on source-code success is used, and an effect alone specifies no Lüders update.

Directed integer trigonometric enclosures select a 41-bit angle grid; exact Walsh identities compile its layers into existing native pulses. The full-register operator error from target angles is at most $11\,2^{-41}$. The emitted word uses at most 23579 pulses, below the unchanged source-word limit. The certificate checks every emitted stored six-dimensional primitive, including negative inverse requests and fixed basis-change angles, against its full ideal matrix and rejects a failed bound. For a passing word with N pulses and $g = 2\,10^{-11}$, put

$$x = Ng, \qquad e = \frac{x}{1-x} + 11\,2^{-41}, \qquad F_N = \frac{Ng^2}{(1-x)^2}.$$

Telescoping the good-code products and retaining every first processor failure gives complete half-diamond error at most $[e(2+e) + F_N]/2 < 10^{-6}$, including arbitrary untouched references. These are bounds on the stored primitive maps; floating propagation arrays are diagnostics. The direct nine-bit compiler executes the tape, while the generic six-bit word evaluator returns its resource stop. The real projector onto $(3|0\rangle + 4|65\rangle)/5$ also has an 80-pulse construction with error below $2\,10^{-9}$, despite lying outside the positive cone of the fixed balanced-analyzer menu. Fixed precision and word length supply neither arbitrary-accuracy dense effects nor universal coexistent refinement. The composition consumes the declared coherent services and Kraus probability semantics; it does not derive those premises from repair counting (`model/native_effect_synthesis`, [1]).

Complete coexistent weak measurements. The same source services give joint noncommuting effects with their disturbance. For a normalized rational-direction vector u, projector $P = |u\rangle\langle u|$ and rational $0 \le s \le 1$, couple u to one existing padding address by a rank-one phase. Complete

source export and a second correcting phase give the ideal surviving contraction $L = I + (s-1)P$ and abort effect $(1-s^2)P$. The abort retains its reference state and terminates; the success remains a quantum input to the next stage. For at most eight such stages, put $A_0 = I$ and $A_j = L_j A_{j-1}$. The complete effects are

$$E_j = A_{j-1}^*(1-s_j^2)P_j A_{j-1}, \qquad E_{m+1} = A_m^* A_m, \qquad \sum_{j=1}^{m+1} E_j = I.$$

The last identity telescopes, including every earlier disturbance. With $u_1 = |0\rangle$, $s_1 = 3/5$, $u_2 = (|0\rangle + |65\rangle)/\sqrt{2}$ and $s_2 = 4/5$, the $(0, 65)$ entry of $[E_1, E_2]$ is $216/3125$. Their coexistence follows from one common complete instrument.

Each stage uses two separately admitted 58392-pulse words in the existing nine-bit work register. Every first processor failure, padding abort and residual outcome is retained, with no intermediate destructive tree readout. Conditional on all exact primitive checks passing, a gain-aware complete half-diamond bound for eight stages is below $1/50000$. The generic evaluator retains its resource stop; the direct compiler and lazy complete instrument have their own stated domain. This bounded family supplies noncommuting coexistent effects but does not implement every effect or discharge the universal operational Born premises (`model/native_coexistent_effects`, [1]).

A prepared phase contrast under fair repair. A separate twelve-bit occupation-register experiment targets the fair source's mixed state $\rho = P/\operatorname{Tr} P$, where $P = \frac{1}{30}\sum_e Q_e$, $Q_e = (I + \mathrm{swap}_e)/2$ and $\operatorname{Tr} P = 56$. Declared coherent services target the ideal successful blocks $q\rho$ and $qD\rho D$, with $D = (-1)^{n_0}$. Their ideal signed single-occupation exchange readouts on ports $(0, 1)$ are $\pm q/168$, $\pm 5q/1008$ and $\pm q/240$ before repair, after C and after E, respectively, where $q = 1 - (368/2048)^{16}$ is the unconditional selector acceptance. The exact difference operator is multiplied by $5/6$ under C and $7/10$ under E, although the two repairs share their population transfer. A separate rational implementation checks the numerical source execution's prepared density blocks with Frobenius tolerance $2\,10^{-10}$, checks outside-carrier mass separately, and compares every outcome with its target. Source failures, selector timeout and all terminal labels are retained. These bounded numerical checks supply no arbitrary-reference channel-error certificate. The mixed preparation differs from the transfer ground; its decaying phase contrast identifies neither a protected sector nor a particle mass. Matching a population transfer alone does not establish a common state-and-instrument dictionary (`model/native_phase_experiment`, [1]).

7.15 Protected charge and its native reference

An explicit composition of the biased native repair and its persistent ordering bit connects protection to a state and a modular clock. For the binary total-two source, let $\theta_0 = \theta_1 = 25$ and $\theta_i = 8$ otherwise. Its reference is $\rho_* = 7505^{-1}\sum_x \prod_i \theta_i^{x_i} |x\rangle\langle x|$. The ordered branch is a weighted partial injection with an ordering flip X when transport crosses the occupied spectator. All branches commute with $Q = I_{66} \otimes X$. Their complete star algebra has commutant $\operatorname{span}\{I, Q\}$. A faithful invariant state makes this the adjoint fixed space, so the stationary densities are exactly

$$\rho_\alpha = \rho_* \otimes [\alpha P_+ + (1-\alpha)P_-], \qquad P_\pm = (I \pm X)/2, \quad 0 \le \alpha \le 1.$$

The weight α is conserved. With the ordering bit prepared in either sharp sector, 1600 unread attempts prepare the source/reference marginal within 10^{-6} of its stationary product; the complete archive is not claimed independent. The proof includes arbitrary untouched references.

For $0 < \alpha < 1$, put $K_\alpha = -\log \rho_\alpha$. A recorded current j on seam (a, b) obeys $[K_\alpha, K_h] = j \log(\theta_a/\theta_b) K_h$. Thus every complete branch is covariant under the same modular clock. The negative exchange sector has the actual two-occupation CAR transport sign. Its closed odd exchange returns to the initial occupation and has zero modular energy change. The two protected sectors differ in mean modular energy by $\log[\alpha/(1-\alpha)]$, set by their conserved reference weight. Protection therefore supplies no nonzero rest energy. The potential commutes with occupations; its unitary flow is not the stochastic motion. These statements concern the composed 132-state source, whose ordering memory is an additional declared service (`model/protected_reference`, [1]).

The fixed-selector raw instrument has the exact weighted reverse $\rho_\alpha^{1/2} K_h^\dagger \rho_\alpha^{-1/2} = K_{\bar{h}}$, where a label involution exchanges each moving outcome with its reverse and fixes the holding and zero slots. This proves Kubo–Martin–Schwinger detailed balance and equality of stationary fixed-length record probabilities under reversed order and labels. At $\alpha = 0, 1$ the identity uses the invariant supported corner. It concerns stationary preparation and this selector; it supplies neither a reversal theorem for arbitrary feedback nor physical time reversal (`model/protected_reference/REVERSAL.md`, [24, 1]).

Complete gauge transport. The same literal quantum-link matrices supply a complete locally covariant operation for each of the five matter blocks, including colored matter. For a block of dimension d, let $A = \sum_{ab} |a\rangle\langle b| \otimes U_{ab}$ and $B = A/\sqrt{d}$. Both the raising coordinates and their conditional rotor completion obey $(A^\dagger A)^2 = dA^\dagger A$. On the two-position edge define

$$T = |x\rangle\langle y| \otimes B, \qquad P_y = T^\dagger T, \quad P_x = TT^\dagger.$$

The endpoint intertwining equations imply that T, P_x, P_y commute with the total local Gauss action. Since $P_x P_y = 0$, the complete directional instrument is $\{T, I - P_y\}$, its reverse is $\{T^\dagger, I - P_x\}$, and $W = T + T^\dagger + I - P_x - P_y$ is a Hermitian unitary. Every blocked flux state is retained. The exact identities cover all sixty link states and all twelve currents; the hard-core spectator contributes the usual CAR sign.

The covariant unitaries $U(c, s) = I + (c-1)(P_x + P_y) - is(T + T^\dagger)$, $c^2 + s^2 = 1$, expose the unselected operation parameter. On a fixed gauge-singlet preparation with a dual charge reference at y, the complete detector $\{P_x, I - P_x\}$ gives probability s^2. The rational choices $(c, s) = (3/5, 4/5)$ and $(5/13, 12/13)$ give $16/25$ and $144/169$ on that same experiment. Covariance and completeness therefore do not determine its evolution. This finite operation uses a declared matter/link extension, without equivalence to the original C/E law or a full collision extension (`model/gauge_defect_transport`, [1]).

The protected reference and this transport admit a specified common edge process. On native seam $(0, 11)$ the move $\{5, 11\} \to \{0, 5\}$ crosses the spectator, so its joint operator is $\widetilde{T} = |\{0, 5\}\rangle\langle\{5, 11\}| \otimes X \otimes B$. The forward and reverse coins have probabilities $25/33$ and $8/33$. Their complete transport/hold instruments preserve Q and the reference $\rho_* \otimes \sigma_\alpha \otimes I_{60d}/(60d)$, where $\sigma_\alpha = \alpha P_+ + (1 - \alpha) P_-$, by exact endpoint detailed balance. They retain the other 64 configurations as holds. This full reference is gauge invariant without support on the joint Gauss kernel. A declared dual reference gives an orthonormal Gauss-singlet pair ψ_x, ψ_y and stationary corner state $(25|\psi_x\rangle\langle\psi_x| + 8|\psi_y\rangle\langle\psi_y|)/33$. In a fixed memory-charge sector one unread attempt replaces every admitted corner input by this mixture, including reference-entangled inputs. Its complete records are retained before taking that marginal. The same modular reference assigns forward energy change $\log(8/25)$. This is a declared one-edge law, with a designated moving internal fiber and separate singlet preparation. It supplies no all-seam gauge network, global mixing theorem, two-particle colored identity or selected coherent Hamiltonian (`model/protected_reference/GAUGE_JOIN.md`, [1]).

The same corner supplies a finite thermodynamic instance. Summing all four outcomes gives the diagonal kernel $P(i,j) = p_j$, $p = (25/33, 8/33)$. With $E_i = -\log p_i$ and multiplier one, the Lean theorem `corner_same_kernel_four_laws` identifies this kernel with the heat bath consumed by the finite four-law core (`Lean/Thermodynamics/ProtectedGaugeCorner.lean`, [1]). It gives the first-law split, relative-entropy contraction and multiplier identifiability. The energy is nonconstant, so the replacement exchanges modular heat for some inputs. Its fixed-spectrum mathematical Gibbs family has zero low-temperature entropy; the fixed native coin supplies no implemented temperature variation, physical calibration or uniform capacity limit.

Gauge-invariant propagation on trees. The configured sixty-state quantum-link Hamiltonian has a complete spatial sector on a finite rooted tree $\mathcal{T}$. Retain all five native matter blocks, fix total fermion number one, and put a static dual reference for one block r of dimension d_r at the root. Every block has nonzero central charge and scalar current commutant. Leaf elimination therefore forces all links off the root-to-fermion path to vacuum; irreducibility and inequivalent central charges leave one contraction along that path. The entire Gauss kernel has an orthonormal basis $|v; r\rangle$, one state for every vertex, including the root.

For the owning literal raising-link hopping of strength J and the electric occupation-projector cost $\epsilon \geq 0$, the exact restriction is

$$H\big|_{\ker G} = J\sqrt{d_r}\ \mathrm{Adj}(\mathcal{T}) + \epsilon\, \mathrm{diag}(\mathrm{dist}(0, v)).$$

The factor $\sqrt{d_r}$ comes from the norm of the contracted flux string; substituting normalized partial-isometry hopping would change the configured action. Position projectors commute with Gauss and form a complete detector, with every position and repeated-position outcome retained. Their readout can change hopping energy, whose exact work balance is accounted for.

On an additionally specified infinite bounded-degree tree, positive ϵ gives a self-adjoint operator with compact resolvent: its diagonal potential tends to infinity and hopping is bounded. For initial support inside radius R_0, compression to radius R gives the complete continuous-time position-record total-variation bound

$$\min\left\{1,\ 2\sum_{n\geq R-R_0+1} \frac{(D_{\max}|J|\sqrt{d_r}\,t)^n}{n!}\right\},$$

where $D_{\max}$ bounds degree and t is the total evolution time across all readouts. The executed Cayley clock has a separate finite-step comparison allowance. These reference-bound dressed levels use a specified graph, link, coupling and clock. They supply no freely propagating asymptotic particles, relativistic dispersion or physical mass (`model/native_gauge_strings`, [1]).

Source coupling readout and its color-action boundary. On one link, the same Gauss-singlet source states give the exact block

$$H_r = \begin{pmatrix} 0 & J\sqrt{d_r} \\ J\sqrt{d_r} & \epsilon \end{pmatrix}, \qquad p_1(t) = \frac{4d_rJ^2}{\epsilon^2 + 4d_rJ^2} \sin^2\left(\frac{t\sqrt{\epsilon^2 + 4d_rJ^2}}{2}\right).$$

Here p_1 is an unconditional endpoint probability; the other outcome and all measurement back-action are retained. For $J \neq 0$, $p_1''(0) = 2d_rJ^2$ and $\epsilon^2 = -p_1''''(0)/p_1''(0) - 2p_1''(0)$. Two finite-time probability intervals at t and $2t$ supply rational Taylor enclosures, avoiding an assumed derivative oracle. The configured source sets $J = \epsilon = 1$ in its model clock, hence determines the clock-independent ratio $\epsilon/|J| = 1$. Reading these specifications is sufficient to fix that model ratio.

The admitted color currents have $\mathrm{tr}_3(T_aT_b) = \delta_{ab}/2$, whereas their selected fifteen-state matter lift has $\mathrm{Tr}_{15}(T_aT_b) = 2\delta_{ab}$. These trace conventions must not be interchanged. The five nonvacuum source link sectors have (C_3, C_2, Y^2) equal to

$$(0,0,1),\ (4/3,0,1/9),\ (4/3,3/4,1/36),$$
$$(0,3/4,1/4),\ (4/3,0,4/9).$$

Their configured electric energies are all one. No coefficients a, b, c give this vector as $aC_3+bC_2+cY^2$: the second and fifth entries force $c = 0$, contradicting the first. The zero-energy link vacuum excludes an added common identity. Thus the source electric law cannot be directly identified with a quadratic gauge Laplacian. This does not exclude an emergent color action after a controlled coarse-graining.

For a separately justified continuum action $S_E = (\kappa/2)\int \mathrm{tr}_3\, \mathcal{F}_{\mu\nu}\mathcal{F}_{\mu\nu}$, with $\mathcal{A} = g_sA$, $\alpha_s = 1/(4\pi\kappa)$ in the specified normalization. The finite gauge-invariance theorem permits every positive κ. The missing source identification is the continuum action coefficient and its renormalization scheme and scale. Running a supplied coupling does not select this boundary value. The exact finite readout and coefficient comparison are in `model/source_strong_coupling` [1].

Complete color-loop readout in the interacting Higgs source. The separate Casimir-weighted action retains its full 438- and 1,596-state Gauss sectors, with empty colored-fermion occupation, three identity family matrices and the stated scalar cutoff. Write $H = eE - m(L + L^*) + hT + gS + sN_H$, where S is the actual Higgs vertex and T includes the hopping sign. Its three colored link blocks have $(d_r, c_r) = (3,4), (6,25/8), (3,3)$, with $c_r = 2C_3 + C_2/2 + (6Y)^2/12$. The vacuum input and its actual reference-neutral addition image are labeled $q = 0, 1$. The complete instrument resolves scalar occupation and both link blocks; all output states and zero-probability records are retained. Summing both scalar occupations in the equal-block loop outcome gives

$$\frac{p_r(t)}{d_r^2} = m^2t^2 - m^2\left(\frac{e^2c_r^2}{3} + 18m^2 + \frac{qh^2}{3}\right)t^4 + O(t^6).$$

The exact full-source moments cancel the Higgs contribution at fourth order; a sixth-order control detects its nonzero interacting contribution. For the two dimension-three blocks, let M be the quadratic coefficient divided by nine and D the difference of quartic coefficients divided by nine. Then $M = m^2$, $D = -7e^2m^2/3$ and $(e/|m|)^2 = -3D/(7M^2)$. This reads a clock-invariant bare ratio when $m \neq 0, e > 0$. Six positive times $t_j = j/1024$, exact polynomial evolution through order 28 and norm-controlled remainders enclose the default value one. The full source, diagonal energies and detector labels are regenerated. The preparation, flux instrument and Hamiltonian clock are declared finite interfaces. No repair-only flux compiler, chiral colored continuum or renormalized strong coupling is inferred (`model/color_loop_readout`, [1]).

A local symmetry-sector excitation. On the actual open eight-site selected spin ground at $J = 1, B = 1/4$, a one-site Z pulse flips the global internal X parity and leaves every initial exterior expectation unchanged. Second quantization includes the minus sign on doubly occupied sites; the full centered Hamiltonian and every complete recorded C/E branch commute with this parity. The protected single-occupancy spin code has identity repair branches with fair coin records. The ideal selected-state energy excess is enclosed by $[1.825861195059, 1.825861195062]$. Complete native preparation and detection execute in 1,090 and 1,092 pulses and certify pointer contrast above 0.67. The local commutator bound and complete source/spin comparison retain contrast

above 2/5 through at most ten repairs and time 1/16 on $c = 12^4, U = 12^8$. This lifetime tape is an analytic construction; evolution spreads the excitation. The parity is a finite symmetry charge, not a topological boundary invariant or a physical mass (`model/defect_localization_readout`, [1]).

Continuous charge, creation work and conjugate fields. The exact full-Fock spin generator induces $Q = \sum_x \epsilon_x X_x/2$ in the Marshall frame. It commutes with the source action and every recorded repair branch. Ground uniqueness and a charge-reversing source automorphism force $Q\Omega = 0$. The complete local instrument $A_q = (Z_0 - qi\epsilon_0 Y_0)/2$, $q = \pm 1$, obeys $[Q, A_q] = qA_q$ and $\sum_q A_q^* A_q = I$. A local Z followed by one X pointer prepares these coherent branches. Measuring every local X and grouping the outcomes gives the charge probabilities but dephases within a sector, so it is a different preparation.

The original weights for $q = -1, +1$ are approximately 0.83547507093 and 0.16452492907. A complete native Pauli estimator and an independent controlled-clock phase measurement recover the unconditioned work $W_q = \langle A_q^*[H_B, A_q]\rangle$. The conditional mean excitation energies are approximately 1.84463098607 and 1.73054620199 at $J = 1$. Both channels have positive spectral variance. Their first return-survival value 0.99 occurs in the certified source-time intervals $[0.1726210202, 0.1726210331]$ and $[0.1044566248, 0.1044566376]$, respectively. Return survival can rebound. A fixed uniformly chosen local-X detector retains original-weight charge contrast $w_q/8 > 1/50$ at every source time. Thus charge conservation, return retention and a particle decay lifetime are distinct assertions. The generator action over an autonomous interval is TE_{exc} while ongoing system power is zero. Scalar bare-repair effects cannot measure these different energies; the coherent controls supply the sensitive readout (`model/source_energy_effort` and `model/charge_channel_lifetime`, [1]).

For the ordered source, define the second staggered configuration

$$Q_{2,p}(F) = p^2 \sum_x \epsilon_x F(px) S_x^2.$$

Append its bounded groups to the existing $Q_{3,p}, P_{2,p}$ histories. Internal rotation symmetry supplies its variance and energy bounds. The equal-time estimate

$$\|[Q_{2,p}(F), Q_{3,p}(G)]\| \le \frac{p^4}{2} \sum_x |F(px)G(px)| = O(p)$$

and the full-history transport bounds construct commuting configuration groups on a common further subsequence. The source U(1) acts there with charged vectors $\Phi_\pm \Omega = (Q_3 \mp iQ_2)\Omega$. An odd microscopic translation combined with the internal axis-two π rotation induces charge conjugation, commuting with the limiting energy and spatial translations. It gives

$$\mu_{\Phi_+\Omega} = \mu_{\Phi_-\Omega} = 2\mu_{Q_3\Omega}.$$

These measures have nonzero positive gapless spectral weight. This ordered-zero-field construction supplies no second uniform canonical momentum; no Lorentz spin, sharp shell or physical antiparticle follows from this internal symmetry (`model/protected_defect_catalog`, [1]).

For a normalized source-created vector $a \in \operatorname{Dom} K$, retain its full complement: $r = (I - |a\rangle\langle a|)Ka$ and $D = K$ compressed to that complement. Then

$$\langle a, (K+z)^{-1} a\rangle = \left[z + \langle K\rangle_a - \langle r, (D+z)^{-1} r\rangle\right]^{-1}.$$

The positive memory term is nonzero in both finite charge channels. Bounded sine preparation and the source resolvent filter construct this response on the same continuum history subsequence

with nonvanishing original probability. Their filtered first two moments converge through bounded spectral integrands. This retains the spectral complement without fitting a mass or assuming a one-frequency field (`model/spin_dynamical_memory`, [1]).

A bounded response test avoids energy-domain assumptions. For an actual source-preparation isometry V and $S = z(K+z)^{-1}$, $z > 0$, put $A = V^*SV$. Then $D = V^*S^2V - A^2 = R^*R$, where $R = (I - VV^*)SV$. The response reduces K exactly when D vanishes. Identifying $A = z(\omega(P)+z)^{-1}$ additionally requires the actual spatial translation action. Small D controls bounded polynomials, but time comparison also needs energy-tail control. The complete native dimer distinguishes a nonreducing preparation from a reducing two-energy span, without identifying either as a particle mass (`model/spin_dynamical_memory/RESOLVENT_CLOSURE.md`, [1]).

There is a constructive hierarchy for this test. For admitted source creators $\psi_a = A_a\Omega$, let $\mathcal{H}_r = \operatorname{span}\{S^j\psi_a : 0 \le j < r\}$ and compress S to T_r there. Splitting powers across the two retained vectors proves equality of source and compressed moments through degree $2r-1$. Bernstein polynomial approximation consequently gives, for a real L-Lipschitz score f and original weight $w = \|\psi_a\|^2$,

$$|\langle\psi_a, f(S)\psi_a\rangle - \langle\psi_a, f(T_r)\psi_a\rangle| \le \frac{wL}{\sqrt{2r-1}}.$$

Coherent Gamma–Cauchy time superpositions prepare $S^\alpha A_a\Omega$ from the same local histories, with time-tail weight at most $2\alpha/(\pi zT)$. Complete failed preparations, Gram floors and window boundary errors remain explicit. The closure of the source cyclic union reduces K; adjoining actual translation orbits also reduces those translations. Finite Ritz nodes are quadrature nodes, not proved source energy levels. The exact dimer resolves energies $9/8, 5/4$ and two finite reflection labels with all failures retained, without a continuum momentum or particle assignment (`model/spin_dynamical_memory/SPECTRAL_BRANCH.md`, [1]).

An isolated pole is not required for every charged physical sector [22]. A controlled infrared edge has a separate readout criterion. For an original subprobability measure ν with edge $\theta = \inf\operatorname{supp}\nu$, assume the source estimate $\nu([\theta, \theta+\epsilon]) \ge a\epsilon^\alpha$ for every sufficiently small positive ϵ. Certified CDF bounds $\nu((-\infty, x_-]) < a\epsilon^\alpha$ and $\nu((-\infty, x_+]) > 0$ imply $x_- - \epsilon < \theta \le x_+$, by interval inclusion. An empty finite bin cannot supply the tail premise. A source-derived relativistic dictionary and inclusive or dressed asymptotic detectors are additional requirements; the theorem assigns no OPH infrared mass (`model/spin_shell_readout/THRESHOLD_READOUT.md`, [1]).

Bounded charge and pair preparation. The finite selected reference can be prepared along a gapped source bond ramp. An exact basis change puts its positive-field action in the Lee–Yang class, with full gap at least $B/8$; a quantitative adiabatic bound and complete native compilation give an explicit preparation error. Interpolating boundary bonds preserves that gap. Source locality then bounds the interior marginal error by a surface factor times decaying functions of boundary distance, uniformly in exterior volume at fixed B. This supplies controlled finite positive-field references, without a zero-field marginal rate or a particle mass. The source substitutions, imported gap and adiabatic theorems, and complete implementation bounds are given in [8] and `model/spin_fixed_field_locality` [1].

A distinct joint family keeps $p = n^{-2}$, $B = Jn^{-5}$, the positive-field ground and its H_B clock together. Its gap controls order and bond fluctuations. Auxiliary transverse and longitudinal phases then control the mixed momentum commutator through complete histories. The field limit has two regular pairs (Q_3, P_2) and (Q_2, P_3), with symplectic coefficient $m\int FG$, $m > 1/10$, and a full internal U(1). Even microscopic translations give common continuous spatial translations. Fixed-scale preparation and local records have one complete error budget. This is a chosen preparation/scaling

prescription, not a theorem selecting it from unread repair or identifying it with the ordered-zero-field history state (`model/spin_joint_field_limit`, [1]). The actual filtered current has norm $O(p^2)$. A sufficient source-norm route to the linear force is a specific residual $o(p^2)$, measured with its original probability and retained complements. This cancellation and a sharp spectral branch are unproved.

The two-site positive ground marginal also gives a strict correction to mean-spin factorization. With transverse bond coefficient κ_B and staggered moment m_B, the source dimer variational bound implies $\liminf(\kappa_B - m_B^2) \geq 61/14400$. Exact ground double commutators then exclude the unfiltered force coefficients $12m_B, m_B$ obtained by replacing each longitudinal spin by its mean. Corrected coefficients and an infrared sharp component are allowed; the result supplies no pole (`model/spin_joint_field_limit/QUANTUM_FORCE_SLACK.md`, [1]).

The actual two-force equations admit a useful static reduction. Here P is canonical field momentum, distinct from spatial translation momentum. For $q = Q\Omega$, $a = P\Omega$, put $h_Q = \|K^{1/2}q\|^2$, $\chi_Q = \|K^{-1/2}q\|^2$ and similarly for P, and $c = \mathrm{Im}\langle q, a\rangle$. With positive susceptibilities,

$$u = c/\chi_P, \quad b = c/\chi_Q, \qquad \delta_Q = h_Q - c^2/\chi_P \geq 0, \quad \delta_P = h_P - c^2/\chi_Q \geq 0.$$

These are exactly the squared $K^{-1/2}$-form norms of $iKq - ua$ and $iKa + bq$. On a positive energy window $[\ell, M]$ their ordinary norms are at most $\sqrt{M\delta_Q}$ and $\sqrt{M\delta_P}$. For $c \neq 0$, substitution bounds the original Q weight outside an ϵ tube about $\omega = \sqrt{ub}$ by $M(M\sqrt{\delta_Q} + |u|\sqrt{\delta_P})^2/(\omega^2\epsilon^2)$, divided by the actual preparation normalization squared. The filtered cross moment must use the same filter; it is not the unfiltered order commutator. Growing preparation normalization can suppress original success even for a nonzero field vector. Uniform source saturation remains unproved. The actual joint dimer has a zero-defect pair and a positive-defect pair on the same sharp finite energy line: polarization mismatch alone can make this sufficient whole-pair test fail (`model/spin_joint_field_limit/WARD_DYNAMICS.md`, [1]). Its actual conserved charge projectors resolve the latter into two zero-defect branches, each with original Q/P preparation weight greater than $1/25$. Their filtered cross moments have opposite signs. This is a complete finite-source resolution, without a continuum species assignment. For the uniform pair on $K_0 = H_B - E_B$, the source identity gives $\delta_P = 0$ and $\delta_Q = M_{-3}\,\mathrm{Var}_\pi(E^2)$, with $M_j = \int E^j\, d\nu$ for its configuration energy measure and $d\pi = E^{-3}d\nu/M_{-3}$. With N sites and $m = N^{-1}\langle\sum_x \epsilon_x S_x^1\rangle$, the joint clock obeys $h_L/\chi_L \leq R = 6/(mn) + n^{-6}$ and $\mathrm{Var}_\pi(E^2) \leq R^2/4$, with energy rescaled to L. Finite boxes with $m > 1/20$ are constructible. The normalized uniform preparation has original weight at least $m/(8N)$ below $\sqrt{2R}$; this supplies no uniform residue or nonzero-momentum shell. Fixed cube spectral lines have positive variance and do not identify the growing limit (`model/spin_joint_field_limit/UNIFORM_MODE.md`, [1]).

Circle projection of $\arctan(X_\theta)/\pi$ gives bounded charges on this joint family. Real even tests in $\delta|k_0|$ caps about $\pm k_0$, $0 < \delta \leq 1/4$, give each charge and direction original probability $1/774400$ in energy $[9|k_0|/774400, 580800|k_0|]$. Inversion pairs the caps using the linear test's support; seven records retain nonlinear complements. A six-site periodic control has thirteen translation/energy records and positive directed variance. Narrow spatial phases give no sharp time phase (`model/spin_joint_field_limit/DIRECTED_CHARGED_BANDS.md`, [1]).

The ordered-zero-field prescription has a separate bound. The rational infrared integral and biased trial $(I + 10M)G$ certify $m > 1/10$. For a unit annular test, $x = \sqrt{r}Q_3(F)$, $y = \sqrt{r}Q_2(F)$, take B_+ as the first circle coefficient of $(x\cos\theta + y\sin\theta)/[1 + (x\cos\theta + y\sin\theta)^2]$ and $B_- = B_+^*$. Their norms are at most $1/2$; the positive square-root complement completes the instrument. Canonical and energy moments give each charge original probability $p_* = 1/1757600$ in $r/2 \leq |P| \leq r$, $3p_*r \leq L \leq r/p_*$. Source quadratures converge on the same ordered history subsequence; no spectral pole follows (`model/spin_charge_dispersion`, [1]).

Since translated configuration fields commute, finite spatial averaging of $A = B_+^* B_+$ gives arbitrarily far rational displacements with $\|\alpha_a(B_+)B_+\Omega\|^2 \geq p_*^2/4$. This uses positivity and translation invariance without clustering. For every fixed n, Jensen's inequality applied to the spatial average of A gives separated preparations of every charge-sign pattern, with original weight at least $p_*^n/4$ and all 3^n records retained. An explicit cube contains suitable profile centres; selecting them effectively also requires the source-state readout bounds. Charge conjugation gives the negative-charge pair. The source energy-form bound gives $\langle L\rangle_{\text{unnormalized}} \leq 27r/4$ for each pair; an upper cutoff $54r/p_*^2$ retains at least $p_*^2/8$ original probability, possibly including zero energy. On the finite cube the pair columns span each entire 28-dimensional charge-± 2 sector, with a complete detector and connected retarded response. Asymptotic factorization, a comparison dynamics and a controlled long-time limit are additional requirements for particle scattering (`model/source_charge_scattering`, [1]).

Local creation, positive transfer and the scattering comparison. The actual canonical translations shift the first configuration while fixing the second. Their strong continuity, the configuration rotation symmetry and Fubini's theorem imply absolute continuity of every finite independent joint configuration law. No second canonical momentum is assumed. Nonzero polynomials in distinct bounded charge creators have strictly positive vacuum norm; their finite Gram matrices are nondegenerate. This does not supply a tensor-product particle inner product.

Cutting off an annular test in physical space gives compactly supported bounded charge creators. A direct canonical-commutator estimate retains original joint-band probability $9/200000000$ for each charge, with $r/2 \leq |P| \leq r$ and $ar \leq L \leq br$, where $d = 3m_0/1000$, $a = 3d^2/2$, $b = 2/d^2$ and $m_0 = 1/10$. Time filtering with the same source clock gives C_+, C_- of norm at most $1/2$ and positive energy transfer in $[ar/2, 2br]$. Analytic Fourier norm bounds fix a scale-independent normalization M; the inner-band probability is at least $p = 9/(200000000M^2)$. The positive square-root complement completes the three-record instrument. Both operators raise energy and are exchanged by charge conjugation; they are not each other's adjoints.

The inherited causal cone bounds their separated commutators and time derivatives. Applying Jensen's inequality to the spatially averaged positive effect, and controlling the noncommuting product by those bounds, gives equal-sign n-charge preparations of original probability at least $p^n/4$. All 3^n records remain. The selected n-charge branch has its entire energy in $[nar/2, 2nbr]$; the complementary records need not lie in that interval. Every nonzero integer internal charge therefore has positive spectrum approaching zero. This statement does not identify each charge with an elementary massless species.

For a proposed comparison dispersion $w(k)$, the actual single-packet derivative norm squared is

$$\int |E - w(k)|^2 |\widehat{f}(k)|^2 \, d\mu_{C\Omega}(E, k).$$

It is independent of time. A fixed broad energy band alone cannot make this defect integrable on an infinite time tail. The exact product derivative estimate separates this single-packet term from the controlled inter-packet commutators. Minimizing over measurable w leaves the conditional energy variance at fixed momentum. Narrowing a filter onto a zero-weight graph loses its original success probability; a preparation with positive variance can nevertheless contain a distinct sharp subbranch. Source spectral branches, comparison-space normalization and long-time estimates are additional scattering inputs (`model/spin_multipacket_comparison`, [1]).

Spatial quantum numbers and a common matter clock. The actual proper cubic site rotations preserve the selected source and act on its charged fields by scalar test-function pullback. The binary-cubic central two-pi element consequently acts trivially. This does not determine a

continuum spin: the harmonic polynomial $x_1^4+x_2^4+x_3^4-3|x|^4/5$ is a cubic singlet in spin four. The local spin creators are even full-source fermion bilinears and commute on disjoint sites. A physical spin-statistics assignment requires their continuous spacetime representation and spectral-sector attachment.

The centered Hubbard ground has interacting local CAR histories. Finite projectors admit algebraic preparation; pressure secants certify infinite-ground scores. At $U/c = 8$ its symmetric one-site density is $\mathrm{diag}(d, 1/2-d, 1/2-d, d)$, $1/90000 < d < 233/1600$. Use its proper-cubic average and actual magnetic translations. The composite fields $b = 2(n_{\mathrm{opposite}}-1/2)a$ give $L_{\mathrm{ret}}V_0 = V_0h + (U/2)V_1$. Three local ground correlations determine

$$C(k) = \kappa h(k), \quad G(k) = [g_0 + 2g_2 \sum_j \cos(2k_j)]I, \quad R^*R(k) = G(k) - \kappa^2 h(k)^2,$$

Here $h = c\sum_j \cos k_j \Gamma_j$ and $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}I$. The same algebra gives $R^*R \geq 18c^2/125$ on $S = \sum_j \cos^2 k_j \leq 3/250$. Finite-range moments and their uniform exponential bounds define a unique spectral measure at every momentum, weakly continuous in local Bloch charts. At $k_* = (\pi/2)^3$, magnetic and proper-cubic symmetries, inherited site inversion and particle-hole symmetry give one even scalar measure on all sixteen components. All eight transported valley probes share it. Its zero residue is $m_0/(\alpha^2+m_0)$, with $m_0 I = R^* 1_{\{0\}}(L_\perp)R$ and $\alpha = U/2$. Write $Q_{xy} = S_x \cdot S_y + \eta_x \cdot \eta_y$ and $r = 6^{-1/2}\sum_{|y|_1=1} a_y$. The full-CAR stars $12Q_{0r} \pm \sum_{j,\epsilon=\pm1} Q_{0,2\epsilon e_j}$ have minima $(-13-\sqrt{577})/4$ and $(-11-\sqrt{673})/4$. Thus $G(k) \leq \Gamma c^2 I$, $\Gamma = (31+\sqrt{577})/4$. At $U/c = 8$, the nodal atom and every packet's ballistic fixed-window limsup are at most $\Gamma/(16+\Gamma)$. The same source has exact local sixth and eighth moment observables, reduced by its state symmetries to 493 and 27965 Hermitian Majorana words. In units $c = 1$, write $A = 16$, $H = A + s$, $s = \|R\|^2$, $t = \|L_\perp R\|^2$ and $u = \|L_\perp^2 R\|^2$, their moments are $M_6 = A(H^2+t)$ and $M_8 = A(H^3+2Ht+u)$. For $u > 0$ the atom is at most $(su-t^2)/(Hu-t^2)$. The selected-ground moment values are undetermined (`model/source_family_compatibility/NODAL_NEXT_MOMENTS.md`, [1]).

On this fixed normalized ground under $c = v/a$, positive nodal atom is equivalent to positive original weight in a bounded physical-energy window for packets with $\|k-k_*\|_1 \leq aK$. The window includes zero; no dispersion follows. For $U/c \to \infty$ with $c = v/a$, original fixed-energy weight vanishes (`model/source_family_compatibility`, [1]).

With this fixed reference, $F_a(p,B) = \nu_{k_*+ap}(aB/v)$ obeys $F_a(\lambda p, \lambda B) = F_{\lambda a}(p,B)$. Set $d\mu_a = dp\,\mathrm{tr}(dF_a)/16$. A full joint limit has $\mu(\lambda A) = \lambda^3\mu(A)$. Local finiteness excludes positive-weight nonzero mass shells in that limit. Whole geometric-refinement convergence only forces discrete mass dilates. Resonances and different source trajectories require separate analysis.

The default cube has $3I_{16}/80 < R^*R < 61I_{16}/1000$, without a species assignment. On the separate weak-contact path $c = v/a$, $0 \leq U = o(c^{2/7})$, eight kinetic nodes yield free Weyl fields with winding $-(-1)^{|b|}$ and opposite conjugate helicities. Period-two masses require supplied coefficients (`model/source_family_compatibility/KINETIC_NODE_MAP.md`, [1]). Complete interference instruments read Lorentzian scores with explicit errors. A native 8194-record dimer execution retains original spectral weight above 0.595 in $[0,1]$; infinite-ground pressure certificates require a strict ground margin [8]. The cube's 256 clock records resolve profiles $f_{k,s}(x) = (-1)^{k\cdot x}/\sqrt{8}$; Table 8 gives original addition weights. Conjugation maps removal to addition at $k \oplus (1,1,1)$. All complements remain; profile labels and conditional means do not identify momentum or rest mass (`model/source_family_compatibility/CUBE_SPECTRUM.md`, [1]).

The signed action also has an invariant aligned-hole Fock sector with bands $\epsilon_\pm(k) = U/2 \pm c\sqrt{\sum_j \cos^2 k_j}$, with the actual π-flux translations and complete vacancy position detectors. A prescribed band-minimum limit gives $cP^2/(2\sqrt{3})$ and the operational position–momentum uncertainty

k_3	Original addition weight	Conditional mean E	Window weight
0	$[0.583370, 0.584440]$	$[0.479760, 0.482820]$	≥ 0.466330
1	$[0.415560, 0.416630]$	$[0.522980, 0.527380]$	≥ 0.335250

Table 8: Cube source energies $E = K - E_0$ at $c = 1/8, U = 1$. The last column bounds original weight in $[0.125, 0.935390]$, including finite clock resolution. Conditional variances are at least 0.009860 and 0.008300. Coherent services are supplied; the full tape is not executed. Phase records are not sharp energy projectors; no physical masses are assigned.

relation. This free nonrelativistic sector does not describe the full interacting ground or identify a physical species (`model/source_family_compatibility/ALIGNED_HOLE_DISPERSION.md`, [1]).

The internal export matches fifteen source weights, including color conjugation, while eighteen comparison channels retain distinct family, defect and mass fields. Shared family blocks and neutrino flavour labels do not establish mass eigenstates. Physical assignments and mass predictions are empty (`model/standard_model_source_map`, [1]).

Particle identification requires preparation, controlled charge transport and complete energy, momentum and interaction readouts on one dynamics. The memory charge, compact texture and gauge–Higgs response have distinct carriers and actions; their separate constructions do not establish that map.

7.16 A quartic interaction on the complete compact band

The spatial source is the cubic operator $h = T + Mn \cdot \sigma$, $\|T\| < 2$, $M \geq 8$. Its direction is the normalization of $(x, y, z^2 - a^2)$ inside the cube of radius $a + 1$ and a constant outside, with zero amplitude at the two cores $(0, 0, \pm a)$. The four-band bound is $r = b^2/(M - 4)$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. The spectral and particle–hole identities hold for this source by the compact-pair construction. Let W contain the four localized modes and set $A = W^\dagger h W$. The isometry $E = W \otimes I_{45}$ embeds 180 annihilators into the complete finite field. The current and pairing intertwiners hold for every complete memory record, with A retained in the quadratic core. Write $P = EE^\dagger$ and choose

$$V = \lambda(N_W - 90)^2, \qquad N_W = d\Gamma(P), \qquad \lambda \geq 0. \tag{12}$$

This is an explicit controller design. The particle–hole identity $C\overline{W} = WS$ ensures that the pairing has no localized/complementary cross term. All terms are even, so $H_r^{\text{full}} = H_{r,W} \otimes I + I \otimes H_{r,\perp}$ on every record. The interaction is parity even and commutes with all twelve native internal currents. In the localized occupation basis it contains the quartic term $2\lambda \sum_{i<j} n_i n_j$. The pairing Hamiltonian does not conserve N_W, and its commutator with V need not vanish. Every record preserves the same localized algebra because V acts entirely on that factor.

Hold the texture fixed throughout the memory averaging interval. The complete record limit has a bound independent of exterior volume. Writing the compact energies as $\{-e_2, -e_1, e_1, e_2\}$ with $e_i \leq r$, the kinetic norm is $45(e_1 + e_2) \leq 90r$. If g is the source feedback gain, the pairing bound and $0 \leq N_W \leq 180$ give

$$\|H_{r,W}\| \leq M_W = 90r + 90(1 + g) + 8100\lambda.$$

The radial two-column zero-mode construction is a separate special case with ninety annihilators, centre 45 and bound $45(1 + g) + 2025\lambda$. The actual memory transition has the reset decomposition

$T = (1-p)R + pS$, and contracts zero-sum operator-valued memory blocks by at most p. At step size t/n, its reduced core state consequently satisfies

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W}\rho_W(0)e^{it\bar{H}_W}\|_1 \le \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M_W^2 t^2}{n}. \tag{13}$$

The estimate includes arbitrary initial classical–quantum memory correlations and a finite untouched reference. The first term vanishes for stationary product memory. The mean Hamiltonian contains the same memory-derived mass matrix and the interaction (12). Uniformity concerns this fixed localized algebra under its source isometries; it gives no uniform continuum estimate for all spatial observables.

Equation (13) compares the unread core. There is a separate estimate for the full native history register. For a fixed initial memory label, purify the core/reference state. Write p_ω for the field-independent history probability, $\psi_\omega = U_\omega\psi$, and $\bar{\psi} = e^{-it\bar{H}_W}\psi$. If δ is the unread trace-norm bound, then $\sum_\omega p_\omega(1 - |\langle\bar{\psi}|\psi_\omega\rangle|^2) \le \delta/2$. The pure-state distance identity and Jensen's inequality bound the full history/core trace distance from $\sum_\omega p_\omega|\omega\rangle\langle\omega| \otimes |\bar{\psi}\rangle\langle\bar{\psi}|$ by $\sqrt{2\delta}$. Partial trace covers mixed inputs. For initial classical memory/core correlations, condition on the initial memory label and retain that tag in the ideal history law; convexity preserves the same bound. The complete final record/core rate is therefore $O(n^{-1/2})$. Opposite qubit phase records saturate this square-root relation. No uniform estimate for rare conditioned histories or intervening field measurements follows.

The interaction is defined through a spectral projector and need not be strictly onsite. For the radial special case, the resolvent estimate at texture coefficient 64 and $W_x^\dagger W_x = \rho_x I_2$ gives, for the projection A_L onto $|x|_\infty < L$,

$$\|(I - A_L)E\|^2 \le \epsilon_L = \frac{3}{2\cdot 62^2}\, 32^{-2(L-1)}.$$

Thus $\|P - A_L P A_L\| \le 2\sqrt{\epsilon_L}$, and finite rank gives $\|N_W - d\Gamma(A_L P A_L)\| \le 360\sqrt{\epsilon_L}$. The radial interaction $\lambda(N_W - 45)^2$ is exponentially localized about the prescribed origin, without being a strictly onsite interaction.

On this radial subspace there is an exact obstruction for the onsite density alternative. On the two-particle sector let $D_g = \sum_x g_x \bigwedge^2 P_x$ and $E_2 = \bigwedge^2 E$. Direct compression gives

$$E_2^\dagger D_g(I - E_2E_2^\dagger)D_g E_2 = \left[\sum_x g_x^2\rho_x^2 - \left(\sum_x g_x\rho_x^2\right)^2\right] I. \tag{14}$$

Writing $c = \sum_x \rho_x^2 < 1$, Cauchy–Schwarz bounds the bracket below by $(1-c)\sum_x g_x^2\rho_x^2$. Every nonzero member of this onsite density class leaks from the source subspace on which it has support. This statement does not exclude a dressed low-energy subspace or other local interaction tensors.

Connected dynamics and spectral shifts. The compact mean pairing operator has 84 charged two-mode blocks and twelve unpaired neutrino modes. Select the two spatial pairs of one charged-lepton singular value m. Fix one occupation in each of the other 82 blocks and fill six neutral modes. Pair creation and annihilation kill these spectators, whose number is 88. Subtracting their fixed energy gives the exact reducing four-mode Hamiltonian

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2a_1 + a_3^\dagger a_4^\dagger + a_4a_3) + \lambda(N_4 - 2)^2.$$

This is an invariant sector of the mean Hamiltonian, not its asserted vacuum or an invariant sector of every finite memory pulse. Pair parities split its sixteen levels into the four even/even values $0, 4\lambda, 2\lambda \pm 2\sqrt{\lambda^2 + m^2}$, four odd/odd values $\pm e_1 \pm e_2$, and eight mixed values $\lambda \pm e_i \pm m$. Consequently

$$E_0 = \min\{2\lambda - 2\sqrt{\lambda^2 + m^2}, -e_1 - e_2, \lambda - m - \max(e_1, e_2)\}.$$

This enumeration is a complete diagonalization by pair parity. For the unique odd/odd ground, a positive-energy orbital has two addition-plus-removal poles $e_1 + \lambda \pm m$, each of residue $1/2$. At $M = 64$, half-separation one and $\lambda = 1/128$, rational coefficient and exterior-volume enclosures place all three inherited charged-lepton slices in this ground sector:

Quadratic coefficient m	Lower pole	Upper pole
0.000002072030404	0.01171632730	0.01172047136
0.0004284302055	0.01128996912	0.01214682953
0.007205207119	0.004513192209	0.01892360645

The unit is the declared source pulse energy. These are slice poles, without a physical mass assignment or a determination of the full interacting vacuum. The compact exterior error δ_L bounds each displayed pole error by δ_L in this certified sector; no complementary mode is discarded.

The full 180-mode ground also has an explicit variational bracket. List the charged singular values, including colour multiplicities, as m_j, $j = 1, \ldots, 21$. Each occurs in two pairing blocks at each spatial energy e_i, with quadratic levels $\{\pm m_j, \pm e_i\}$. The twelve neutral modes contribute $-3(e_1 + e_2)$, giving

$$E_{\mathrm{quad}} = -2\sum_{j=1}^{21}[\max(m_j, e_1) + \max(m_j, e_2)] - 3(e_1 + e_2).$$

Positivity of V makes this a lower bound. A quadratic product ground has mean occupation 90 and number variance one per even pairing block, zero per odd block. If b blocks are even, its trial energy is $E_{\mathrm{quad}} + \lambda b$. A completely blocked trial state has exact occupation 90, zero interaction energy and kinetic energy $-45(e_1 + e_2)$. Thus, retaining every localized mode,

$$E_{\mathrm{quad}} \le E_{\mathrm{full}} \le \min\{E_{\mathrm{quad}} + \lambda b, -45(e_1 + e_2)\}.$$

The specified coefficient witness has $b = 28$. These bounds constrain the full vacuum energy without identifying its excitation spectrum.

The radial control sets $e_1 = e_2 = 0$. Its ground energy is $2\lambda - 2\sqrt{\lambda^2 + m^2}$, and its single-mode spectral poles and weights are

$$\omega_\pm = 2\sqrt{\lambda^2 + m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2 + m^2}}{2}.$$

In this radial control the cross-pair density cumulant is $-\lambda/(4\sqrt{\lambda^2 + m^2})$, while its normal and anomalous cross-pair two-point contractions vanish. Hence positive λ gives a non-Gaussian connected interaction. These spectral shifts preclude identifying the quadratic mass coefficients with the interacting excitation energies without further calculation.

A common finite energy and modular response. For the same full localized Hamiltonian choose $\rho_\beta = Z^{-1}e^{-\beta\bar{H}_W}$, $\beta > 0$. This state is faithful. A global spectral reset with Kraus operators $\sqrt{p_i}|i\rangle\langle j|$ prepares it and preserves total probability. That reset is a declared service, not an implementation by the original site repair. For $H(s) = \bar{H}_W + sX$, Duhamel differentiation gives

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X\rangle)\rho_\beta^{1-u}\, du.$$

The positive logarithmic-mean kernel yields $F' = \langle X\rangle$, $F'' = -\beta C_{\rho_\beta}(X, X) \le 0$, and $S' = \beta(E' - \langle X\rangle)$, where C_ρ is the Bogoliubov–Kubo–Mori covariance. The noncommuting choice $X = (N_4-2)^2$ is checked against an independent matrix-exponential derivative. These are finite identities for one nonquadratic Hamiltonian. They supply no geometric modular flow, local relativistic stress, area normalization or geometric entropy stationarity, and therefore imply no Einstein dynamics. Exact characteristic polynomials, complete finite spectra, record controls and the onsite leakage calculation accompany the proof.

Intervening observations. The bounds above extend to adaptive experiments. On the compact 180-mode core at $\lambda = 1/128$, every record's localized Fock generator obeys $\|H_{r,W}\| \le M_W \le 7869/32$. Between m evolution intervals, apply arbitrary complete quantum instruments to the field and any auxiliary observer register, chosen from every retained native and detector record, with all failure outcomes kept. Replacing actual interval channels by ideal mean-dynamics channels one at a time bounds the complete-history distance by $\sum_j \min\{2, \sqrt{2\delta_j}\}$, where δ_j is the unread bound of interval j. The native memory continues under its actual transition law, since conditioning does not justify resetting it to a stationary product. Exact resource schedules and a complete finite execution check the correspondence. The bounds concern a declared temporal refinement, and adapting at every microscopic step needs its own budget.

7.17 Work on the moving compact interacting band

A deformation must act on the same Hamiltonian used to identify the matter states. On the compact opposite-degree texture, vary the actual hopping by $h_s = sT + MV$, with $1/2 \le s \le 3/2$, $M \ge 8$ and $\|T\| < 2$. The texture, memory coefficients and quartic coupling are fixed. The four-dimensional spectral band has energies in $[-3, 3]$ and separation at least $M - 6$ from its complete complement. Let P_s be its projector, C the source particle-hole matrix, and K_ω the internal pairing kernel on a complete memory record ω. The same source law is

$$N_s = \mathrm{d}\Gamma(P_s \otimes I_{45}), \qquad H_{\omega,s} = \mathrm{d}\Gamma(h_s \otimes I_{45}) + \mathrm{pair}(\mathsf{C} \otimes \overline{K_\omega}) + \lambda(N_s - 90)^2.$$

Its exact force conjugate to the hopping control is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P'_s \otimes I_{45})\}. \tag{15}$$

The second term follows by differentiating the noncommuting square. Freezing the localization projector omits this contribution to source work.

Proposition 7.2 (Localized force with the moving band retained)**.** *The band admits a unitary comparison frame W_s in which its pairing form is constant and its intrinsic force is*

$$J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45}), \qquad \|J_s\| \le 360, \qquad \|P'_s\| \le \frac{6}{M-6}.$$

For two even localized states with the same even complement reference, their contrast of (15) *equals their contrast of J_s. The bounds are independent of the containing finite cube.*

Proof. For band vectors p and complementary eigenvectors q, $\langle q, P_s' p\rangle = \langle q, Tp\rangle/(E_p - E_q)$ and the diagonal blocks of P_s' vanish. Hence $\|P_s'\|^2 \le \|P_s'\|_{\mathrm{HS}}^2 \le 8\|T\|^2/(M-6)^2 < 36/(M-6)^2$. Set $L_s = [P_s', P_s]$ and solve $U_s' = L_s U_s$. Then $P_s = U_s P_1 U_s^*$ and $W_s = U_s W_1$. Particle-hole covariance gives $\mathsf{C}\overline{U_s} = U_s\mathsf{C}$, so the pairing form is fixed. Differentiating $A_s = W_s^* h_s W_s$ gives $A_s' = W_s^*(T + [h_s, L_s])W_s = W_s^* T W_s$: h_s is block diagonal and L_s off diagonal. Number and pairing therefore have no derivative in this transported frame. The 180-mode second-quantization norm is at most $180\|T\| < 360$. In an even product state the cross-band terms have zero expectation; subtracting the same complement reference cancels its remaining energy. □

At fixed s, a complete-memory trace-norm error δ consequently gives a force-readout error at most 360δ. The comparison transporter is an explicit mathematical identification; Hamiltonian evolution with changing s does not implement it automatically. For the supplied Gibbs reference of the same mean Hamiltonian at fixed $\beta > 0$, Duhamel differentiation gives

$$F'_{\mathrm{thermal}} = \mathrm{Tr}(\rho_s J_s), \qquad S' = \beta\big(E' - \mathrm{Tr}(\rho_s J_s)\big).$$

The complete compact source and a reducing sixteen-state interacting slice check these derivatives, retaining the spatial splitting and work performed by the hopping control. The hopping amplitude is a declared control. Its work derivative supplies no identification with a metric variation or conserved Lorentz stress, and this finite Gibbs identity supplies no regional geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

7.18 A finite gauge–Higgs response with complete detector records

A separate two-vertex control joins leptonic fermion hopping, two oppositely oriented native 60-state gauge links and the charged-lepton Higgs tensor. There are three families and two spin components. The scalar at vertex zero has vacuum or one conjugate-doublet quantum; colored fermions have zero occupation and the up/down vertices are disabled. Colored link sectors are retained. The conserved quantity is $Q_* = N_F - 2N_H$, and Gauss constraints imply $N_L = N_H$, $N_e = Q_* + N_H$ and bulk hypercharge $6Y = 6Q_*$. A reference of charge $-6Q_*$ makes the complete state neutral. The scalar cap defines this finite model, rather than an invariant truncation of an unrestricted bosonic field.

For F families the complete physical dimensions are

$$d_0 = 6 + 48F^2, \qquad d_1 = 16F + 64F^3 - 20F^2.$$

At $F = 3$ these are 438 and 1596. Exact Gauss kernels, independent representation counts and the full operator images verify completeness and zero complement leakage. The charged scalar-vacuum subspace has dimension 48, with its entire 1548-dimensional occupied complement retained. Writing $A = PHP$, $C = QHP$ and $D = QHQ$ gives

$$P(z-H)^{-1}P = [z - A - \Sigma(z)]^{-1}, \qquad \Sigma(z) = C^\dagger (z-D)^{-1} C.$$

For $\mathrm{Im}\, z > 0$, $-\mathrm{Im}\,\Sigma(z) \succeq 0$. In the declared unit-strength control, $\|\Sigma(1+0.5i)\| \simeq 1.793879153064$ and the hopping/Higgs commutator is nonzero. This verifies interaction in the finite system.

Let $H_0\Omega = E_0\Omega$ be its prepared neutral ground and $b_j = c_j^\dagger d_{\mathrm{ref}}^\dagger$ the six family/spin creation maps. The odd reference makes b_j even. Each is a partial isometry, and the full signed response is

$$C_{jk}(t) = \langle\Omega| b_j^\dagger e^{-i(H_1 - E_0)t} b_k |\Omega\rangle = \sum_n e^{-i(E_{1n} - E_0)t} \langle b_j\Omega|n\rangle\langle n|b_k\Omega\rangle.$$

Every spectral row and zero residue is retained. For the specified family matrices, unit dynamical strengths and scalar energy, and zero fermion/reference energies, the first three distinct numerical addition frequencies are $2.58004293348 \times 10^{-8}$, 0.001006653510723 and 0.452014700447579. Exact-input 50/80-digit residual refinement supports the tiny positive first difference; it supplies no interval-certified enclosure or ordering.

These frequencies enter a complete measurement, not a postselected creation curve. On the direct sum of the two sectors set

$$G_j = \begin{pmatrix} 0 & b_j^\dagger \\ b_j & 0 \end{pmatrix}, \qquad W_{s,j} = I - G_j^2 - isG_j, \qquad s = \pm 1.$$

Since $G_j^3 = G_j$, $W_{s,j}$ is unitary. With $U(t) = e^{-it(H_0 \oplus H_1)}$, a detector ancilla compares the forward paths $W_{+,j}U(t)$ and $U(t)W_{s,k}$. Their overlap is $F_{s,jk} = A_{jk} + sC_{jk}$, where $A_{jk} = \langle a_j | e^{-i(H_0 - E_0)t} | a_k \rangle$ and $a_j = (I - b_j^\dagger b_j)\Omega$ retains blocked creation. The two detector bases give

$$p_\pm^X = \tfrac{1}{2}(1 \pm \operatorname{Re} F_{s,jk}), \qquad p_\pm^Y = \tfrac{1}{2}(1 \pm \operatorname{Im} F_{s,jk}), \qquad C_{jk} = \tfrac{1}{2}(F_{+,jk} - F_{-,jk}).$$

The maps $(W_{+,j}U \pm UW_{s,k})/2$ and $(W_{+,j}U \mp iUW_{s,k})/2$ retain both outcomes and their back-action on arbitrary inputs, including purifications. The controlled ancilla, quarter-turn probes and ground preparation are supplied services.

For independent prepared shots at times t_i, compare H_1 with $H_1 - \delta\Pi$, where Π is a spectral projector. The two pre-readout pure states have overlap $z = 1 + (e^{i\delta t} - 1)w/2$, $w = \|\Pi b_k \Omega\|^2 \le 1$, hence $1 - |z|^2 \le \delta^2 t^2/4$. Multiplication of fidelities and contraction under the complete measurement give the record-only bound

$$\mathrm{TV} \le \min\left\{1, \frac{|\delta|}{2}\sqrt{\sum_i t_i^2}\right\}.$$

The supplied schedule has 24×4096 shots at each $t = i/10$, $0 \le i \le 128$: 12,681,216 retained bits. For $|\delta| \le 3 \times 10^{-8}$ the bound is strictly below $1/2500$. An observer who reads the simulator specification can have further information excluding that comparison Hamiltonian. Applying this bound to an exactly vanishing smallest line also requires an enclosure absent from the numerical precision ladder.

Finally, $\epsilon_F N_F + \epsilon_H N_H = \epsilon_F Q_* + (2\epsilon_F + \epsilon_H)N_H$. Changing (ϵ_F, ϵ_H) by $(\delta, -2\delta)$ shifts addition frequencies if the reference energy is fixed. A common gauge-potential shift changes the compensating reference by $-\delta$ and cancels exactly in the complete Hamiltonian. This is a reference-calibration distinction, not freedom to fit intrinsic masses. The graph, scalar cap, couplings, preparation and physical clock are not selected by these finite identities. Full certificates and record instruments are in the repository package `model/interacting_extension` [1].

Native state and carrier bounds. The 66-dimensional occupation factor ρ_* of Section 7.15 has eigenvalues 625/7505, 200/7505 and 64/7505, with multiplicities $1, 20, 45$. Its ideal modular commutator, divided by $\log(25/8)$, obeys $D(D^2 - I)(D^2 - 4I) = 0$. Any injective target-algebra map into this source that intertwines the two derivations must preserve that polynomial identity. This is a condition on an exact common clock, without a numerical exclusion of the Higgs response lines. The full ρ_α also carries its reference-dependent memory gap.

An isometric image of ρ_* retains its 66 nonzero eigenvalues, so it has trace distance at least $1 - 625/7505 = 1376/1501$ from every pure target. A pure field subsystem is possible if other retained quantum memory carries the mixedness; its preparation and records must then be supplied.

Separately, if an entire quantum channel crosses a single 66-dimensional carrier, arbitrary classical records cannot raise its entanglement fidelity above $66/D$ on a D-dimensional input. With no additional quantum path or entanglement assistance, its half-diamond distance from the identity is therefore at least $62/73$ for $D = 438$, or $255/266$ for $D = 1596$. The bound concerns all inputs and references, rather than one prepared ground-state experiment. The named observer's coherent work register is an additional quantum resource. These exact tests rule out direct identifications while leaving explicit active preparations and larger encodings to their own comparison theorems (`model/source_state_bridge`, [1]).

7.19 Compensated probe uncertainty and collision records

The six creation maps above also supply an exact uncertainty relation on the complete finite carrier. For any one map b, set

$$P_0 = b^\dagger b, \quad P_1 = bb^\dagger, \quad P = \mathrm{diag}(P_0, P_1), \quad D = \mathrm{diag}(P_0, -P_1),$$

$$X = \begin{pmatrix} 0 & b^\dagger \\ b & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -ib^\dagger \\ ib & 0 \end{pmatrix}.$$

Each normalized b is a signed partial permutation of rank 384. Consequently $X^2 = Y^2 = D^2 = P$, $[X, Y] = 2iD$ and $\{X, Y\} = 0$. For either $A = X$ or Y, the spectral projectors $(P - A)/2$, $I - P$ and $(P + A)/2$ have ranks 384, 1266 and 384. The zero outcome retains 54 neutral blocked states and 1212 charged states outside the range. These identities are checked in exact rational complex arithmetic for all six source maps.

For any density supported on the neutral sector, including a reduced density entangled with an arbitrary reference, let $w = \mathrm{Tr}(\rho P_0) = 1 - \langle n_j \rangle$. Both means vanish, both variances equal w, and

$$(p_{-1}, p_0, p_{+1}) = (w/2, 1 - w, w/2), \qquad \Delta X \, \Delta Y = w = \tfrac{1}{2} |\langle [X, Y] \rangle|.$$

Thus the supplied finite density saturates Robertson's inequality [23]. The tracial neutral state gives $(32/73, 9/73, 32/73)$. The complete maps $\rho \mapsto E_a \rho E_a$ preserve the untouched reference marginal when outcomes are summed. These are exact projective instruments, distinct from the two-outcome Ramsey readout; a native pulse compiler for them is not asserted. The density and complex measurement semantics are those of the specified finite Higgs source.

These even charge-compensated quadratures do not identify physical position and momentum. Finite matrices satisfy $\mathrm{Tr}[Q, P] = 0$, hence $\|[Q, P] - i\hbar I\| \geq |\hbar|$ for nonzero $\hbar$. The source's separate fluctuation and ordered-spin constructions obtain canonical field limits on their stated states and domains; the number of independently prepared fluctuation copies is not the physical memory capacity. The ordered field's Gram form constructs a position conjugate to its existing spatial translation generator, on an explicit field-vector domain. A complete extension includes a no-position outcome. Declared coherent source services implement complete finite detectors, and finite Gram compressions give ordered convergence on fixed states and references. This gives no effective source-size modulus or moment-tail bound. A larger response space, generated by all time-evolved field vectors, admits a covariant position measure and a time-invariant dark complement. It need not be a particle sector. Protected total-load coordinates are scalar and cannot supply that position. Finite uncertainty alone selects no Planck scale or particle interpretation [8]. The proof, full instruments and source bindings are in the `model/particle_identification` package [1].

Capacity and complete local experiments. The native interpreter has an exact record-capacity result. Under its `export_oldest` policy, capacities one through eight give the same complete experimental channel after reversible resegmentation of archive and queue, with carrier, commands, parameters and evaluator limits fixed. Each adaptive selector reads only the last retained record; export preserves it. Induction over raw branches therefore gives zero channel distance, including arbitrary references and resource-stop outcomes. The archive is part of the output; readable capacity specifications and resource costs differ. With the `halt` policy, two holds at capacities one and two instead have disjoint timeout/completion statuses. This theorem concerns record slots, not physical global capacity or a capacity-dependent particle action; its proof and exact controls are in `model/particle_identification/NATIVE_CAPACITY.md` [1].

For the specified full-Fock Weyl/contact action at fixed spacing, nested local buffers give a stronger comparison than agreement of a few frequencies. Let ϵ_0 bound the initial trace distance of the joint largest-buffer, observer and reference state. At stage i, let δ_i be the complete operator locality error, η_i the unitary splitting error, and ζ_i the half-diamond error of the complete instrument, uniform over incoming states, references and previous records. For target (t_i, c, U) and simulator (s_i, c', U'), let E_i, V_i count buffer edges and sites. Then

$$a_i = E_i|t_i c - s_i c'| + V_i|t_i U - s_i U'|, \qquad D_{\mathrm{final}} \le \min\left\{1, \epsilon_0 + \sum_i (\delta_i/2 + \eta_i + a_i + \zeta_i)\right\}.$$

Here D_{final} is trace distance on the entire retained local quantum state and classical record. Each edge has two hopping terms of norm $c/2$; each site has a contact projector. Duhamel's formula bounds the local action change by a_i, and complete-channel contraction telescopes all stages, including adaptive instruments and failure records. Durations are fixed by stage. The bound has no ambient-volume factor, but comparison across physical capacities requires a common local identification and controlled couplings, clock, joint input and instruments. No such capacity-to-action map follows from volume independence. Nor does convergence at fixed time imply control of arbitrarily small spectral gaps.

Spatial scattering and physical identification. The contact source has complete two-body lattice wave operators and a unitary scattering matrix on its absolutely continuous subspace. Critical tuning of the moving-contact family gives a nontrivial two-body continuum limit in three dimensions, with all eight edge valleys retained. This limit is nonrelativistic. Its separated bound pairs prevent identifying the empty additive Fock state as a ground vacuum. The accompanying packet construction controls incoming and outgoing time tails, packet truncation, preparation, finite volume, splitting and native readout, and separates a retained interaction event from its matched free law by more than 3/10. That fine-spacing finite schedule is constructive; the fine-grid incoming-packet experiment is unexecuted. A separate four-mode contact echo at $c = 1/8$, $U = 1$, $t = 4$ executes 22,531 native pulses and yields pointer signal about 0.046915; the ideal-word bound exceeds 0.0361998125, compared with zero for the matched free control. Floating and hardware errors are separate from that rational bound. The executed five-qubit control retains all 32 source/reference rows. These interaction echoes are not executed asymptotic collisions. The respective packages are `model/contact_scattering`, `model/moving_contact_scattering` and `model/scattering_packet_readout` [1].

A collider identification requires one source-selected state, spatial action, clock and detector family to join these ingredients. Separated incoming packets, their interacting propagation and complete outgoing bins must share quantitative errors, including soft and failed events. The effective theory must control its relevant relativistic symmetries, regulator sectors and resolution window.

An infinite ultraviolet limit is not required merely to compare a finite-resolution experiment, but a free dispersion relation cannot replace the interacting record law. Stable neutral shells, charged inclusive thresholds, resonance line shapes and scheme-dependent confined parameters then require their respective physical definitions, as described below.

7.20 Quantum electric sectors and their stationary freedom

The configured native link has dimension 60: an invariant vacuum and all five spaces $V_r^* \otimes V_r$, with $d_r = 1, 2, 3, 3, 6$. Its matter and static dual reference admit ten exact Gauss singlets $a_r = A_r/\sqrt{d_r}$ and $b_r = B_r/d_r$. Direct action of the ambient Hamiltonian and all twelve current generators gives

$$H_r = \begin{pmatrix} 0 & \sqrt{d_r} \\ \sqrt{d_r} & 1 \end{pmatrix}, \qquad U_r = (I + ihH_r/2)^{-1}(I - ihH_r/2), \quad h = 1/8.$$

This is the declared native Cayley service. Its complete position instrument has operators $P_a U, P_b U$, retaining both outcomes and arbitrary reference inputs. On this invariant carrier the actual weak Casimir gives $P_{1/2} = |b_2\rangle\langle b_2| + |b_6\rangle\langle b_6|$ and $P_0 = I - P_{1/2}$. These are weak SU(2) electric sectors.

The adjoint image $U^* P_{1/2} U$ has nonzero off-diagonal entries and therefore does not lie in $\mathrm{span}(P_0, P_{1/2})$. The states $(a_2 \pm b_2)/\sqrt{2}$ have identical initial electric probabilities and distinct next electric records. Even diagonal states in the same $j = 0$ sector can have different exits. Retaining the family and endpoint labels instead gives the exact closed kernels

$$K_r = \begin{pmatrix} 1 - p_r & p_r \\ p_r & 1 - p_r \end{pmatrix}, \qquad p_r = \frac{1024 d_r}{(d_r + 256)^2 + 256}.$$

The full stationary quantum family is $\bigoplus_{d=1,2,3,6} M_d \otimes I_2/2$: the $d = 1, 2, 6$ matrices are nonnegative scalars, M_3 is any positive 2×2 matrix on the two distinct $d = 3$ copies, and $\sum_d \mathrm{tr}\, M_d = 1$. All twenty-five fixed-block equations are checked exactly. The fixed operator space has dimension seven, including the surviving copy coherences. The centre balance is

$$\langle (-1)^{2j} \rangle = 1 - w_2 - w_6 \in [0, 1].$$

It is $3/5$ for a declared tracial preparation and tends to zero for the native a_6 preparation. These are conditional inputs to the imported scale formula $v = E_{\mathrm{cell}} b \langle (-1)^{2j} \rangle$; the channel selects neither the family weights nor a physical coupling. The exact calculation is in `model/quantum_sector_measure/PROOF.md`.

7.21 Conditional neutrino mass norms with complete phase records

The calibrated rank-two Weinberg kernel is $M_\nu = \overline{U}\, \mathrm{diag}(0, m_2, m_3) U^*$. Its actual weak-current mass frame supplies the weights $|U_{ei}|^2$. Consequently

$$m_\beta^2 = (M_\nu^* M_\nu)_{ee} = \sum_i |U_{ei}|^2 m_i^2, \qquad m_{ee} = |(M_\nu)_{ee}|.$$

For the retained normal-ordering packet,

$$\begin{aligned} m_\beta &= 8.8993443986 \times 10^{-12}\ \mathrm{GeV}, \\ m_{ee} &\in [1.4941387506, 3.7470789224] \times 10^{-12}\ \mathrm{GeV}, \\ \sum_i m_i &= 5.8791468982 \times 10^{-11}\ \mathrm{GeV}. \end{aligned}$$

The printed zero Majorana phases give $m_{ee} = 3.2748590233 \times 10^{-12}$ GeV. Writing $a = m_2 s_{12}^2 c_{13}^2$ and $b = m_3 s_{13}^2$, $m_{ee}^2 = a^2 + b^2 + 2ab\cos(\alpha_{31} - \alpha_{21} - 2\delta)$. The two phase endpoints are attained explicitly, while m_β and the oscillation generator remain unchanged.

The actual 180-dimensional zero-momentum Nambu symbol has an invariant six-coordinate neutral block $H_\nu = \left(\begin{smallmatrix} 0 & M_\nu^* \\ M_\nu & 0 \end{smallmatrix}\right)$. The Takagi frame proves that $C = H_\nu/m_3$ is a Hermitian contraction. With $F = \sqrt{I - C^2}$, the block matrix $\left(\begin{smallmatrix} C & F \\ F & -C \end{smallmatrix}\right)$ is unitary. Six success operators $|z\rangle\langle z|C$, one failure F and the identity on the full 174-coordinate spectator complement form a complete eight-record logical instrument. On an electron input,

$$P(\text{success}, \overline{\nu}_\alpha) = \frac{|(M_\nu)_{\alpha e}|^2}{m_3^2}, \qquad P(\text{failure}) = 1 - \frac{m_\beta^2}{m_3^2} = \frac{633281684707}{653906250000}.$$

The complete isometry retains reference correlations; forgetting its records need not give the identity channel. This apparatus reads two conditional mass norms, without a nuclear matrix element, decay-rate calculation, native pulse compilation or source selection of the calibrated masses and phases. The construction is in `model/neutrino_mass_observables/PROOF.md`.

7.22 A coherent native-family word with nonzero CP contrast

Retain the unchanged positive native family coefficient Y and the actual geometric rank-one projectors P_2, P_3, P_{10}. Predeclare the chronological word $(2, 3, 10)$ and pulses $T_k = I + (-i-1)P_k$. The parent's coherent feedback service realizes these pulses on a prepared $X = +1$ auxiliary state. Set

$$V = T_{10}T_3T_2, \qquad Y_u = Y_e = Y, \qquad Y_d = VY.$$

Transport enters the down Yukawa vertex in both Higgs components; it is not a basis change of only half of the weak doublet. All seventy-two internal Higgs Ward identities, the full neutral 45-coordinate kernel and the 90-mode antisymmetric CAR pairing are retained. The native positive singular values remain approximately 0.00206415351504, 0.06365982535695 and 0.93427602112801. Their squared polynomial has strictly positive exact discriminant D. Thus the three families are nondegenerate, and

$$J = \frac{\operatorname{tr}[Y^2, VY^2V^*]^3}{6iD} = -0.0241058965648816\ldots$$

is an exact nonzero element of $\mathbb{Q}(\sqrt{5})$. Complex-conjugating every pulse reverses its sign. All forty words of length zero through three in the declared alphabet are retained as controls; no empirical mass or mixing target selects the primary word.

There is also an exact one-pulse shortcut. In Y's real mass eigenframe, write a normalized native rank-one projector as pp^T and set $a = p_0^2$, $b = p_1^2$, $c = p_2^2$. Direct expansion gives

$$J\big(I + (e^{-i\phi} - 1)pp^T\big) = -2abc\sin\phi(1 - \cos\phi).$$

For the actual port-two projector, the positive product abc is computed without fitting eigenvectors:

$$abc = \frac{\det[\operatorname{tr}(P_2Y^{i+j})]_{i,j=0}^2}{\operatorname{disc}\det(xI - Y)} > 0.$$

Its quarter-turn pulse therefore has nonzero $J = -2abc$ and agrees exactly with the commutator calculation. The identity word has zero contrast, so one pulse is minimal within the prescribed alphabet. This simplifies the declared coherent control without selecting physical quark masses or the measured CKM matrix.

For the ordered real eigenframe F of Y, the complete source mass frame gives $V_m = F^T V F$ in each of the three colour copies of the actual weak current. Its eighteen-record two-current instrument has Kraus rows

$$K_{rs,\pm} = \frac{1}{\sqrt{2}}((V_m)_{r0}(V_m)_{s1},\ \mp i(V_m)_{r1}(V_m)_{s0}).$$

All nine output family pairs and both path-Y outcomes are retained; $\sum K^*K = I$ and $P(01,-) - P(01,+) = J$ on the plus path input. The result supplies a concrete coherent flavour architecture with nonzero CP response. Its unchanged up/down singular spectra do not supply physical quark splittings, a CKM fit or a GeV calibration. The word is a declared addition to the static coefficient service, not an output forced by the unchanged real memory. Exact invariants, full-source frames and complete-record formulas are in `model/native_flavour_transport/PROOF.md`.

7.23 Finite spectroscopy and conditional mass scales

The calibrated two-momentum quadratic source admits a complete thirty-two-outcome phase detector which measures charge, color Casimir, helicity and momentum without first measuring the energy or family label. For the exact common source projectors P_g, a coarse class a and pointer outcome y, its Kraus operator is

$$K_{a,y} = \sum_{g\in a} A_y(\phi_a(E_g))P_g, \qquad A_y(\phi) = \frac{1}{32}\sum_{j=0}^{31} e^{2\pi i j(\phi - y/32)}.$$

The declared phase window is $[1/4, 1/2]$; the range and coherent evolution parameter retain the parent calibration. The complete instrument has 8,010 original probe/detector records, including 6,533 zero weights, the vacuum and the full many-excitation complement. It preserves whole color fibers and coherence between unresolved families. Exact-family lookup before the phase pointer would instead change an unresolved neutrino/reference output by half trace distance approximately 1/2.

The six-bin confidence half-width gives ideal coverage at least 65/72 for every one-excitation input to the phase detector after the original probe, including inputs entangled with untouched references. On the full detector input space, let P_{one} project onto that sector. The spectral failure effect satisfies $0 \le F_{\mathrm{bad}} \le (7/72)P_{\mathrm{one}}$, so bad phase-record weight is at most 7/72 times the original one-excitation weight. Vacuum and many-excitation terminal outcomes carry no mass interval and are retained without conditioning. Empty confidence sets are inconclusive records. For original probe Kraus operators B_b, the complete pre-probe failure effect obeys $\sum_b B_b^* F_{\mathrm{bad}} B_b \le (7/72)\sum_b B_b^* P_{\mathrm{one}} B_b$. The stable identity $E - 1 = m^2/(\sqrt{1+m^2}+1)$ and rational mass-frame intervals avoid binary64 cancellation of neutrino splittings. For all seventeen target types the receipt retains conditional GeV coefficients, their calibration ancestry and sufficient ideal precision plans. The parallel-pointer neutrino plans use $80, 85, 80$ pointer qubits, with a largest coherent exposure approximately $2.431 \times 10^{26}\,\mathrm{GeV}^{-1}$; these plans are not executed. Only the thirty-two-outcome instrument is evaluated, and it does not resolve the three neutrino species. Source-parameter errors, physical clock attachment and hardware precision remain separate from the ideal confidence theorem. See `model/finite_particle_spectroscopy`.

The complete phase instrument also admits an exact sequential realization with one reusable quantum qubit and all n original classical bits retained. Within a fixed coarse class set $U = e^{2\pi i \phi_a(H)}$. Read low bits first, and let $r = \sum_{\ell<j} b_\ell 2^\ell$. The next binary source Kraus operator is

$$K_{b_j|r} = \frac{1}{2}\left[I + (-1)^{b_j} e^{-2\pi i r/2^{j+1}} U^{2^{n-1-j}}\right].$$

For a full word $y = \sum_j b_j 2^j$, the polynomial identity $\prod_{\ell=0}^{n-1}(1 + z^{2^\ell}) = \sum_{k=0}^{2^n-1} z^k$ gives $\prod_j K_{b_j|r_j} = A_y(U)$ exactly. Thus every original source/reference output is preserved, including its quantum coherence; no history is erased or selected. Known-bit feedback recycles the measured pointer while retaining that bit in the archive. All 112×32 source-energy/outcome amplitudes are checked against the full pointer. The executed equivalent law uses one qubit, five bits and five sequential source services. An n-bit precision plan can likewise replace n simultaneous pointer qubits by one reusable qubit and n retained bits, without reducing its cumulative coherent exposure or physical-control requirements. The large precision plans remain unexecuted.

Preserving quantum archives at small dimension. The actual prepared lepton instrument has outcome-conditioned environment ranks $1, 2, 1, 2$. Exact nonzero-minor certificates imply complete one-, two- and four-round rank counts $(2, 2, 0)$, $(4, 8, 4)$ and $(16, 64, 176)$. The four-round 48-qubit padded archive admits a reversible representation in at most two quantum archive qubits plus all eight original classical bits. Every source vector, family/reference correlation and history weight is reconstructed. Twenty-eight small complete orthogonal gates and native address services implement the compression, costing at most 1,767,798 additional native pulses. No unknown archive is erased and no autonomous reset or thermodynamic work identity is inferred. See `model/lepton_archive_compression`.

Paired hadron scales and bare Wilson GeV coordinates. The retained thirty-two stream-deletion chains propagate the critical and ratio-tuned hopping parameters together with the potential and correlated hadron fits. The inverse spacing is $1.400072 \pm 0.019676\,\mathrm{GeV}$ under the empirical rho anchor, $1.077787 \pm 0.005391\,\mathrm{GeV}$ under the conventional $\sqrt{\sigma} = 0.440\,\mathrm{GeV}$ assignment, and $1.162478 \pm 0.004107\,\mathrm{GeV}$ under an explicit $r_0 = 0.5\,\mathrm{fm}$ benchmark. Separate crossed-fit allowances are 0.028178, 0.095230 and 0.028464 GeV, respectively. These allowances are not confidence intervals or complete model uncertainties. The rho/string scale ratio is 1.299025 ± 0.017142 statistically, with a separate 0.129481 fit allowance; this is a conditional scale-compatibility result, not a discrepancy significance with full physical errors.

For each original hopping coordinate and paired deletion, the bare Wilson parameter is $am_q = (1/\kappa - 1/\kappa_c)/2$. Multiplication by the chosen inverse spacing gives its value in GeV. With the rho anchor, the coordinates $\kappa = 0.16, 0.165, 0.1663, 0.1675$ give, respectively, 0.236947, 0.104364, 0.071199, 0.041041 GeV. At the pion/rho-ratio-tuned coordinate the value is $0.00483755 \pm 0.00006981\,\mathrm{GeV}$ statistically, with a separate 0.00013899 GeV fit allowance. These are bare action parameters, not renormalized MSbar masses or identified physical quark masses. External anchor errors and quenched, volume and continuum effects remain unbounded. The full paired covariance and all fit variants are retained in `model/paired_hadron_quark_scale`.

8 Comparison with the measured spectrum

8.1 Observables and conventions

At finite volume the spectral measure of a prepared operator O is

$$d\mu_O(\omega) = \sum_n |\langle n|O|\Omega\rangle|^2 \delta(\omega - (E_n - E_\Omega))\, d\omega.$$

A visible eigenvalue requires nonzero residue in the actual detector channel, and the physical dictionary must transport O, the state, the Hamiltonian, current charges and the clock together.

Localized wavefunctions are not momentum eigenstates, and a Fourier transform of a pinned mode is no moving-defect dispersion experiment.

In a justified relativistic regime a stable massive excitation has a common rest-energy and propagation relation. An unstable particle requires a resonance convention and width [20]. Charged sectors coupled to a massless photon require an infrared treatment rather than an assumed isolated sharp pole [22, 25], and color confinement prevents reading a free color-mode frequency as an observable quark or gluon mass. Running charged masses and mixing data must share scale and scheme [18, 19]. The finite source spectra of Section 5 have no certified running-to-pole, confinement, or laboratory-clock map. The calibrated-source results and the separate rejected comparison packet are distinguished below.

8.2 Calibrated native energy and flavour records

The electron anchor is $m_e = 0.00051099895069\,\mathrm{GeV}$. The charged analyzer then gives $m_\mu = 0.105659414496$ and $m_\tau = 1.776984970813\,\mathrm{GeV}$, conditionally on its free Dirac realization and phase $2/9$. The ratio residuals are about $+9.83$ and $+30.9$ ppm; no radiative correction is calculated here that removes them. The four-input native hadron source has

$$(m_p, m_n, m_{\pi^\pm}) = (0.93827208943, 0.93956542194, 0.13957039)\,\mathrm{GeV}, \qquad |g_A| = 1.2753.$$

All four numbers are calibration inputs. The strong/free neutron endpoint after subtracting the electron unit is $0.78233355931\,\mathrm{MeV}$. The full weak source has filled-reference shifts and multiple neutron energy lines, so this endpoint is not an interacting physical pole difference.

The neutral apparatus instead calibrates six oscillation parameters and declares normal ordering with lightest mass $0.001\,\mathrm{eV}$. It gives $(0.001, 0.008712060606, 0.050139804547)\,\mathrm{eV}$, with active and sterile coordinates of each Dirac mass and both antiparticle/helicity branches retained. Its complete free-flight comparison error below 4.1×10^{-9} is an implementation bound against the calibrated oscillation law, not an uncertainty on those empirical inputs. The common 192-mode semileptonic source has an exact electron threshold and certified nucleon energy windows, while physical pole and continuum identifications remain distinct.

The seven-input hyperfine member fits the same proton, neutron, charged-pion, axial and deuteron data together with neutral-vector and Delta energies. Complete nuclear thresholds and the shared dimensional attachment are tested on that action. Its neutral spin-zero comparison and charge radii fail their unused experimental tests. The mass-only and four-input actions retain their distinct failed controls; their spectra are not merged into the seven-input construction. None of these finite calibrated energies supplies a complete observed spectrum.

8.3 A rejected twelve-value comparison control

The comparison branch is specified by a boundary packet: pixel coordinate $P = 1.630971465307864$, unified coupling $\alpha_U = 0.04112426184$, external unit $E_* = 1.22089 \times 10^{19}\,\mathrm{GeV}$, vacuum expectation value $v = 246.60010\,\mathrm{GeV}$ and scale $Q_0 = 91.58958701\,\mathrm{GeV}$. Its W and Z couplings use one-loop supersymmetric running, its Higgs and top values use imposed ultraviolet criticality, and its quark Yukawa couplings are the six-power texture

$$Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1), \qquad Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2}).$$

No proof here selects these values from finite repair. Multiloop outputs come from the pinned Standard Model in Dimensional Regularization code [17], with two corrections to its electroweak

decoupling routine, the threshold-function operator and the placement of the strong-coupling assignment. A replay reproduces the five light quarks and the W, Z, Higgs and top to maximum relative discrepancy 5.85×10^{-11}. Table 9 gives the result against the 2026 Particle Data Group tables [21].

Table 9: The twelve nonzero comparison values, in GeV. All twelve lie outside their printed intervals. No covariance-aware joint significance or total theory-error bound is asserted.

	Output	Reference	Residual	Convention
e	0.000510962911	0.00051099895069	−70.53 ppm	Pole premise
μ	0.105650931342	0.1056583755	−70.45 ppm	Pole premise
τ	1.77680479525	1.77693	−70.46 ppm	Pole premise
u	0.00668144352	0.00216	+209.33%	$\overline{\text{MS}}$, 2 GeV
d	0.00665065487	0.0047	+41.50%	$\overline{\text{MS}}$, 2 GeV
s	0.239423082	0.0929	+157.72%	$\overline{\text{MS}}$, 2 GeV
c	1.56844907	1.2729	+23.22%	$\overline{\text{MS}}$ self-scale
b	6.59077908	4.186	+57.45%	$\overline{\text{MS}}$ self-scale
t	170.611817	172.1	−0.865%	Pole, cross-section comparator
W	79.8563905	80.3625	−0.630%	Breit–Wigner
Z	90.5845168	91.1879	−0.662%	Breit–Wigner
H	124.842549	125.13	−0.230%	Pole

For positive outputs x_i and intervals $[l_i, u_i]$, one common factor is feasible exactly when $\max_i l_i/x_i \le \min_i u_i/x_i$. The intersection is empty for all twelve values, for the charged triple alone and for W, Z alone. This is an exact statement about the printed rectangle.

8.4 Charged leptons

The generation register of Section 2.6 fixes the ratios from its own read counts, $\rho = \sqrt{2}$ and $\delta = 2/9$, and gives $m_\mu/m_e = 206.770316$ and $m_\tau/m_e = 3477.4728$ (Table 4). The twelve-value table uses a separate target-informed packet that also supplies absolute values.

The lepton circulant of that packet is chosen near the measured pole-mass Koide coordinates. Its response constants $50, 31, 310, 512, 77, 21, 27, 135$, its determinant exponent and the pole reading are target-informed: with the measured electron or muon mass the law returns $m_\tau = 1776.93\,\text{MeV}$, within 2×10^{-3} of the world-average uncertainty, and this agreement carries no evidential weight. Absolute values use the external unit E_*. Summing the three cosines and their squares proves $Q = \sum_j m_j/(\sum_j \sqrt{m_j})^2 = (1 + e^{-2\chi})/3$, and the packet has $Q = 0.66666446$, $\rho = \sqrt{2}e^{-\chi} = 1.41420889$ and $\delta = 2/9 + \zeta = 0.22222476$, against the same coordinates of the measured pole triple to nine digits.

Reading the outputs as pole masses is an adopted modeling premise. Under it the discrepancy is almost entirely a common scale of about −70.5 parts per million. The muon/electron ratio residual is +0.0725 parts per million and the tau/electron ratio residual is +0.0662 parts per million. A common normalization cannot remove the first, and the electron and muon intervals admit no shared factor. The exact Koide point $\rho = \sqrt{2}$, $\delta = 2/9$ gives m_μ/m_e off by +9.83 parts per million and m_τ/m_e off by +30.9 parts per million; its tau ratio lies in the printed rectangle and its muon ratio does not. Assigning the same outputs running masses at Q_0 and applying the retained one-loop matching relation instead shifts the pole comparisons by $(4.854, 2.870, 1.820)$ percent. Proximity to pole coordinates selects neither scheme.

8.5 Quarks

The declared texture has the transported light ratios $m_s/m_d \simeq 35.99993$ and $m_u/m_d \simeq 1.00463$, in the modified minimal subtraction scheme with quantum chromodynamics and electrodynamics, four active quarks and three charged leptons at 2 GeV. The rectangle of the two quoted strange and down intervals is approximately $[19.33, 20.22]$, and the directly quoted m_u/m_d interval is $[0.442, 0.482]$. A common multiplier cancels in each ratio, so no energy unit can repair the light-quark shape, and this texture is rejected. Precise identification with the reference quantum-chromodynamic masses also needs an electromagnetic scheme conversion.

For a coefficient $m = A\,6^{-n}$ and ratio $r =$ output/reference, holding the transport factor fixed gives $n_{\text{required}} = n + \log_6 r$. The u, d, s, c, b coordinates are approximately 6.63023, 6.19375, 4.52837, 3.11653 and 2.25334. Self-scale masses and threshold dependence require a coupled rerun for an actual changed boundary, so these are diagnostic coordinates and no selected replacement texture. Matched five-flavor running gives $\alpha_s(91.1879\,\text{GeV}) = 0.11944066$ for the stipulated branch; a reference-ratio estimate of 0.11945742 is a different operation.

8.6 Weak bosons, Higgs and top

The W, Z, H, t residuals depend on the renormalization convention and attachment scale of the supplied vacuum expectation value. The source value agrees with the reference model's $v(172.4\,\text{GeV}) = 246.60030\,\text{GeV}$ within one part per million, while the branch attaches it at Q_0, where the reference value is 248.38564 GeV. Table 10 substitutes reference inputs at Q_0 to diagnose this sensitivity. A substitution derives no attachment, and the numerical equality supplies no scale-selection theorem. The controls omit the stored finite-light shifts of 1.7 and 13 keV.

Table 10: Percent residuals under reference substitutions at Q_0.

Inputs replaced by reference values	W	Z	H	t
None	−0.630	−0.662	−0.230	−0.865
Gauge couplings	−0.717	−0.698	−0.241	−0.899
v	+0.076	+0.047	+0.436	−0.200
v and gauge couplings	−0.012	+0.011	+0.425	−0.233
v, y_t, λ	+0.063	+0.029	+0.110	+0.245
All running inputs	−0.025	−0.008	+0.099	+0.210

The weak vectors use the stated Breit–Wigner convention, and the top comparison uses the 172.1 ± 0.6 GeV cross-section pole extraction, distinct from a direct event-generator mass. Loop-order truncation, reference-input uncertainties and physical model error are distinct from numerical replay precision, and criticality itself is sensitive to loop order: at two loops the replayed boundary has $\lambda = 2.3 \times 10^{-8}$, and at three loops it has $\lambda = 9.7 \times 10^{-4}$.

8.7 Massless and neutral sectors of the comparison control

The photon and gluon zeros are conditional action coefficients. The minimal neutrino action in the twelve-value control gives three zero masses and zero oscillation gaps, contrary to measured oscillations. The massive calibrated Dirac source of Section 6.9 is a distinct construction; its lightest mass is chosen, and its six oscillation parameters are inputs. The specified unshifted amplitude-to-mass attachment is excluded by Proposition 9.3. No particle abundance or early-Universe population is computed by this branch.

These are conditional comparisons, including target-informed choices. None of these constructions is a derivation of the measured spectrum. The finite gauge and matter results alone select no continuous mass spectrum, mixing matrix or cosmological population history. Explicit calibrated sources give the displayed finite laws without turning their calibration data into independent predictions. A joint physical test requires one source law, its state and complete record map, compatible mass conventions and declared experimental and theoretical uncertainties. Exact equality to measured central values is not an experimental theorem; equality of observable laws is the stronger mathematical target.

9 What records and symmetries exclude

Each statement below is exact on its stated domain and removes a tempting route to a spectrum. None excludes OPH as a whole; together they locate the missing selection.

9.1 Coherent phases and family averaging

The native event rates $p = 1/12$, $q = 1/6$ satisfy $2\sqrt{p/q} = \sqrt{2}$ and $p + q - 2pq = 2/9$, the base coordinates of the charged circulant.

Proposition 9.1 (Classical records select no coherent phase)**.** *For a finite instrument with Kraus operators K_{ra} and arbitrary real θ_{ra}, replacing K_{ra} by $e^{i\theta_{ra}}K_{ra}$ leaves every outcome map, every repeated composition and every record-dependent feedback experiment unchanged, on the full matrix space and with arbitrary entangled references. A well-defined function of the instrument maps therefore cannot depend on these phases.*

A coherent sum $\sum_{ra} c_{ra}K_{ra}$ thus does not follow from the instrument alone, and setting a probability equal to an angle in radians is an additional rule. An explicit same-rate counterfamily, separate from the native 128-record instrument, has identical classical instruments and different coherent spectra. A source-selected Stinespring isometry with an accessible coherent reference can carry the missing information, and such a reference is outside this exclusion.

Proposition 9.2 (Real cyclic averaging forces a repeated eigenvalue)**.** *For a real regular three-cycle R and $P = (I + R + R^2)/3$,*

$$\tfrac{1}{3}\sum_{j=0}^{2} R^j X (R^T)^j = aP + b(I - P), \qquad X = X^T \in M_3(\mathbb{R}).$$

The two-dimensional rotation plane forces the repeat, so cyclic averaging of a real symmetric kernel cannot select three distinct family values. An equivariant construction from real inputs has real outputs, and a nonreal chiral response needs an input or selection rule supplying the oriented complex structure.

9.2 The passive family metric

For the native core-biased reference and nonnegative seam weights, stationarity gives $\sigma^T(I-T)\sigma = (157/90000)W_{\mathrm{cross}}$, which forces all eight core-to-outer weights to vanish. Every retained branch conserves a nonzero core-charge coordinate. The harmonic extension U of the native passive Schur calculation obeys $B^TU = GI$ and

$$Z = U^TU/G = I + (U - B)^T(U - B)/G \geq I.$$

The Schur mass matrix has the conserved family direction as a null vector, and the maximal 22-seam mixture attains rank two. Positive normalization is an invertible congruence and preserves rank, so it cannot generate the third charged family coefficient. A different reference or active recovery changes the premises of this obstruction.

9.3 Neutral hierarchy

Proposition 9.3 (Neutral squared-gap bound)**.** *Put $z = e^{-ih}$, $U = I+(z-1)P_2$, $V = I+(z-1)P_3$ on the bright plane of the two native rank-one family projectors, which have $\mathrm{tr}(P_2P_3) = 1/5$, and let $F = \left(\begin{smallmatrix} (1-p)U & U \\ pV & 0 \end{smallmatrix}\right)$ be the recurrent transfer. For $0 \le p \le 1/12$, $0 < h \le 1/2$, its two roots of modulus above $1/2$ have phase coefficients $\omega, 1-\omega$ with*

$$0 \le \omega \le a = \tfrac{1}{2}\left(1 - \tfrac{95}{96}\sqrt{\tfrac{653}{848}}\right).$$

Under the common-scale attachment $(m_1, m_2, m_3) = M(0, \omega, 1-\omega)$, the adjacent squared-gap ratio is at most $a^2/(1-2a) < 0.004988$.

The determinant of F reduces to a four-dimensional calculation, with the other ten memory inputs retained as a transient block of induced norm at most p. Substituting $\lambda = e^{-ih/2}x$ and $y = x - p/x$ reduces the quartic to $y = (1-p)\cos(h/2) \pm i\sqrt{C_0}\sin(h/2)$, with $C_0 = (1-p)^2 + 4p/5$, and elementary bounds on the slow root give the estimate. The measured normal-ordering ratio is 0.03108, with printed rectangle $[0.03011, 0.03208]$, which excludes this attachment. Arbitrary neutral actions, offsets and other mass readouts are outside the proposition, and it fixes no absolute neutrino mass.

9.4 Residues, references and mass readouts

The complete instrument determines which spectral values an observer can read. Let

$$T = \begin{pmatrix} 1-p & 1 \\ p & 0 \end{pmatrix}, \qquad \pi = \frac{(1,p)}{1+p}, \qquad U_j = \exp(-ihX \otimes P_j),$$

with $0 \le p \le 1/12$. On the bright plane B, choose its antisymmetric area form J and put $A = |+\rangle\langle -| \otimes J$, $O = A + A^*$, $P_B = I_2 \otimes B$. Then $U_j A U_j^* = e^{-ih}A$ and $O^2 = P_B$. The three projectors $(P_B + O)/2$, $(P_B - O)/2$ and $I - P_B$ give a complete readout with values $+1, -1, 0$. Starting from π and the stationary bright, dark or tracial response state $P_B/4$, $(I - P_B)/2$ or $I_6/6$ gives, respectively,

$$C_B(n) = \cos(nh), \qquad C_D(n) = 0, \qquad C_{\mathrm{tr}}(n) = \tfrac{2}{3}\cos(nh).$$

The same transfer frequency therefore has a nonzero or a zero measured residue depending on the supplied state and detector, with the first measurement operation included.

Density records also cannot fix absolute field frequencies. Adjoin a reference line and extend each branch by $V_j(\delta) = 1 \oplus e^{-i\delta h}U_j$. All complete record probabilities and conditional density matrices confined to the original response sector are independent of δ, while coherences with the added reference acquire that phase, $F_\delta = e^{-i\delta h}F_0$.

A coherent family potential need not be a scalar mass. For the supplied Dirac symbol $\alpha_i = Z \otimes \sigma_i \otimes I$, a potential $I_4 \otimes A$ has zero projection onto the spin-invariant terms anticommuting with all α_i. For Hermitian A, M and $H_h(k) = \alpha \cdot k + X \otimes I_2 \otimes M + hI_4 \otimes A$,

$$\mathrm{Tr}[(H_h(k)^2 - |k|^2 I)^2] - \mathrm{Tr}[H_h(0)^4] = 16h^2|k|^2\,\mathrm{Tr}(A^2),$$

so a nonzero potential in this truncated Hamiltonian is no collection of constant relativistic masses. Finally, cyclic trace gives $\mathrm{Tr}([K,P]H) = \mathrm{Tr}(K[P,H])$. If $[P,H] = 0$, $[K,P] = iH$ and H is Hermitian, then $\mathrm{Tr}(H^2) = 0$ and $H = 0$. Exact global finite translation and boost identities therefore require a different domain from a nonzero relativistic Hamiltonian; controlled sectors and infinite limits carry their own error bounds.

9.5 Comparing capacities

Finite tests can compare whole record laws without enumerating histories.

Proposition 9.4 (Sufficient criterion for equal record laws)**.** *Let A_y, B_y be nonnegative outcome matrices of two finite classical instruments with column-stochastic sums, and C a nonnegative column-stochastic map. If $CA_y = B_yC$ for every record y and $Cp = q$, then all finite joint record laws coincide. With $\epsilon_0 = \|Cp - q\|_1$ and $\delta = \max_j \sum_{y,i} |(CA_y - B_yC)_{ij}|$, the record total-variation distance after n steps is at most* $\min(1, (\epsilon_0 + n\delta)/2)$.

Induction gives the exact case, and a telescoping sum of ℓ_1 contractions gives the approximate one. Equality of unread channels is insufficient: the one-state instruments $A_0 = 1, A_1 = 0$ and $B_0 = 0, B_1 = 1$ have the same unread channel and disjoint records. Determinism supplies neither the intertwiner nor a compatible initialization.

10 The catalogue and what a source-selected spectrum must supply

10.1 Source assignments and their physical meaning

A spectrum assertion includes the action, reference, preparation, current and energy instrument. The following source members therefore have different meanings even when they use the same native matter module.

Source	Specified data	Established records
Charged analyzer	Phase 2/9, complete Dirac compiler and electron anchor	Three charged rest gaps, with explicit finite preparation and detector errors
Four-input hadron action	Positive native-current squares fitted to $p, n, \pi^{\pm}, \lvert g_A\rvert$	Complete baryon/meson spectra and calibrated native-current matrix element
Neutral-current apparatus	288-mode Dirac completion; six oscillation inputs and m_1	Three massive neutrino/antineutrino branches, both helicities and active/sterile-coordinate records
Common semileptonic action	192 modes, charged/neutral tensors, strong action and full cross-current	Exact electron threshold, six colourless branches and bounded nucleon energy windows
Gauge/scalar action	Canonical one-doublet action, g_i, λ_H, v	Conditional symmetry breaking and quadratic coefficients on its own bosonic source
Minimal comparison control	Powers of 1/6 for quarks and zero neutrino coefficients	A rejected mass/mixing attachment; no measured quark or oscillation fit

These assignments do not form a complete catalogue of observed species. The common semileptonic source does not include the separate photon, gauge/Higgs dynamics, meson outgoing apparatus or quantum geometry merely because their labels coincide. The neutral apparatus switches from

its specified production instrument to free flight; it is not the full autonomous contact evolution. Interacting baryon windows are not sharp physical poles, and none of these finite comparisons establishes a physical neutron lifetime. The compact four-mode band also has a nonzero $A = W^{\dagger}hW$ that contributes to its energy. Its quartic interaction changes spectral poles, so its electromagnetic feedback cannot be read from a zero-mode or quadratic mass table.

10.2 The required attachments

A constructed lattice sector and its refinement boundary. For the declared staggered-field spin source, an onsite frame gives $\widetilde{H}_B = (J/4)\sum_{xy}(X_xX_y - Y_xY_y - Z_xZ_y - I) - (B/2)\sum_x Z_x$. Its normalized bond perturbation has grouped norm at most $3J/B$. The simple onsite excitation is separated from all multiparticle sums, so sufficiently small positive J/B gives an isolated analytic lattice particle band [33]. Exact full-source perturbation, including ground subtraction, gives

$$E(k) = B + 3J + \frac{J^2}{2B}\left[3 - \left(\sum_{j=1}^{3}\cos k_j\right)^2\right] + O(J^3/B^2).$$

The local cluster construction gives a uniform momentum remainder for sufficiently small coupling; no numerical admission threshold or remainder constant is asserted. Opposite sublattices carry conjugate internal charges with equal energies. The complete local creator and its failure branch preserve original fermion grading, so this is no single-fermion assignment. On an eight-site cube at $J = 1, B = 400$, the product reference has Rayleigh residual squared 3 and excited-spectrum separation at least 394, hence ground half-trace error at most $1/197$. A complete 167-pulse native preparation and charge instrument retains all 608 first-failure slots, with full-output error below $1/100$ from the exact-ground instrument. This finite certificate does not evaluate the infinite-volume smallness threshold or isolate its particle-band weight. The source calculations and their complete instrument are given in `model/high_field_particle` [1].

The same ground generator has gap at least $B/8$ and a field-independent Lieb–Robinson speed at most $72J$. The isolated-shell velocity bound [34] therefore implies, at physical spacing a and common positive energy conversion s,

$$sE_{\min} \le M_{\max} \quad \Longrightarrow \quad v_{\text{physical}} \le 72asJ \le 576a(J/B)M_{\max}.$$

For bounded J/B, the physical dispersion has a vanishing Lipschitz constant as $a \to 0$. A nonconstant relativistic finite-rest-energy limit with local observable weight is excluded on this clock and spatial map. This result does not exclude a fixed-cutoff measurement or the distinct ordered family with $B/J \to 0$. Its pole concentration and physical labels require their own source results.

A sufficient concentration test on that positive-field source uses a packet ψ away from zero spatial momentum and the joint spectral ratio $X = K_B/[J\sqrt{2\sum_j(1-\cos P_j)}]$. Write $w = \|\psi\|^2$, $h = \langle\psi, X\psi\rangle$ and $\chi = \langle\psi, X^{-1}\psi\rangle < \infty$. For $c, \epsilon > 0$,

$$D_c = h - 2cw + c^2\chi = \int \frac{(x-c)^2}{x}\, d\nu_\psi(x) \ge 0, \qquad \nu_\psi(|X-c| \ge \epsilon) \le \frac{c+\epsilon}{\epsilon^2} D_c.$$

This needs no upper energy cutoff. A complete preparation ψ/Λ retains the original probability factor $1/\Lambda^2$. Exact source commutators determine h from static bond and order correlations, while equal-time correlations determine w. The remaining dynamical statistic χ has bounded joint-score enclosures using the positive-field gap. Its required saturation on the growing source is unproved; the finite certificate does not identify a physical speed or particle.

A comparison of one sector has fewer dependencies than the complete spectrum. Fix one admitted source, prepared ground, clock and accessible operator menu. For two neutral channels the vacuum-relative Euclidean correlations have the form

$$C_i(t,p) = \langle O_i(p)\Omega, e^{-t(H-E_0)}O_i(p)\Omega\rangle = \int_0^\infty e^{-tE}\, d\mu_{i,p}(E).$$

The complete source readouts determine the correlation data; a persistent isolated contribution with nonzero original weight must be established with volume and cutoff errors. If its dispersion is relativistic with common speed v, its rest energy $\Delta_i = m_i v^2$ gives $m_1/m_2 = \Delta_1/\Delta_2$. A common positive clock rescaling cancels from this ratio. A finite box level or a decay threshold alone supplies no stable particle identification. Matching the physical symmetry labels and detector response is a separate obligation, and target masses cannot select the source parameters. A finite-resolution comparison needs controlled errors over its measurement window; neither the full Standard Model catalogue nor Einstein dynamics is a prerequisite.

This reduction reuses the exact bound-pair poles, complete local charged responses and controlled one-dimensional massive dispersion. Their respective reference states and scaling laws cannot be exchanged. In particular, the bound-pair rest/curvature ratio differs from the relativistic value, and the one-dimensional running family admits arbitrary positive limiting mass through the same initial source point. Those exclusions restrict the interpretation; recomputing their spectra does not select physical masses.

Table 11 gives the additional attachments for the complete spectrum. Measured inputs calibrate the quantities they fix; an independent spectrum prediction requires the listed source derivation.

Table 11: Attachments between the finite stack and a physical spectrum.

Attachment	**Required result**
Charged-family source law	Physical scalar Yukawa eigenvalues, phase and flavor frames selected without measured-target fitting
Gauge, Higgs and top parameters	Gauge content, running and matching prescription, vacuum scale and the criticality premise derived from the same source
Neutral masses and mixing	A nonzero-gap mechanism and complete readout outside the excluded minimal and unshifted-phase branches
Physical records	Matched preparation, instruments, clock and estimator, retaining all failure outcomes, with a complete error budget; the reversible seam construction supplies a finite gauge-flux response on its own declared source
Capacity and refinement	A full-history equivalence or controlled limit for the physical source, beyond the four-address encoding
Early-Universe distribution	The relevant state, finite-temperature dynamics, interactions and population observables from the same source

These combine into one joint map,

(same source, vacuum, clocks, instruments)
$\longrightarrow$ (total defect energies, current residues, dispersion, vertices)
$\longrightarrow$ (one response kernel and its pixel fixed point).

Its first row must supply an autonomous finite-energy defect process in place of an externally fixed texture. Its second must retain all spatial regulator modes until an actual decoupling or

continuum theorem controls them. The third must use the first two, including thresholds and physical normalization. These are composition requirements on one permitted design; uniqueness among all possible simulator designs is not a prerequisite.

10.3 Interaction, chiral continuum and responsive geometry

Two controls separate spatial propagation from electric motion of a chiral subspace. The complete free three-dimensional overlap Hamiltonian on a flat gauge fiber has $E(p) \to |p|$ and modified axial weight tending to one under refinement; all spatial momenta and internal components are retained. It is vectorlike. A flat fiber has zero Haar measure and supplies no normalizable vacuum for a dynamical gauge field. A separate four-edge compact Abelian flux ring retains all five native charges, with electric energy, matter hopping J and a loop term κ. For initial flux support $|\ell| \le K_0$, cutoff $K > K_0$, total time T and $r = \lceil (K - K_0 + 1)/\max(1, |q|) \rceil$, a Dyson bound for complete position records, including untouched auxiliary references, gives

$$\mathrm{TV}(\text{complete records}) \le \min \left\{ 1, 2 \sum_{n \ge r} \frac{[(2|J| + |\kappa|)T]^n}{n!} \right\}.$$

At $J = \kappa = 1$, $K = 48$, $K_0 = 0$, $T = 1/2$, this is below 0.000248607 uniformly for the five charges. It controls the stated finite-time preparation and complete observations, not an unbounded-flux ground state without a tail bound.

For the source Euclidean overlap projector $P(\theta)$ on a local admissible charge-six patch, electric differentiation must preserve its range. With $C = [P, P']$, $\nabla = \partial_\theta + C$ obeys $\nabla P = 0$; parallel transport has generator $-C$. For $VV^\dagger = P$, $V^\dagger V = I$, the compressed parent kinetic energy is

$$V^\dagger(-\partial_\theta^2)V = -(\partial_\theta + A)^2 + \Phi, \qquad A = V^\dagger V', \qquad \Phi = V'^\dagger(I - P)V' = V'^\dagger V' + A^2 \succeq 0.$$

Omitting Φ changes the parent model. A nine-point Dirichlet control keeps all 576 parent components and its 288-dimensional chiral subspace. For positive parent kinetic matrix K and $H_\Lambda = K + \Lambda Q$, let a_j be the eigenvalues of PKP. The entire complement is controlled by

$$0 \le a_j - \lambda_j(H_\Lambda) \le \frac{\|PKQ\|^2}{\Lambda - a_k}, \qquad j \le k, \quad \Lambda > a_k.$$

The differential scalar Φ is positive; the finite-grid difference from a polar-link connection-only discretization need not be. Local frame covariance supplies neither a global anomaly-compatible chiral measure nor physical axial conservation. These spatial, flux and electric constructions are separate controlled models [1].

The quartic result gives a connected interaction on a fixed localized algebra, with a bound uniform in the surrounding radial volume. Its spatial tails and fixed number of localized modes do not define an interacting local 3+1 field theory. A controlled common field limit must also establish the intended chiral measure, anomaly realization, local gauge constraints, surviving interactions and regulator census. Finite internal anomaly sums are necessary representation checks; a chiral lattice measure has additional locality and integrability requirements [26, 27, 28].

The same matter energy and stress must then enter the geometric response. The companion Einstein implication uses compatible clocks and metric, conserved stress, a geometric modular normalization and controlled fixed-proper-volume generalized-entropy variations [6]. The finite Gibbs first law proved for the interacting localized Hamiltonian is a common-operator identity, and it supplies no geometric area term, small-ball remainder, Ward/Bianchi map or universal Newton normalization.

11 Finite selection tests for particle readout

Several supplied field controls admit exact tests against the finite source. The following calculations retain the stated source, complete records and operator domains. They determine which coefficients follow from those data and which require an additional selection law. No measured particle mass enters their construction. Transport to another carrier requires an identification preserving its state, currents and clock; equality of spectra alone supplies no such identification.

For the coherent three-cycle R, write $C = aI + rb(e^{i\theta}R + e^{-i\theta}R^2)$ and $Y = C^2/3$, where $a = \sqrt{5}/6$, $b = a/\sqrt{2}$ and $0 \le r \le 1$. We use the common nonnegative-root chamber $|\theta| \le \pi/12$. The exact source calculation gives

$$\operatorname{Tr} Y = \frac{5}{36}(1+r^2), \qquad \operatorname{Tr} Y^2 = \frac{25}{3888} + \frac{25r^2}{648} + \frac{25r^4}{2592} + \frac{25\sqrt{2}\,r^3}{1944}\cos 3\theta.$$

With a declared neutral Higgs number $h > 0$, the charged free-CAR sea energy at kinetic norm p is $-2\sum_j \sqrt{p^2 + h^2 y_j^2}$, where y_j are the eigenvalues of Y. At $p = 0$ it equals $-2h \operatorname{Tr} Y$, independent of phase. Convexity in each root proves that external energy minimization sets $r = 1$ for every positive-weighted finite momentum set. Rational bounds at $p = 1/256$, $h = 1$ prove an interior phase minimum exists; changing p can reverse a phase comparison. The momentum prescription is supplied.

The actual finite quantum scalar has vacuum and two doublet states. For its gauge-invariant pair-creation vertex $W(Y)$, the conserved $N_F - 2N_H = 0$ sector has dimension $1 + 2\binom{18}{2} = 307$. The vacuum and $W(Y)$ applied to it span an invariant two-dimensional subspace; the scalar cutoff forbids another creation step. With common number energies and $d = 2\epsilon_F + \epsilon_H$, its levels are

$$E_\pm = \frac{d \pm \sqrt{d^2 + 16 \operatorname{Tr} Y^2}}{2}; \qquad 305 \text{ retained dark levels have energy } d.$$

The lower level is the whole-sector ground. Minimization selects $\theta = 0, r = 1$, whose action ratios are $1 : 1 : (34 + 24\sqrt{2})$. This sector calculation excludes spatial hopping and supplies neither the global interacting ground nor a Higgs expectation value. A dynamical phase register requires its own mixing or preparation law and common energy reference.

The finite record-budget response uses the literal reset memory, event probabilities $p_C = 1/12, p_E = 1/6$ and declared C-enabled coherent pulses. Its first-derivative coefficient Y_n has trace $1/12$, rank two at one attempt and rank three at every $n \ge 2$. Exact invariant bounds give, for ordered positive eigenvalues,

$$\frac{\lambda_2(Y_n)}{\lambda_1(Y_n)} < \frac{1171058}{6633} < 177, \qquad \frac{\lambda_3(Y_n)}{\lambda_1(Y_n)} < \frac{8616}{11} < 784.$$

The complete direct archive has 24^n positive histories and capacity $n \log 24$ nats. Varying this budget cannot reproduce observed lepton ratios by a common linear mass conversion. These coefficients are derivatives of a noisy finite response; no interacting mass identification follows. Pixel capacity and archive capacity require a separate attachment.

An independent record-clock calculation uses all sixty seam/coin outcomes on the binary total-two carrier. Its transition P, occupation matrix B and graph Laplacian L satisfy $\mathsf{P}B = B(I - L/60)$. Consequently the occupation correlations decay at exact rates $e_\lambda = -\log(1 - \lambda/60)$ for $\lambda = 5 - \sqrt{5}, 6, 5 + \sqrt{5}$. Common attempt blocking and common energy conversion cancel from their ratios. These are record-relaxation rates and reconstructed $-\log \mathsf{P}$ gaps; the charged action has no proved identification with them, and GeV units require an energy reference.

The finite band and canonical dressing give two further operator connections. Uniform sampling of the twelve incidence-defined near rotations has displacement cost $2I - \sum_g R_g/6$ on the full 36-state pair. Its strictly least positive eigenvalue is $(5 - \sqrt{5})/3$, of rank three, with both zero-cost states retained. This identifies the declared cooling band without a dimension window; selection of that controller and its physical family attachment are extra premises.

For two CAR modes and every level of the 60-state quantum link, let $S = c_x^\dagger U c_y + c_y^\dagger U^\dagger c_x$ and $W = e^{i\pi S/4}$. The operators $b_i = W c_i W^\dagger$ satisfy exact CAR and retain all endpoint gauge transformations. Their canonical one-particle projector has rank 60, including 59 dark directions beyond the rank-one edge projector. Its sign construction satisfies the Ginsparg–Wilson identity but has 118 dark zero columns; it supplies no relativistic kinetic regulator. A local interacting chiral limit requires a selected kinetic and complement Hamiltonian, compatible quantum electric transport, uniform locality bounds and a positive physical state on the same clock.

11.1 Complete interacting records on the specified finite source

The finite observer construction uses that same scalar/full-CAR vertex, with $G = 4\,\mathrm{Tr}(Y^\dagger Y) > 0$ and $\Delta = \sqrt{d^2 + 4G}$. It resolves both bright eigenprojectors and the entire 305-dimensional dark projector. A phase-shifted vertex pulse, with the number term paused or refocused, prepares the neutral-sector ground by a unitary that fixes the dark inputs. Complete occupation and energy instruments retain every outcome, including outside-code outcomes when the finite sector is embedded in its larger carrier. The finite coherent-instrument compiler supplies implementation and error bounds for these declared services; execution of its full pulse tape is not claimed.

Let $J_{01}(t)$ be the original joint probability of an empty scalar record followed by an occupied record after evolution for time t. For any reproducible input with empty weight $a_0 > 0$,

$$J_{01}(t) = a_0 \frac{4G}{\Delta^2} \sin^2(\Delta t/2), \qquad \frac{J_{01}(2t)}{4J_{01}(t)} = \cos^2(\Delta t/2).$$

On $0 < \Delta t < \pi$ this determines the gap without ground-state preparation or renormalizing selected outcomes. The declared test coefficient $Y = \mathrm{diag}(1, 2, 3)$ and $d = 1$ gives $G = 56$, $\Delta = 15$ and levels $-7, 8, 1$. At $t = \pi/45$, the probability ratio is exactly $3/4$. These are model energies with no measured mass input.

A faithful Gibbs state of this same Hamiltonian provides a common modular clock. For the chosen $\beta = \log 2$, its per-state spectral weights are $32768/71809$, $1/71809$ and $128/71809$, the last repeated 305 times. Thus $-\log \rho = \beta H + \log Z\, I$ and $\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H}$. The full field density is gauge invariant, though its support includes nonsinglet field states. Its canonical purification with a conjugate charged reference is jointly gauge invariant. This is a declared computable preparation, without an assertion that passive repair prepares a thermal vacuum.

Charged probes use the same operator $H = W + W^\dagger + dN_H + \epsilon_F Q^*$, where $Q^* = N_F - 2N_H$. The entire 36-state removal sector has energy $d - \epsilon_F$. In the 1650-state addition sector, the literal pair map W_1 has Gram matrix $S = GI - \gamma$, with blocks $\gamma_L = YY^\dagger$ and $\gamma_e = 2(Y^\dagger Y)^T$ in the all-left creation convention. For squared singular values s_j of Y, its eigenvalues are $G - s_j$ with multiplicity four and $G - 2s_j$ with multiplicity two. Each produces energies

$$E_{\lambda,\pm} = \epsilon_F + \frac{d \pm \sqrt{d^2 + 4\lambda}}{2},$$

with 1614 dark states at $\epsilon_F + d$. Addition and removal operators coupled to conjugate probe references form a complete instrument by the CAR identity. Their original spectral weights are

computed from the prepared neutral ground. Changing ϵ_F at fixed d leaves all neutral records invariant while changing charged gaps: neutral measurements do not determine this independent coefficient. These lines have no asserted spatial dispersion or physical lepton identity.

When both addition lines of a resolved channel have nonzero original probe weight, their separation δ_j and the neutral gap give $s_j = (\Delta^2 - \delta_j^2)/(4\kappa)$, with $\kappa = 1$ for a doublet channel and $\kappa = 2$ for a singlet channel. The additive ϵ_F offset cancels, and a common clock scale cancels from the resulting coefficient ratios. Exact reconstruction gives $1 : 2 : 3$ for the declared diagonal fixture and preserves the repeated pair of the coherent zero-phase source. This is an operational measurement of source coefficients, not a charged-lepton rest-mass identification or a selection of the target-fitted phase.

On the positive-C branch, the source identity $\sum_j \sqrt{3y_j} = 3a$ removes a common coefficient scale. Thus $r^2 = 3\sum_j y_j/(\sum_j \sqrt{y_j})^2 - 1$, and the two trace identities above determine $\cos 3\theta$. The unordered spectrum retains the equivalence $\theta \mapsto \pm\theta + 2\pi k/3$; at $r = 0$ the phase is unobservable. Complete spectral records therefore reconstruct the configured controller within that branch. They do not select its setting independently of a target fit.

11.2 The full scalar ground and its occupation records

For the declared test action $Y = \mathrm{diag}(1, 2, 3)$, $d = 1$, $\epsilon_F = 0$, a complete-carrier calculation covers all $3 \cdot 2^{18} = 786432$ inputs. Splitting the scalar-empty and occupied spaces gives $H = \left(\begin{smallmatrix} 0 & A^\dagger \\ A & I \end{smallmatrix}\right)$. Six signed three-mode cells reduce the full Gram comparison to matrices of dimension at most $3^6 = 729$. Exact positive-integer Perron witnesses and Sturm root counts give a unique largest Gram eigenvalue λ at the largest root of

$$x^6 - 244x^5 + 23737x^4 - 1171722x^3 + 30638880x^2 - 394771968x + 1881169920.$$

The unique full ground has $E_0 = (1 - \sqrt{1 + 4\lambda})/2 \simeq -7.94630530665$ and is separated from every other state by more than 0.04690056. It has $Q^* = 6$, zero hypercharge and singlet weak-gauge and source-spin actions. A controlled preparation from a six-fermion singlet seed has certified infidelity below 2.6×10^{-11}. The closed pair action cannot reach this sector from empty $Q^* = 0$; the seed and controller work are part of the specified experiment.

For the full occupation apparatus, of ranks 2^{18} and 2^{19}, put $\Delta = \sqrt{1 + 4\lambda}$, $\alpha = (1 + 1/\Delta)/2$ and $\beta_b = 1 - \alpha$. Original two-time probabilities are

$$(J_{00}, J_{01}, J_{10}, J_{11}) = (\alpha(1 - T), \alpha T, \beta_b T, \beta_b(1 - T)), \qquad T = \frac{4\lambda}{\Delta^2} \sin^2(\Delta t/2).$$

Exact intervals enclose these records at ticks $1/32$ and $1/16$. The occupation transition $\Delta \simeq 16.89261061331$ differs from the full isolation gap. These source energies are not assigned rest masses; the neutral-sector charged residues do not describe this other prepared ground. Family singular-value decomposition transports the block method to general Y, but the displayed root and numerical certificate belong only to the specified test action.

A complete charged calculation uses this same full ground, action and clock. Compensated CAR probes have effects $I - N_F/18$ and $N_F/18$; their sum is identity. All twelve family/multiplet/addition-or-removal channels retain their original weights and conjugate references. Exact reducing blocks, of dimensions at most 240, contain every source image of the probe. Dyadic eigenframe proposals are admitted by exact integer Gram and residual bounds, including degenerate projectors and the preparation error. First-family doublet removal and addition have lines near 0.0710 and 0.4873 in the same source-energy units, with many other retained lines. They are not one-particle mass shells.

Adding $\epsilon_F Q^*$ shifts addition/removal frequencies by $+\epsilon_F$ and $-\epsilon_F$ about this same reference, leaving their weights and conditional variances unchanged. For the first-family doublet, the addition-minus-removal variance lies in $[-0.12243, -0.08993]$. No common offset therefore makes even those normalized spectral shapes equal. This is a statement about the specified reference, not physical CPT violation or an exclusion of every offset-dependent vacuum. The global ground must be recomputed outside its certified stability window. The two-line coefficient inverse above belongs to the separate $Q^* = 0$ preparation.

11.3 Gauge-coordinate cutoffs and recorded evolution

A separate declared field program uses the retained native representation on a cubic lattice and the coordinate Hilbert space $L^2(G, \mathrm{Haar})$ on each link. A bounded gauge-covariant polynomial of the Wilson kernel, full Dirac fermion occupations and positive electric Casimirs define one self-adjoint interacting Hamiltonian. Coordinate-word degree defines finite gauge-invariant truncations. If the initial word degree is K_0, the interaction increases degree by at most d_c, and its integrated norm is X, the complete record-law cutoff error is bounded by $2\sum_{n\geq r} X^n/n!$, with $r = \lfloor (K - K_0)/d_c \rfloor + 1$. Instruments must preserve the filtration or have their degree included in this accounting.

An executed one-cell quantum-flux/spinor control retains all flux outcomes at two times and has exact initial current six in its declared units. Gaussian-integer Taylor evolution computes unnormalized rational record weights, with an explicit operator remainder. At flux cutoff $K = 30$, the complete numerical half-ℓ^1 error is below 1.155×10^{-35}; representation-cutoff error is bounded separately. This supplies a controlled finite interacting experiment. An interacting chiral continuum, its vacuum, physical particle dispersion and GeV matching are additional dynamical statements, independent of whether architectural choices are unique.

The same-field locality estimate uses even interactions and graded fermion commutators on the uncut coordinate field. Onsite spectral clipping treats the unbounded electric generators; its uniform bounded-observable estimate passes to the strong unitary limit. At fixed polynomial order and spacing, if the weighted local interaction sum is L, disjoint homogeneous operators satisfy

$$\|[\alpha_t(A_X), B_Y]_{\mathrm{gr}}\| \leq 2\|A_X\|\|B_Y\||X|\, 2^{-\operatorname{dist}(X,Y)}(e^{2L|t|} - 1).$$

Local complete instruments inherit this bound. Comparing finite boxes uses the same initial box/reference marginal and gives a volume-error modulus independent of the exterior size. The declared lowest-order control bounds a remote detector contrast at distance 32 and duration $1/64$ below 10^{-7}. These are fixed-spacing estimates, without uniform relativistic refinement. Total-representation-degree compression is nonlocal: compressed distant shifts can have commutator norm one. Its complete-record cutoff error must therefore be added to parent locality estimates, rather than asserting exact cutoff locality.

A finite coordinate preparation uses the normalized gauge average of $\prod_e [(1+\mathrm{Re}\,\mathrm{Tr}_{15}\, U_e/15)/2]^L$ on a periodic cubic carrier. Haar covering and positive polynomial bounds control normalization and the probability of a Wilson gap below $1/2$. The state lies in a finite representation-word code; its electric energy has a Casimir bound. A complete Householder preparation acts as identity on the entire orthogonal complement. The conservative gap planner uses large codes and gives an extremely short finite-time overlap comparison, without a uniform-volume vacuum theorem. An executed one-cell, degree-one character preparation has 1728 original electric records, including 397 zero complements, with mean electric energy 99/9100. This smaller seed alone does not certify a useful Wilson gap. The coordinate Hilbert space is an analytic parent for these finite executable preparation and evolution codes.

A distinct large-plaquette-cost configuration admits all-time gap control. The represented plaquette admissibility theorem implies $\mathbf{1}_{\mathrm{bad}} \le 48000S$ for $S = \sum_p (1 - \mathrm{Re}\,\mathrm{Tr}_{15}\, U_p/15)$. Energy conservation for $H_E + K_m + \kappa S$, with $H_E \ge 0$, yields

$$\Pr_t(\mathrm{bad}) \le 48000 \left[\langle S \rangle_0 + \frac{\langle H_E \rangle_0 + 2\|K_m\|}{\kappa} \right].$$

Compression preserves this positive-operator inequality on finite codes containing the preparation. The compressed effect is not a claim about the spectral gap of a compressed Wilson matrix. An explicit configuration has polynomial-versus-exact-sign complete final-record error below 0.000208 on the periodic parent at time one, with very large declared coefficient and code size. Finite-cutoff, preparation, compiler and finite-volume errors are additional. This finite energy argument does not supply a physical gauge coupling, vacuum or uniform interacting refinement.

A single prepared field experiment composes these bounds on all sixty fermion modes per cell. At spacing one and polynomial order zero, a periodic side-21 gauge preparation and filled origin induce the exact radius-five whole-box marginal, including boundary charge correlations. A finite conjugate purification prepares that marginal. With initial word degree K_0, cutoff $K = K_0 + 12$, the same large κ throughout, and one fixed rational duration, the complete local occupation pointer has all 61 outcomes. The comparison retains untouched external references and traces the evolving exterior; it is not a global-state approximation. Periodic Wilson-gap control is not applied to the open-boundary Wilson matrix.

Six native hopping bonds give exactly $\|P_{59} H_K \psi\|^2 = 2880/289$ on the occupied-origin input, independent of its prepared link state. A unitary Taylor bound gives a strictly positive finite transfer probability. At the conservative duration $T \simeq 1.23896 \times 10^{-152}$, its enclosure is $[1.3779, 2.4609] \times 10^{-303}$. Parent sign, box and cutoff errors sum to less than 1.05065×10^{-149} before additional implementation errors. This bound is too coarse to resolve that rare event in the exact-sign parent. The tiny duration and the large declared coefficient provide a finite controlled experiment, without a uniform physical observation interval, interacting chiral limit or species map. In this planner, κ grows with the preparation's total electric energy, while the locality rate contains 192κ. For side length proportional to box radius R, even fixed polynomial degree gives $\kappa \gtrsim R^3$; the resulting exponential-in-κT bound does not decay against 2^{-R} at fixed positive T. This is a limit of the estimate, not an exclusion of a relativistic phase. A local energy/admissibility or low-mode bound, with uniform interacting correlations, is needed for that refinement argument.

A distinct mixed preparation removes the global normalization cost from the local-density argument. Normalize each degree-L link polynomial independently, tensor the full maximally mixed Fock density, and twirl the joint density under site gauge transformations. Gauge-invariant electric and plaquette means are unchanged by this trace-preserving twirl. A canonical conjugate reference retains the source's nonsinglet charges and makes the joint preparation invariant. This is not a projection onto a field-only singlet vacuum.

For $L = n^2$, the squared link norm is at least $(59/60)^2(1+8n)^{-450}$. An exact dyadic tail bound and a four-link plaquette estimate give volume-independent initial densities e_E, s_0. Translation symmetry and the same conserved action imply

$$\Pr_t(\text{rough plaquette } p) \le 48000 \left[s_0 + \frac{e_E + 2b}{\kappa} \right], \qquad \|K_m\| \le bV.$$

The finite compressed roughness effects obey the same inequality, with both outcomes and the reference retained. At order 128 and unit spacing, the exact planner fixes $L = 2^{64}$ and $\kappa \simeq 2.98009 \times 10^{50}$ for every volume. Its all-time bound is $11567/17179869184 < 6.733 \times 10^{-7}$. These

large declared parameters have no observed-coupling identification. They supply uniform local roughness, not a global Wilson gap or a physical mass gap.

The full Wilson decomposition also yields a local spectral estimate. On the actual $60V$ one-body space, $H_W^2 = I + P + E$ with $P \geq 0$ and $\|E\|_1 \leq 20 \sum_p \|I - R(U_p)\|_1$. For $\Pi = \mathbf{1}_{|H_W|<1/2}$ this implies $\operatorname{Tr} \Pi \leq 400\sqrt{6VS}$. Translation symmetry and Jensen's inequality give

$$\langle \operatorname{Tr}(P_X \Pi P_X) \rangle \leq 400|X|\sqrt{6\langle S\rangle/V} \leq 2\sqrt{5\eta}\,|X|$$

when the same source density obeys $\langle S\rangle/V \leq \eta/48000$. Spectral calculus then bounds the state-averaged squared local Hilbert–Schmidt error of the one-body sign approximation by $60|X|\epsilon_m^2 + (2+\epsilon_m)^2\langle \operatorname{Tr}(P_X \Pi P_X)\rangle$. Compression gives a complete two-outcome finite-code effect, without identifying it with a truncated-kernel projector. This controls local spectral weight, not low-mode spatial decay or the interacting detector response to the sign operator.

The same mixed preparation has exact compatible local marginals: tracing exterior registers leaves only the independent Haar variables incident to retained links. At fixed spacing, polynomial order and local coefficients, locality therefore makes complete local record laws Cauchy as volume grows at any fixed finite duration. A word-cutoff Dyson bound then gives finite simulators for that limit. The declared order-zero, time-one example has total volume-plus-cutoff error at most 2^{-23}, with enormous radius and code-size bounds rather than a materialized simulation. Complete number pointers preserve the filtration; other apparatus must include its degree cost. Global canonical purifiers are not identified across volumes. Correlated untouched references require compatible joint marginals or a common local preparation operation.

This state has a nonzero local dynamical diagnostic: $\langle \dot{n}_x^2 \rangle = 1440/289$ and $\partial_t^2 \langle (n'_x - n_x)^2 \rangle|_{t=0} = 2880/289$ for the complete two-time occupation law. These are exact native current identities, not a particle dispersion. The resulting polynomial field has a controlled fixed-spacing thermodynamic record limit. The local spectral estimate alone does not control the sign operator's dependence on the dynamical electric/link variables in a many-body Duhamel comparison. A different declared regulator supplies a direct local-interaction construction, without claiming equivalence to the discontinuous action.

11.4 Electric sign response and a smooth interacting action

An exact native gauge family exposes the distinction between spectral weight and electric response. For $0 < \delta < 1/2$, write $c = (1-\delta)/2$, $s = \sqrt{1-c^2}$, $\cos\theta = c$, and take weak links $U_1(u) = \exp(i(\theta + u)\sigma_1)$, $U_2(u) = \exp(i(\theta - u)\sigma_2)$, $U_3 = I$, with trivial color. On all sixty native Dirac/internal rows at $u = 0$, the squared Wilson kernel has eigenvalues and ranks

$$(\delta^2, 16), \qquad (1, 28), \qquad (3 + 2\delta, 16).$$

The source electric metric gives the unit tangent $X = \frac{1}{4}\partial_u$, and

$$\|X \operatorname{sign} H_W\| = \frac{1-\delta}{4\delta}, \qquad \|X \operatorname{sign} H_W\|_{\mathrm{HS}}^2 = \frac{(1-\delta)^2}{\delta^2}.$$

A near-critical packet with original weight δ can have vanishing weighted low-mode count and diverging weighted derivative cost. Each smooth packet has finite electric energy; no uniform energy bound or identification with the configured mixed preparation is asserted. This disproves an inference from low-mode weight alone. It neither excludes the exact-sign theory nor proves failure of that preparation.

A smooth alternative uses the same gauge-coordinate carrier and clock units, with declared positive smoothing δ and spacing a:

$$h_\delta = \frac{\beta + \sqrt{1+\delta^2}\, H_W (H_W^2 + \delta^2 I)^{-1/2}}{a}, \qquad H_\delta = H_E + d\Gamma(h_\delta) - \kappa \sum_p \frac{\operatorname{Re}\operatorname{Tr}_{15} U_p}{15}.$$

The flat-source normalization sets zero free energy at zero momentum and leading Dirac velocity one; it uses no measured mass. Since $\|H_W\| < 8$, set $D = 64+\delta^2$, $q = 64/D < 1$ and $Z = (64I - H_W^2)/D$. The binomial expansion

$$\sqrt{1+\delta^2}\, H_W (H_W^2 + \delta^2 I)^{-1/2} = \frac{\sqrt{1+\delta^2}}{\sqrt{D}} \sum_{n\geq 0} \frac{\binom{2n}{n}}{4^n} H_W Z^n$$

converges uniformly over every gauge configuration and volume. The nth term has range $2n+1$ and norm at most $(1+\delta)q^n$. Rooting its full second quantization at a cell costs at most the native rank sixty. Exponentially weighted interaction norms are therefore summable. The onsite unbounded electric generator is handled in the interaction picture; no derivative of the sign function is needed.

The resulting graded locality estimate controls a local Duhamel comparison between H_δ and its finite polynomials. Complete instrument histories, all occupations, and untouched references are retained. With the compatible mixed-state marginals, the same estimate gives a thermodynamic local-record limit and explicit finite-box and representation-cutoff approximations. In that volume comparison, volume-specific canonical purifiers are traced; independent untouched references or compatible references from a common local preparation may be retained. For $\delta = a = T = 1$, the declared plan allocates 2^{-20} to each of these three errors, giving total half-trace distance at most $3/2^{20}$. Its degree, box and cutoff are enormous certified finite resources, not a materialized simulation. Preparation, apparatus compilation and numerical errors have separate budgets. The roughness estimate is recomputed for this action rather than transferred from a different polynomial.

The free symbol has one zero corner and $h_\delta(p) = \boldsymbol{\alpha}\cdot p + a|p|^2\beta/2 + O(a^2)$ at fixed p, δ. Other corners have energy at least $2/a$. This is a vectorlike free statement. The interacting physical refinement, chiral matter, particle poles and masses are not consequences of the fixed-spacing locality theorem. Sending $\delta \to 0$ is required only to reproduce the discontinuous action. The allowed $\delta \to \infty$ endpoint is the ordinary finite-range Wilson action $(\beta + H_W)/a$; it may instead be declared directly. Neither choice inherits an overlap chirality theorem.

11.5 Position and momentum on the referenced field

The actual periodic field has translations $T_S(b)$ moving every matter and gauge-link register. Their fermionic action is the signed exterior lift. They commute with the uniform action and coordinate-word cutoff, and normalize the local gauge group. This algebra applies to polynomials and the uniform smooth functional calculus above. On the mixed preparation's canonical purification, the admitted translation also moves the conjugate reference: $T_J(b) = T_S(b) \otimes \overline{T_S(b)}$. It preserves the joint Gauss-zero subspace; source-only translation need not do so. Character orthogonality gives complete conserved crystal-momentum projections $\Pi_k = V^{-1}\sum_b e^{-2\pi i k\cdot b/s} T_J(b)$ on $V = s^3$ cells, $s \geq 2$. The initial purification has joint crystal momentum zero. These are whole-field/reference records, without a single-particle identification. The momentum instrument is global; the local-record theorem does not assert a thermodynamic limit of these global projectors.

A complete position instrument has effects $F_x = n_x N^{-1} \mathbf{1}_{N>0}$ and $F_{\mathrm{vac}} = \mathbf{1}_{N=0}$, with square-root Kraus maps. It samples the site of a uniformly chosen occupied mode and retains vacuum. This menu is also global because N^{-1} depends on the whole field; the local-record theorem instead applies

to local occupation pointers. Canonical purifications are volume-specific, without an asserted cross-volume identification. For the actual maximally mixed sixty-mode-per-cell fermion marginal, define

$$A = \frac{1 - 2^{-60V}}{V}, \qquad C = \mathbb{E}\left[\frac{\sqrt{n_x n_y}}{N}\mathbf{1}_{N>0}\right] \quad (x \neq y).$$

At preparation time the complete sequential Born law is

$$p(x,0) = \frac{A + (V-1)C}{V}, \qquad p(x,k\neq 0) = \frac{A - C}{V}, \qquad p(\mathrm{vac},k) = 2^{-60V}\mathbf{1}_{k=0}.$$

Here $A - C = \frac{1}{2}\mathbb{E}[(\sqrt{n_x} - \sqrt{n_y})^2/N; N > 0] > 0$. Position measurement thus changes conserved total-momentum records, with no selected number sector or discarded outcome. This explicit initial-state formula is not claimed for the subsequently interacting state, whose complete records use its evolved density.

Finite Fourier momentum and the position instrument's first moment obey the existing finite Robertson theorem on these actual operators. The variance of the recorded unsharp position is at least the variance of that first-moment operator. In a separate complete two-site $\mathbb{Z}_2$ control these variances are $15/64$ and $11/64$, respectively, while $\operatorname{Var} P = 3\pi^2/16$ and $\langle [X,P]\rangle = i\pi/4$ for its stated phase preparation. This control illustrates the instrument algebra; it is not the native sixty-mode experiment. Finite-dimensional trace excludes $[X,P] = i\hbar I$. A universal physical uncertainty normalization and particle momentum require their own continuum identification.

11.6 A propagation estimate uniform in physical refinement

The physical-distance construction fixes positive smoothing and declares $0 \leq \kappa(a) \leq \kappa_0/a$. The weighted interaction bound is then $W_a \leq W_*/a$. Arbitrary nonnegative onsite electric coefficients are removed in the interaction picture; their magnitude does not enter this propagation estimate. For actual supports separated by assigned distance $d = aD$, the graded commutator chain gives

$$\|[\alpha_{a,t}(A_X), B_Y]_{\mathrm{gr}}\| \leq 2\|A_X\|\|B_Y\||X| \exp\left[-\frac{\log(\lambda)d - 2W_*|t|}{a}\right].$$

The weighted row sum includes every destination in Y, without a factor growing with the remote support. For the directly declared ordinary Wilson action, each native full-Fock bond has norm $30/a$, giving $W_* = 360\lambda + 48\kappa_0\lambda^2$. These constants retain all sixty native modes per cell. Neither action is identified with a chiral physical theory by this estimate.

Let $0 < a \leq 1$ and $\lambda > 1$ in the declared units. Let the observed supports fit inside a fixed physical cube of half-width R, and let $k(a) \leq Ka^{-p}$ complete local interventions occur by duration T. Deferred measurement and the actual cell count give a half-trace bound

$$\min\{1, Ca^{-(p+3)}e^{-g/a}\}, \qquad C = 2(2K+1)(2R+3)^3, \quad g = \frac{\lambda - 1}{\lambda}d - 2W_*T > 0.$$

It retains every local record and untouched reference. The remote apparatus is private, with no record sent directly to the observer. The envelope increases with a for $a \leq g/(p+3)$, so an exact dyadic planner certifies every smaller real spacing. With Wilson $\lambda = 2$, $\kappa_0 = 1/100$, $R = 1/10$, $T = 1/1000$, $d = 4$ and $K = 2, p = 0$, the bound is at most 2^{-41} for every $a \leq 1/128$. This is a conservative source cone, not a measured light speed. Separate complete implementation errors on both experiments add to it; choosing each at most a preserves vanishing influence outside the cone. A convergent interacting state and observable family is an additional premise, not a consequence of this causal bound.

11.7 An interacting thermal preparation with complete flux records

The same-source thermal construction uses the unchanged charge-six control: sixteen Dirac occupations times thirteen electric fluxes, with all 208 states and $-4I \le H \le 22I$. Its declared target is $\rho_b = e^{-bH}/Z$, $b = 1/64$. Put $A = b(22I - H)/2$ and $Q_M = \sum_{j=0}^{M} A^j/j!$. The positive Gaussian-rational state $\rho_M = Q_M^2/\operatorname{Tr} Q_M^2$ has a canonical purification prepared deterministically by a joint gauge- and parity-preserving Householder unitary. No success outcome is selected. For $x = 13b < M + 2$, its half-trace error to the target purification is at most

$$\epsilon_{\mathrm{prep}} = \frac{2x^{M+1}}{(M+1)!}\frac{1}{1 - x/(M+2)}.$$

At $M = 12$ this is below 3.27×10^{-19}. The source density is stationary; its purification need not be stationary under $H \otimes I$.

All 169 initial/final flux labels are retained. A rational Cayley unitary with J substeps approximates the original duration t with operator error at most $|t|^3 22^3/(12J^2)$. The complete original weights are $\|P_g U_{t,J} P_f Q_M\|_F^2/\operatorname{Tr} Q_M^2$. For measurements at 0 and $1/64$, $J = 4$ gives allowance $1331/6291456$, added to ϵ_{prep}, including subnormalized outcome states and the untouched reference. This is a compiled approximation to the same clock. The first flux measurement has its full backaction.

The ideal target supplies the finite Gibbs, thermal correlation and entropy identities for this actual H, with heat capacity $b^2 \operatorname{Var}(H)$. Both the target and ρ_M are passive under cyclic source unitaries. Exact thermal correlation identities at b are not asserted for ρ_M, and source passivity does not cover arbitrary operations consuming the purifying reference. The temperature and preparation service are declared; autonomous cooling and a spatial thermodynamic limit are separate constructions.

The spatial degree-zero Legendre mixed preparation has a concrete ground-state obstruction. On a periodic carrier with side at least two, fill all thirty negative-β modes at each site while retaining its gauge density. The filled state is a gauge singlet because the native representation has determinant one. Both preparations have zero intersite coherence and identical electric and magnetic means. The onsite coefficient $25\beta/(17a)$ therefore lowers the total energy by $750V/(17a)$. The mixed state is not a finite ground; the comparison neither proves that this Slater alternative is the interacting ground nor excludes every continuum interpretation of the mixed family.

11.8 A fixed source, preparation, reference and clock

The declared finite sequence uses open cubes of side $L_n = 2^{2n+1}$, spacing $a_n = 2^{-n}$ and per-link word degree $\ell_n = n + 1$, with all sixty CAR modes at each site. The finite-range Wilson action has electric coefficient 105 and magnetic coefficient $1/7$, so its gauge and matter principal speeds agree. Each site owns three outgoing links. Zero external charge is imposed also at their ghost endpoints; the electric terms remain, while hopping bonds and plaquettes require interior support. Parent operator products are formed before product-word compression, with the original energy zero. These are architectural choices without measured-mass inputs.

The physical preparation uses $P = \prod_x P_x$, the commuting local Haar projectors. The nonabelian Lie components at a single site need not commute. Since $[P, H] = 0$, the positive polynomial

$$Q = P\left[T_m\left(\frac{\beta(U - H)}{2s}\right)\right]^s, \qquad \rho = \frac{Q^2}{\operatorname{Tr} Q^2}$$

is supported on the physical carrier. The entropy bound below fixes β, m, s from ground-energy tolerance 2^{-n-4}. An exact algebraic Householder service maps the empty-Fock, constant-link blank

to $\operatorname{vec} Q/\sqrt{\operatorname{Tr} Q^2}$. The conjugate reference is retained and separately Gauss invariant, with idle Hamiltonian zero. This finite specification does not assert execution of its enormous spatial matrix or a compiled preparation tape. A finite implementation requires its own error bound.

The clock is $U(t) = e^{-itH}$ in source units with $\hbar = 1$. The marginal ρ is stationary, while its purification need not be stationary under $H \otimes I$. On the physical support, $K = \log(\operatorname{Tr} Q^2) I - 2 \log Q$; it is not replaced by the exact Gibbs modular clock. Computable source bounds, preparation and quadratic characteristic results are regenerated from these same inputs before admission. Distinct scalar, warm-flux and ground-trial controls have their own preparations and cannot supply this spatial state's detector law.

The regional restriction also consumes this exact polynomial state. A certified $c > 0$ gives $\rho \geq cP$ and $\rho_R \geq c \operatorname{Tr}_{\bar{R}} P$. Its logarithm is defined on this physical regional support; full ambient rank is unnecessary. Boundary charge sectors remain in the observable algebra. Their representation multiplicities enter regional entropy; no geometric area or local diamond-energy kernel is inferred from them. The specified simultaneous volume, spacing and word-degree sequence requires a distinct convergence proof.

11.9 Physical boundary charges and complete regional readout

For the same prepared spatial state, the boundary-state construction takes the origin cell and its three outgoing links as a region. Its cut vertices are e_1, e_2, e_3. A regional charge pairs with the opposite outside charge in every physical state; zero external charge at the box's ghost endpoints does not impose zero charge on this regional factor. The positive polynomial is faithful on the full Gauss kernel, and $\rho \geq cP$ implies $\rho_R \geq c \operatorname{Tr}_{\bar{R}} P$. Thus the prescribed deterministic preparation includes the boundary correlations without a further acceptance condition or charged seed.

Use integer-normalized hypercharge $Q_6 = 6Y$. The right-endpoint charge spectrum on the complete word space V_ℓ is $S_\ell = \{a_1 + \cdots + a_\ell : a_j \in \{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 6\}\}$. Every sum occurs: a product of corresponding diagonal matrix entries is nonzero at the identity. For the three commuting endpoint generators,

$$E_{\boldsymbol{q}} = \prod_{j=1}^{3} \prod_{r \in S_\ell \setminus \{q_j\}} \frac{Q_j - rI}{q_j - r}, \qquad \sum_{\boldsymbol{q} \in S_\ell^3} E_{\boldsymbol{q}} = I.$$

The neutral record unitary $\sum_{\boldsymbol{q}} E_{\boldsymbol{q}} \otimes S_{\boldsymbol{q}}$ uses cyclic pointer shifts and commutes with all source Gauss constraints. Its complete menu has $11^3 = 1331$ labels at degree one, including zero-probability labels. An exact control retains the entire 119-dimensional link space, since $1 + 2\sum_r d_r^2 = 119$. Hypercharge tuples can combine different nonabelian representations; their full multiplicities are not replaced by this central charge measurement.

The native character dimensions and charges are $(q, d) = (-4, 3), (-3, 2), (1, 6), (2, 3), (6, 1)$. For each representation and its conjugate, a normalized elementary Wilson-plaquette character is an actual physical vector at every selected level. Across the origin-cell cut it has charges $(-q, q, 0)$ and d^2 equal Schmidt weights. If Π_r projects onto that regional Schmidt span, positivity therefore gives

$$\Pr_\rho(-q, q, 0) \geq c > 0, \qquad \rho_R \geq \frac{c}{d^2} \Pi_r.$$

Together with the vacuum these are eleven distinct positive-weight sectors, with $c = 2^{-582407133}$ a conservative bound for the first selected state. They are boundary sectors, not particle counts or predicted abundances. This supplies a finite physical preparation and complete regional charge apparatus on the specified clock; it does not assert execution of the full spatial matrix or compatibility of states under simultaneous volume, spacing and cutoff refinement.

11.10 Locality of the complete physical boundary record

The local influence theorem applies directly to this source and its selected state. Product link compression preserves every interaction's owner-cell support and cannot increase its norm. A full-Fock Wilson bond has norm $30/a$; a plaquette has norm at most $1/(7a)$. The electric term $105C/(2a)$ is onsite in the link-ownership convention. Its interaction picture preserves the supports and norms, even when its finite-cutoff norm grows. With spatial weight 2, six bonds and at most twelve plaquettes incident on a cell give

$$\sup_x \sum_{Z \ni x} |Z| \|\Phi_Z\| 2^{\operatorname{diam} Z} \le \frac{W_*}{a}, \qquad W_* = 720 + \frac{192}{7} = \frac{5232}{7}.$$

This estimate is uniform in volume and word degree. The ambient commutator bound restricts to the source Gauss kernel because both the Hamiltonian and the admitted instruments preserve it. No tensor factorization of the physical Hilbert space is assumed.

Compare the complete boundary detector above with and without a remote unitary e^{iB_p}, where $B_p = P_\ell \operatorname{ReTr}_{15}(U_p) P_\ell/15$ on an actual interior plaquette. The selected canonical purification, clock and elapsed time T are identical in the two experiments. If its owner support has graph distance D from the origin cell, the output half-trace distance obeys

$$D_{1/2}(\mathrm{out}_1, \mathrm{out}_0) \le \min\left\{1, 2^{1-D}\left[e^{2W_* T/a} - 1\right]\right\}.$$

Every boundary-charge outcome and the unchanged full conjugate reference are included. The remote field is traced out; directly delivering a remote record defines a different experiment. At $T = 0$ the outputs agree exactly, even for entangled input. At nonzero time this is a propagation bound with tails, not exact spacelike commutativity.

For $T = 1/10000$ and a plaquette anchored at $(L_n - 1, 0, 0)$ in the second and third directions, rational bounds give respectively $2^{-2}, 2^{-7}, 2^{-30}, 2^{-124}$ at levels $0, 1, 2, 3$. At level 6, a plaquette anchored at $(64, 0, 0)$ instead fixes assigned separation $d = 1$ and gives 2^{-21}. These are analytic bounds for complete finite experiments; the full spatial histories are not executed. The cone does not prove convergence of the states or records, Lorentz covariance, or an interacting particle spectrum.

One reconstruction domain. The joint field and spectrum contract uses this fixed source, physical preparation and complete local experiment as finite inputs. The bounded filtered algebra below supplies one common state and multitime correlation subsequence with positive source energy. A physical field reconstruction additionally requires surviving fields, relativistic transformations, chiral species attachment, physical spectral and scattering laws, and an independent calibrated comparison. A nonfree interaction witness must survive in a fixed generating-field and current dictionary; connected correlations of nonlinear composites of free fields are insufficient. The ninety-coordinate conditional gauge–Higgs action is an explicit extension of the sixty-mode primary, with no supplied state, operator or clock embedding. Shared internal currents do not transfer the primary preparation or its locality theorem to that action. The separate pure compact-simple gauge branch must instantiate the same source and reconstruction requirements, together with dense vacuum correlation decay and finite-energy noncollapse. A massless photon precludes a positive global gap in the full matter theory, and selecting a Lie-algebra factor does not prove an invariant strong-sector gap.

11.11 Gap-free low-energy preparation and signed detector records

The full spatial preparation retains sixty CAR modes per cell and a tensor product of native coordinate-word spaces, each of degree at most ℓ on one link. This cutoff differs from a single

total-degree restriction. For V cells, $D \le 2^{60V} 451^{3V\ell}$. Compressing the declared Wilson action gives an explicit finite Hamiltonian with certified spectral interval $[L, U]$. The finite Gibbs entropy identity and its log-dimension bound imply

$$\operatorname{Tr}(\rho_b H) - E_0 \le \frac{\log D}{b}, \qquad b > 0.$$

Indeed $S = \log Z + b\langle H\rangle \le \log D$ and $Z \ge e^{-bE_0}$. No positive gap or ground-energy oracle is required. The finite Lean bound reuses the existing entropy normalization and cardinality theorems.

A scaled positive Taylor polynomial in $b(U-H)/2$ prepares a canonical purification with half-trace error at most η. Its source density is stationary and has mean ground-energy excess

$$\epsilon \le \frac{\log D}{b} + (U-L)\eta.$$

Temperature and polynomial order are chosen from a finite tolerance. The joint preparation retains the conjugate gauge reference and selects no success branch. Its resources can be exponentially large. Gauge invariance of this full-carrier density does not put it entirely in a source-only Gauss-singlet sector. Exact Wilson matrix entries and controlled smooth-action approximations have their separate compilation bounds; stationarity refers to the Hamiltonian actually used to prepare the polynomial state.

For every bounded probe A, stationarity and excess at most ϵ give the approximate ground inequality

$$\omega(A^*[H, A]) \ge -\|A\|^2 \epsilon.$$

The positive term is $\operatorname{Tr} \rho A^*(H - E_0)A$; the other term is bounded using $[\rho, H] = 0$. This finite premise supplies no convergence of states across physical spacings.

At fixed spacing and fixed per-link cutoff, the Wilson interaction has finite local factors and a volume-independent incident norm bound J. For a probe supported on s_0 cells, $\|\operatorname{ad}_H^n A\| \le \|A\|(2J)^n \prod_{j=0}^{n-1}(s_0 + 3j)$, giving uniform local dynamics for small time and then all finite times by composition. Cooling growing cubes to excess $1/R$ and taking a local-state subsequence passes stationarity and the ground inequality to the quasi-local limit. Spatial averaging can give a translation-invariant ground state. This is a ground construction for the Wilson quantum-link action at fixed regulators; it selects no computable subsequence and proves no spacing or link-cutoff continuum limit.

The 208-state spectral control retains the unchanged charge-six Hamiltonian and its complete compensated CAR addition/removal instrument. An exact charge conjugation preserves H and reverses the source charge, pairing the $N = 1$ and $N = 3$ spectra. Rational inertia comparisons bound its $N = 2$ ground and the conjugate charged thresholds. These finite source energies have no physical particle or GeV assignment.

Write p_i for the stationary preparation weights and w_{ij} for original spectral-event weights, with $w_{ij} \ge 0$ and $\sum_j w_{ij} \le p_i$. For every $\delta > 0$,

$$\sum_{E_j - E_i \le -\delta} w_{ij} \le \frac{\sum_i p_i (E_i - E_0)}{\delta} \le \frac{\epsilon}{\delta}.$$

Every downward event requires $E_i - E_0 \ge \delta$. Summing this inequality proves the bound for complete contraction instruments without a channel-count factor. The finite Lean proof retains raw weights rather than conditional success probabilities. Its zero-excess case has no negative-energy events.

For the selected primary family, the same inequality applies to every physical source operator A_n with $\|A_n\| \le C$ and the stationary source density of excess $\epsilon_n \le 2^{-n-4}$:

$$F_n(t) = \omega_n(A_n^* \alpha_t^{(n)}(A_n)) = \int e^{itE}\, d\mu_n(E), \qquad \mu_n((-\infty, -\delta]) \le \frac{C^2 \epsilon_n}{\delta}.$$

This concerns the source algebra; its idle-reference purification is not assumed stationary. Charged operators mapping out of the physical carrier need their own compensating source and accessible ground floor.

11.12 One filtered observable algebra and complete finite records

The common algebra construction uses the actual selected sequence. At centered rational locations $x_n(c) = a_n(c - L_n/2)$, take the integer-normalized link-charge projections and the normalized Hermitian Wilson plaquettes. They preserve every Gauss constraint and have norm at most one. Cells outside the interior at early levels and absent charge outcomes give zero operators. Every fixed location lies in the interior for sufficiently large n; no translation invariance of the state is inferred.

For each countable probe label k, choose a fixed positive rational T_k, and set

$$B_{n,k} = \int f_{T_k}(s) \alpha_s^{(n)}(A_{n,k})\, ds, \qquad f_T(s) = (1 - |s|/T)_+ / T.$$

The bounds $\|B_{n,k}\| \le C_k$ and $\|\alpha_h^{(n)}(B_{n,k}) - B_{n,k}\| \le 2C_k|h|/T_k$ hold uniformly in n. Let $\mathcal{B}$ be the unital C* algebra of bounded matrix sequences generated by these probes, their adjoints and all rational-time translates. It is separable and invariant under every real source-time translate. The group is point-norm continuous by the displayed bound and polynomial approximation. The norm-null sequence ideal $\mathcal{I}$ is closed, two-sided and invariant, so $\mathcal{A} = \mathcal{B}/\mathcal{I}$ inherits that group.

Diagonal extraction on a countable dense set of star polynomials makes $\mathrm{Tr}(\rho_n b_n)$ converge simultaneously for every $b \in \mathcal{B}$. The limit is a positive normalized stationary state annihilating $\mathcal{I}$. Its canonical GNS implementation satisfies $U(t)\Omega = \Omega$ and $U(t) = e^{itL}$. Every cyclic vector is represented by a uniformly bounded physical section. The negative-transfer bound, together with continuous convergence of its ordered correlations, implies that its spectral measure is supported on $[0, \infty)$. Density of these vectors gives $L \ge 0$. The GNS null space is a left ideal, not a second C* quotient ideal. No exact finite inter-level embedding or effective subsequence algorithm is assumed.

The same source cone supplies operator commutation in this quotient. For supports with assigned separation d, polynomially bounded cell counts and rounding error $O(a_n)$, the filtered commutator vanishes when

$$d/2 > 2W_*(|t| + T_k + T_j), \qquad W_* = 5232/7.$$

This conservative cone includes both temporal filter widths. It proves neither boost covariance nor an observed light speed.

For a filtered charge projection set $E = B$; for a normalized Hermitian loop set $E = (I + B)/2$. Continuous functional calculus gives the complete pair $\sqrt{E}, \sqrt{I - E}$ in the same sequence algebra. Every fixed finite ordered instrument history K_r therefore has a convergent original probability $\omega_n(K_r^* K_r)$, and repeated completeness preserves their sum one. Its finite pointer entries $\omega_n(K_s^* K_r)$ converge in trace norm. The same subsequence carries all these histories, including complementary outcomes and disturbance. This pointer-marginal theorem does not give a limit of the growing canonical reference, a complete infinite charge PVM without charge tightness, or a nontrivial physical field representation.

The finite filtered apparatus implements coherent positive-weight quadrature in the same source clock. For M midpoint cells and spectral width $U-L$, its successful Kraus operator obeys $\|B_M - B\| \le (U-L)T/M$. Every failure output is retained. Cayley source echoes have certified error $|t|^3\|H\|^3/(12r^2)$, and a complete Ramsey instrument measures the original real and imaginary ordered correlations. A linear-combination success map B_M has effect $B_M^*B_M$; it is distinct from the square-root effect instrument above. Positive quadrature gives an effect E_M; if $\|E_M - E\| \le \delta$, square-root continuity bounds the complete output half-trace error by $\min(1, \sqrt{2\delta}+\sigma)$ for arbitrary reference, including a compiler allowance σ. Finite history errors add by composition. These are finite computable constructions, with no efficient full spatial execution or outcome postselection claim.

Continuous spatial translations on the same regular probes. The spatial construction uses the identical local source on an infinite comparison lattice at each fixed regulator. In its gauge-invariant algebra, every local Haar projector P_x is central. The quotient by the closed ideal generated by $I - P_x$ is nonzero: invariant empty-Fock and constant-link states satisfy all these constraints. Integer shifts act on this physical quotient and commute with its homogeneous Wilson time group. The actual finite state extends with invariant exterior blanks because its ghost endpoints have zero charge. This is a state extension, without an asserted isometric embedding of finite physical observable algebras.

Deleting boundary-crossing interactions leaves the original finite Hamiltonian on its owned factors, including dangling electric links. There are $6L^2$ crossing bonds and $12L^2 - 3L$ crossing plaquettes. The source cone therefore bounds the bulk clock difference for a contraction with s owner cells and collar r by

$$\|\alpha_t^{\infty}(A) - \alpha_t^{\mathrm{box}}(A)\| \le \min\left(2, \frac{2544 s L^2 |t|}{7a} e^{-r\log 2 + 2W_*|t|/a}\right).$$

At fixed physical support and time this tends to zero along the selected sequence: the centered collar grows as 4^n, while $1/a_n = 2^n$. No stationary infinite-volume extension is assumed before the limit.

For an integrable spatial test function f, form $A_n(f) = \sum_z (\int_{a_n(z+[0,1)^3)} f(x)\,dx) A_{n,z}$. Then $\|A_n(f) - A_n(g)\| \le \|f-g\|_1$. Rounded lattice shifts $\lfloor y/a_n \rfloor$ induce the exact action $\beta_y A(f) = A(f(\cdot - y))$ on the regular norm-null quotient. Their finite rounding defect vanishes there, and translation continuity in L^1 makes this an action of $\mathbb{R}^3$ continuous in operator norm on each observable. It commutes with the source time action. One diagonal extraction retains both the original point menu and this regular extension; the bulk estimate identifies all limiting word moments and their cyclic GNS representations. Bounded averaging may give only scalar or zero connected responses. Spatial invariance of this selected state is not proved. Only with that invariance do canonical spatial unitaries fix the GNS vacuum and supply commuting momentum generators; spatial Fourier labels alone are not measured momenta.

A quantitative test for a surviving excitation. For a filtered spectral measure with mass at most M_0 and first absolute moment at most M_1, let a complete experiment bound $F(0) - \mathrm{Re}\, F(t) \ge d_0$. If $|t| \le t_0$ and its negative weight below $-\delta$ is at most η, the positive-window estimate gives, for $0 < \delta < R$,

$$\mu([\delta, R]) \ge \max\left(0, \frac{d_0 - M_0(\delta t_0)^2/2 - 2\eta - 2M_1/R}{2}\right).$$

For the actual filtered primary probe, $M_0 = C^2$, $M_1 = 2C^2/T$ and $\eta \le C^2 2^{-n-4}/\delta$. An eventual strictly positive lower bound on one fixed compact interval would exclude zero-energy collapse. The finite primary witness below uses level-dependent times and intervals and does not prove this uniform bound. A pole, interaction and relativistic species require further source observables even if this test succeeds.

11.13 Finite excitation weight, rest energy and physical units

The finite primary witness uses the actual normalized Hermitian plaquette $A = P_\ell \,\mathrm{ReTr}_{15}(U_p) P_\ell/15$ at the centered interior location, for $n \ge 1$. Let b be the physical empty-Fock, constant-link vector. It is a matrix-element witness, not a replacement for the selected density. The five native irreducible character loops and their conjugates are ten orthogonal vectors in the cutoff carrier. Their trace-metric Casimirs are $4/5, 9/20, 21/20, 7/10, 3/10$. The matter commutator annihilates b; the magnetic commutator does too, because both parent multiplication operators commute and their images of b lie inside the product cutoff. The complete electric action then gives

$$\|[H, A]b\|^2 = \frac{K}{a^2}, \qquad K = \frac{4949}{20}.$$

The span of these vectors is not asserted invariant under the full Hamiltonian. For the actual physical density floor $\rho \ge cP_G$, $c = 2^{-b_n}$, the original spectral measure has second moment at least cK/a^2. Here b_n is a density-floor exponent, not a cosmological capacity.

Let W_n be the certified full source spectral width and choose the level-dependent triangle width and observation time $T_n = t_n = 1/W_n$. Writing $q = (23/24)^4$, the sine and cosine remainder bounds give

$$F_B(0) - \mathrm{Re}\, F_B(t_n) \ge \frac{11qKc}{24a_n^2 W_n^2} > 0.$$

The exact polynomial preparation is nonincreasing in source energy. For a Hermitian probe this implies that every positive-frequency set has at least the weight of its reflected negative-frequency set. Consequently, with $\delta_n = (8/a_n)2^{-\lceil b_n/2 \rceil}$,

$$\mu_{B_n}([\delta_n, W_n]) \ge \frac{(qK - 64)c}{2a_n^2 W_n^2} > 0.$$

The complete finite quadrature and Cayley apparatus has a precision schedule resolving a positive contrast and retaining positive original weight in the same band. Every failure and the idle quantum reference are included. Its gate count can be enormous; no full spatial matrix or gate tape is executed by these bounds. The lower endpoint and proved weight tend to zero, the upper endpoint diverges, and T_n shrinks. Thus this witness proves finite excitation, without a uniform fixed-band particle response.

The empty and fully filled Fock sectors are exact physical reducing sectors, with gauge action $H_E + H_B$. Their combined selected weight is at most $7/(627\, 2^{8n+7})$. A physical trial filling the two negative Dirac rows gives ground upper bound $-90V/a$, whereas these extreme sectors have energy at least $-P/(7a)$, with P the plaquette count. Conditioning on either sector discards that original probability cost and does not supply a surviving pure-gauge particle sector of this preparation.

Spatial and internal channels of the finite response. The complete spatial Fourier instrument coherently selects M physical plaquettes and retains all $2M$ address and complementary outcomes. For unsmeared normalized loops, each blank Fourier label has original weight at least $2^{-b_n}/(90M)$ on the selected state. Shifting the open aperture by one address has operator phase

error at most $2/M$; these records are not a conserved momentum PVM. The exact source channels include five conserved occupation counts N_q, with $(q, d_q) = (-4, 3), (-3, 2), (1, 6), (2, 3), (6, 1)$ and $\sum_q qN_q = 0$ on the physical carrier. Plaquette instruments change none of them. Ghost Gauss constraints fix dangling links to constants, so proper cube rotations act on the physical interior graph and preserve the prepared density. Averaging the selected plaquette's 24-point orbit gives a distinct cubic scalar probe with $\|[H, A_{A_1}]b\|^2 = 4949/(480a^2)$ and positive finite excitation weight. This is an operator channel, without a scalar-vacuum or Lorentz-spin assignment. The product-vacuum expansion separates $aH = H_0 + \lambda V_{\mathrm{pert}}$, with the primary source at $\lambda = 1$. The onsite/electric anchor has physical gap 6 and the filled negative-Dirac-row vacuum. Its exact finite energy expansion is

$$\widehat{E}(\lambda) = -90V - \lambda^2 \left[\mathcal{B} \sum_r \frac{d_r}{6 + 105C_r/2} + \mathcal{P} \sum_r \frac{2}{210^3 C_r} \right] + O(\lambda^3),$$

where $\mathcal{B} = 3L^2(L-1)$, $\mathcal{P} = 3L(L-1)^2$ and C_r are the five trace-metric Casimirs above. Orthogonal hopping and loop packets give these coefficients on the actual carrier. No remainder bound reaches $\lambda = 1$, and the anchor vacuum does not replace the selected polynomial state. These finite channel and perturbation results do not supply a momentum-resolved particle pole or a physical mass.

Interacting endpoint bounds and joint spectroscopy. At $\lambda = 1$, a one-link electric Schur inequality retains all Fock occupations and the quantum link. Assign onsite cost $\mu = 3/d_{\max}$ to each bond endpoint, where $d_{\max} = 3$ for $L = 2$ and 6 otherwise. The exact two-site fermion spectrum gives $\Delta(\mu) = 60(\sqrt{\mu^2 + 1/4} - \mu)$. The nonconstant link block has positive floor $63/4 - \Delta(\mu)$, and Haar contraction in native block r has full-occupation norm at most $2(d_r + 1)$. Summing the resulting operator inequalities before Gauss restriction gives

$$E_0(H) \geq \frac{-90V - \mathcal{B}c(\mu) - \mathcal{P}/7}{a}, \qquad c(\mu) = \sum_r \frac{2(d_r + 1)}{105C_r/2 - \Delta(\mu)}.$$

An exact eleven-vector Rayleigh calculation uses the physical anchor and its five hopping and five real-loop packets. An occupation-sensitive Schur estimate retains the native quartic's vacuum mean d_r, variance one and four-defect off-diagonal image. At $n = 0$ it gives $-727.498195 \leq E_0 \leq -723.7678199$ in source units, of width below 3.731. Exact parent Haar and CAR contractions bound the full-space trial residual by 5.231016. The parent product has word degree at most two; the degree-one source projection contracts its norm. These bounds retain every occupation and omitted state. They do not isolate a particle branch or replace the selected polynomial preparation.

Retaining 2/5 of the onsite defect penalty, with the full electric coefficient in the link clusters, proves $\mathrm{Tr}\,\rho D < 5$ on this member. Here D counts positive-beta occupations and negative-beta holes, rather than topological species. For the sixteen matched local probes $B_j = c^\dagger_{j,+} c_{j,-}$ in the one-dimensional native charge-six block, $\sum_j B_j^\dagger B_j \geq 16 - D$ gives averaged original success at least 11/16. The probes themselves are gauge neutral and even. Their actual interacting commutators satisfy $\|[H, B_j] - 6B_j\|^2 \leq 3$ at each corner, including overlapping neighbor vectors in the full four-mode CAR calculation. Hence

$$\frac{1}{16} \sum_{j,u,v} p_u |(B_j)_{vu}|^2 (E_v - E_u - 6)^2 \leq 3.$$

The original transfer measure has weight at least $11/16 - 3/16 = 1/2$ in $[2, 10]$. Selected cooling with excess 1/16 and initial-excitation cutoff 1/2 gives final excitation in $[2, 21/2]$ with weight at least 3/8. Complete finite clocks incur their stated tails and implementation errors. No successful-record

normalization or exact-ground substitution enters the bound. The common retained fraction used for this local estimate fails the next member's admission. A separate block-sensitive bound below controls larger sources. A resolved energy–momentum branch and physical channel attachment are required for a rest mass.

Concentrated response and its unresolved spectral type. Retaining only the native charge-six onsite fraction $r_6 = 4/5$ in the full-occupation electric inequality sharpens the selected bound to $\langle D_6\rangle < 1.840$. The actual corner residual squared is at most $3/2 + \sqrt{2} < 102/35$. The original local-pair measure therefore satisfies

$$\mu([4,8]) \geq \frac{3}{20}, \qquad \operatorname{Re} F(1/2) < -\frac{3}{8}, \qquad F(t) = \int e^{it\omega}\, d\mu(\omega).$$

Cooling gives final excitation weight at least $7/80$ in $[4,9]$. Two complete energy clocks retain original reported weight at least $9/64$ in $[63/16, 129/16]$. A separate complete Ramsey instrument measures the signed correlation, retaining both failure signs and the idle reference. These statements use the original polynomial state. They permit a single line or several transitions; neither a threshold onset nor a pole is isolated.

An independent twelve-vector A_{1u} trial separates scalar-pair electric packets with relative denominators $63/4$ and $87/4$ and squared norms 3 and $21/2$. Its full-sector Rayleigh upper bound is -717.8547086510 source units, rounded outward. The difference from a ground-trial upper bound is not an excitation bound. An exact local moment calculation instead bounds its full outside-space residual by 5.365163597. The charge-six second-action Haar Gram, keeping all 32 CAR modes and independent link characters, is

$$\begin{pmatrix} 93/2 & 123/2 \\ 123/2 & 348 \end{pmatrix} \quad \text{for } YP_{D_6=2}Y\phi,\ YP_{D_6=4}Y\phi.$$

Combining it with the other native and magnetic moments puts an A_{1u} eigenvalue in

$$[-723.219872248, -712.489545054].$$

Its lower endpoint exceeds the certified ground upper. With the same source's ground lower bound, some strictly excited A_{1u} energy lies in $[0.54794769, 15.00864937]$. This is neither a global gap nor an isolated spectral line. The trial is not substituted for the preparation. Optimizing the retained onsite/electric fractions within this Schur-bound family cannot certify isolation against the authenticated ground upper: two rational cases bound every resulting complement floor below -724, whereas that upper is greater than -724. This excludes that estimate, not a physical gap or a different resolvent calculation.

The selected coherent scalar outcome is $K = T/(16\sqrt{2})$, with $\|K\|^2 = 9/64$ and $[K^\dagger, K] = (16 - D_6)/512$. In beta particle/hole variables the hopping residual splits into a pair-creation term and a number-preserving term. Their full-Fock form bounds are respectively $48(1 + D_6/2)$ and $36D_6$. Young's inequality therefore gives

$$([H,K] - 6K)^\dagger([H,K] - 6K) \leq \frac{624 + 858D_6}{3584}.$$

On the unchanged selected state this proves original transfer weight greater than $5/1024$ in $[3/4, 45/4]$ and final excitation weight greater than $1/2048$ in $[3/4, 53/4]$. This is the coherent apparatus's original normalization, not a normalized successful subensemble. Overlap with the trial enclosure does not identify its transitions with that eigenvalue.

Positive physical occupation thresholds. The full-space sector calculation uses the five conserved occupations in native charge order $(-4,-3,1,2,6)$. Relative to the filled-beta occupations, the increments $\pm(0,2,0,0,1)$, $\pm(1,0,0,2,0)$ and $\pm(0,0,0,3,-1)$ have explicit local gauge-singlet witnesses. The first two pairs are CAR odd; the third is even. Exact kernels of all twelve actual gauge generators verify the weak and color contractions. On each reducing sector, $D_j \geq |\Delta N_j|$, so the full-occupation electric Schur bound retains a positive sector-specific onsite penalty. Together with exact interacting trial quotients and the same full-source ground bracket, this encloses their sector bottoms above the ground in, respectively, $[1.5467, 12.7132]$, $[0.5664, 12.7244]$ and $[3.5668, 15.7156]$ source units. The lower endpoints bound every sector energy; the upper endpoints bound sector minima. No isolated-level assumption is used. Two idle physical memories prepare opposite increments together; a graded exchange and parity correction retain the displaced source and its arbitrary reference correlations. Combined native occupations and fermion parity are preserved. A complete clock gives excited-window probability at least $255/512$. The original polynomial state's probability in either signed sector instead obeys the cooling upper bound $(1/16)/(L_\Delta - U_0)$; it is not replaced by the prepared experiment. No relativistic rest energy is identified by these finite thresholds.

Conjugate source sectors. The full source charge conjugation combines the finite CAR transformation $c \mapsto Mc^\dagger$, with $M = \gamma_2$ in the actual Clifford chart, and the linear link-function pullback $g \mapsto \overline{g}$. The native exterior representation obeys $R(\overline{g}) = \overline{R(g)}$ and $\det R(g) = 1$. Exact multiplication gives $-M^T\beta^T M^* = \beta$ and $-M^T S_i^T M^* = S_i^\dagger$ for every forward Wilson block. Thus this unitary preserves the Gauss projector, product cutoff, original Hamiltonian and polynomial state. It sends each occupation deviation $N_q - 2d_q V$ to its negative, proving exact spectral equality of opposite sectors at every finite capacity. Conjugate complete instruments have equal original energy-record probabilities on the same clock. Proper binary cube rotations commute with this transformation; spatial inversion obeys $C\mathcal{I}C^{-1} = (-1)^{N_F}\mathcal{I}$. The canonical-reference comparison uses $C \otimes \overline{C}$, without an active reference gate. The centered local number changes sign, so the two signs of each local phase probe have exactly equal original energy laws on the selected state. No Lorentz spin, physical antiparticle identity or relativistic CPT theorem is inferred.

Local multiplets and actual exchange. The full minimal gauge kernels allow both beta spin rows. Their ambient dimensions are $12, 90, 40$ for the three occupation types above, and their gauge kernels have dimensions $6, 6, 8$. The actual Dirac coefficient generators give Casimirs $3/4$ and $15/4$ with multiplicities two and four in both odd families, and $2, 6$ with multiplicities three and five in the even family. Restricting the exact exterior action to all 48 binary-cube elements, with a uniform eight-site orbital preparation, gives $G_{1g} \oplus H_g$ for the positive odd increments, $G_{1u} \oplus H_u$ for their negatives, and $T_{1u} \oplus E_u \oplus T_{2u}$ for both signs of the even increment. Here G_1 and H are the two- and four-dimensional cube representations; the local spin coefficient algebra is not a continuous symmetry of H. Every component has a positive full-sector floor and a complete prepared symmetry/energy event of weight at least $255/512$.

Replacing each kernel vector by its signed quasiparticle monomial gives a gauge-invariant operator F_v with $[N_r, F_v] = \Delta_r F_v$. The trilinears anticommute at distinct sites, including against adjoints; their local anticommutator need not be the identity. This proves finite graded exchange and actual current labels, without assigning Lorentz spin, chirality or Standard Model particle names. The calculation explicitly distinguishes the ascending-mask gauge chart from the lexicographic exterior-coordinate chart and intertwines them before applying a non-diagonal link matrix.

Full interacting composite transport. For normalized localized odd singlets ϕ_x, the source mean is $b = -711$. Occupation transfer forces $\langle \phi_y, H^j \phi_x \rangle = 0$ for distinct sites and $j < 3$. The exact full-source third moment is $-3/4$ between adjacent sites, times the local six-state spin overlap. Only three directed hops on one link contribute. Gauge invariance removes that link coordinate pointwise, and the beta-preserving Wilson blocks give $3!(-1/2)^3 = -3/4$. First-action admission on both sides proves the identity after the actual degree-one link compression; no invariant eight-state subspace is assumed. If $R \geq \max_{z \in \{x,y\}} \|(H-b)^2 \phi_z\|$, the full-space Taylor remainder gives

$$0 < |t| \leq \frac{3}{2R^2} \quad \Longrightarrow \quad |\langle \phi_y, e^{-itH} \phi_x \rangle|^2 \geq \frac{|t|^6}{256}.$$

For the weak-pair/singlet one-spin-row witnesses, the certified interval reaches about 2.77×10^{-6} source time units using the fourth-moment bound below. The complete detector tests the full bare target and retains its complement. Coherent cube characters have different third energy moments, despite equal means and variances. Exact weak-interaction effective hopping coefficients have a certified coupling radius below 0.009; the selected interaction value one is outside it. Neither those coefficients nor the moment matrix is a physical dispersion spectrum.

For the specified weak-pair/singlet Slater composite, an exact parent fourth-moment calculation gives $\langle (H-b)^4 \rangle \leq M_4 = 9738478674383/18007500$ after the actual link compression. Conditional Slater traces keep all native-block cross terms; Haar integration retains the 120 backtracking four-step cube walks and removes the 48 plaquette walks. This supplies a full-complement resolvent enclosure at interaction coefficient one. Write $\langle \phi, (H_\Delta - z)^{-1} \phi \rangle = (-711 - z - \Sigma(z))^{-1}$ on the reducing occupation sector of the composite. With $L = -722.22108$, $v = 132301/735$, $d_3 = 369869271/42875$, $\mu = -711 + d_3/v$ and variance cap $\tau = M_4/v - v - (d_3/v)^2$, its full-complement self-energy obeys, for $z < L$,

$$\frac{v}{\mu - z} \leq \Sigma(z) \leq \min \left\{ \frac{v[(L-z)(\mu - L) + \tau]}{(L-z)[(\mu - L)(\mu - z) + \tau]}, \frac{v(-711 - L)}{v + (L-z)(-711 - L)} \right\}.$$

The complement contains every other physical state in that sector. For example, $\langle \phi, (H_\Delta - L + 8)^{-1} \phi \rangle$ lies between 0.060458 and 0.070342. An independent cubic polynomial of $H + 711$ certifies prepared weight above $1/26$ in $[L, -714]$; a complete clock retains more than $1/27$ in the widened report. These bounds use the explicitly prepared composite, retaining the displaced original state and reference. They also imply finite original-state positivity: $\rho \geq c P_G$ and $A\Omega = \phi$ give

$$A \rho A^* \geq c |\phi\rangle\langle\phi|.$$

Applying the same positive complete clock effect yields weight above $c/27$ for fixed A, or $c/(108V)$ for its one retained row in the uniform four-spin/address instrument. The clock error is included before this multiplication. This is a final-energy event with a deteriorating faithfulness floor, without an initial-energy or transfer-law assertion. It supplies neither an isolated pole nor a refinement-uniform residue.

Odd creation on the selected state. A complete trilinear probe creates the two weak entries in one positive-beta spin row and the charge-six singlet in either spin row. A loaded physical memory supplies the compensating occupations, with failure and reference retained. The source input is the original polynomial density. Averaging the four spin choices and cells, charge conjugation and vacancy counting give

$$p \geq 1 - \frac{\langle D_{-3} + D_6 \rangle}{4V}, \qquad \|[H, A] - 9A/a\|^2 \leq \frac{3 z_x}{2a^2},$$

where z_x counts actual neighbors. After factoring out $\sqrt{z_x/2}/a$, the residual is a sum of three exterior creation monomials on six orthonormal modes with full-CAR squared norm exactly three. The full link/Gauss compression preserves the bound. An occupation-sensitive Schur inequality gives $\langle D_{-3} + D_6\rangle/V < 3.41$ uniformly. Thus the original transfer measure satisfies

$$\mu_0([3,15]) > \frac{23}{32}, \qquad \mu_n([1/(2a_n), 35/(2a_n)]) \geq \frac{1}{50}.$$

Complete two-clock reports retain at least $1/64$ in the corresponding enlarged band. At member zero the original final excitation has weight above $21/32$ in $[3,16]$. No success normalization or substitute vacuum enters these bounds. The uniform moving band proves that these unscaled local-composite measures are not uniformly energy tight: at least $1/50$ escapes every fixed upper energy. A separate finite-energy component can coexist with this weight; its existence and particle interpretation are not established by this estimate.

Spin-resolved records and finite filtering. The complete six-row creation instrument resolves its two-dimensional G1 and four-dimensional H probe multiplets on the original selected state. Exact full-CAR occupation inequalities and residual bounds give complete reported weights above $121/512$ and $61/128$, respectively, in $[11/4, 61/4]$ at member zero. Uniformly in capacity, their reported weights exceed $1/200$ and $3/200$ in $[1/(4a), 71/(4a)]$. The labels transform as the actual source cube representations; the mixed input and final state are not assigned those total-spin eigenvalues. Coherent occupation-transfer meters distinguish the native increments without measuring the initial absolute occupations.

The coherent spatial instrument also gives positive original response in each of the eight cube Walsh rows. For a chosen flat phase, set $A_{f,j} = \sum_x f_x F_{x,j}$ with $|f_x|^2 = 1/8$, and retain all six creation operators and their adjoints, divided by $\sqrt{6}$, together with the full failure. Exact local anticommutators give completeness and the original vacancy floor. Four colors pair opposite cube vertices with disjoint residual stars, bounding the total signed centered second moment by $110/3$. The actual polynomial state supplies a stronger paired-adjoint balance: for positive transfer ω,

$$e^{-\beta\omega}\, d\mu_A(\omega) \leq d\mu_{A^*}(-\omega) \leq e^{-\beta_{\mathrm{low}}\omega}\, d\mu_A(\omega), \qquad \beta_{\mathrm{low}} > 22271.$$

This uses the actual selected polynomial, not a substituted Gibbs state. Combining the bound with the signed centered moment gives original creation weight above $1/3$ in the reported interval $[1/4, 71/4]$; the G1 and H parts exceed $5/48$ and $11/48$. In one complete eight-row Walsh instrument, each row therefore has weight above $1/24$, with G1 above $5/384$ and H above $11/384$. All 96 signed successes and eight address failures remain present. The adjoint records annihilate excitations; charge-conjugate creation is a different operation. These are populated spatial analyzers, without an isolated conserved-momentum dispersion branch.

The actual total cube and native-number projectors commute with H, ρ and every energy-clock kernel. The cube group has 96 elements: 48 proper binary rotations and a separately implemented inversion. Complete initial and final measurements of these labels preserve the original classical energy marginal, while generally changing its reference coherence. Tensor products allow 28 G1 and 44 H total-cube transitions. Positive original weight implies at least one populated allowed pair, not that every permitted pair is populated. For the filled-beta native occupations $N_0 = (48, 32, 96, 48, 16)$ and $\Delta = (0, 2, 0, 0, 1)$, explicit physical witnesses and the actual floor $\rho \geq cP_G$ prove original initial-label/final-energy branches

$$(N_0, A_{1g}) \longrightarrow (N_0 + \Delta, G_{1g}) \text{ or } (N_0 + \Delta, H_g),$$

with weights greater than $85c/4096$ and $85c/2048$ in the certified final-energy window. Charge conjugation gives the corresponding $N_0 - \Delta$ branches. These very small bounds do not use an initial energy condition or replace the selected state by the witness. Two orthogonal physical vectors in each indicated final sector have energy means -711 and -699, so a symmetry label cannot identify one energy. The means are not eigenvalues or particle masses. More directly, the actual first hopping action on the filled-beta witness has residual norm squared $9/4$, orthogonal to its created component. The original source floor therefore gives

$$\operatorname{Tr}\rho\,([H,A_f]-\omega A_f)^*([H,A_f]-\omega A_f)\geq c[(9-\omega)^2+9/4].$$

Each normalized original transfer measure has variance at least $9c/4$ and cannot consist of one spectral line. This tiny bound gives no useful line separation, no resolution claim for the current clocks, and no exclusion of an isolated pole accompanied by other spectral weight.

Cube symmetry gives an exact reduction of the original energy-transfer kernel after summing the components of either probe multiplet. Its entry between vertices x, y is a real spectral measure m_d, where d is their Hamming distance; off-diagonal measures can be signed. The four Walsh eigenmeasures are

$$\begin{pmatrix}\Lambda_0\\ \Lambda_1\\ \Lambda_2\\ \Lambda_3\end{pmatrix}=\begin{pmatrix}1 & 3 & 3 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & -1 & 1\\ 1 & -3 & 3 & -1\end{pmatrix}\begin{pmatrix}m_0\\ m_1\\ m_2\\ m_3\end{pmatrix}.$$

Their multiplicities are $1, 3, 3, 1$. The matrix squares to $8I$; the selected-phase measure is $\Lambda_s/6$, and one row of the complete apparatus has measure $\Lambda_s/48$. Thus four measures determine all eight spatial rows. This diagonalizes a probe spectral kernel, not the source Hamiltonian. Nonzero off-diagonal measures give unequal joint responses; unequal intensities alone do not imply different normalized energy profiles or dispersion. The component-resolved and cross-multiplet kernels require their full transformation laws.

Original density shells and recorded mean contrast. For $n_x = N_x - 30I$ on the same eight-cell source, a complete apparatus has successes $K_q = \sum_x (-1)^{q\cdot x} n_x/(30V)$ and failures $\sqrt{I-(n_x/30)^2}/\sqrt{V}$, $V = 8$. The exact density Ward sum is

$$M^{(1)}=\frac{\tau}{2}(3I-\mathrm{Adj}_{\mathrm{cube}}),\qquad \tau=-\frac{\operatorname{Tr}(\rho H_{\mathrm{hop}})}{12}\geq 0.2373480904.$$

Its complete-row signed first moments are $s\tau/7200$, where $s = |q| = 0, 1, 2, 3$ has multiplicity $1, 3, 3, 1$. The constant mode is exactly elastic; each nonconstant mode has positive inelastic weight. The $1 : 2 : 3$ ratios concern unnormalized energy-weighted records, not particle masses or success-conditioned energies. The actual finite clocks give per-row mean error below $\gamma/14400$, with $\gamma = 0.2373480904$ replaced by its exact rational lower bound; thus their reported means resolve this contrast.

Native sublattice parity reverses each block's hopping T_r. The strict polynomial slope and the actual Haar/CAR first action give $-\operatorname{Tr}(\rho T_r)/12 \geq 2d_r c\beta_{\mathrm{low}} > 0$ for each $d_r = 3, 2, 6, 3, 1$. These are even density channels, not created fermions. The complete success weight is at most $\langle D\rangle/(30V) < 0.0208$, from $n_x^2 \leq 30D_x$. An explicitly conditional signed mean is also bounded, but neither conditioning nor the shell contrast isolates a rest-mass pole.

An explicit complete filter uses a triangle of half-width $T = a/6$, analyzer frequency $9/a$ and 1024 selector cells. The actual centered commutator bounds its quadrature error by $1/2048$; algebraic

source clocks and two energy clocks retain original blank probability above $3/5$ in $[11/4, 61/4]$ at member zero. Summing every selector outcome exactly restores the original creation energy law. Demodulation selects a response without shifting source energies. A fixed-width zero-centered filter has success limsup at most $1/9$, with no positive fixed-energy residue bound. Dressing the complete odd output by critical-density phase operations and source waits requires at least $[(a^{-1/2}-20)/450]_+$ such operations to reduce its original excitation weight above energy one to $1/200$. This is a necessary depth for that operation class, without excluding other dressed fields.

Prepared scalar-odd spectroscopy. An alternative source-determined rational combination of the same twelve packets has full-space residual at most 4.947619. Its exact residual form encloses some actual A_{1u} excitation in $[1.0666, 14.6923]$ above the same ground. The coefficients change the trial, not the source action. A complete preparation makes this vector operational: prepare it by an algebraic Householder unitary in an identical idle physical carrier, then apply the graded exchange with the source. The displaced original state and all its reference correlations remain in that memory. Both operations preserve physical Gauss support and combined fermion parity; the source marginal changes by this explicit intervention. The declared memory allocation is at most 696 qubits, without an efficient implementation claim. With prepared mean $c \simeq -717.75357$, variance $\sigma^2 < 49/2$ and radius $r = 23/4$,

$$\Pr_{\psi}\{H \in [c-r, c+r]\} \geq 1 - \sigma^2/r^2 > 137/529.$$

The whole interval lies above the ground upper bound. One complete source-energy clock, with resolution $1/64$ and tail/implementation allowances $1/1024$ each, retains reported probability greater than $137/529 - 1/512 > 1/4$. All original source information, memory and reference are retained. This is a prepared excited window; no single line or original mixed-state detector residue is assigned to it.

An exact checkerboard cancellation. The checkerboard scalar outcome uses the unchanged selected state with

$$T_w = \sum_{x,\alpha} (-1)^{x_1+x_2+x_3} B_{x\alpha}, \qquad K_w = T_w/(16\sqrt{2}).$$

The opposite signs on every bond cancel the pair-creation block of $[H, T_w] - 6T_w$. The remaining trace-zero number operator has full-Fock squared form bound $36D_6$, giving

$$\int (\omega - 6)^2 \, d\mu_w < 207/1600, \qquad \mu_w([3,9]) > 17/1280.$$

Cooling gives original final-excitation weight greater than $23/5120$ in $[3, 10]$. The actual hopping matrices also commute with $i(T_w - T_w^\dagger)$; the separately completed Hermitian quadrature has signed first energy moment greater than $531/1600$. The finite operator type is A_{2g}. It specifies neither a final-state irrep nor physical spin or momentum, and its checkerboard label does not stay at fixed physical momentum under refinement.

A populated scan of spatial phase. The phase-scanned analyzer multiplies the complete eight-cell Walsh transform by $e^{-i\kappa\cdot x}$. All 64 spatial/internal success and failure records are retained. Its scalar outcome is $K(q) = \sum_{x,\alpha} e^{-iq\cdot x} B_{x\alpha}/(16\sqrt{2})$, with $q = \kappa + \pi k$. The actual Wilson pair-creation residual has Hilbert–Schmidt norm squared $16\sum_i \cos^2(q_i/2)$; the number remainder has full-Fock squared form at most $36D_6$. This gives positive original energy bands for every phase

and a populated neighborhood of the checkerboard cancellation. At $\kappa = (\pi/16)^3$, the eight disjoint scalar energy-band events have total original weight above 0.06646. Their complete reported weight exceeds $1/16$; their final-excitation union has weight above $15/256$ using the scalar subinstrument bound $I/4$. Exact cube covariance compares the corresponding complete experiments. The phase labels specify the finite aperture; differing interval widths are neither a measured dispersion curve nor a species count.

Admitted spectral improvement and exact symmetry reduction. The two-link source trial adds 30 mutually orthogonal second-hop packets to the existing 13 coordinates. Distinct-link actions remain inside the degree-one cutoff; their exact CAR/Haar norm squares sum to 59220. Bounding only their magnetic quadratic form gives a rational full-sector Rayleigh upper $U_\Delta = -715.100189896245\ldots$, improved from $-714.7851\ldots$. This is an upper bound, not a computed eigenvalue or a mass. The odd native sector carries only binary-cube G1, G2 and H irreps, so each energy has multiplicity at least two. A rank-one complement in that entire sector necessarily retains a ground vector. The exact source matrix units

$$E^{\lambda}_{ab} = \frac{d_\lambda}{48} \sum_g \overline{D^{\lambda}_{ab}(g)} U_g$$

commute with H and reduce it to $I_{d_\lambda} \otimes h_\lambda$. All eight diagonal projectors are retained. A rank-one Feshbach test inside one multiplicity space therefore avoids the compulsory partners; its complementary spectral bound remains uncomputed.

The original polynomial slope and global ground lower bound L_0 give $\rho \geq c_{\mathrm{low}} 1_{H \leq U_\Delta}$ on physical source support, with $c_{\mathrm{low}} = 2^{-552953}$. This is an energy-local bound, not a floor on the retained pure reference state. One complete final-energy clock reports the odd sector interval, enlarged by $1/64$, with original probability at least $2c_{\mathrm{low}}(1-\delta)$ per occupation sign, $\delta < 1/512$. The clock loss is applied to each eligible eigenvector before multiplying the small density floor. This event has no initial energy measurement or creation-probe interpretation.

An isolated finite gauge transition on the original source. The empty-matter reduction commutes with the actual member-zero Hamiltonian. Matter terms vanish there, leaving the native product-group electric and magnetic action. Gauss support has no dangling nonconstant edge. On the cube, its least nonconstant electric states are the twelve oriented charge-six face loops, at energy 63; every other electric excitation is at least $189/2$. The actual magnetic norm is at most $6/7$, so the vacuum and this twelve-state cluster remain isolated. The cluster's cubic A_{1u} component is one-dimensional. Exact native Haar moments and Temple bounds give a transition from the sector ground g to that state u:

$$63 - \frac{59}{1351350} \leq E_u - E_g \leq 63 + \frac{1}{45675}, \qquad |\langle u, Ag\rangle|^2 > \frac{4}{49}, \qquad A = \frac{1}{12i} \sum_p (\chi_6(U_p) - \overline{\chi_6(U_p)}).$$

This is an actual finite transition, not a Rayleigh estimate. The original polynomial density has $p(E_g) \geq 2^{-32406376}$. Complete initial and final clocks with resolution $1/64$, the contraction AP_{empty} and its retained failure give original event weight above $2^{-32406376}/13$, in the displayed interval widened by $1/32$. No conditional empty-sector state replaces the preparation. The selected refinement makes the entire empty-sector weight vanish. Its ground is also far above the full-source ground. Thus this line establishes neither a physical-vacuum particle mass nor a pure Yang–Mills continuum gap.

Static loop fusion and its complete records. Two sequential outward charge-six face loops on adjacent faces cancel their common edge, leaving a six-edge electric loop. Opposite faces retain eight edges. The full-source fusion calculation counts all sixteen hexagons and six weak-character squares, isolating an actual 44-level cluster in $[1311/14, 1335/14]$. Adjacent-loop creation from the exact empty-sector ground has weight above $99/100$ there; opposite-loop creation has weight above $4/5$ in $[124, 128]$. The complete 30-pair, 90-outcome menu and two unchanged clocks retain original probability above $2^{-32406376}/40$ for each recorded pair. Relative to twice the isolated A_{1u} transition, every fused-cluster energy has a static deficit in

$$\left[\frac{429}{14} - \frac{118}{1351350}, \frac{453}{14} + \frac{1}{45675}\right].$$

The connecting flux explains the energy difference: cancellation removes two excited links. These are static creation records on the full source, not an asymptotic binding mass or an executed collision of moving packets. Their original rare-sector weight vanishes under refinement.

Candidate names and finite threshold ratios. The candidate dictionary compares exact internal gauge weights and source rotation labels with named particle families. The charge-six regional spinor is a charged-lepton-singlet-type lead; the opposite endpoint has conjugate charge. It does not distinguish electron, muon and tau. Neutral LLe^c and $u^c(d^c)^2$ singlet composites motivate, respectively, neutral-lepton and neutral-antibaryon hypotheses, conditional on their physical current and Lorentz-spin attachment. Protected source-object IDs remain separate from these excitation IDs.

The source identity tests use conserved occupations and detector covariance before assigning names. If $[N_r, O_\delta] = \delta_r O_\delta$, with $[N_r, H] = [N_r, \rho] = 0$, then the normal spectral kernel between O_δ and O_ϵ vanishes for $\delta \neq \epsilon$. The anomalous kernel instead requires $\delta + \epsilon = 0$. For the six signed singlet-composite channels, $S = N_{-4} + N_2 + N_6$ has increments $\pm 1, \pm 2, \pm 3$. A coherent radix-seven difference counter distinguishes these interventions and failure without measuring the unknown initial S. It retains the source/reference histories and their energy probabilities; multiple excitations outside this alphabet can alias.

On the local negative-β CAR modes, the spin contraction

$$\Phi_a = \frac{l_{a0}e_1 - l_{a1}e_0}{\sqrt{2}}, \qquad D_s = \sum_a l^\dagger_{as}\Phi_a$$

has a finite source realization. Here Φ has weak anti-doublet weight $Q_6 = -3$ and scalar source-spin transformation. The dressed D is a weak singlet with $Q_6 = -6$ and the same native occupation increment as an e^c hole. This permits a charged channel that bare cross-representation correlators cannot supply. The construction supplies neither a condensate nor a physical Higgs field, Lorentz chirality or a lepton mass.

With $h_s = \sum_u \epsilon_{su} e_u$, the full local CAR identity

$$\sum_s h^\dagger_s D_s = \sqrt{2}\sum_a \Phi^\dagger_a \Phi_a \geq 0$$

gives a positive original-state overlap. Physical comparison uses the Wilson-linked compensated rows $J_{ps} = \chi P_0 c^\dagger_p h_s/2$ and $K_{ps} = \chi P_0 c^\dagger_p D_s/3$, each completed by failure. Their coherent nine-record comparison has summed-success plus/minus probability contrast at least $2\sqrt{2}c/3$, where $\rho \geq cP_G$ is the actual source-marginal floor. All reference and failure records are retained. This is a common physical source-channel overlap, without a resolved energy interval or a single charged-particle mass.

A sharper full-occupation comparison attaches a reported energy window. With $E_s = D_s - \sqrt{2}h_s$, exact block spectra give $\sum_s E_s^\dagger E_s \le 3(4 - N_L)/2$. On the complete 24-edge, 96-success menus, write $A_{xyps} = J_{xyps}/\sqrt{24}$ and $B_{xyps} = K_{xyps}/\sqrt{24}$. The original mean squared difference from rA, $r = 2\sqrt{2}/3$, is at most $\delta/48$, where the selected weak-plus-singlet defect bound is $\delta < 4.99118$. The same complete source-energy clocks and source/reference purification give

$$\sqrt{\Pr_B(J)} \ge r\sqrt{\Pr_A(J)} - \sqrt{\delta/48}, \qquad J = [77/8, 271/8], \quad \Pr_A(J) > 43/160.$$

The exact rational margins imply $\Pr_B(J) > 109561/4000000 > 1/40$. The triangle inequality compares only matched success histories; both distinct failure maps are retained. No additional clock error is subtracted. This is a populated original pair-transfer window at member zero, with the connecting flux and opposite endpoint included. Its spatial dispersion, capacity survival and single-particle mass require separate estimates.

The time-filtered pair calculation instead starts from the ideal bare transfer weight above $37/120$ in $I = [39/4, 135/4]$. The same occupation residual gives dressed ideal weight above $160801/4000000$. A normalized triangular time filter of width $T = 1/128$ multiplies that measure by $\operatorname{sinc}^4(T\omega/2)$ and retains weight above $39/1000$. Positive midpoint weights, coherent control of the complete instrument and the declared Cayley clocks give original reported probability above $3/80$ in $[311/32, 1081/32]$, after each implementation error is subtracted once. The retained-origin construction admits this bounded two-cell seed to the common regular energy/momentum algebra. Its bulk-aperture recipe is inactive at member zero, so the local positive weight is not a lower bound on that aperture or on a limiting particle residue.

The separated endpoint construction uses all shortest cube paths of length $l = 1, 2, 3$, with $N_l = 24, 48, 48$. Uniform endpoints and link incidence give original bare success $w_l > 1 - l(1 - 993/1250)$. The exact onsite/electric commutator center is $\lambda_l = 6 + 63l/4$. Orthogonal constant-to-character path blocks bound selected-link hopping by $\sqrt{40l}$; the remaining endpoint and magnetic terms contribute at most $2 + 2(l + 1)/7$. Thus the full centered second moments are below $(80, 140, 199)$. Radii $(20, 30, 45)$, the same dressing residual $\delta/48$, and the complete triangular-filter clocks prove

l	1	2	3
reported interval	$[55/32, 1337/32]$	$[239/32, 2161/32]$	$[263/32, 3145/32]$
original probability	$> 4/25$	$> 2/25$	$> 7/250$.

These probabilities refer to fixed apparatus settings; a uniform retained choice divides each joint weight by three. Coherent charge counters inside each temporal control retain the endpoint increments and every failure. The bands overlap. Their centers are apparatus moments, without a static-potential or endpoint-mass identification: pair data alone preserve $E(R) = 2m + V(R)$ under $m \mapsto m + d$, $V \mapsto V - 2d$.

The complete composite spatial experiment uses all forty signed components. For a normalized local contraction A of parity p, its neighboring graded first action is $G = A_x^\dagger[H, A_y] - (-1)^p[H, A_y]A_x^\dagger$. Exact two-cell CAR and Haar calculations give a physical witness with $\|G\phi_x\|^2 \ge \gamma = 1/972$ in every component. Let $\rho \ge cP_G$, $c = 2^{-582407133}$, and $R \ge \|H\|$. At $t = c\gamma/(64R^2)$, the complete success/failure experiment followed by native-occupation and final-energy clocks has neighbor-minus-nonneighbor probability at least $ct^2\gamma/32 > 0$ for each fixed component and apparatus setting. Uniform retained component and setting labels divide the joint weight by forty and two, respectively. The final-energy interval lies above the full ground enclosure. This extremely small original-state bound concerns a final energy, without an initial-energy measurement; it supplies neither a transfer mass nor the signed original two-point first moment. No source preparation is replaced by the witness.

The original neutral two-point bound resolves a different observable from the positive transport witness. For either the G_1 or H weak-pair/singlet multiplet of dimension d, the exact occupied-pair derivative satisfies $0 \le J \le dN/2$. The selected occupation bound gives $z > 27d/32$ and $|m_1| \le d\delta/64$, with $\delta < 5$. The four normalized Walsh centroids therefore obey

$$e_s = \frac{m_0 + (3-2s)m_1}{z}, \qquad \frac{16}{3} < e_s < \frac{38}{3}, \qquad \max_s e_s - \min_s e_s < \frac{5}{9}, \qquad \frac{\max_s e_s}{\min_s e_s} < \frac{53}{48}.$$

The local full-CAR residual supplies the stated common energy enclosure. A complete two-energy-clock estimator of m_1 has deterministic bias below $1/2048$; its original histogram and statistical error require execution. The bound permits $m_1 = 0$. Centroids of broad measures, finite Walsh labels and their ratios supply no mass pole or conserved momentum identification.

For the same local composite, put $S = \sum_a \Phi_a^\dagger \Phi_a$, $A = S/2$ and $h = 6 - N_L - N_e$. The full radial calculation gives $0 \le 2I - S \le h$ and $0 \le I - A^2 \le 3h/4$. The complete site menu $A_x/\sqrt{V}$, with failure retained, has total original probability at least $3481/160000$ for every source member, and greater than $383/500$ at member zero. This includes elastic response. The actual corner witness also obeys $\|[H,A]\Omega\|^2 = 9/(16a^2)$ and $\|[A,[H,A]]\Omega\|^2 = 27/(256a^2)$. For $Q_\tau = (I + i\tau H/2)^{-1}(I - i\tau H/2)$, $\tau = 1/(64aR^2)$, use the corresponding member's bounds $\rho \ge cP_G$ and $R \ge \|H\|$. The complete two-record commutator instrument $[A, Q_\tau^\dagger A Q_\tau]/2$ has original success probability greater than $c\tau^2/(256a^2)$, with zero success at zero delay. This finite dynamical response supplies no resolved scalar mass or capacity-uniform inelastic residue. A physical Higgs interpretation requires the scalar branch and its vector and fermion interaction records.

A frequency-cut bound separates strictly positive radial transfer from elastic detection. At every site the original operator witness gives $\|[H, A_x]\| \ge d_0 = 3/(4a)$. The normalized kernel $f_\lambda(t) = \lambda \operatorname{sinc}^2(\lambda t/2)/(2\pi)$ has triangular Fourier transform. Filtering with $2f_{2\lambda} - f_\lambda$ leaves all frequencies below λ unchanged and bounds the filtered commutator by 15λ. For $\lambda = 1/40$, the complementary part has norm at least $d_0/(2D)$, where D is the full source spectral width. The actual density floor $\rho \ge cI$ and decreasing polynomial preparation then give

$$\mu_{A_x}([\lambda, D]) \ge cq, \qquad q = \frac{9}{128a^2D^2}.$$

Averaging the complete site instrument loses no volume factor. Two finite clocks, each with tail and implementation allowance $cq/16$, retain original weight at least $3cq/4$ in $[3\lambda/4, D + \lambda/4]$. The auxiliary infinite-time kernel is used only in the proof. Executable dyadic clock resources are finite but enormous. The same radial menu has elastic weight at least $3481/160000$, and above $7/10$ at member zero. The positive cutoff is fixed across capacities; its upper endpoint grows and the proved weight decreases, so no compact limiting branch, Higgs mass or global mass gap follows.

The exact local commutant of the gauge generators and every Wilson spin coefficient has dimension five, with one scalar per native gauge block; there is no extra onsite three-generation factor. A single scalar of sharp native occupation increment also cannot neutralize all three bare Yukawa vertices: $Le^cH^\dagger$, $Qd^cH^\dagger$, and Qu^cH require, respectively, $\Delta_{H^\dagger} = -e_L - e_{e^c}$, $-e_Q - e_{d^c}$, and $e_Q + e_{u^c}$. All have $Q_6 = -3$ but different conserved source occupations. These constraints leave differently dressed fields and explicitly constructed symmetry-breaking states to be tested. A zero gauge-covariant scalar one-point function does not exclude gauge-invariant Higgs-phase responses.

The isolated gauge cluster has one copy each of $A_{1g}, E_g, T_{1u}, A_{1u}, E_u, T_{1g}$ under the actual cube symmetry. Twelve compressed loop contractions and their failure give a complete detector menu. Each component has squared ground-to-cluster matrix element above $4/49$ and original clocked probability above $2^{-32406376}/13$. Inversion-even scalar and vector probes have zero A_{1g}-to-A_{1u} amplitude, even with finite energy-clock resolution when the exact symmetry records are retained.

These are total cube representations in a rare reducing sector; intrinsic Lorentz spin, physical gauge-boson identity and a surviving particle mass do not follow.

The common-ground comparison uses sector energy intervals $[L_i, U_i]$ and one ground interval $[L_0, U_0]$, rather than assigning independent energy zeros. Positive thresholds $D_i = E_i - E_0$ obey

$$\min_{e\in\{L_0,U_0\}} \frac{L_i - e}{U_j - e} \le \frac{D_i}{D_j} \le \max_{e\in\{L_0,U_0\}} \frac{U_i - e}{L_j - e}.$$

All ratios among the three conjugate sector families are below 21.16; those among the two odd families are below 15.31. Thus these particular member-zero thresholds cannot directly reproduce the charged-lepton hierarchy under a common scale. The observed muon/electron ratio is 206.7682827(46)[4], used only for comparison. The exclusion applies neither to other excitations nor to other capacities. These are rigorous finite threshold ratios, not physical rest-mass ratios.

Original odd sum rules and the chirality constraint. For local trions F_{xj}, combine their creation kernel with the reflected, transposed adjoint-annihilation kernel to form S_{xy}. Exact CAR gives

$$\int dS_{xy} = \delta_{xy} \operatorname{Tr}\rho \sum_j \{F^*_{xj}, F_{xj}\}, \qquad \int \omega\, dS_{xy} = \operatorname{Tr}\rho \sum_j \{F^*_{xj}, [H, F_{yj}]\}.$$

The actual source sum rules make all flat spatial rows equally weighted. Only onsite and nearest neighbor first moments remain; their Walsh values are $[m_0+(3-2s)m_1]/48$. The signed expectation m_1 requires calculation. Independently, literal CAR/Haar algebra proves that the neighboring graded operator has positive squared norm in the original faithful state. A complete two-order Cayley interference instrument has strictly larger neighboring than nonneighboring probability at an explicit finite source time. This is a four-point propagation record, not a two-point dispersion calculation.

The tested trions have $Q_6 \cdot \Delta = 0$ and are local gauge singlets. Native occupation labels therefore do not identify charged leptons. At every capacity, the exact physical inversion acts by site reflection, link reversal and the Dirac matrix β. It exchanges $\Gamma_\pm = (I \pm \gamma_5)/2$ and preserves H, the selected ρ and each internal gauge representation. Complete Wilson-dressed Γ_+ and reflected Γ_- experiments have identical energy laws. These projectors are not established Lorentz chirality and can couple to the same massive excitation. A parity-equivariant physical assignment cannot simply omit the paired response. A one-handed sector needs a specified parity-breaking state or selection, a changed source, or a proved different current/field dictionary.

Controlled spatial work at growing capacity. A complete local phase experiment retains every address X, both signs and both energy records:

$$U_{X,\sigma} = \exp\left(i\sigma\pi s a \sum_y f_b(y - X)(N_y - 30) \right), \qquad K_{X,\sigma} = U_{X,\sigma}/\sqrt{2V}.$$

Here f_b is a product of finite tents with fixed physical half-widths $b_i \in \mathbb{N}$ and $s \in \mathbb{Q} \setminus \{0\}$. Only the apparatus address list wraps; the source bonds remain open. Writing $h_n = -a \operatorname{Tr}(\rho H_{\text{hop}})/V$, the selected energy estimate sharpens to $h_n \ge h_* > 0.0424$, while $h_n < 90$. Exact sign and address averaging gives $W_n = h_n \bar{k}_n / a$. Finite tent sums and the cosine remainder then prove, on a common subsequence with $h_n \to h$,

$$W_n \longrightarrow \frac{\pi^2 h}{6} \int_{\mathbb{R}^3} |\nabla(s f_b)|^2\, dx, \qquad \int |\nabla f_b|^2 = \frac{8}{9} \sum_i \frac{b_j b_k}{b_i}.$$

[4] 2022 CODATA, https://physics.nist.gov/cuu/Constants/Table/allascii.txt.

The spatial discretization error is $O(a^2)$ and the phase remainder $O(a^4)$, with explicit rational bounds. Complete finite-clock reported means have an additional $O(a^2)$ error and the same limit. Ratios cancel the unknown h_n. Starting from the unit-width, amplitude-one cube, doubling amplitude, all widths, or one width gives limiting ratios 4, 2, or 3/2. This is a nonzero continuum limit of measured work, not a continuum field action. Rare ultraviolet records can retain positive work while the probability limit is elastic; fixed-energy survival and uniform integrability remain required for a particle conclusion.

The same original work spectrum uses $\omega = E_j - E_i$ and $\kappa = \log p_i - \log p_j$, where $\rho = p(H)$ is the actual polynomial density. Sign reversal and adjoint paths give $P_{\text{reverse}} = e^{-\kappa}P$ and $\mathbb{E}e^{-\kappa} = 1$, with $\beta_{\text{low}}W \le \mathbb{E}\kappa \le \beta W$. The energy-index transition kernel is symmetric and doubly stochastic, instantiating the finite fluctuation theorem with uniform reference. Its entropy production $\log p_i - \log q_j$, $q = Tp$, differs from κ by the relative-entropy term in $\mathbb{E}\kappa = H(q) - H(p) + D(q\|p)$.

The selected-state moment calculation uses $D_\pm = U_\pm^* H U_\pm - H = C \pm S$. Exact sign averaging gives $M_2 = \langle C^2 + S^2\rangle$ and

$$m_3 = \tfrac{1}{2}\langle [C,[H,C]] + [S,[H,S]]\rangle + \langle C^3 + 3SCS + [S,[S,C]]\rangle.$$

The actual electric expectation is at most $91V/a$. Native link derivatives preserve the word cutoff, and their Casimir form bounds the electric double commutators without a link-degree factor. Reverse-path population bounds control the negative moments. Positive-part Cauchy–Schwarz then closes a quadratic bound on M_2, yielding

$$M_2 \le C_n = 4000000/a, \qquad \mu_n([w_n/4, 4C_n/w_n]) \ge \frac{w_n^2}{4C_n},$$

where $w_n > 0$ is the uniform original work lower bound. The same complete clocks retain more than half the finite floor in the widened band. Its lower weight scales as a and its upper endpoint as $1/a$. This improves the actual finite readout without proving uniform integrability or a nonzero fixed-energy limit. The remaining quantity is the selected expectation of the current double commutators; a uniform bound would close the same quadratic estimate uniformly.

Equal work does not determine the spectrum. A distinct apparatus on the actual source uses local parity $P_x = (-1)^{N_x}$ and a retained coin. Its mean parity work is $-4\,\mathrm{Tr}(\rho H_{\text{hop}})/V$. The source-defined probability $p = \bar{k}/4$ therefore matches the critical phase work exactly, with every inactive branch retained. For the unit tent, $p_0 = 1$ and $p_n \le 45a/32$ for $n \ge 1$. Its transfer law is

$$\mu_n = (1 - p_n)\delta_0 + p_n \nu_{\text{parity}}, \qquad M_2 = \Theta(a^{-1}), \qquad W_n = W_{\text{critical},n} \ge w_* > 0.$$

Every fixed positive-energy probability vanishes. Recorded outputs, including arbitrary references, approach the matching idle experiment within p_n in half trace distance; unchanged complete clocks retain the positive mean work. Every fixed finite set of profile-work settings can be matched eventually. This leaves the original critical apparatus's spectrum undetermined: its selected current correlations or positive joint detector weight must be calculated. A work form alone does not establish a surviving field or its mass.

Regional charged spinors with retained compensation. The same-source regional construction uses $B_{xy,ab} = c^\dagger(p_{xa})c(m_{yb})\chi_6(U_{xy})P_{0,xy}$. The inside factor is CAR odd, has charge increment $+6$ and carries native binary-cube G_1; its outside conjugate carries -6. The complete pair is even and gauge invariant. Coherent scalar/Pauli combinations transform as $A_1 \oplus T_1$. Their full

97-outcome apparatus has original scalar energy-band weight above $5/64$ and three-vector weight above $15/64$ in $[69/8, 279/8]$. A separate raw-spin basis resolves charge and native spin-projection increments together: each of four rows has original weight above $1/16$ in $[77/8, 271/8]$, with increments $\pm 1/2$ at the endpoints. Finite coherent counters cancel unknown initial occupations; every source, boundary, reference and failure output is retained. The two apparatus bases are not simultaneously assigned incompatible outcomes. These are populated regional spin and pair-energy records. The transfer center $87/4 = 6 + 63/4$ includes link electric energy, so neither halving it nor retaining only its onsite part gives a lepton mass. A controlled charged threshold or branch, dispersion, physical spin and statistics remain required on the same state and clock.

Physical charged endpoints and energy records. For a directed interior bond, the native weak/color singlet character $\chi_6(U_{xy})$ dresses the pair

$$B_{xy\alpha} = c^\dagger(p_{x\alpha})c(m_{y\alpha})\chi_6(U_{xy})P_{0,xy}.$$

It preserves Gauss support and global occupations while changing local integer charges by $+6$ and -6. Its CAR parity is even; these endpoint labels do not identify a single electron. The Haar projection P_0 is an intervention with a retained failure, not a replacement preparation. The complete 49-outcome apparatus uses the 48 directed-bond/component successes $B_b/\sqrt{48}$ and their complementary root. Retaining onsite fraction $7/10$ for the charge-six block and electric fraction $8/45$ in the full-Fock inequality gives original success probability greater than 0.79446654. The exact onsite/electric transfer center is $87/4$; full commutators obey

$$\|[H, B_b] - (87/4)B_b\| \le \sqrt{40} + \sqrt{2} + 4/7 < \sqrt{70}.$$

The selected-bond hopping terms have orthogonal input and output blocks; the four other incident bonds give two disjoint CAR raising pairs. Consequently original transfer weight exceeds $3/10$ in $[39/4, 135/4]$. Cooling puts weight at least $19/80$ in final excitation interval $[39/4, 139/4]$. Complete finite clocks report $[77/8, 271/8]$ with original weight greater than $1/4$. A separate initial/final local-charge apparatus obeys the endpoint selection rules but disturbs the input state; these energy bounds are not transferred to that charge-dephased history.

A coherent charge-transfer meter avoids this initial dephasing. For local integer charge Q_z, let S_z be a cyclic shift on a five-state meter and define $W = \sum_{\mathbf{q}} P_{\mathbf{q}} \otimes \bigotimes_z S_z^{q_z}$. Every success or failure K_b has a definite transfer vector d_b; the failure has $d_b = 0$. Coupling with $W^\dagger$ before the probe and W immediately after it gives

$$W(K_b \otimes I)W^\dagger = K_b \otimes \bigotimes_z S_z^{d_{b,z}}.$$

No meter is measured between the couplings. The known transfer alphabet $\{0, +6, -6\}$ has distinct residues $\{0, 1, 4\}$ modulo five. The meter's dependence on initial charge cancels; source coherences are preserved and the intervention's charge transfer is readable. Placed between the two unchanged energy clocks, this adds a deterministic transfer label to each original source/reference map. The successful reported energy event retains probability greater than $1/4$ and has opposite endpoint-transfer records. It does not assert that the final energy-resolved state has permanently localized charges. An optional terminal absolute-charge measurement preserves earlier record probabilities when summed, with its own postmeasurement disturbance.

Symmetry-resolved neutral records. In the actual beta chart define

$$C_{x,a} = \frac{1}{\sqrt{2}} \sum_{u,v} (\sigma_a)_{uv} c^\dagger_{x,+u} c_{x,-v},$$

where $\sigma_0 = I$ and the other three matrices transform as the source spin vector. Each is a full-Fock contraction satisfying $C^\dagger_{x,a} C_{x,a} - C_{x,a} C^\dagger_{x,a} = I - D_{6,x}/2$. The physical cubic action and beta inversion give local operator types $A_{1u} \oplus T_{1u}$. On all eight cells the 32 coefficients decompose, in each parity, as $A_1 \oplus A_2 \oplus E \oplus 2T_1 \oplus 2T_2$. A complete coherent channel instrument retains 32 successes and 32 failures. An isotypic projector of rank d has aggregate original commutator at least $11d/512$. For the actual selected polynomial, $0 < -\partial_E \log p(E) \le \beta$; this controls positive frequencies near zero without replacing the state by a Gibbs density. Together with the centered second-moment cap six it gives original transfer weight at least $11d/1024$ in $[11/(2048\beta), 40]$. These are operator channels; neither a mixed invariant preparation nor a cubic irrep fixes the spin of an outgoing particle. A Walsh label is not a conserved momentum eigenvalue of the open source.

Populated responses across the finite family. The block-sensitive electric estimate uses an exact full-occupation charge-six two-site bound and a tiled physical eight-cell variational vector. Cross-tile hopping and plaquette expectations vanish by fixed tile occupations and Haar integration. The selected state is unchanged; the tiled vector supplies an energy upper bound. The resulting total defect density obeys $\langle D \rangle / V < 97/10$ on every primary member. In the native dimension-six $Q_6 = 1$ block, the neutral collective probe

$$C_x = \frac{1}{\sqrt{42}} \sum_{u=1}^{2} \sum_{i=1}^{6} c^\dagger_{x,+ui} c_{x,-ui}$$

is a contraction and has $[C^\dagger_x, C_x] = (12 - D_{1,x})/42$, with $D_{1,x}$ counting that block's beta defects. Its averaged commutator is at least $1/20$. The full-Fock neighbor calculation gives centered moment at most $M_6(L)/a^2$, where

$$M_6(L) = \frac{48B - 66(L-2)^3}{7L^3} \le \frac{3141}{224}, \qquad B = 3L^2(L-1), \quad L = 2^{2n+1}.$$

The interior residual is a sum of 24 independent CAR-pair raising operators; its auxiliary spin decomposition computes a norm, not physical particle spin. Polynomial population monotonicity then gives original transfer weight at least $3/100$ in $[1/(100\beta_n), 46/a_n]$. Cooling with cut $100\epsilon_n$ gives final excitation weight at least $1/50$ in $[1/(100\beta_n), 46/a_n + 100\epsilon_n]$. These bounds precede finite-clock implementation errors. Their probability floors are uniform, while their bands approach zero and infinity. They do not establish an isolated mass or a fixed finite-energy continuum response. GeV conversion uses a declared tick; identifying rest energy requires the same channel's controlled dispersion.

The physical beta-filled constant-link vector Ω provides the exact check

$$\|([H, C_x] - 6C_x/a)\Omega\|^2 = \frac{\deg(x)}{7a^2}.$$

These images survive every selected link cutoff. Thus a uniform operator-norm residual for this bare probe is impossible as $a \to 0$. This is not a selected-state weighted lower bound: dressing, spatial smearing or a direct fixed-window estimate could retain physical energy weight. Such a result must prove its original selected overlap.

The same selected preparation has strictly nonzero hopping energy: $-a\langle H_{\mathrm{hop}}\rangle/V > 0.0345$ uniformly in the primary family. This follows from the tiled ground upper bound, nonnegative onsite defects and electric energy, and the full magnetic lower bound. For total matter number N_x, every non-hopping term commutes with $A_f = \sum_x f_x(N_x - 30)$, giving the exact sum rule

$$[A_f, [H, A_f]] = -\sum_{b=(x,y)} (f_x - f_y)^2 H_{\mathrm{hop},b}.$$

Use cosine and sine weights along one axis, with $\Lambda = 30V$, success operators $A_c/\Lambda, A_s/\Lambda$, and the positive-root failure. Physical cubic symmetry equates the three directional hopping means. The sum of their original signed first moments is

$$\int \omega \, d(\mu_c + \mu_s) = -\frac{1 - \cos\theta}{3\Lambda^2}\langle H_{\mathrm{hop}}\rangle > 0 \quad (1 - \cos\theta > 0).$$

For the fixed spatial phase label π, with phase step $\theta = \pi a$, the actual hopping norm and selected cooling bound nevertheless imply $\mu((\delta, \infty)) \le (a/(6V) + \epsilon)/\delta \to 0$ for every fixed $\delta > 0$. The complete volume-normalized density readout therefore loses fixed positive-energy weight. This conclusion concerns that apparatus normalization, not every local or dressed field of the source.

Bounded local phase response across capacities. The local phase apparatus uses $R = 1/a$, a product tent f_R of physical radius one, and

$$U_{X,\sigma} = \exp\left(i\sigma\pi a \sum_y f_R(y - X)(N_y - 30)\right), \qquad \sigma = \pm 1.$$

Uniform retained address and sign give a complete instrument of unitaries, with total original spectral weight one. Only the apparatus coefficients wrap around the finite address list; the source has open boundaries. The exact cosine-gradient average and selected hopping estimate give signed first moment greater than $(16/9)h_* > 0.06149$ and absolute first moment at most $14179/32$, uniformly across capacities. Thus original energy measures are tight and their subsequential limits have continuous characteristic functions and nonnegative support. Restricting the retained address to the common bulk cube changes complete record laws by at most its omitted volume fraction, which tends to zero. The limiting measure can be purely elastic: $(1-a)\delta_0 + a\delta_{1/a}$ has positive first moment and the same probability tightness but tends to δ_0. A uniform-integrability estimate or a populated fixed positive-energy window is required to establish an inelastic surviving field.

The energy-form bound joins these unfiltered phase operations to the existing filtered algebra. For $F_X = \pi a \sum_y f_R(y - X)(N_y - 30)$ and $K = H - E_0 \ge 0$, the actual tent gradients give $\|[F_X, [H, F_X]]\| \le 2025/2$, uniformly in capacity. Positivity of $\langle e^{itF_X}\psi, Ke^{itF_X}\psi\rangle$ and its bounded second derivative imply

$$\|K^{1/2}e^{\pm iF_X}\psi\| \le \|K^{1/2}\psi\| + (45/2)\|\psi\|.$$

For a filtered contraction B, the resolvent formula gives $\|[K^{1/2}, B]\| \le 2\sqrt{\|[H, B]\|}$ without a gap assumption. Thus every fixed mixed word has a uniform selected-state energy moment. Its weighted time modulus, together with the finite downward-record bound, constructs one common positive-energy time representation and complete fixed-history limits for both menus. This is strong continuity in the representation, not point-norm continuity of unfiltered source sequences. Spatial momentum for these additional fields and a nonelastic limiting response require separate estimates. The physical beta-filled anchor has an averaged commutator norm squared at least $160/a$, excluding

a uniform absolute commutator bound for this apparatus; that anchor is not the selected state and gives no lower bound on its weighted variance.

The same phase apparatus has a finite source-marginal modular response. For the actual polynomial state, the modular slope defect satisfies $\delta \le y/(m+y) \le 1/3$, since its cooling construction has $m \ge 1$ and $0 < y \le 1/2$. Finite passivity gives $\beta(1-\delta)W(U) \le D(U\rho U^\dagger\|\rho) \le \beta W(U)$. Retaining every origin and sign makes the classical–quantum relative entropy the original weighted sum of these source-marginal entropies. Consequently $0.04099 < D_{CQ}/\beta \le 14175/32$ uniformly in capacity. The idle reference is retained by the apparatus and traced out for this entropy comparison; its purified relative entropy is not substituted. This finite response supplies neither a geometric temperature nor a regional area law, and does not pass an unbounded energy mean to the limit.

The normalized-density protocol bound restricts a tempting continuum shortcut. For $Q_X = \sum_y f_R(y-X)(N_y-30)/(30R^3)$, the actual source obeys $\|[Q_X,[H,Q_X]]\| \le 9a^4/80$. A complete coherent protocol of calls $\exp(i\pi r_j Q_X \otimes A_j)$, $A_j = A_j^\dagger$ and $\|A_j\| \le 1$, has final mean excess at most $(\sqrt{\epsilon} + 3a^2\sum_j |r_j|/4)^2$, including every filter failure and retained reference. Thus total strength $o(a^{-2})$ loses every fixed positive-energy transfer band. The original Weyl call has strength $30/a^2$ and is outside this obstruction. Uncontrolled waiting-time comparisons also lose contrast in the excluded class; controlled clock calls can retain an apparatus ground-energy phase, which cannot be removed by shifting the source energy convention.

Relative observations and canonical momentum. The retained-origin construction adjoins an independent uniform classical origin X with zero Hamiltonian. The same X controls every intervention in a complete history. For $F_n = \{b_n, \dots, L_n - b_n - 1\}^3$, take $b_n = \min\{\lfloor (L_n - 1)/2 \rfloor, 2^{\lfloor 3n/2 \rfloor}\}$. The relative map is the direct sum $j_n(A) = \bigoplus_{x\in F_n} \beta_x(A)$, a homomorphism; an arithmetic operator average or a fresh origin at each stage would change histories. The source state, reference and time generator remain unchanged. For an integer displacement m, the unit-norm expectation defect is at most

$$2\left[1 - \prod_i \left(1 - \frac{|m_i|}{L_n - 2b_n}\right)_+\right].$$

It tends to zero for fixed physical displacements. The physical collar $a_n b_n$ diverges while its volume fraction vanishes, controlling the finite-to-infinite boundary comparison for fixed spatial and time smears. The common relative-observation limit is stationary and translation invariant. Its canonical $H \ge 0$ and three momentum generators strongly commute and annihilate the cyclic vector. This does not prove invariance of the frozen fixed-origin state or a nonzero momentum response. The actual defect-density bound also gives a populated retained-origin moving band at level seven; its unsmeared response is not silently identified with a nonzero regular limiting field.

A complete coherent-offset detector attaches finite records to the relative representation. For a bounded physical seed with fixed spatial and temporal smears, retain the common origin and all coherent offset, Fourier and failure labels. The distinguished success approaches

$$B_{q,\nu} = \frac{1}{(2Q)^3\, 2S} \int_{[-Q,Q]^3} d^3y \int_{-S}^{S} dt\, e^{-i(q\cdot y + \nu t)} \alpha_t \beta_y(A).$$

Its canonical vector is

$$B_{q,\nu}\Omega = \operatorname{sinc}\big(S(H-\nu)\big) \prod_{j=1}^{3} \operatorname{sinc}\big(Q(P_j - q_j)\big) A\Omega.$$

The squared norm is the original response, not a probability conditioned on an analyzer outcome. Positive finite quadrature, source-norm Cayley errors, regular outer-grid errors and the bulk boundary estimate are controlled separately. This is a detector-to-generator connection on the common subsequence; it supplies no effective state-convergence rate, nonzero joint weight, finite exact momentum PVM or relativistic dispersion.

The complete joint instrument records initial energy, a spatial Fourier outcome, and final energy on a specified primary member with an admitted aperture. For the unchanged source clock, define

$$F_j(H) = \frac{1}{J}\sum_{r=0}^{J-1} e^{2\pi ijr/J} e^{-ir\tau H}, \qquad L_{j,b,k,l} = F_l(H) K_{b,k} F_j(H).$$

All $2MJ^2$ records are retained. Their original probabilities are $\sum_{uv} p_u |(K_{b,k})_{vu}|^2 |F_j(E_u)|^2 |F_l(E_v)|^2$. The initial energy instrument preserves the stationary source marginal and changes its correlations with the retained idle reference. A signed energy chart with $\tau = 1/(4\|H\|_{\mathrm{bound}})$ avoids aliasing; finite clock tails and complete implementation errors are bounded. If the reported transfer lies in $[l,u]$ with original weight at least w, energy resolution η, per-clock tail t_J, complete error e_J covering both clocks and other event errors, and initial-excitation cutoff $r_0 > 0$, then final excitation lies in $[\max(0, l-2\eta), u+2\eta+r_0]$ with weight at least $\max(0, w-2t_J-e_J-2^{-n-4}/r_0)$. Authenticated lower and upper ground bounds also convert a final-energy interval directly, without an initial cooling-tail subtraction. Every bound is recomputed at the apparatus's member; a small-box ground bound is not transferred to a larger aperture. These energy and spatial-phase records define an operational structure factor. A resolved branch and its physical momentum attachment require their own error-controlled record evidence.

From transfer energy to final excitation. The mass-readout transport bound uses a common eigenbasis of H and ρ. The positive weights $w_{ij} = p_i |B_{ji}|^2$ couple source transfers $\omega_{ij} = E_j - E_i$ to final excitations $z_{ij} = E_j - E_0$. For $\|B\| \le C$ their original masses agree and

$$\sum_{ij} w_{ij} |z_{ij} - \omega_{ij}| \le C^2 \operatorname{Tr} \rho(H - E_0) \le C^2 2^{-n-4}.$$

No success normalization, initial energy measurement or reference counter-evolution is used. A fixed-filter first-moment bound gives $\int z\, d\nu \le 2C^2/T + C^2 2^{-n-4}$, so the transfer and final-excitation laws have the same weak limit on the common subsequence. If a complete transfer readout gives $\mu([l,u]) \ge w$, then

$$\nu([l, u+r]) \ge \max(0, w - C^2 2^{-n-4}/r), \qquad r > 0.$$

For the broad finite witness band, $z \ge \omega$ and $z \le W_n$ give the same positive lower weight in $[\delta_n, W_n]$ directly, without the subtraction. This identifies finite excitation energy, not zero-momentum rest energy.

An invariant state and commuting energy and spatial generators make a mixed-state joint response measure transfers $(\Delta E, \Delta p)$. Cooling alone does not fix the initial momentum. A controlled initial momentum radius and tail, together with a surviving common relativistic branch, are required before an interval for $M^2 = E^2 - v^2 p^2$ is a rest-energy interval. A coordinate or Fourier label cannot supply those premises. In source action units, declaring one tick to be τ seconds permits the explicit conversions

$$\text{metres per source coordinate} = c_{\mathrm{SI}}\tau/v, \text{GeV per source energy} = \hbar_{\mathrm{GeV\,s}}/\tau.$$

Such a scale is an allowed comparison input. It does not create a primary pole, derive its species, or establish agreement with measured masses. The primary source has no derived rest-mass value in GeV.

Capacity and scaling. The capacity bounds distinguish ambient dimension, physical dimension and prepared entropy. The exact degree-one link space has dimension 119. With $V = L^3$, the selected family satisfies

$$4V \log 2 \le \log \dim \mathcal{H}_{\text{phys}} \le \log \dim \mathcal{H}_{\text{ambient}} \le V(60 + 27\ell) \log 2.$$

The lower bound uses sixteen independent local singlet occupations and constant links; it does not identify an invariant low-energy sector. The smallest ambient member has approximately 447.41 nats of capacity. Its full conjugate reference doubles the ambient log dimension, and apparatus requires further allocation. Every source Hamiltonian term has the factor $1/a_n$, so $H_n = \widehat{H}_n/a_n$. At a fixed unit convention, finite nonzero rest energy would require the relevant dimensionless excitation to scale as a_n. Volume, spacing and link cutoff vary together; these bounds alone do not give a mass-capacity law.

11.14 Accessible charged thresholds and finite-volume cutoff removal

The original charged response of the 208-state control has nonzero weight at its lowest opposite-charge thresholds. Exact inertia counts isolate the ground and rank-two charged targets. A rational ground trial has half-trace error below 1.065×10^{-7}. For each charge sign the original threshold weight lies in $[0.4985578116, 0.4985579794]$, at source energy approximately 1.107713183. A complete collision apparatus retains all forty outcomes, including failures; its original weights sum to one. A separately bounded Fourier apparatus gives a positive threshold-window guarantee. These finite directional records do not identify a spatial particle mass.

At fixed finite spatial volume and spacing, write the uncut Wilson source as $H = E + B$, with positive electric operator E of compact resolvent and bounded B. The nested product-word projections commute with E and converge strongly to identity; native matrix coefficients separate group points. Their union is a form core. Rayleigh–Ritz ground energies decrease to the uncut minimum. Cooled states with bounded electric expectation have trace-norm subsequences supported in the uncut ground. The same argument restricts to the commuting physical Gauss projection.

The same-clock comparison uses $E + P_\ell B P_\ell$: its propagators converge strongly, uniformly on compact time intervals. Fixed bounded Kraus menus are compressed with the missing contraction weight retained as an explicit failure outcome. Complete finite histories converge along the state subsequence, including the retained canonical references. This is not uniform diamond-norm convergence or a joint thermodynamic and spacing limit. For a complementary electric block with floor $b > L$ and cross-block norm at most c, the finite bound is $H \ge L - c^2/(b - L)$. Native rotor cutoffs six and twelve have disjoint certified finite ground intervals, demonstrating a real cutoff effect. A finite-cutoff residue therefore needs its own continuation bound.

11.15 The common quadratic gauge characteristic

The native gauge calculation uses the full fifteen-state representation with Hermitian generators, $\mathrm{Tr}(T_a T_b) = I_a \delta_{ab}$, $I_a = 2$ for weak and color generators, and $I_Y = 10/3$ for $Y = q_6/6$. The inherited electric Casimir uses the inverse trace metric. With $H_E = e_0 C_{\text{tr}}/(2a)$ and $H_B = -\kappa_0 \sum_p \mathrm{Re}\mathrm{Tr}\, U_p/(15a)$, the ordered plaquette expansion gives, with dimensionless momentum $k_i = a p_i$,

$$H_B^{(2)} = \frac{\kappa_0}{30a} \sum_{p,a} I_a F_{p,a}^2, \qquad \omega_g^2(k) = \frac{e_0 \kappa_0}{15a^2} 4 \sum_i \sin^2(k_i/2).$$

The electric inverse metric cancels I_a in Hamilton's equations. All twelve gauge directions share this quadratic speed. The lattice curl retains one longitudinal null and two transverse modes at nonzero momentum, with its zero-momentum flat coordinates treated separately.

The Wilson matter symbol has unit principal speed. The declared relation $e_0\kappa_0 = 15$ matches the gauge characteristic on the same clock; $e_0 = 1$, $\kappa_0 = 1/7$ instead gives squared gauge speed $1/105$. This is an allowed architecture choice without measured-mass calibration. Rescaling the hypercharge coordinate changes electric and magnetic coefficients inversely and leaves the characteristic unchanged. A common quadratic cone does not establish Lorentz covariance of the interacting state, identify physical photon or gluon states, or derive a running coupling or chiral spectrum.

11.16 Spatial strain and pressure from the chosen action

The coframe response declares a constant spatial matrix V with $\det V > 0$, $G = V^T V$ and $v = \det V$. On its periodic finite-range Wilson source, the Dirac coefficient is V^{-T}, the Wilson Laplacian coefficient is G^{-1}, and electric and based-plaquette Gram coefficients are G/v. The magnetic subtraction retains the original flat potential and energy zero. Products are formed before coordinate-word compression. Every term is gauge invariant; the six symmetric strain derivatives retain the complete Fock space. Constant frame rotations have their explicit Dirac spin lift, without an inferred Lorentz boost.

For the open primary, a shift is a partial isometry and $(I - T_i)^\dagger (I - T_i) = 2I - T_i - T_i^\dagger - \Pi_i$, where $\Pi_i = I - T_i^\dagger T_i$. Direct substitution into the periodic coframe gives $H_{\text{primary}} - H_{\text{naive}} = d\Gamma(\beta_D \sum_i \Pi_i)/(2a)$, of physical norm $45L^2/a$. The boundary-completed coupling instead uses

$$L_{ij} = \frac{(I - T_i)^\dagger (I - T_j) + (I - T_j)^\dagger (I - T_i)}{2} + \delta_{ij}\Pi_i.$$

Then $L_{ii} = 2I - T_i - T_i^\dagger$. Absent plaquette planes contribute zero to the magnetic Gram form, whose diagonal subtraction counts only present planes. The resulting $H(V)$ obeys $H(I) = H_{\text{primary}}$ exactly, with the same energy zero and all Gauss constraints. Its six symmetric strain operators K_S have explicit full-carrier norm bounds R_S. The complete effects $(I \pm K_S/R_S)/2$ with scores $\pm R_S$ have original mean $\operatorname{Tr} \rho K_S$, retaining the reference. For the zero direction $S = 0$, both effects are $I/2$ with zero score. Isotropic dilation gives $K_I = -H - H_W$, including the completed open Wilson term. This fixes one declared spatial coupling; it does not identify proper geometry, a symmetric Lorentz stress or an equation of state. The same-source modular comparison uses the actual polynomial density, whose modular slope lies between $\beta(1 - \delta)$ and β, with $\delta < 1.116\,10^{-14}$ on member zero. For a unitary U on the physical Gauss carrier, finite passivity gives

$$\beta(1 - \delta)W(U) \le D(U\rho U^\dagger \| \rho) \le \beta W(U).$$

Writing $K = \beta H + cI + R$ gives an oscillation bound $\operatorname{osc} R \le \beta\delta(E_{\max} - E_{\min})$. Modular flow at parameter s and source flow at time $-\beta s$ have complete-output half-trace error at most $|s|\beta\delta(E_{\max} - E_{\min})/2$, with an idle reference retained. For a Gauss-preserving tensor-local unitary, regional restriction bounds the corresponding marginal entropy response by the same work. The charge-six beta flip has $18 < W \le 144$ source units on member zero. These are source-marginal statements; the relative entropy of distinct pure retained-reference states is infinite. Proper geometry, the diamond modular kernel and fixed-volume area response are separate hypotheses.

These constant coframes retain a uniform causal estimate. The three outgoing link factors belong to their tail cell, so the positive electric quadratic form, including cross terms, remains onsite. Its self-adjoint generators therefore admit the same interaction-picture argument. Put

$C = G^{-1}$ and $A = G/v$. With $\kappa(a) \le \kappa_0/a$, a full-Fock interaction majorant is

$$W(V) = 120\lambda \sum_i \max\{\sqrt{C_{ii}}, C_{ii}\} + 1080\lambda^2 \sum_{i<j} |C_{ij}|$$
$$+ \kappa_0\lambda^2 \left(16 \sum_i A_{ii} + 64 \sum_{i<j} |A_{ij}| \right).$$

It reduces to the preceding Wilson bound at $V = I$. The conservative cross coefficient also covers collapsed periodic cells. The rational certificate uses $\sqrt{C_{ii}} \le \max\{1, C_{ii}\}$. On a compact family $\sigma_{\min}(V) \ge m > 0$, $\|V\|_{\mathrm{op}} \le L$, its supremum is finite. The declared coframe distance is at most L times the coordinate path length, giving speed bound $2L \sup_V W(V)/\log\lambda$. The lower singular bound also controls growing physical-region support counts. Periodic carriers use quotient distance across their seams. This joins the mechanical and causal constructions; it supplies neither proper geometry nor convergence of their interacting states.

Write H_{reg} for the Wilson Laplacian energy. At fixed bare regulator length, isotropic strain $V(s) = e^s I$ gives

$$H(s) = e^{-s}(H_E + H_D + H_B) + e^{-2s} H_{\mathrm{reg}}, \qquad K_{\mathrm{iso}} = -H_E - H_D - H_B - 2H_{\mathrm{reg}}.$$

The Wilson term's extra coefficient is necessary. Along isolated driving, $d\langle H\rangle/dt = \langle \partial_t H\rangle$. For separately prepared Gibbs targets at fixed b, the same source instead gives $F' = \langle K\rangle$ and $S' = -b^2 \operatorname{Cov}(H, K)$. These different experiments are not identified by assuming thermalization during unitary evolution. The declared volume fixes the mechanical pressure $p = -\langle K_{\mathrm{iso}}\rangle/(3v)$.

The 208-state control shares exactly the thermal preparation's Hamiltonian, target, prepared state and clock. Its directional reduction sets the magnetic coefficient to zero and has $K_{\parallel} = H_E - H_D - 2H_{\mathrm{reg}}$ and $K_{\mathrm{iso}} = -H_E - H_D - 2H_{\mathrm{reg}}$. Each has norm at most 28, so effects $(I \pm K/28)/2$ and labels ± 28 give a complete mean-reading instrument. They do not assert that K has only two eigenvalues. The pressure error at declared $v = 1$ is at most $56\epsilon_{\mathrm{prep}}/3$. This control is an algebraic reduction with transverse classical links fixed; it is not a normalizable reducing subspace of the full three-dimensional quantum-link field.

Flat records do not determine the metric coupling: adding $\xi \log(\det V) I$ changes no flat Hamiltonian or fixed-flat record, but shifts strain by $\xi \operatorname{Tr}(S) I$ and pressure by $-\xi/v$. Declaring one complete family fixes this response without a uniqueness theorem. Identifying its coframe and stress with the observers' proper geometry, and supplying the same-state regional modular and area estimates, are additional Einstein premises. This finite mechanical pressure is not a cosmological equation of state.

11.17 Finite holonomy and shared event meaning

An explicit link-register extension on the same icosahedral incidence illustrates defect creation with work accounting. Give each edge a sign Z_e and each face the holonomy $B_f = \prod_{e \in \partial f} Z_e$. An edge flip toggles its two incident face signs. Complete reversible controls exchange a work quantum $2J$ for two negative faces, move one endpoint without changing the cost, and return the work upon annihilation. They preserve $H_{\mathrm{cost}} = J \sum_f (1 - B_f)/2 + 2Jn$ on every input. For a region R, internal edges cancel and $Q_R = \prod_{f \in R} B_f = \prod_{e \in \partial R} Z_e$. Updates with even intersection with its boundary preserve Q_R; if $Q_R = -1$, those updates cannot make the entire region flat. This protects regional parity under the specified operations. Gauge invariance alone supplies no permanent individual defect, and a sphere has no nontrivial flat global $\mathbb{Z}_2$ holonomy. The declared link extension is distinct from the native protected-memory program and is not assigned a physical species or mass.

An explicit coupled construction makes this protected parity operationally visible. For either actual regional Q_R, or the separate ordered-memory Q commuting with every native repair branch, declare

$$H_Q = \frac{3I - Q}{2} \otimes V, \qquad V = W(Y) + W(Y)^\dagger, \quad Y = \mathrm{diag}(1,2,3), \quad d = 0.$$

The full 307-state scalar sector and all native raw records are retained. At $t = \pi/(6\sqrt{56})$, a scalar vacuum gives occupied probabilities $1/4$ and $3/4$ in the positive and negative protected sectors. The geometric birth experiment has original joint no-pair/empty, no-pair/occupied, pair/empty and pair/occupied weights $(27, 9, 16, 48)/100$. Protected transport commutes with the coupled action. The strengths one and two are declared coefficients, not consequences of protection or a measured mass hierarchy.

Birth commutes with H_{cost} but not H_Q. Its scalar mean energy stays zero on the vacuum input while its second moment changes from G to $73G/25$. A separate two-time energy experiment records controller work $0, +\sqrt{G}, -\sqrt{G}$ with weights $9/25, 8/25, 8/25$. That initial energy measurement disturbs the scalar preparation and is not the occupation experiment above. The uncoupled defect work counter does not supply the whole coupled experiment's energy balance. Neither protected memory nor regional parity is identified with scalar occupation, spacetime spin or a physical particle species.

The observer endpoint is also an explicit finite calculation. A complete four-bin instrument writes a pointer; controlled permutations copy only its orthogonal record labels. Two observers use different local codes and dictionaries to compare event meanings. For a chosen phase-analyzer experiment the original probabilities are $(1/2, 1/4, 0, 1/4)$. A deliberately swapped dictionary causes disagreement with probability $3/4$; correcting it changes no detector probability. Failure records and untouched quantum references are retained. Compatible event functions on a finite cover glue to a common function, and deterministic interpretation cannot increase total variation between complete event laws. These are simulation-generated observables. Smooth spacetime and field equations are effective mathematical descriptions with their own domains and approximation bounds; the program generates no additional objective continuum state. Neither agreement nor finite gluing proves the interacting relativistic limit.

11.18 Energy conservation and the geometric premises

The cubic-covariant Wilson energy partition allocates onsite matter to its cell, half of each interior electric and bond term to each endpoint, and one quarter of each plaquette to each of its four vertices. Ghost Gauss constraints make dangling electric terms zero on the physical carrier. Thus $\sum_x h_x = H$ with the original action and energy offset retained, and $R h_x R^* = h_{Rx}$ under the actual proper cube rotations. The physical even operators $J_{xy} = i[h_x, h_y]$ are Hermitian and antisymmetric, and vanish for owner-cell distance greater than four. Their explicit caps are

$$\|h_x\| \le K_n = \frac{180 + (1827/2)\ell_n^2 + 3/7}{a_n}, \qquad \|J_{xy}\| \le 2K_n^2.$$

The reused finite continuity theorem gives $\partial_t h_x(t) + \sum_y J_{xy}(t) = 0$. Every regional internal pair cancels; integrated crossing currents supply the coarse-energy theorem's premise. Complete effects $(I \pm J/M)/2$, with $M \ge \|J\| > 0$, record labels $\pm M$ with unbiased mean $\langle J \rangle$. A zero current uses one zero label. Reference registers and complementary outcomes remain in the output. Endpoint and flux readouts use separate copies of the same preparation, so a sampled flux record does not assume an undisturbed sequentially measured state.

At fixed preparation, coefficient controls $H_f = \sum_x f_x(t) h_x$ obey $\partial_t \langle H_f \rangle = \sum_x \dot{f}_x \langle h_x \rangle$. They transform as scalar controls under the cube. Their identification with proper lapse or a metric stress tensor is a separate premise. The difference from the three-owner plaquette and owned-electric allocation is an exact nearest-edge divergence, preserving total energy and the source clock. Its norm need not vanish under refinement. Charge conjugation fixes these energies and their currents.

A covariant regional cut retains the matter in R and links with both endpoints in R, assigning crossing links to the complement. The actual cube action then factors across the cut, making the partial trace, supported modular operator and complete boundary records covariant. If b links cross the cut, the actual native representations give

$$S_{\text{extended}} = S_{\text{algebra}} + \sum_\lambda p_\lambda \log d_\lambda, \qquad 0 \le \sum_\lambda p_\lambda \log d_\lambda \le b\ell \log 6.$$

Recording the existing central label leaves the regional entropy unchanged, although it can disturb global coherences. On the two-site, one-interior-link cut, native Wilson characters give the positive lower bound $4c \log 108$, with the actual source floor $\rho \ge cP_G$. The whole contained action $H_{\text{int}}(R)$ belongs to this regional algebra; its difference from $\sum_{x \in R} h_x$ is a crossing operator of norm at most $b[609\ell^2/4 + 15 + 3/14]/a$. Neither this boundary comparison nor the representation entropy identifies a proper area, regional modular-energy kernel or finite Newton coefficient.

The same cut admits an exact source information identity. Let $\mathcal{D}\rho = \sum_\lambda \Pi_\lambda \rho \Pi_\lambda$ record the full boundary representation, $S_{\text{edge}} = \sum p_\lambda \log d_\lambda$ and $C_{\text{center}} = D(\rho \| \mathcal{D}\rho)$. With unweighted block trace in the regional observable algebra,

$$I_{\text{ambient}} = I_{\text{algebra}} + 2S_{\text{edge}} + C_{\text{center}}.$$

This is mutual information between the source region and its complement, not the entropy of their retained purification. Ten actual native or conjugate plaquette characters connect the physical empty-Fock, constant-link vector to distinct nontrivial boundary sectors, each with Hamiltonian matrix element $-\gamma$, $\gamma = 1/(210a)$. The source floor $c = 2^{-N}$ and strict slope imply

$$C_{\text{center}} \ge 10(c\beta_{\text{low}}\gamma)^2 > 0, \qquad W_{\text{record}} := \operatorname{Tr} H(\mathcal{D}\rho - \rho) \ge 20c\beta_{\text{low}}\gamma^2 > 0.$$

These follow from full-source commutator norms, without assuming that the eleven witness vectors span an invariant space. Pinching also gives

$$\beta_{\text{low}} W_{\text{record}} \le C_{\text{center}} + D(\mathcal{D}\rho \| \rho) \le \beta W_{\text{record}}.$$

The boundary apparatus therefore supplies positive source work despite leaving each regional marginal unchanged. All eleven populated sectors have probability at least c, giving $I_{\text{algebra}} \ge 10Nc \log 2$ and $I_{\text{ambient}} \ge 10Nc \log 2 + 8c \log 108 + 10(c\beta_{\text{low}}\gamma)^2$. These finite floors decay with capacity and establish no proper-area law.

An additional physical Schur-block comparison retains both original regional marginals, center weights and singlet pairing, replacing only multiplicity correlations by products. This nonlinear comparison state σ is not a preparation channel. Write $X = S(\sigma) - S(\rho) = J + C_{\text{center}}$, where J is the weighted within-sector mutual information. For the actual polynomial, $-\log \rho = \beta H + kI + R(H)$ with $0 \le R \le r$. Relative entropy and cancellation of wholly regional terms yield

$$X \le \beta \operatorname{Tr}[(\sigma - \rho)V_{\text{cut}}] + r \le 2\beta \|V_{\text{cut}}\| + r, \qquad \|V_{\text{cut}}\| \le \frac{30b + p_{\text{mix}}/7}{a}.$$

Only crossing hops and mixed plaquettes enter this interaction; electric terms belong wholly to one regional factor. With $p_{\mathrm{mix}} \le 2b$, the crossing-link representation bound gives

$$I_{\mathrm{ambient}} \le \ell b(\log 451 + 2\log 6) + \frac{2\beta(30b + p_{\mathrm{mix}}/7)}{a} + r.$$

A finite polynomial state can correlate noninteracting regions, so r cannot generally be dropped. The displayed area-normalized upper coefficient diverges along the trajectory; this neither proves actual entropy divergence nor supplies a proper-area or Einstein law.

Write $\rho \ge 2^{-N} P_G$ for the selected source floor. Its regional density obeys $\sigma_R \ge m_R P_R$, where $P_R = \operatorname{supp} \operatorname{Tr}_{\bar{R}} P_G$ and $m_R = 2^{-(N+b_R)}$, $b_R = (60 + 27\ell_n)|R|$. For a physical local unitary U_R and $\sigma_s = (1-s)\sigma_R + sU_R\sigma_R U_R^*$, put $X = U_R\sigma_R U_R^* - \sigma_R$. The actual supported modular operator $K_R = -\log \sigma_R$ gives

$$S(\sigma_s) - S(\sigma_R) = s \operatorname{Tr}(XK_R) - D(\sigma_s \| \sigma_R), \qquad 0 \le D \le s^2/m_R.$$

This is the source marginal after mathematically tracing its retained coin and reference. The coin-plus-region entropy is instead $H_2(s) + S(\sigma_R)$, with singular derivative at zero. Neither entropy is identified with proper area or Einstein generalized entropy, and the finite support floor is not uniform under refinement.

Actual controlled energy-current response. For the cube-covariant local energies $H = \sum_x h'_x$, set $A = h'_0$ and $B = i[H, A] = -\sum_y J'_{0y}$. A literal neutral source witness has six orthogonal one-hop components and $\|[H, A]\phi\|^2 \ge 27/(2a^4)$. The original polynomial slope therefore gives

$$q = \operatorname{Tr} \rho[A, [H, A]] \ge q_0 = 2^{-N} \beta_{\mathrm{low}} \frac{27}{2a^4} > 0.$$

With $R \ge \|H\|$, $K \ge \|A\|$, $C = 2RK$, $G = R + K$ and $\tau = q_0/(16G^2C)$, the finite Cayley pulse for $H + A$ followed by complete effects $(I \pm B/C)/2$ changes the original output law by at least $d_0 = \tau q_0/(4C) > 0$ in total variation. Its cyclic source work is at least $2d_0^2/\beta$. These floors are finite and not uniform under refinement. For the declared control $H_f = \sum_x f_x h'_x$, the actual Ward identity is

$$\frac{d}{dt}\langle f_x h'_x \rangle + \sum_y f_x f_y \langle J'_{xy} \rangle = \dot{f}_x \langle h'_x \rangle.$$

Subtracting the integrated applied work supplies the finite energy-Ward consumer's endpoint term. A source-relative Duhamel derivative follows on the same dynamics. Neither this driven identity nor that derivative identifies a metric coupling, proper stress tensor or Einstein response.

On the scalar source with $d = 1$, the preparation $(\Omega + ib)/\sqrt{2}$ has exact component exchange and a complete current detector with ranks $(1, 1, 305)$. Changing d at fixed state and clock gives $\partial_d p_{\mathrm{occupied}}(\pi/30) = 2\sqrt{14}/3375$. This is response to an action coefficient; geometric stress requires an independently specified metric coupling. The regional Einstein consumer additionally requires actual curvature, proper clocks, the same-state modular diamond kernel, fixed-proper-volume area response and controlled small-ball errors. Energy conservation alone supplies neither the missing stress components nor these geometric identities. The existing matched-regional theorem needs only ten future timelike probes, with sharp decoder norm 17/8. For a common proper radius ℓ, it bounds the component Einstein residual by $255(e_{\mathrm{bulk}} + \eta e_{\mathrm{area}} + e_{\mathrm{stationarity}})/(32\pi\eta\ell^4)$. Its tensor inversion is reused; the task is to realize the three regional estimates on one geometric state, with errors $o(\ell^4)$ and one positive spacetime-independent limiting η. The separate free-Weyl modular

and stress construction cannot supply those inputs for an interacting state without an identified map.

The actual regional restriction of the faithful $Q^* = 0$ scalar Gibbs state supplies a distinct finite modular calculation. One family's six modes have a 22-dimensional positive support, with spectral ranks $(1, 6, 2, 13)$ and 42 retained zero outcomes. Its modular Hamiltonian includes $\log(p_{\mathrm{dark}}/p_{\mathrm{pair}})B_f$, where B_f projects onto two pair states. For the first family the coefficient is $\log(13440/41873)$. A gauge-invariant pair/dark probe detects this induced interaction; no polynomial of local number reproduces it. The source-admitted tangent $B_f/2 - D_f/13$ has that entropy slope and an explicit relative-entropy remainder bound. The tracial regional conditional expectation exists but cannot preserve the interacting Gibbs state because it removes nonzero cross-center coherence. These are internal regions. Their modular operators are not a spatial diamond kernel, and global modular time does not restrict to their algebras as the same Hamiltonian evolution.

Reproduction and mathematical scope

Every statement in this paper has a public executable certificate, proof text or formal module in the project repository [1], including exact characteristic polynomials, rational and integer enclosures, complete line tables, adversarial controls and the source-pinned multiloop replay. They distinguish exact algebra and rational interval proofs from floating spectral checks and empirical comparisons. A source hash identifies a calculation; its mathematical scope is the stated theorem and domain.

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AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.